



Tekscan

I-Scan[®] & High Speed I-Scan[®] User Manual

Tactile force & pressure measurement system



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v. 6.0x

Tactile force & pressure measurement system

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WELCOME

ISO

Tekscan is registered to the following standard(s):

- ISO 9001: 2000
- ISO 13485: 2003

DECLARATION OF CONFORMITY

The Tekscan ***I-Scan*** System has been tested and conforms to the following standards for a 'Tactile Sensor System':



Europe

EN55011, EN50082-1, IEC801-2, IEC801-3, IEC801-4, IEC801-5

USB Handle Conformity

Class B equipment

	US	Canada	Europe	World
Safety	UL61010B-1	CSA61010	N/A	N/A
EMC	N/A (FCC Part 15.103)		EN55011: Emissions EN61000-6-2: Immunity Or EN61326: Immunity	CB

Test Standards/Methods:

BS EN61326:1998	- EMC for Laboratory Test and Measurement Equipment
BS EN60601-1-2:2002	- EMC for Medical Electrical Equipment
CFR 47, Part 15	- Emissions for Unintentional Radiators (15.107, 15.109)
EN61000-4-2:1995	- Electrostatic Discharge Immunity
EN61000-4-3:1996	- Radiated Electromagnetic Field Immunity
EN61000-4-4:1995	- Electrical Fast Transient Immunity
EN61000-4-5:1995	- Surge Immunity
EN61000-4-6:1996	- Conducted Disturbances RF Immunity
EN61000-4-8:1993	- Power Frequency Magnetic Immunity
EN61000-4-11:1994	- Voltage Dips and Dropouts
EN61000-3-2:2000	- Harmonic Current Emissions
EN61000-3-3:1995 w/corr. July 1997 - Voltage Fluctuations and Flicker	

Test Parameters:

BS EN 61326:1998	- Class B Emissions Limits (Default to Class A)
EN55011:1998	- Group 1, Class B (Default to Class A)
CFR 47, Part 15	- Class B (Default to Class A)
EN61000-4-2:1995	- ±6kV Contact, ±8kV Air
EN61000-4-3:1996	- 3V/m, 80MHz to 2.5GHz, w/ 80% 1kHz AM
EN61000-4-4:1995	- ±2kV Mains, ±1kV I/Os
EN61000-4-5:1995	- ±2kV Common Mode, ±1kV Differential Mode
EN61000-4-6:1996	- 3Vrms, 150kHz to 80MHz, w/ 80% 1kHz AM
EN61000-4-8:1993	- 50Hz and 60Hz
EN61000-4-11:1994	- <5% 0.5 Cycles, 40% 5 Cycles, 70% 25 Cycles, <5% 5 Sec.
EN61000-3-2:2000	- A14 Class A Harmonic Emissions Limits
EN61000-3-3:1995 w/corr. July 1997	- Voltage Fluctuation and Flicker Limits

**Warnings (Industrial):**

1. The use of accessories and cables other than those specified by the manufacturer as replacement parts may result in increased emissions or decreased immunity of the equipment or system.
2. Only use Tekscan supplied battery packs and power sources to avoid damaging the system.
3. Do not use or attach any components that are not explicitly stated within this manual.
4. Do not connect any additional multiple portable socket outlet(s) or extension cord(s) to the system.
5. EMC (Electro-Magnetic Charge) can interfere with the system. If this occurs, or if there is a high level of noise on your display screen, try moving to a location that is not in proximity to other electrical devices (such as Televisions, radios, and cell phones).
6. ESD (Electro-Static Discharge) can halt the system. If the system stops functioning, shut down the system by turning the power switches on all attached parts off. Also shut down the software. Then turn on the system and restart the software. If problem persists, make sure the humidity in the room is >30%. If you are still having difficulty in operating the system, contact your local Tekscan representative.
7. Dispose of applied parts in accordance with Federal and State guidelines pertaining to computer equipment.
8. Sensor Replacement/Disposal: Dispose of sensors in any waste container. Sensors are not biohazardous waste.
9. Protection against electric shock: Internally powered equipment.
10. If using parts other than those explicitly stated within this manual, always follow the manufacturer's cleaning instructions.
11. Do not disinfect the computer.
12. No user-serviceable parts. Do not try to service or take apart any Tekscan hardware. Consult with your Tekscan representative if a component is not working correctly, or is not working as it should.

Guidance Tables

Table 206 – Recommended Separation Distances between portable and mobile RF Communications equipment and the EH-1
Equipment and Systems that are NOT Life-supporting

Recommended Separations Distances for the EH-1			
The EH-1 is intended for use in the electromagnetic environment in which radiated disturbances are controlled. The customer or user of the EH-1 can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF Communications Equipment and the EH-1 as recommended below, according to the maximum output power of the communications equipment.			
Max Output Power (Watts)	Separation (m) 150kHz to 80MHz D=(3.5/V1) (Sqrt P)	Separation (m) 80 to 800MHz D=(3.5/E1) (Sqrt P)	Separation (m) 800MHz to 2.5GHz D=(7/E1) (Sqrt P)
0.01	0.11	0.11	0.27
0.1	0.36	0.36	0.86
1	1.16	1.16	2.72
10	3.68	3.68	8.60
100	11.66	11.66	27.22

Table 201 – Guidance and Manufacturer's Declaration – Emissions
All Equipment and Systems

Guidance and Manufacturer's Declaration - Emissions
The EH-1 is intended for use in the electromagnetic environment specified below. The customer or user of the EH-1 should ensure that it is used in such an environment.

Emissions Test	Compliance	Electromagnetic Environment – Guidance
RF Emissions CISPR 11	Group 1	The EH-1 uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. The EH-1 is suitable for use in all establishments, including domestic, and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
RF Emissions CISPR 11	Class B	
Harmonics IEC 61000-3-2	Class A	
Flicker IEC 61000-3-3	Complies	

**Table 203 – Guidance and Manufacturer’s Declaration – Emissions
Life-Supporting Equipment and Systems**

Guidance and Manufacturer’s Declaration – Emissions
The EH-1 is intended for use in the electromagnetic environment specified below. The customer or user of the EH-1 should ensure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
Conducted RF IEC 61000-4-6 Radiated RF IEC 61000-4-3	3 Vrms (Outside ISM) 10Vrms (In ISM Bands) 150 kHz to 80 MHz 10 V/m 80 MHz to 2.5 GHz	(V1)Vrms (V2)Vrms (E1)V/m	Portable and mobile communications equipment should be separated from the EH-1 by no less than the distances calculated/listed below: $D = (3.5/V1) (\text{Sqrt } P)$ $D = (12/V2) (\text{Sqrt } P)$ $D = (3.5/E1) (\text{Sqrt } P)$ 80 to 800 MHz $D = (7/E1) (\text{Sqrt } P)$ 800 MHz to 2.5 GHz where P is the max power in watts and D is the recommended separation distance in meters. Field strengths from fixed transmitters, as determined by an electromagnetic site survey, should be less than the compliance levels (V1 and E1). Interference may occur in the vicinity of equipment containing a transmitter.

**Table 204 – Guidance and Manufacturer’s Declaration – Emissions
Equipment and Systems that are NOT Life-supporting**

Guidance and Manufacturer’s Declaration – Emissions	
The EH-1 is intended for use in the electromagnetic environment specified below. The customer or user of the EH-1 should ensure that it is used in such an environment.	

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	(3)Vrms	Portable and mobile communications equipment should be separated from the EH-1 by no less than the distances calculated/listed below: $D = (1.16) (\text{Sqrt } P)$ $D = (1.16) (\text{Sqrt } P)$ 80 to 800 MHz $D = (2.33) (\text{Sqrt } P)$ 800 MHz to 2.5 GHz where P is the max power in watts and D is the recommended separation distance in meters. Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey, should be less than the compliance levels in each frequency range. Interference may occur in the vicinity of equipment containing a transmitter.
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2.5 GHz	(3)V/m	

WARRANTY INFORMATION

Tekscan, Inc. Limited 1-Year Warranty

1. **WARRANTY.** Tekscan, Inc. warrants to the original purchaser of this product that should it prove defective by reason of improper workmanship and/or materials:
 - A. **Tekscan Systems and Components:**

For one year from the date of original purchase at retail, Tekscan will repair or replace, at our option, any defective part without charge for the part or labor if an inspection proves the claim. Parts used for replacement may be used or rebuilt, and are warranted for the remainder of the original warranty period.
 - B. **Tekscan Sensors:**

Tekscan will replace any Tekscan Sensor which fails due to manufacturing defect if an inspection proves the claim. Claims must be made within 30 days of purchase.
2. TO OBTAIN WARRANTY SERVICE, call Tekscan at 1-800-248-3669, (617) 464-4500 in MA, for further instructions. Should you be asked to deliver your product to Tekscan, Inc. in Boston, MA, shipping expenses are the purchaser's responsibility. Proof of purchase is required when requesting warranty service.
3. THIS WARRANTY DOES NOT COVER defects caused by modification, alteration, repair or service of the enclosed product by anyone other than Tekscan or an authorized Tekscan service center, physical abuse to, misuse of, the product or operation thereof in a manner contrary to the accompanying instructions, or shipment of the product to Tekscan or an authorized Tekscan service center for service. This warranty also excludes all costs arising from installation, cleaning or adjustments of user controls. Consult the operating manual for information regarding user controls.
4. ANY EXPRESS WARRANTY NOT PROVIDED HEREIN, AND ANY REMEDY FOR BREACH OF CONTRACT WHICH, BUT FOR THIS PROVISION MIGHT ARISE BY IMPLICATION OR OPERATION OF LAW, IS HEREBY EXCLUDED AND DISCLAIMED. THE IMPLIED WARRANTIES FOR THE MERCHANTABILITY AND OF FITNESS FOR ANY PARTICULAR PURPOSE ARE EXPRESSLY LIMITED TO A TERM OF ONE YEAR. SOME STATES DO NOT ALLOW LIMITATIONS ON HOW LONG AN IMPLIED WARRANTY LASTS, SO THAT THE ABOVE LIMITATION OR EXCLUSION MAY NOT APPLY TO YOU. THE WARRANTIES SET FORTH HEREIN ARE IN LIEU OF ANY AND ALL OTHER WARRANTIES EXPRESS OR IMPLIED INCLUDING THE WARRANTY OF MERCHANTABILITY AND FITNESS. THE BUYER ACKNOWLEDGES THAT NO OTHER REPRESENTATIONS WERE MADE TO HIM OR RELIED UPON BY HIM WITH RESPECT TO THE QUALITY AND FUNCTION OF THE GOODS SOLD HEREIN. NO PERSON, FIRM OR CORPORATION IS AUTHORIZED TO ASSUME FOR US ANY LIABILITY IN CONNECTION WITH THE SALE OF THESE GOODS.
5. UNDER NO CIRCUMSTANCES shall Tekscan, Inc. be liable to purchaser or any other person for any special or consequential damages, whether arising out of breach of warranty, breach of contract, or otherwise. Some states do not allow the exclusion or limitation of incidental or consequential damages, so that the above limitation or exclusion may not apply to you.

08/11/03 – FORM-200-057-B

GETTING ASSISTANCE

Tekscan, Inc. will provide technical assistance for any difficulties you may experience using your *I-Scan* system for 90 days from the system shipping date. After 90 days, Tekscan offers annual Technical Support and System Maintenance Plans or customer support at our standard rates per incident. An incident is defined as one single issue or problem.

Contact Tekscan for additional *I-Scan* sensors. Standard sensors are available in a variety of pressure ranges. Contact us for current pricing and availability.

The flexible manufacturing process used to make sensors allows us to design custom sensors for applications in which standard sensors are not suitable. Custom pressure-sensitive materials can be formulated to produce a sensor whose sensitivity is well matched to a particular application. Contact your Tekscan representative to discuss custom sensors for your special applications.

Write, call or fax us with any concerns or questions. Our knowledgeable support staff will be happy to help you. Comments and suggestions are always welcome.

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Microsoft Windows and **MS-DOS** are registered trademarks of Microsoft Corporation.

INTRODUCTION

The Tekscan Industrial Sensing (***I-Scan***) System is a complete package, which converts an IBM-compatible PC into an advanced pressure distribution measurement system. Using patented Tekscan sensors in your normal operational environment, the system can sample pressure data as it happens (in *real-time*), present the information as a color-coded real-time display, and record the information (as a 'movie') for later review and analysis. The system is so versatile, you can copy pressure data and paste it into other applications, save it as a text (ASCII) file and import it into other programs, or print it out to any Windows-compatible color or grayscale printer.

The ***I-Scan*** system is comprised of the Microsoft (MS) Windows™ -based ***I-Scan*** software, the associated data acquisition hardware, and a sensor(s).

Tekscan sensors use a resistive-based technology. The application of a pressure to an active sensor results in a change in the resistance of the sensing element in inverse proportion to the pressure applied. After a simple calibration is performed, this force can be displayed on the screen in the measurement units that you choose, such as PSI or mmHg.

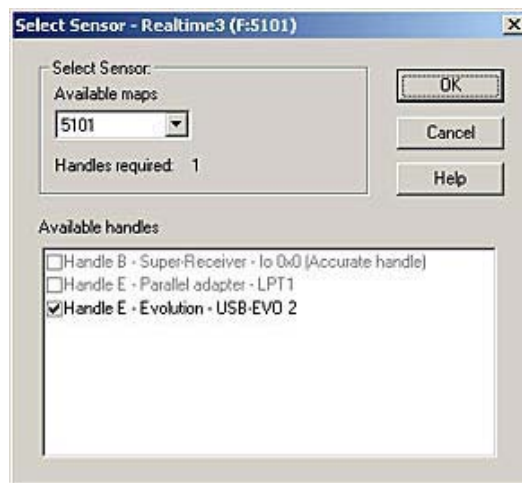
This document provides a thorough description of all of the system's capabilities. Follow the "Quick Start" section as a guide, and refer to specific sections for more detailed instructions on how to use each feature.

QUICK START

This section is a quick look at how to use your ***I-Scan*** system. This **Quick Start** procedure should be followed as a general outline; it will give you the basics on how to view sensor data in a real-time window, record this data, play the recording back, and analyze the data. However, you must read the entire manual before designing your application. *Familiarity with MS Windows is assumed.*

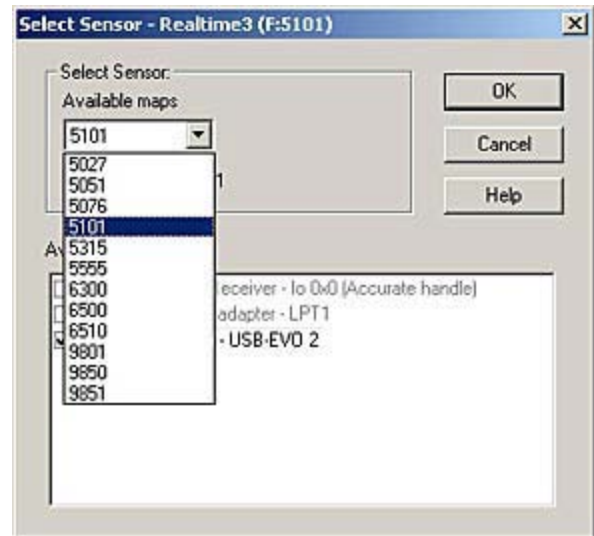
Note: *This procedure assumes that the I-Scan software has been successfully installed on your system.*

1. Make sure the sensor(s) is inserted correctly into the handle(s). Click on the **Start** button at the bottom left of the screen, select **Programs**, then double-click the ***I-Scan*** icon to run the program.
2. The **Select Sensor** dialog box will appear. The currently available maps and handles will be displayed. Select (highlight) the correct map number for your sensor under '**Available maps**'.



Under '**Available handles**', a list of all handles currently attached to the system will be displayed. Select the correct handle(s) for the sensor you are using (a check mark will be placed next to your selection). The type of hardware configuration is listed next to the handle information. Your options are *Evolution*, *USB*, *Super Receiver*, or *Parallel adapter*, depending on your configuration.

Note: Only one sensor map may be selected for each real-time window, but some maps require multiple handles. The number of handles required for the selected map is shown in the dialog box.



Once the sensor map and handle(s) have been selected once, the software will set them as defaults, and the dialog box will not be displayed at start-up. These options can later be changed in the **Options** pull-down menu, under **Select Sensor**.



3. Click <OK>. A new real-time window appears.

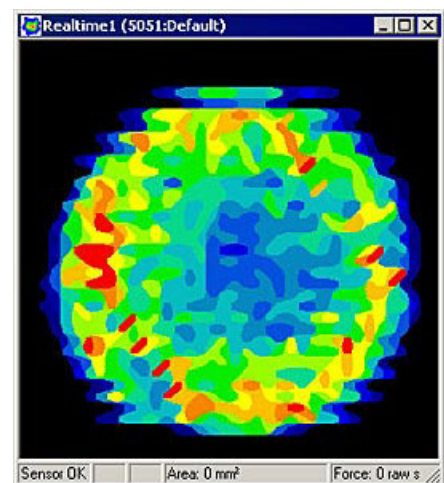
The status bar at the bottom left of the real-time window should say '**Sensor OK**'. If it says '**MISALIGNED!**' remove and reinsert the sensor into the handle. The size of this window will vary, since MS Windows cascades new windows. If the window is too small to view comfortably, enlarge it by dragging an edge of the window with the cursor.

Note: In order to get the largest display possible, the software may rotate a window (90° clockwise) that has a large aspect ratio. In this case, the origin (0,0) will be in the upper right hand corner of the rotated window. If you get confused at any point, put the cursor over a sensel and look at the status bar to get the coordinates.

4. Apply a load to the sensor. The forces on the sensor will be displayed in the real-time window as color-coded pressure information.

The *I-Scan* system enables you to present the pressure data in many different display modes. Utilize one or more of the following View options to display the data as desired:

- 2-D (default)
- 2-D wireframe
- 3-D reverse
- 2-D contours
- 3-D contours

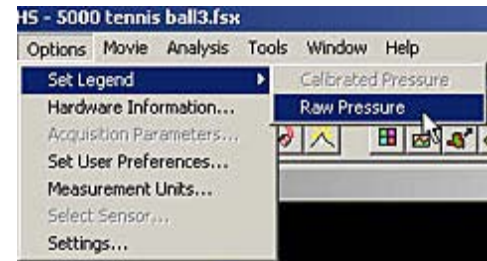


You may also select one of the following View options to help analyze the real-time data:

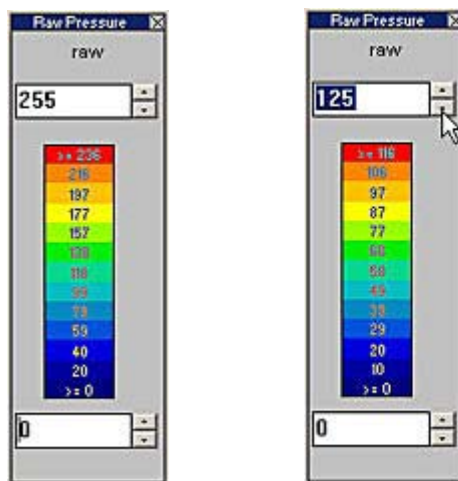
- Averaging
- Fixed Area Averaging
- COF (Center of Force)
- Movie Averaging
- Movie Contact Averaging
- COF Trajectory
- Max Area Frame
- Peak/Phase

Note: Refer to the 'Main Menu' section for detailed descriptions of each View option.

5. Select **Set Legend** from the **Options** pull-down menu, and then select **Raw**. This will place a raw **legend** (pressure scale) in the Main Window for the Real-time window. The legend will show the pressure range that corresponds to each of the 13 possible colors on the screen. Before the sensor has been calibrated, the pressure readings in the window are relative, not absolute. Therefore, the pressure units displayed at the top of the legend will be Raw.



Click on the up or down arrows next to the top value of the pressure range on the Legend, if necessary, to adjust the colors displayed on the screen.



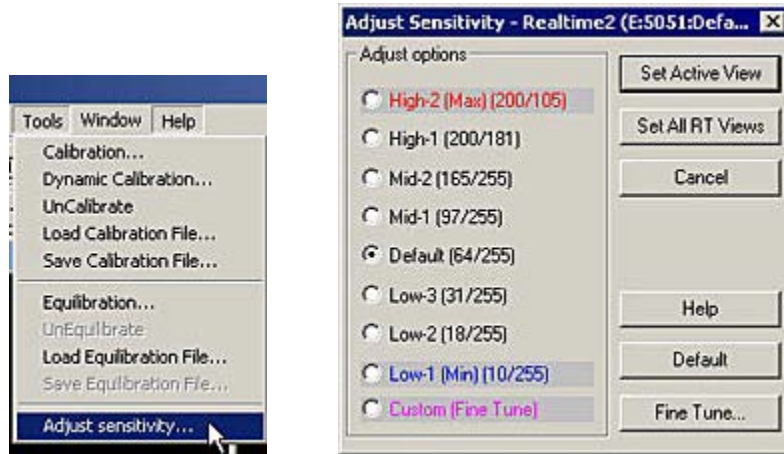
There should be a good variation of colors, with orange and red occasionally appearing in the display in only small areas. If too much red appears, the upper limit needs to be raised. If there is not enough variation in color, the upper limit should be lowered. Experiment to find the optimal settings.

6. **Apply the actual test load to the sensor. Important!** Follow all of the guidelines and warnings in the "Calibration and Equilibration" section.

Place the cursor over a loaded sensel (colored area) in the real-time window. Note that the cursor coordinates ('Row, Col') and the pressure at that point ('Load') are shown on the Main Status Bar. Verify that this point falls into the correct pressure range on the Legend (with the Legend's lower limit set to zero).

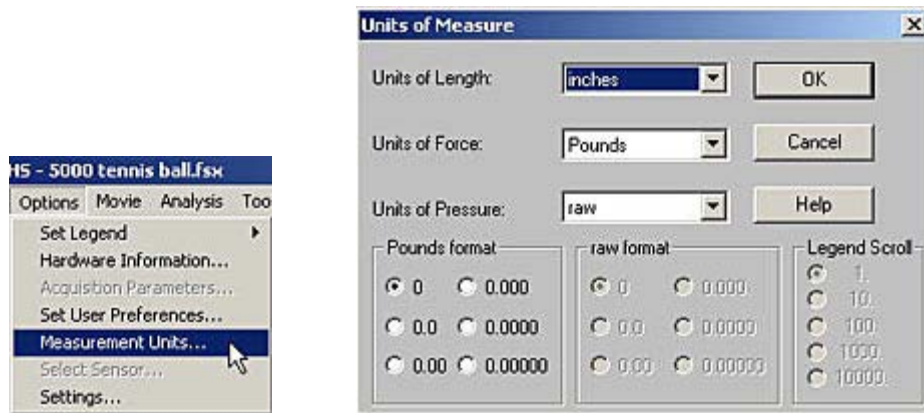
The applied **Force** and contact **Area** are displayed in the Real-time Status Bar. The **force** is displayed as a *Raw Sum* until a calibration is done, and then it is displayed in the desired units.

7. **Manually adjust the sensor handle *sensitivity* (adjustable gain).** This is done by selecting **Adjust Sensitivity** from the **Tools** pull-down menu, then choosing one of the Low, Mid, or Hi settings (or default). The sensitivity adjusts the usable force range of the sensor. As an optional feature, you may also “fine tune” your sensitivity setting by entering actual values.



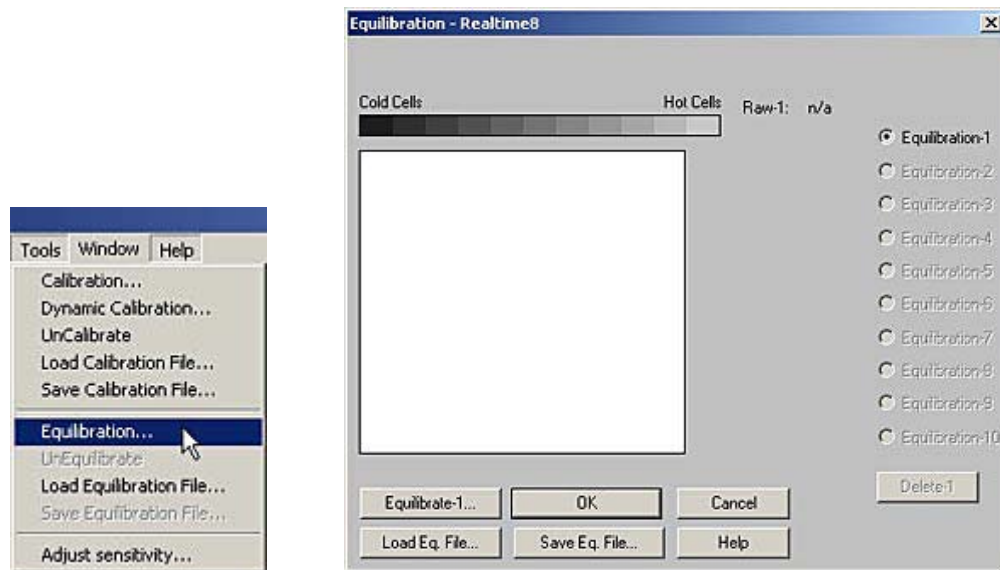
Refer to the “Calibration and Equilibration” section for more information on adjusting the sensitivity. *The sensitivity adjustment is very important in ensuring that your data is accurate, and must be performed before the equilibration/calibration.*

8. **Set the desired measurement units, and the number of decimal places for force and pressure units.** Select **Measurement Units** from the **Options** pull-down menu (or click on the icon in the Toolbar), then select the desired units of length, force, and pressure in the dialog box. You should select the units that you intend to use in the calibration of the sensor.



The ‘**Measurement Unit**’ dialog box may also be accessed from the “Calibration” dialog box, or by clicking the right mouse button with the cursor over the real-time window, then clicking **Units**. The selected units will then be saved when a new real-time window is opened or when a movie is saved.

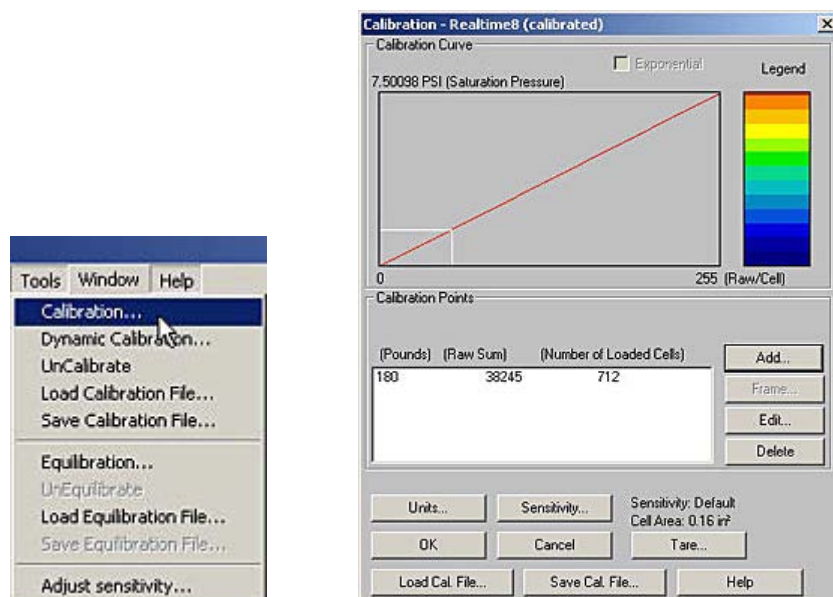
9. **Equilibrate the sensor.** Equilibration is optional, but strongly recommended for accurate results. Load the sensor (uniformly), and then perform the equilibration procedure, as described in the “Calibration and Equilibration” section. To access this option, select **Equilibration** from the **Tools** pull-down menu.



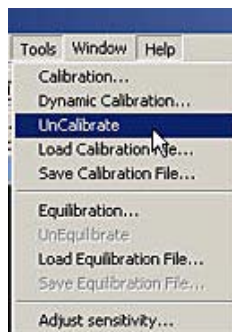
Once the equilibration is complete, the display will show any adjustments that were necessary for each sensel. If you make any mistakes, select **UnEquilibrate** from the **Tools** pull-down menu to remove the equilibration data.

Important! Your data will be inaccurate if the sensor is not loaded properly during equilibration.

10. **Calibrate the sensor.** Load the sensor with the test weight, and then perform the calibration, as described in the “Calibration and Equilibration” section. To access this option, select **Calibration** from the **Tools** pull-down menu, and then click on the **Add** button to add a new calibration point.



If you make any mistakes, select **UnCalibrate** from the **Tools** pull-down menu to remove the calibration data.



You can also choose to perform a calibration on a Movie window, after a recording has been taken, using the 'Movie (Frame) Calibration' feature. When a successful calibration is done, a calibrated Legend, with the selected pressure units, will appear on the screen, and the Force displayed in the real-time Status Bar will show the selected force units.

Important! Your data will be inaccurate if the sensor is not loaded properly during calibration.

11. **Record a movie.** Select **Record** from the **Movie** pull-down menu (or select the corresponding icon from the Toolbar) to begin recording the real-time window data.



If you want to record only a single frame of data, select **Snapshot** from the **Movie** pull-down menu, or click its icon in the Toolbar. Once you have completed recording, you can select **Append** from the **Movie** menu to add additional frames of data to the end of the movie.



The sensor handle may also be used to start a recording. Pressing the long blue button on the top of the sensor handle does this. This function (known as 'Remote Recording') may not be used to stop a recording.

Note: *Remote recording control is an option in the I-Scan® system. It may not be available on your system.*

Refer to the "Taking a Recording" section to learn how to change the recording (data acquisition) parameters, such as the number of frames to record and the duration of the recording, using the Options pull-down menu. The default recording consists of 100 frames.

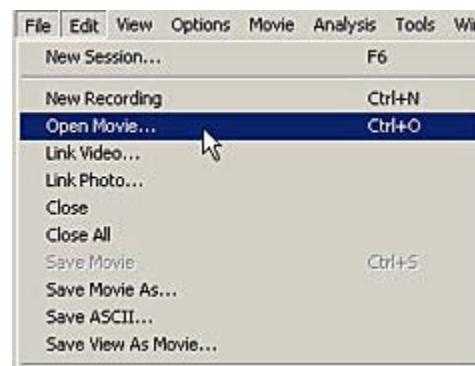
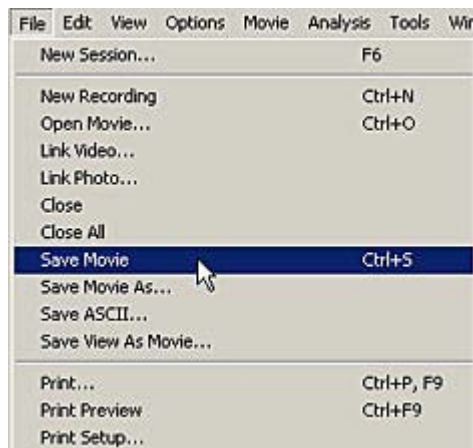
Note: *It is recommended that the sensor be Equilibrated and/or Calibrated before an actual recording is made. They will help to ensure that your recorded data is accurate.*

12. **Review the movie.** Play back your recording by selecting **Play Forward** from the **Movie** pull-down menu, or by clicking on the **Play Forward** icon in the Toolbar. Select any of the other playback options in the Movie pull-down menu, or click on their icons, to play the movie forward or backward, move one frame forward or backward, move to the first or last frame, or stop the movie. You can also press the <SHIFT> key at the same time as **Play Forward** or **Play Backward**, to play the movie forward or backward in a continuous loop. These options are described in more detail in the "Taking a Recording" section.

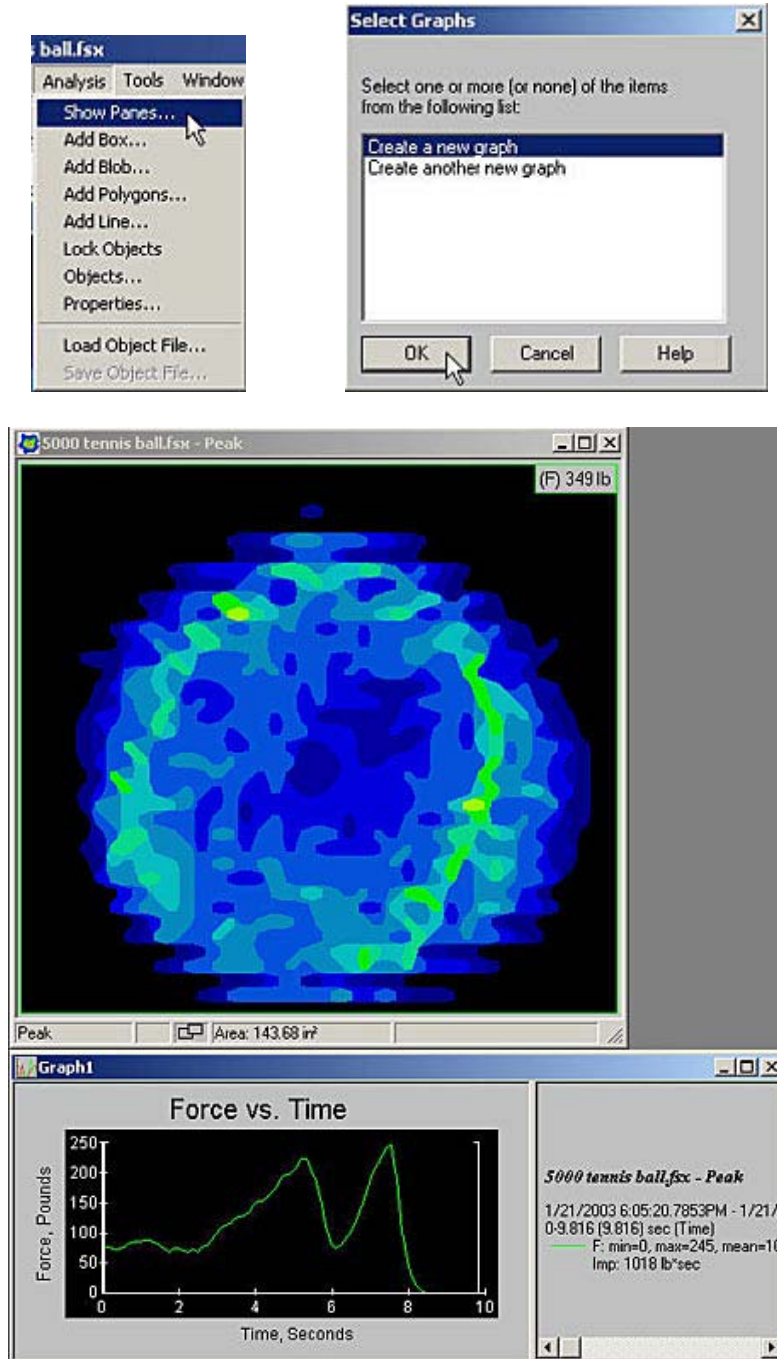
- COF Trajectory
- COF
- Max Area Frame
- Averaging
- Peak/Phase
- Fixed Area Averaging
- Movie Averaging
- Movie Contact Averaging

Note: *Refer to the "View Menu" section for detailed descriptions of each of these View options.*

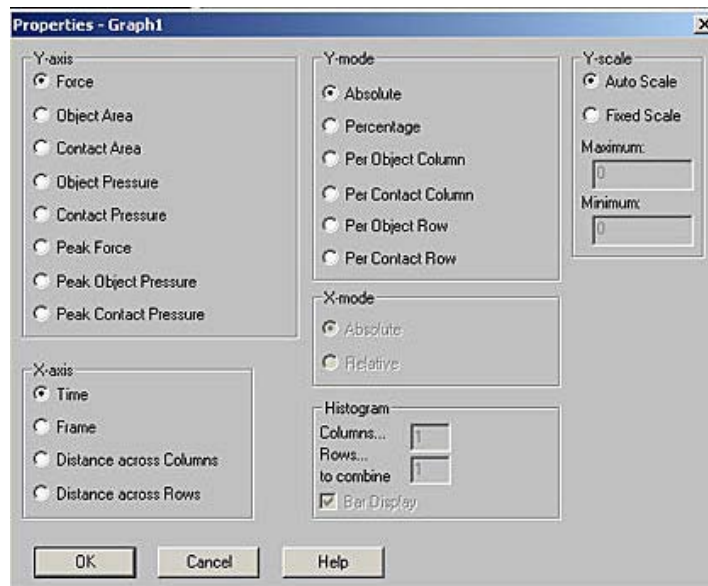
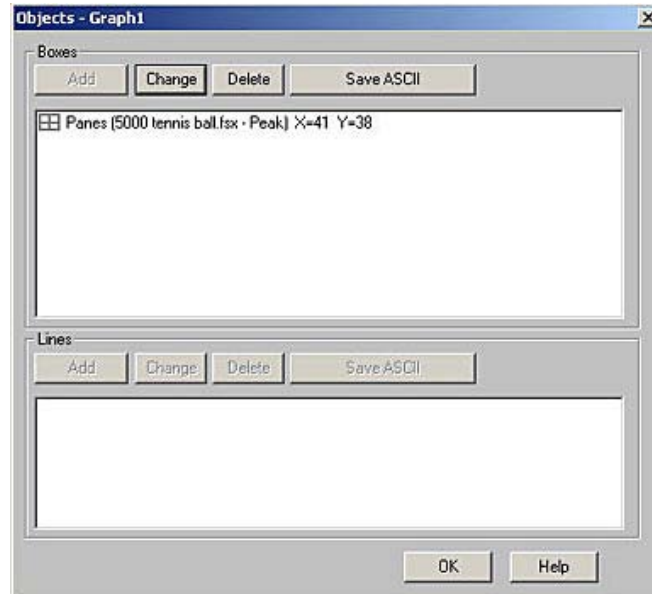
13. **Save the movie.** To save your movie as a file that can be opened later, select **Save Movie** from the **File** pull-down menu. When the 'Save As' dialog box is displayed, enter the desired file name (default is 'Movie.fsx') and destination (path) where you want to save your recording. Recordings are saved as **I-Scan** Movie files (extension *.fsx). Movies can be opened at a later time by selecting **Open Movie** from the **File** pull-down menu.



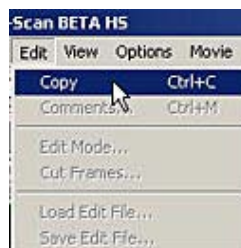
14. Edit the movie. The *I-Scan* software provides you with the tools necessary to remove or average cells (or groups of cells), or to remove specific movie frames, if desired. Refer to the “Editing a Movie” section for more information.
15. **Analyze (and graph) specific data in the movie.** As an example, select **Show Panes...** from the **Analysis** pull-down menu. When the ‘**Select Graphs**’ dialog box appears, with ‘**Create a new graph**’ highlighted, click <OK>. A colored box is placed in the movie window, and ‘**Graph 1**’ is opened. This graph will contain a colored trace that represents the data inside the box. In this way, the data inside the colored box can be studied separately from the rest of the window data.



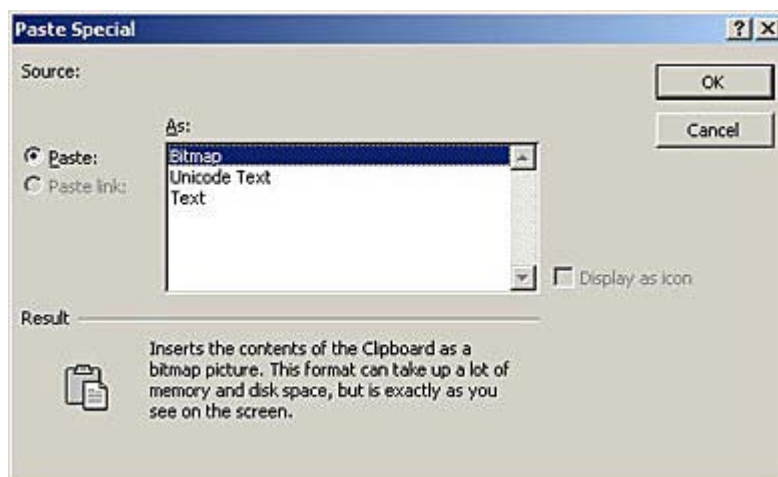
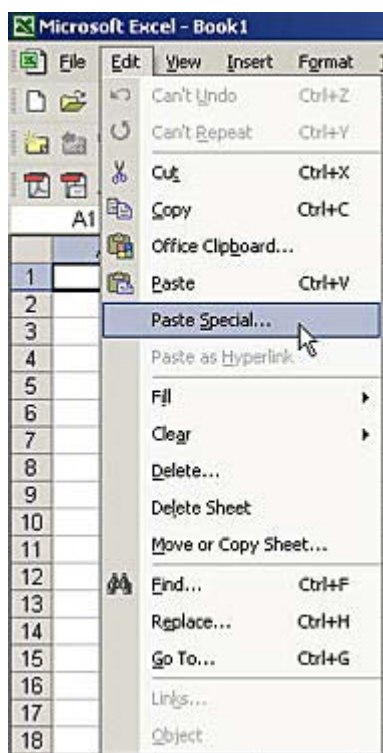
The **Objects** and **Properties** options, in the **Analysis** pull-down menu, are used to change which data is displayed in each object and graph. Refer to the “Analyzing Pressure Data” section for more information.



16. **Copy the pressure data from a Movie, Real-time, or Graph window, or a Legend.** Select **Copy** from the **Edit** pull-down menu (or click on the corresponding icon in the Toolbar), or click the right mouse button while the cursor is above a window (or the Legend), and select **Copy**.

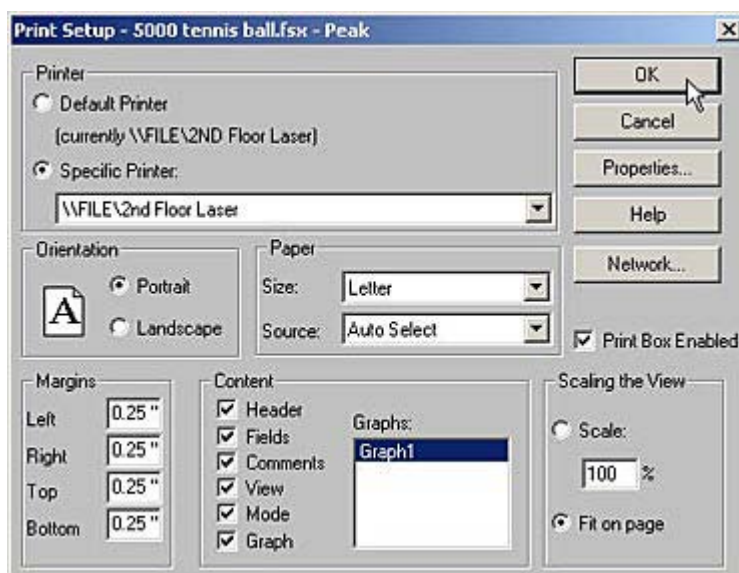


The active window pressure data will be copied to the MS Windows clipboard as both text (ASCII) and a graphic (bitmap), and will be available to be pasted into other Windows applications.



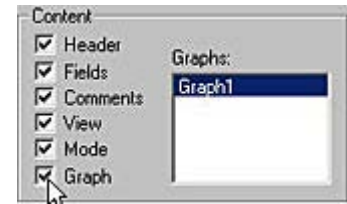
If a box ('Analysis' menu item) is active, and Copy is selected, only the pressure data inside that box will be copied to the clipboard. Refer to the description of the "Edit" menu in the 'Main Menu' section for more details.

17. **Print the active window.** Select **Print Setup** from the **File** pull-down menu, select all your desired printing options, and then click <OK>.



Then select **Print** from the **File** pull-down menu, and click <OK>. *I-Scan* gives you many printing options, including printing graphs on the same page as a movie or real-time window, and printing in grayscale.

Note: *A Graph window can only be printed by selecting 'Graph' in the Print Setup options.*



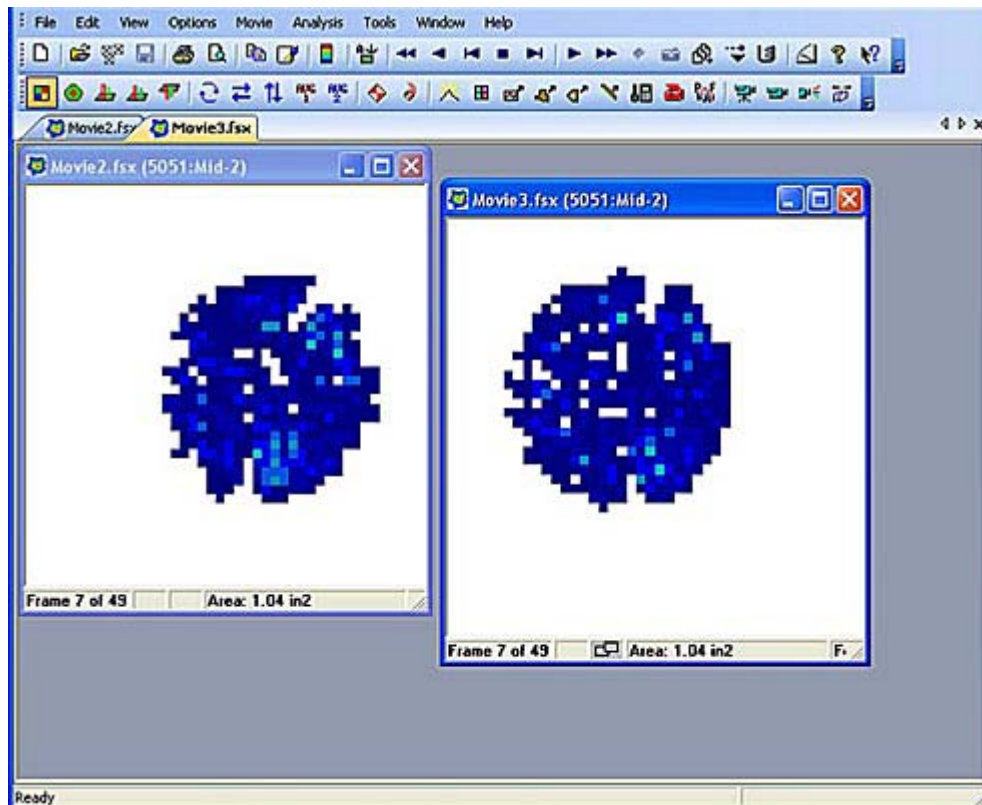
You have now completed the Quick Start section. You should now be familiar with *I-Scan*, and see how easy it is to view, save, and analyze the pressures on the sensor. The rest of the manual will give you detailed instructions on how to get accurate and meaningful data, and how to best use it for your specific application.

NEW GRAPHICAL USER INTERFACE (GUI)

The *I-Scan* software has a new Graphical User Interface. The functionality of the software, as well as the location of the menu commands and toolbar icons have not changed. This is a purely cosmetic change to enhance the look and feel of our software. The few software differences are noted below.

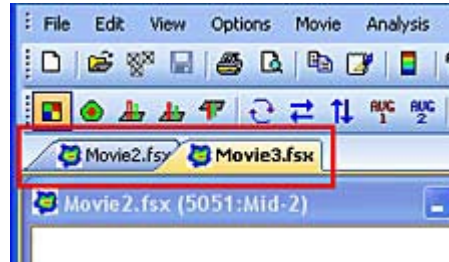
Note: *The documentation (both user manual and help file) may not yet be updated with this new Graphical User Interface. As a result, the images within the documentation may not always correspond to what you see on-screen. We are working toward updating all imagery and hope to have the new look integrated into our documentation soon.*

The following displays the Main window with the New Graphical User Interface:

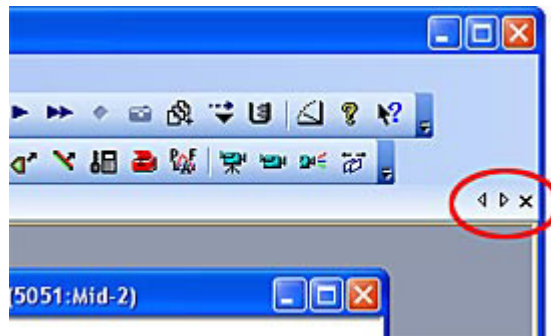


Tab Bar

Directly under the Toolbars is a new **Tab Bar**. Each separate window (Real-Time, Movie, or Graph) has its own associated tab. When a window has focus (its Title bar is highlighted), the tab moves to the front of the tab stack and the tab's font is boldened. You can make a window active (and give it focus) by clicking the Window's Tab. In addition, each tab will have its own color. Note that the **Farm Tread 7100.fsx** movie is located at the top of the layer stack in the image above and below.



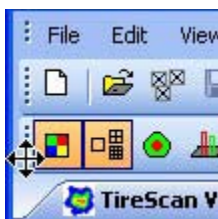
On the right side of the Tab Bar, there are three icons. The left and right arrows are used in situations where the tabs cannot fit in the width of space provided along the Tab Bar. If you have more tabs than can be accommodated along this Tab Bar, use the arrows to move the Tab Bar left or right and scroll through the tabs. The "x" icon is used to close the currently active movie (the movie that is highlighted and has focus).



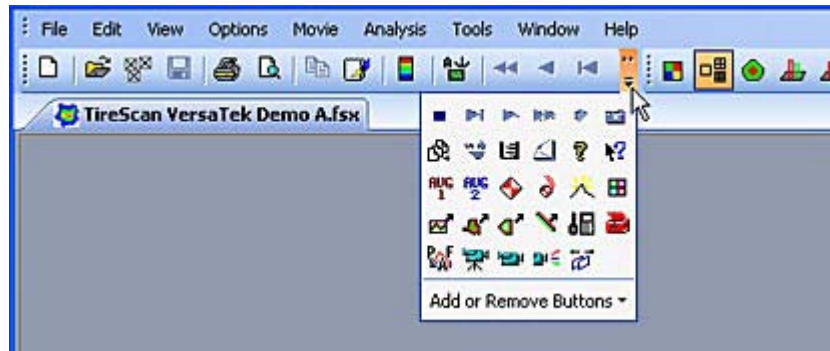
Toolbars

The upper (Main) Toolbar and lower (Advanced) Toolbar are still "docked" to the top side of the application window above the Tab Bar and below the Menu Bar by default.

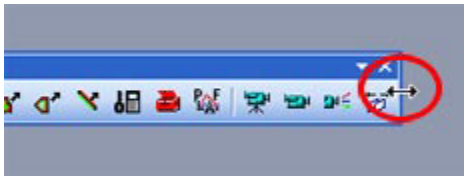
The Main and Advanced Toolbars can still be moved independently of each other. For example, if you want to have a single toolbar, grab the bottom toolbar by the dotted line on the left side. Note that the cursor turns into a four-way arrow (shown below left). Then move it to the right of the toolbar above (shown below right).



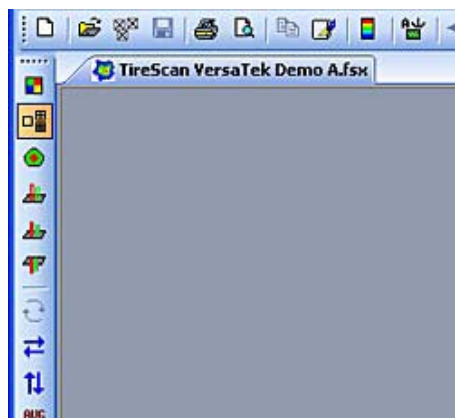
If, as in the image above left, you do not have enough space to display all the toolbar icons (due to the width of the application window being too narrow), a small drop-down arrow is displayed to the right side of the toolbar. Click this arrow and the remaining icons open on-screen (shown below). You can then select the icon of your choice.



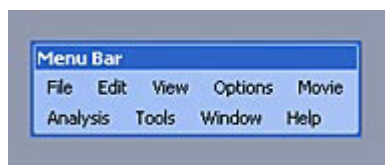
To “undock” a Toolbar, move it away from the top side of the application window towards the center of the application window. The Toolbar will gain a Title Bar, and can then be moved around the screen by grabbing this Title Bar. You can also turn the single row Toolbar into a double-row Toolbar. To do this, hover your mouse over the left or right side of the Toolbar. When you see the cursor turn into a two-way arrow (shown below left), click and drag the side inward until it becomes two rows of icons (shown below right). To turn the Toolbar into a single row of icons again, reverse this process, pulling the side of the Toolbar outward until it becomes a single row.



To dock the Toolbar to a different side of the window, move it close to the side you want until the Toolbar “snaps” into place (shown below).



Note: The Menu Bar can also be moved, sized, docked, and undocked from the application in the same way as the Toolbars.

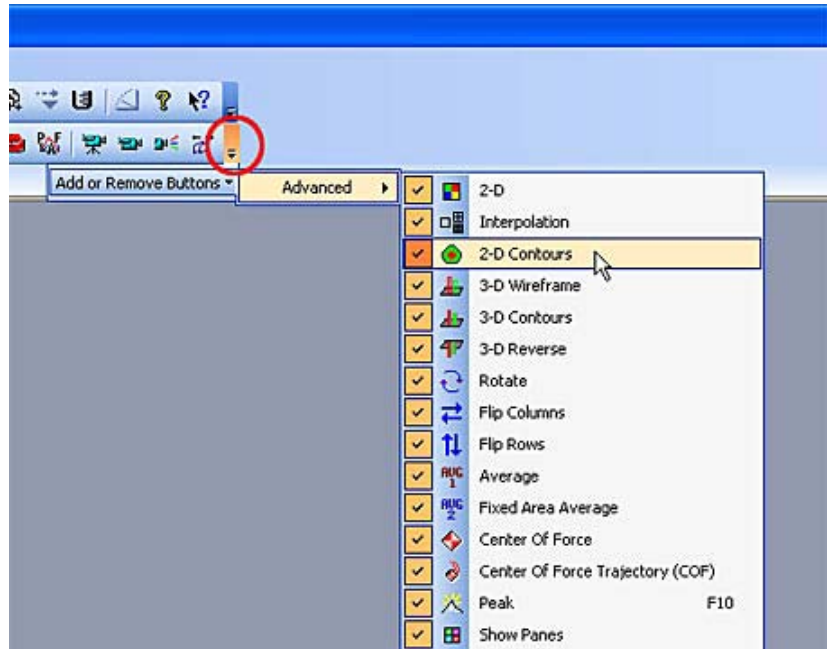




Showing/Hiding Icons on the Toolbar

A new feature with the Toolbars is the ability to show/hide icons on the Toolbars. This gives you the freedom to tailor the Toolbar to your needs. If you never use an icon, you can simply hide it. If you always use a command or icon, you can keep it displayed.

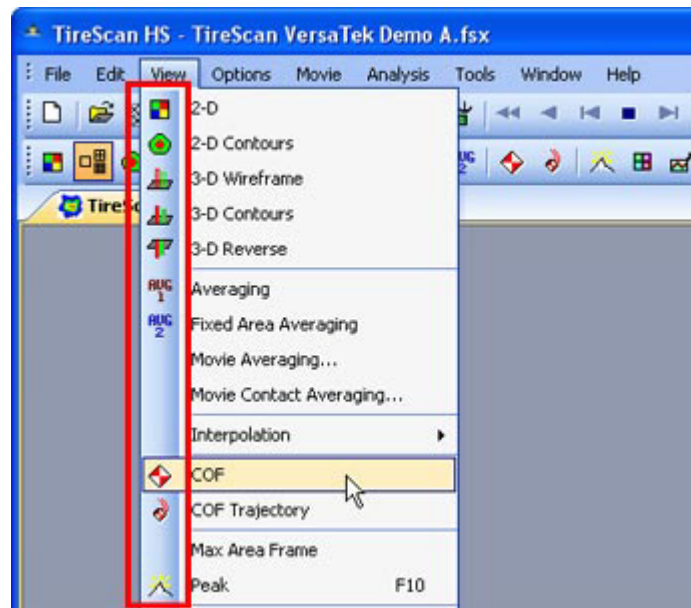
To show or hide an icon, click on the drop-down arrow located to the right of the Toolbar. Then hover and hold your mouse over the “Add or Remove Buttons” command. This opens the “Advanced” command. Slide your mouse over this command and all the icons located on the “Advanced” toolbar are displayed in a menu (shown at right). By default, they are all checked (displayed). Click on any icon in the menu in order to hide it (remove it from the Toolbar). Repeat this process to remove any further icons.



To add the icon back to the toolbar, go back into the Advanced menu of icons, and click the icon once again to check it. The icon is now displayed on the Toolbar.

Icon Display within Menus

Icons are now displayed directly next to their text command under all menus.



Note: Not all menu commands have an associated icon. In this event, there will be no icon displayed next to the menu command.

HARDWARE AND INSTALLATION

This section presents instructions for hardware installation, and also provides the system requirements necessary to support the *I-Scan* pressure measurement system.

TEKSCAN COMPUTER REQUIREMENTS

For your Tekscan system to function properly, your computer must meet or exceed the following requirements:

Suggested Minimum Computer Requirements (desktop or laptop) for all Tekscan Systems

- Intel Pentium 600 MHz or higher processor¹
- 128 MB RAM (512 MB RAM recommended)
- 1 GB hard drive
- CD ROM drive
- Windows 2000 (SP4), XP (SP2), Vista, or 7 operating system (32-bit versions)²

Computer Interface Requirement (one of the following options, depending on your system):

- 1 USB port/handle for ELF or Evolution systems
- 1 USB port for Mobile systems
- 1 USB 2.0 port for VersaTek systems

Requirements for Video Package (in addition to the above requirements)

- 20 GB Hard drive (ATA-33, 7200 RPM speed)
- Firewire (iLink or IEEE1394) port
- MiniDV (digital video) format camcorder with Firewire (iLink or IEEE1394) port³

¹ HR Mat™ Walkway systems that require scan rates higher than 60 Hz. require a computer with a 2 GHz or higher dual core processor.

² Evolution I-Scan and T-Scan systems are compatible with both 32- and 64-bit operating systems. ELF software is not yet supported with the Windows 7 operating system.

³ Currently, MiniDV format is the only video format supported for capturing video directly using Tekscan software. Cameras that use other formats are not compatible with our video capture and playback package. For video cameras not-using MiniDV formats, a third party software must be used to transfer video from the camera to the computer.

VERSATEK SCANNING ELECTRONICS

The Industrial VersaTek Scanning Electronics is an optional configuration for the I-Scan system. Industrial VersaTek Scanning Electronics should be used whenever [Cross-Handle Scanning](#) (multiplexing among any number of VersaTek Handles) or High-Speed scanning (above 100 Hz.) is required.

Cross-Handle Scanning greatly reduces the number of electronic connections or Handles required to address large arrays of sensels. In addition, Cross-Handle Scanning improves the scan rate of Sensors that require 3 or more Handles (refer to the [Scan Rate](#) section for further information). Finally, it enables previously unattainable configuration options (a 3-handle configuration, for example).

Industrial VersaTek Handles can be used with both “T” and “CH-T” type Sensors. “T” type Sensors such as the 5101 or 5027 can be used individually or in Parallel. When used individually, one handle is connected to one sensor, and is mapped in the software. When used in parallel, eight individual sensors with eight different maps can be used simultaneously, for example. If eight to sixteen Handles are to be operated in parallel, two 8-port Hubs are used and the scanning is automatically synchronized.

Since Industrial VersaTek Handles works with both “T” and “CH-T” Sensor types, there is no need for multiple Handle types or multiple electronics. For example, a CMP customer can now use two VersaTek handles and a 6030N sensor to measure overall polishing pad uniformity and then use those same Handles with a 5051 sensor to measure the carrier ring “edge effect.” Working with this combination of sensors used to require two different types of Handles (a pair of Dual Handles and an Evolution handle). Refer to the [I-Scan Sensor Chart](#) for a full listing of all “T” and “CH-T” style I-Scan Sensors.

The VersaTek Handles are connected to a 2- or 8-port VersaTek Hub, which is then connected to a USB 2.0 port on the computer. This eliminates the need for Magma boxes and PCI slots.

The following sections provide information on the VersaTek Scanning Electronics, and procedures for installation and setup.

Note: The [Evolution](#) Scanning Electronics is standard for I-Scan systems

VersaTek I-Scan Configuration

<i>I-Scan System</i>		<i>Components</i>
I-Scan VersaTek Cross Handle Scanning System	IVCHS	• (1) VersaTek 8-Port Hub • (2) VersaTek Handles • (1) Hub Power Supply (with US AC Cord) • (1) USB Cable • (1) I-Scan Software CD with 1 I-Scan Sensor Map • (1) System Carrying Case • (1) System Manual <i>Note: No Sensors included</i>

VersaTek Component Identification

(1) VersaTek 8-Port Hub (Part #V8PH-1) 	(1 or more) VersaTek Sensor Handles with attached CAT5E with RJ45 Connector Cable (number of handles will vary, depending upon application) 	(1) 15 ft. USB Cable  Note: Extending the Hub's USB Cable is not recommended
(1) 8 Port VersaTek Hub Power Supply (12V 60W Universal VPS-2) with US AC Cord  <i>Note: The 2-Port and 8-port VersaTek Hub power supplies are not interchangeable.</i>	(1) User Manual 	(1) Tekscan Software CD 
(1) System Carrying Case 	Note: Tekscan Sensors are not included with System. Sensors are purchased separately and are dependant upon your application. VersaTek Handle CAT5E Cable Extension Handle cable length can be extended up to 100 ft. with the following components: • (1) Shielded RJ45 modular coupler (per cable extension) • (1 or more) Shielded CAT5E cable(s) (multiple cable extensions can be used, but total cable length should not exceed 100 ft.) <i>Note: Tekscan does not provide components for extending your VersaTek Handle cables.</i> <i>Note: Ensure all Handles have an equal total cable length (i.e. Handle A and Handle B each have the same total cable length).</i>	

Versatek Component Descriptions

VersaTek 8-Port Hub

The **VersaTek 8-port Hub** acts as a gateway between the VersaTek Handle (and attached sensor) and the PC or laptop computer. It is powered by a standard AC adapter, which plugs into a wall outlet or power bar. On one end, the Handle cable is connected, which leads to the VersaTek Handle, and on the other end a single USB Cable leads to the PC or laptop computer.

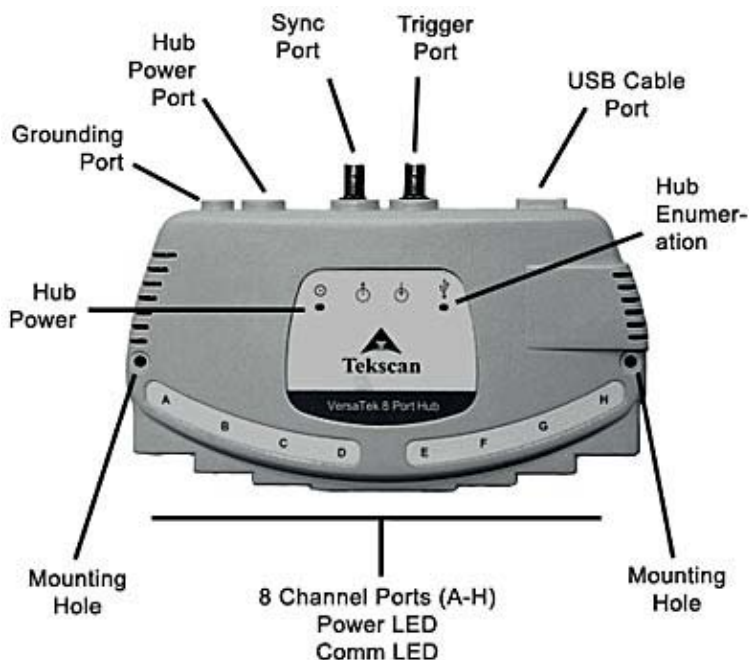
Users who want more portability can connect their own 12V DC supply to the Hub Power Port, capable of sourcing 5 amps through a Switchcraft 760 style plug with center pin positive.

One to eight VersaTek Handles can be connected to an eight port Hub. Depending on the system, VersaTek Handles can operate in parallel, for example, with eight individual sensors with eight different maps. If eight to sixteen Handles are to be operated in parallel, two Hubs are used and the scanning is automatically synchronized. Alternately, up to eight Handles can be used together ([cross-handle scanning](#)).

The VersaTek Hub includes built-in support for Triggering (The trigger switch is an optional component) and Synchronization capability. Synchronization and trigger connectors are always present on the Hub, and a software option is required to enable external Trigger and Synch operation.

Note: To minimize noise, the VersaTek 8-port Hub should only be used with VersaTek Handles. VersaTek Cuffs should not be used with the VersaTek 8-port Hub.

The following shows the VersaTek 8-port Hub (front view):



The image below displays the VersaTek 8-port Hub labels printed on the back of the Hub.



Note: Please refer to the [Declaration of Conformity](#) section for an explanation of the symbols found on the label.

The following provides information on the LED lights found on both VersaTek Hubs:

- **Hub Power:** A green LED indicates the Hub is receiving power and is turned on. If this LED is off, but the Hub is connected, this means either there is no power moving through the Hub, or it is not recognized by the operating system (Under the Device Manager in the "System" dialog – found under the Windows Control Panel).
- **Hub Enumeration:** A green LED indicates the Hub is communicating with the PC or laptop.
- **Channel Ports:** A green LED indicates the Handle is powered and connected to the Hub. A second orange communication LED indicates the Handle unit is communicating with the Hub. The flashing occurs at a frequency that reflects the current communication traffic. This light only flashes when the software is running.

VersaTek Handle

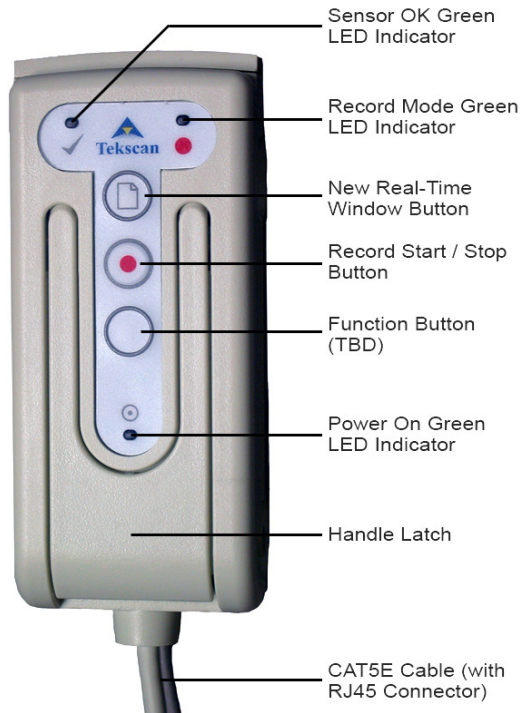
The **VersaTek Handle** gathers the data from the sensor and processes it so that it can be sent easily to the computer. The handle may also be used to start or stop a recording. This function is referred to as "Remote Recording."

The Handle's attached 15-foot (4.5 meter) cable is a shielded CAT5E Cable with RJ45 connector which attaches to one of the Hub's Channel Ports. It is easy to get and adapter for longer cable lengths between the Handle and Hub. RJ45 adapter cables can be used to achieve 100 feet (30 meters) cable length with no degradation of performance. Communication between Handle & Hub is LVDS (low voltage differential signal). The VersaTek Hub is used to convert the communication protocol to USB (connected to the host computer).

The Handle Cable can be extended up to 100 feet, if required for the application. If an extension is required, an RJ45 cable can be purchased along with an RJ45 coupler. Both the cable and the coupler must be shielded. Also, please bear in mind the longer the extension, the more the system may be subjected to ambient noise.

The VersaTek Handle has a latch on its topside. In the "Up" position, the latch retracts the contact pins inside the handle to allow insertion of the sensor tab. The sensor tab is placed into the sensor handle. The handle's attached USB cable is then connected directly to your computer via the USB port.

The following displays the buttons and their functionality for the VersaTek Handle:



The following shows an image of the VersaTek Handle with attached CAT5E with RJ45 Connector Cable:



The following displays a diagram of the label on the back of the VersaTek Handle:



Note: Please refer to the Declaration of Conformity section for an explanation of the symbols found on the label.

- **Sensor OK Green LED Indicator:** A green light here indicates that the sensor is correctly inserted into the handle and a Real-time window can be opened.
- **Record Mode Green LED Indicator:** A green light here indicates that the sensor is recording pressure data and transferring that data to your computer.
- **New Real-Time Window Button:** This will open a new Real-time window in the software, so that you can begin recording pressure data.
- **Record Start / Stop Button:** Use this button to start a recording or stop a recording that is in progress.

- **Function Button (TBD):** This button is yet to be determined.
- **Power On Green LED Indicator:** A green light indicates that the handle is receiving power and is communicating properly with the Hub.

VersaTek Handle CAT5E Cable Extension

Handle cable length can be extended up to 100 ft. with the following components:

- (1) Shielded RJ45 modular coupler (per cable extension)
- (1 or more) Shielded CAT5E cable(s) (multiple cable extensions can be used, but total cable length should not exceed 100 ft.)

Note: Tekscan does not provide components for extending your VersaTek Handle cables.

Note: Ensure all Handles have an equal total cable length (i.e. Handle A and Handle B each have the same total cable length).

VersaTek Hub Power Supply (VPS-2) with US AC Cord

The AC adapter that ships with the VersaTek system is used to power the Hub (shown at left). The 8 Port Versatek Hub Power Supply is a 12V 60W Universal VPS-2 unit. The Power Supply ships with a US AC Cord to plug the supply into a wall outlet. **IMPORTANT: The 2-Port and 8-port VersaTek Hub power supplies are not interchangeable.**



15 ft. USB Cable

The 15-ft. (4.5 meter) USB cable connects the Hub to the PC or laptop computer. This is a standard 2.0 USB cable (shown at right).



Note: Extending the Hub's USB Cable is not recommended

Software CD with sensor map

I-Scan software is a 32-bit application. For software compatibility, see the [Hardware & Installation](#) section (specifically the [Computer Requirements](#) section). The software CD includes the sensor map(s) purchased. The sensor map(s) included depend upon your application requirements.



I-Scan Sensor

The sensor(s) included depend upon your application requirements. Refer to the Sensors section for further information.

System Carrying Case

The system carrying case is used to store the VersaTek Handle system components. This case can be used to transport the system from one location to another (shown at right).



System Manual

Each system ships with a full User Manual. It is strongly advised that you read the manual and any documentation that shipped with your system before operating any of the system components or taking any trials. In addition, a full Help File is included with all shipments which can be accessed from the Start Menu on your computer or from within the software under the "Help" menu.



Cross-Handle Scanning

Cross-Handle Scanning (CHS) is multiplexing among any number of VersaTek handles. CHS greatly reduces the number of electronic connections or Handles required to address large arrays of sensels. For example, with Cross-Handle scanning, four VersaTek Handles can now address the same number of sensing elements that previously required eight Dual Handles (we chose to call this feature “Cross-Handle Scanning” as opposed to “Multiplexing,” to avoid confusion with the row and column multiplexing that we do within all our handles and sensors). The I-Scan Cross-Handle Scanning Systems replace the Dual System.

The benefit of Cross-Handle Scanning is outlined below:

- Less equipment (Handles) are required when scanning more than 9,152 sensels.
- For three or more handles, the advantage increases exponentially with the number of handles.
- Odd numbers of Handles can now be used in custom sensor designs. For example, you can use three VersaTek Handles on a Sensor where previously four Dual Handles (two Dual sets) would be required.

In a Dual Handle system, two Scanning Handles are connected together and communicate to each other when sensing. When another set of Dual Handles are connected, those handles talk to each other, but do not communicate with the previous two Scanning Handles. Since each set of Dual Handles in a system can scan up to 9,152 Sensels, an 8-handle Dual system can sense a maximum of 36,608 Sensels ($4 \times 9,152 = 36,608$). With the introduction of VersaTek, all Handles can talk to each other. This means that with eight Handles, you can scan four times as many sensels ($8^2 \times 2288 = 146,432$) than with eight Dual Handles.

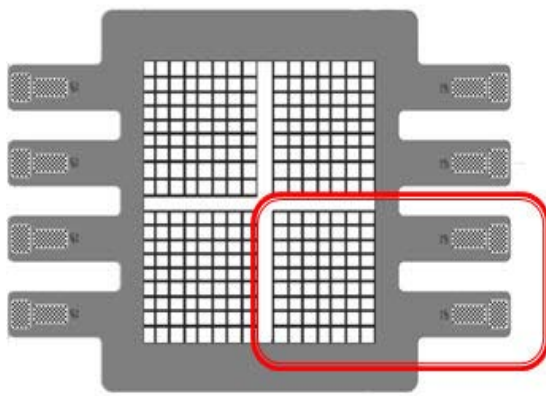
The following chart outlines the differences between Dual Handles and VersaTek Handles with Cross-Handle Scanning capability.

Number of Handles		Sensing Area (Rows x Columns)		Number of Sensels	
Dual	VersaTek	Dual	VersaTek	Dual	VersaTek
N/A	1	N/A	44 x 52	N/A	2,288
1 Set (2 Handles)	2	88 x 104	88 x 104	9,152	9,152
N/A	3	N/A	132 x 156	N/A	20,592
2 Sets (4 Handles)	4	Two 88 x 104	176 x 208	18,304	36,608
N/A	5	N/A	220 x 260	N/A	57,200
3 Sets (6 Handles)	6	Three 88 x 104	264 x 312	27,456	82,368
N/A	7	N/A	308 x 364	N/A	112,112
4 Sets (8 Handles)	8	Four 88 x 104	352 x 416	36,608	146,432

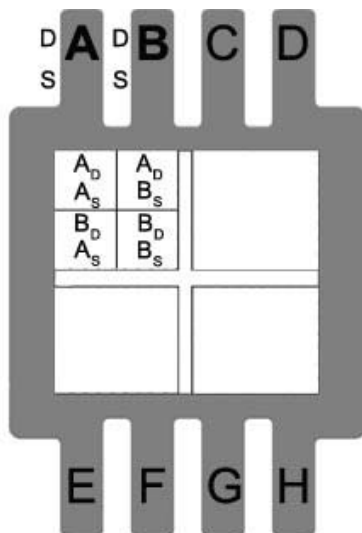
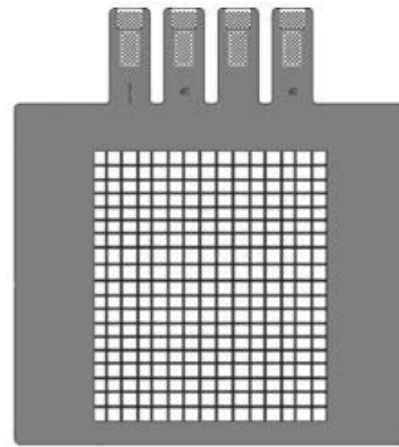
As an example, we will compare the 8050 Sensor with Dual Handles to the 8400 Sensor with VersaTek Handles. With the 8050 Sensor, 4 Sets of Dual Handles (8 Handles in total) are required to scan the same area as the 8400 Sensor with 4 VersaTek Handles. The following illustrations outline this concept.

- Each set of Dual Handles addresses up to 88×104 rows and columns
- Up to 9,152 Sensels per Dual pair
- Only two Handles communicate with each other
- Map #8050 is four of these individual Sensors
- Up to 176×208 rows and columns
- Up to 36,608 Sensels
- All Handles communicate with each other

4 Sets of (8) Dual Handles
(#8050)



4 VersaTek Handles
(#8400)



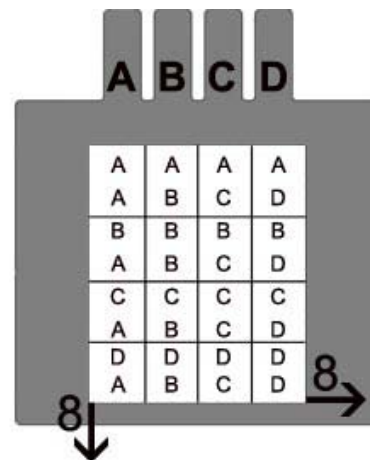
DUAL

$A \neq C, A \neq D$

$8_4 \rightarrow 8 \times 2288$

D=Driven=52 Max

S=Sensed=44 Max



VersaTek

$8_4 \rightarrow 8^2(2288)$

VersaTek Specifications

The following table outlines the specifications for the VersaTek Scanning Electronics.

Mechanical	
<i>VersaTek USB Cable</i>	
Length ft (mm)	10 (3048)
Weight lbs (kgs)	0.40 (0.18)
Connection Type	USB 2.0 A-B Connections
<i>VersaTek Handle</i>	
Size LxWxH in (mm)	5.43 x 2.25 x 1.32 (137.9 x 57.2 x 33.5)
Open Lever Height in (mm)	4.30 (109.2)
Weight Oz (g)	5.5 (155)
Weight (with attached Cable) Oz (g)	10.8 (305)
LVDS Cable Length ft (mm)	15 (4572) 24 AWG CAT5E RJ45 Connection
<i>VersaTek 8-Port Hub</i>	
Size LxWxH in (mm)	7.63 x 4.79 x 1.38 (193.8 x 121.7 x 35.05)
Weight Oz (g)	13.1 Oz (370 g)
<i>VersaTek 8-Port Hub Power Supply</i>	
Size LxWxH in (mm)	4.00 x 2.50 x 1.38 (101.6 x 63.5 x 45.7)
Weight Oz (g)	13.6 (386)
Power In	100-240V, 5A 50-60Hz, 1.2A
Power Out	12V, 5A
DC Cable Length ft (mm)	6 (1829)
Hub Input Power Plug	Switchcraft 760 Plug, center positive or equivalent
AC Socket	IEC320
Ambient Operating Conditions	
Temperature: °F (°C)	14 to 131 (-10 to 55) Prolonged use at high Temperatures should be avoided
Humidity: %	0 to 90 (non condensing)
Pressure: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)
Storage and Transport Conditions	
Temperature: °F (°C)	-4 to 131 (-20 to 55) Short-Term. 41 to 104 (5 to 40) Long-Term.
Humidity: %	0 to 90 (non condensing)
Pressure: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)

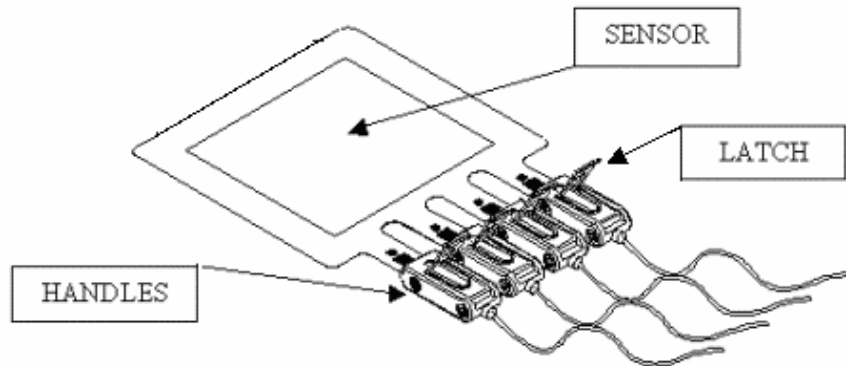
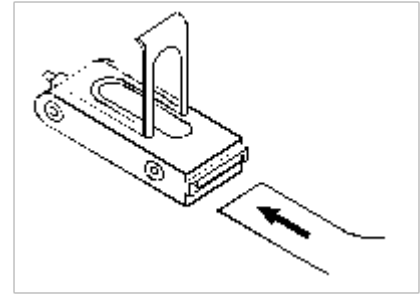
VersaTek Hardware Installation

The VersaTek scanning electronics hardware is simple to setup and run. Follow the instructions below for correct hardware setup.

Note: the following setup procedure uses an 8-port VersaTek Hub. The procedure is the same for the 2-port Hub.

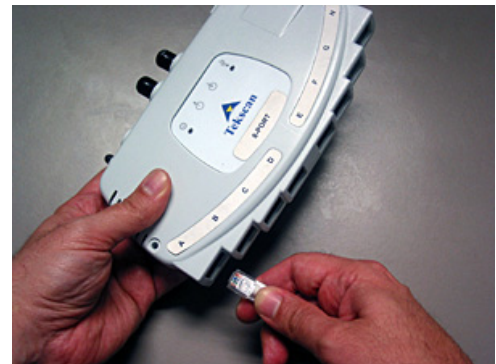
System Assembly

1. Lift the latch on the top of the handle.
2. Slide the sensor tab with the "This Side UP" legend facing the raised handle latch (see figure at right).
3. Slide the tab in until it reaches its mechanical stop. *Do not force the tab into the handle.*
3. Close the latch completely.



Important! Be sure to treat the sensor with care, not to bend or fold the sensor or sensor neck, as this may result in creasing or crinkling and can damage the sensor.

4. Connect the CAT5E Cable to the Hub's CH1 (channel 1) cable port (shown at right). Connect the other end of the cable to the Handle. Connect the other Cables to their respective ports in the same way; for as many as is required for your system.



5. Connect the USB cable to the USB Cable port on the Hub (shown at right). Connect the other end to the USB port on the PC or Laptop.



6. Plug the included AC adapter into a wall outlet. Connect the AC adapter cable into the hub unit (shown at right). Make sure the green Power LED light on the hub illuminates. The green CH1 (channel 1) Port LED should also illuminate.



Starting the software

After starting the Tekscan software, the message below the Real-time window should be “Ready.” If this message is “MISALIGNED!” remove and reinsert the sensor into the Handle. If the “Sensor OK” message still does not appear, consult the [Troubleshooting](#) section.

When the hardware and software are properly installed, a sensor is properly inserted into the Handle, and the correct map (sensor type) is selected, the orange communication LED light is displayed on the Hub unit. This indicates the sensor is communicating with the software on the PC or laptop.

The software is now ready to display pressure detected by the sensor in the Real-time window.

VersaTek Maintenance and Care

- The VersaTek system components cannot be autoclaved.
- Do not let any liquid drip onto the electronics inside the Handles or Hub. If this occurs, the components can stop working and must be allowed to dry for 24 hours. You can use your air syringe, however, to significantly reduce this drying time. Do not attempt to dry out the Handles or Hub using any other method, or you may destroy the delicate electronics.
- To properly clean the device, wipe down the Handles or Hub with a 70% Isopropyl Alcohol solution. To do this, slightly dampen a cloth with the alcohol solution, careful not to soak or saturate the cloth. Then wipe the components after use.
- Do not connect the VersaTek Hub or Handles to the Ethernet port of a PC, laptop or wall outlet. Do not loop a cable from Channel to Channel on the Hub. Doing so may damage the VersaTek System.

VersaTek 8-Port Hub Synch and Trigger Ports

Note: Use of the Synch and Trigger Ports require the optional “External Trigger Synch” software. This software is an optional purchase for Clinical software users, but comes standard in Research software. For more information about the Synch and Trigger Ports, refer to “The External Trigger Synch Software” appendix within your software Help File.

VersaTek 8-Port Hub Trigger Port

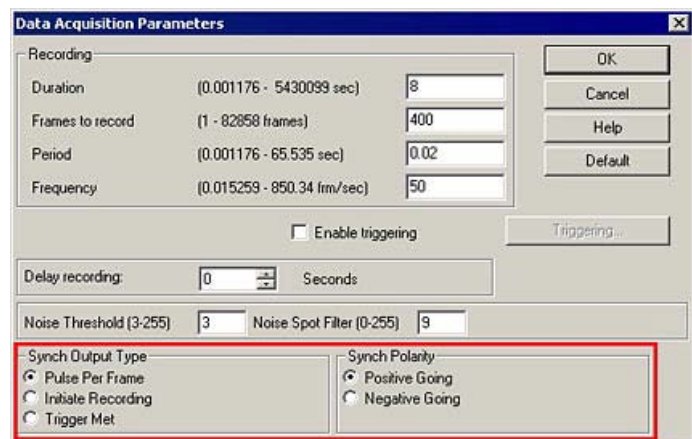
The Trigger Port responds to the change from an open to a closed circuit that will trigger starting a Tekscan recording. With the “Stop Triggering” method set to “External,” it can also stop a recording. Tekscan offers the optional trigger switch (Part #ZCAB-TRIGGER-ASSY-VERSATEK), however, any electronic device that can open or close a circuit can be used.

To use the Trigger Port with the Tekscan Trigger switch, plug it into the connector. Tekscan’s Trigger switch has momentary behavior. When pressed, it is closed, and when released, it is open. Using Tekscan’s “Start Triggering” and “Stop Triggering” settings in the software, this Trigger Switch can be used to start and stop a Tekscan recording.

VersaTek 8-Port Hub Synch Port

The VersaTek 8-Port Hub Synch Port is a BNC connector port. It offers the capability to trigger an external signal. The parameters that govern how the signal operates are determined by the “Synch Output Type” and “Synch Polarity” options, found in the Data Acquisition Parameters dialog (under the “Options” menu).

Note: These parameters are only visible if the optional “External Trigger Synch Software” is installed.



EVOLUTION SCANNING ELECTRONICS

The following section provides information on the Evolution Scanning Electronics hardware, and procedures for installation and setup.

Evolution Component Identification

The following table provides a view of all the components that ship with your Evolution System:



Evolution Handle with attached USB cable



Installation CD



User Manual

Optional:

USB Cable length can be extended to 30 or 45 ft. with the following optional components:

- (1) or (2) 4-Port USB 2.0 Powered Slim Hub for PC (Part # **ZUSB-4PORT-HUB-USBG-4PU2SI**)
- (1) or (2) additional 15 ft. USB Cable(s) (Part # **ZCAB-15'-USB-CABLE-340465**)

Optional:

One USB Hub is shipped as a standard component in system configurations where three or more Evolution Handles are required.

USB Cable length can be extended to 30 or 45 ft. with the following optional components:

- (1) or (2) 4-Port USB 2.0 Powered Slim Hub for PC (Part # **ZUSB-4PORT-HUB-USBG-4PU2SI**)
- (1) or (2) additional 15 ft. USB Cable(s) (Part # **ZCAB-15'-USB-CABLE-340465**)

Refer to the [Evolution Handle](#) section for more information on extending Evolution Handle cable length.

Also Included:

Tekscan I-Scan “family” Sensor(s) -- Depends upon application.

Sensor Carrying Case (Part **ZCAS-4850**)

Evolution Handle

The Evolution Sensor handle gathers the data from the sensor and processes it so that it can be sent easily to the computer. The buttons on the sensor handle may also be used to start or stop a recording. This function is referred to as **'Remote Recording'**.

A single Evolution Handle connects to a single USB 2.0 port on your Computer or Laptop via the attached USB Cable. You can connect as many Handles as you have ports on the computer. If you do not have enough USB 2.0 ports on your computer, wish to use only one of your computer's USB ports to connect multiple Evolution Handles, or wish to extend the cable length between your Evolution Handle and the computer, an additional 4-port USB 2.0 powered slim hub is required (Part # **ZUSB-4PORT-HUB-USBG-4PU2SI**) and additional 15-foot cables may be required for cable extension (Part # **ZCAB-15'-USB-CABLE-340465**).



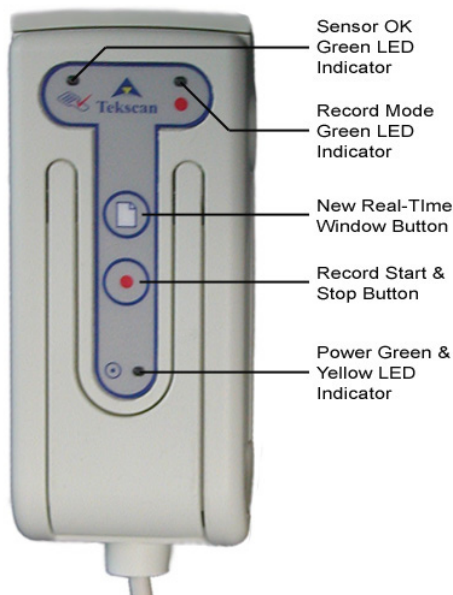
One USB Hub is shipped as a standard component in system configurations where three or more Evolution Handles are required.

The handle has a latch on its topside. In the 'Up' position, the latch retracts the contact pins inside the handle to allow insertion of the sensor tab. The sensor tab is placed into the sensor handle. The handle's attached USB cable is then connected directly to your computer via the USB port.

Note: the handle is shipped with the latch in the "open" position. It is advised that you keep the handle latch in the "open" position when the handle is not in use.

The image at right displays the Evolution Handle with attached USB cable.

The following displays the buttons and their functionality for the Evolution Handle:



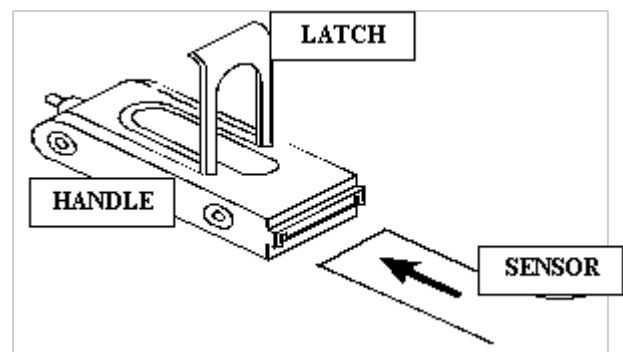
The following label is printed on all Evolution handles displaying the compliancy, model number, warnings and parameters:



- **Sensor OK Green LED Indicator:** A green light here indicates that the sensor is correctly inserted into the handle and a Real-Time window can be opened.
- **Record Mode Green LED Indicator:** A green light here indicates that the sensor is recording pressure data and transferring that data to your computer.
- **New Real-Time Window Button:** This will open a new Real-Time window in the software, so that you can begin recording pressure data.
- **Record Start & Stop Button:** Use this button to start a recording or stop a recording that is in progress.
- **Power Green & Yellow LED Indicator:** When yellow, this light indicates that the handle is receiving power, but is not yet initialized. When Green, this light indicates that the handle is receiving power and has been initialized by the computer (i.e.: the device shows up under the Windows device manager).

Inserting a Sensor into the Evolution Handle

1. Lift the latch on the handle.
2. Slide the sensor tab with the **This Side UP** legend facing the raised handle latch. Slide the tab in until it reaches its mechanical stop. *Do not force the tab into the handle.*
3. Close the latch.



Once the sensor is inserted into the handle, connect the attached USB cable to the computer's USB port. Your computer should automatically detect the new hardware and configure it for your system. If the driver is not found, insert the Tekscan software CD that came with your system and have the computer locate the driver on your CD. Then follow the on-screen instructions.

After following this procedure and starting the *I-Scan* software, the message below the real-time window for the sensor should be 'Sensor OK'. If the message is 'MISALIGNED', re-insert the sensor using the above procedure. If the 'Sensor OK' message still does not appear, consult the 'Troubleshooting' section.

Caution! Do not allow the handle to hang from the sensor. The handle may become damaged, resulting in a misaligned sensor.

Extending the Handle's USB Cable Length

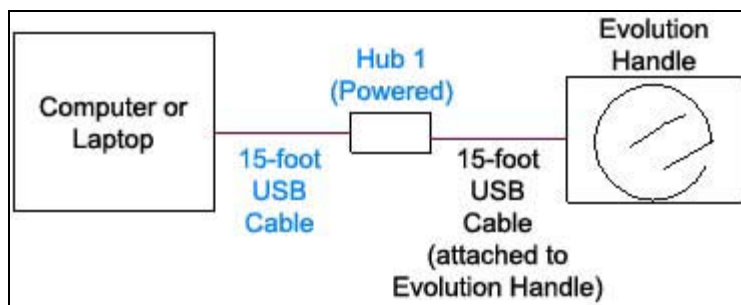
The Evolution Handle interfaces to a PC with a standard attached 15-foot USB cable. This cable can be extended either 30 or 45 feet, if desired, using one or two USB hubs and one or two 15-foot USB cables, respectively.

Please Note the following:

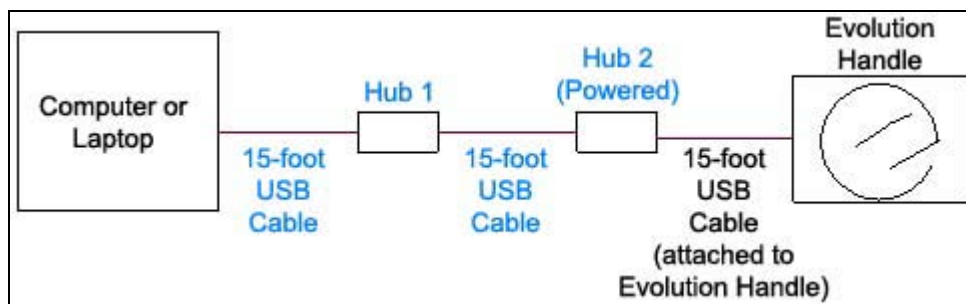
- Do not exceed 15 feet of cable between any two devices (Hub, Handle, and Computer/Laptop).
- Do not exceed 45 feet in total cable length. Longer than 45 feet, please consult the factory.
- The Hub closest to the Evolution Handle must be powered.
- All Hubs purchased should have the USB logo (shown at right)



Example showing a 30-foot USB cable extension setup between the Evolution Handle and Computer/Laptop. Components in blue are optional:



Example showing a 45-foot USB cable extension setup between the Evolution Handle and Computer/Laptop. Components in blue are optional:



Evolution Handle Maintenance and Care

- The handle cannot be autoclaved.
- Do not let any liquid drip onto the electronics inside the handle. If this occurs, the handle will stop working and must be allowed to dry for 24 hours. You can use your air syringe, however, to significantly reduce this drying time. Do not attempt to dry out the handle using any other method, or you may destroy the delicate electronics.
- To properly clean the device, be sure to wipe down the handle with a 70% Isopropyl Alcohol solution. To do this, slightly dampen a cloth with the alcohol solution, careful not to soak or saturate the cloth. Then wipe the handle after each use.

Evolution Handle Specifications

COMMUNICATION/DATA ACQUISITION:	
COMMUNICATION PROTOCOL TO HOST COMPUTER	USB 1.1 or 2.0 Compatible, 12 Mbps
SCAN SPEED	Up to 100 Hz.
DIGITAL PRESSURE RESOLUTION	8 BIT
ELECTRICAL:	
POWER SOURCE:	Host Computer's USB BUS
POWER CONSUMPTION:	200mA, 1W
MECHANICAL:	
USB CABLE:	
LENGTH in (mm)	180 (4572) - Standard
WEIGHT lbs (kgs)	0.40 (0.18)
HANDLE ENCLOSURE:	
SIZE LxWxH in (mm)	5.43x2.25x1.32 (137.9x57.2x33.5)
OPEN LEVER HEIGHT in (mm)	4.30 (109.2)
WEIGHT lbs (kgs)	0.40 (0.18)
AMBIENT OPERATING CONDITIONS:	
TEMPERATURE: °F (°C)	14 to 131 (-10 to 55) Prolonged use at high Temperatures should be avoided
HUMIDITY: %	0 to 90 (non condensing)
PRESSURE: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)
STORAGE AND TRANSPORT CONDITIONS:	
TEMPERATURE: °F (°C)	-4 to 131 (-20 to 55) Short-Term 41 to 104 (5 to 40) Long-Term
HUMIDITY: %	0 to 90 (non condensing)
PRESSURE: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)

Evolution Hardware Installation

The Tekscan Evolution handle does not require an additional interface card or parallel box in order to be connected to your computer. Most computers now come equipped with at least 2 USB connectors. These handles can be connected directly to your computer via the handle's USB cable. When inserted into your computer, the computer's operating system will automatically detect and configure the hardware for use.



USB SCANNING ELECTRONICS

The following section provides information on the USB Scanning Electronics, and procedures for installation and setup.

USB Component Identification

The following table provides a view of all the components that ship with your USB System:



USB Handle with attached USB cable



Installation CD



User Manual

Also Included:

Tekscan I-Scan "family" Sensor(s) -- Depends upon application.
Sensor Carrying Case (*Part ZCAS-4850*)

Type CF equipment

F-type applied part which provides a higher degree of protection of electric shock than that provided by type BF applied parts.



USB Handle

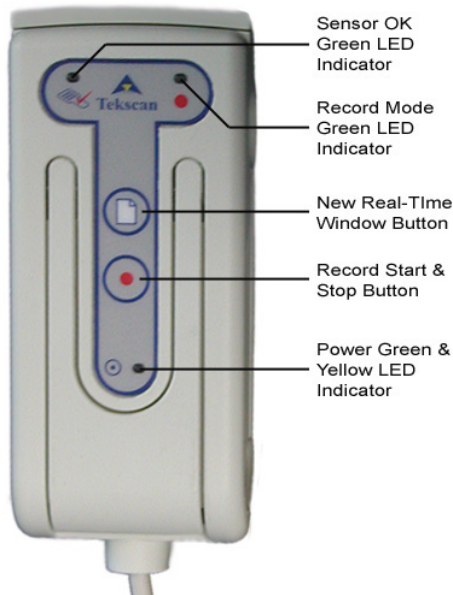
The USB Sensor handle gathers the data from the sensor and processes it so that it can be sent easily to the computer. The buttons on the sensor handle may also be used to start or stop a recording. This function is referred to as '**Remote Recording**'.

The handle has a latch on its topside. In the 'Up' position, the latch retracts the contact pins inside the handle to allow insertion of the sensor tab. The sensor tab is placed into the sensor handle. The handle's attached USB cable is then connected directly to your computer via the USB port.

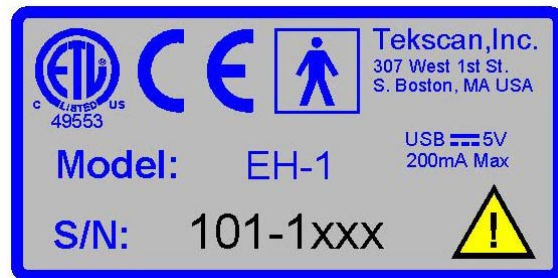


Note: the handle is shipped with the latch in the "open" position. It is advised that you keep the handle latch in the "open" position when the handle is not in use.

The following displays the buttons and their functionality for the USB Handle:



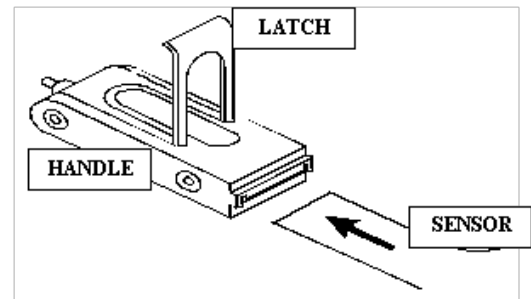
The following label is printed on all USB handles displaying the compliancy, model number, warnings and parameters:



- **Sensor OK Green LED Indicator:** A green light here indicates that the sensor is correctly inserted into the handle and a Real-Time window can be opened.
- **Record Mode Green LED Indicator:** A green light here indicates that the sensor is recording pressure data and transferring that data to your computer.
- **New Real-Time Window Button:** This will open a new Real-Time window in the software, so that you can begin recording pressure data.
- **Record Start & Stop Button:** Use this button to start a recording or stop a recording that is in progress.
- **Power Green & Yellow LED Indicator:** When yellow, this light indicates that the handle is receiving power, but is not yet initialized. When Green, this light indicates that the handle is receiving power and has been initialized by the computer (i.e.: the device shows up under the Windows device manager).

Inserting a Sensor into the USB Handle

2. Lift the latch on the handle.
3. Slide the sensor tab with the **This Side UP** legend facing the raised handle latch. Slide the tab in until it reaches its mechanical stop. *Do not force the tab into the handle.*
4. Close the latch.



Once the sensor is inserted into the handle, connect the attached USB cable to the computer's USB port. Your computer should automatically detect the new hardware and configure it for your system. If the driver is not found, insert the Tekscan software CD that came with your system and have the computer locate the driver on your CD. Then follow the on-screen instructions.

After following this procedure and starting the *I-Scan* software, the message below the real-time window for the sensor should be '**Sensor OK**'. If the message is '**MISALIGNED**', re-insert the sensor using the above procedure. If the '**Sensor OK**' message still does not appear, consult the '**Troubleshooting**' section.

Caution! *Do not allow the handle to hang from the sensor. The handle may become damaged, resulting in a misaligned sensor.*

USB Handle Maintenance and Care

- The handle cannot be autoclaved.
- Do not let any liquid drip onto the electronics inside the handle. If this occurs, the handle will stop working and must be allowed to dry for 24 hours. You can use your air syringe, however, to significantly reduce this drying time. Do not attempt to dry out the handle using any other method, or you may destroy the delicate electronics.
- To properly clean the device, be sure to wipe down the handle with a 70% Isopropyl Alcohol solution. To do this, slightly dampen a cloth with the alcohol solution, careful not to soak or saturate the cloth. Then wipe the handle after each use.

USB Handle Specifications

COMMUNICATION/DATA ACQUISITION:	
COMMUNICATION PROTOCOL TO HOST COMPUTER	USB 1.1 or 2.0 Compatible, 12 Mbps
SCAN SPEED	Up to 9.84 Hz.
DIGITAL PRESSURE RESOLUTION	8 BIT
ELECTRICAL:	
POWER SOURCE:	Host Computer's USB BUS
POWER CONSUMPTION:	200mA MAX at 5V
MECHANICAL:	
USB CABLE:	
LENGTH in (mm)	180 (4572) - Standard
WEIGHT lbs (kgs)	0.40 (0.18)

HANDLE ENCLOSURE:	
SIZE LxWxH in (mm)	5.42x2.25x1.88 (137.7x57.2x47.6)
OPEN LEVER HEIGHT in (mm)	4.30 (109.2)
WEIGHT lbs (kgs)	0.40 (0.18)
AMBIENT OPERATING CONDITIONS:	
TEMPERATURE: °F (°C)	14 to 131 (-10 to 55) Prolonged use at high Temperatures should be avoided
HUMIDITY: %	0 to 90 (non condensing)
PRESSURE: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)
STORAGE AND TRANSPORT CONDITIONS:	
TEMPERATURE: °F (°C)	-4 to 131 (-20 to 55) Short-Term 41 to 104 (5 to 40) Long-Term
HUMIDITY: %	0 to 90 (non condensing)
PRESSURE: psi (kPa)	1.7 to 14.7 (11.6 to 101.3) (sea level to 50,000 ft)

USB Hardware Installation

The Tekscan USB handle does not require an additional interface card or parallel box in order to be connected to your computer. Most computers now come equipped with at least 2 USB connectors. These handles can be connected directly to your computer via the handle's USB cable. When inserted into your computer, the computer's operating system will automatically detect and configure the hardware for use.



DUAL SYSTEM

The following section provides information on the Dual System hardware, and procedures for installation and setup.

Component Identification

The following table provides a view of all the components that ship with your system:



PC Interface Board (SR-2)



Data Handle with Cable



Installation CD



User Manual

Also Included:

Tekscan Sensor(s) -- Depends upon application.

*Note: System components can differ depending on your order.

PC Interface Board (SR-2)

The interface is how the sensor pressure data is brought into the computer so that it is available to the software. The *PC Interface Board (SR-2)* is inserted into a 16-bit PCI expansion slot inside your computer. Each card can accommodate 4 Dual Handles. This card is displayed below:



Dual Handles

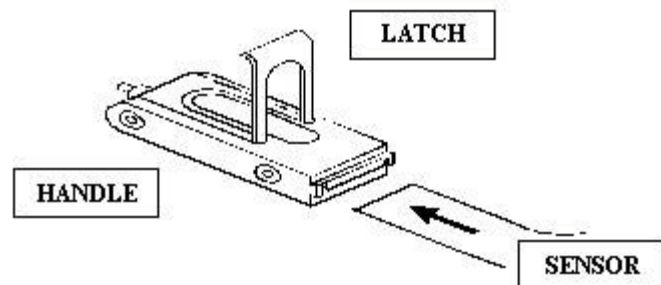
The Dual Handle gathers data from the sensor and processes it so that it can be sent easily to the computer. The sensor tab is placed into the sensor handle. The handle then connects to the PC Interface Board inside your computer.

The sensor handle may also be used to start a recording. Pressing the long blue button on the top of the sensor handle does this. This function is referred to as '**Remote Recording**', and may not be used to stop a recording.

The handle has a latch on its top side. In the "Up" position, the latch retracts the contact pins inside the handle to allow insertion of the sensor tab.

To insert a sensor into the handle:

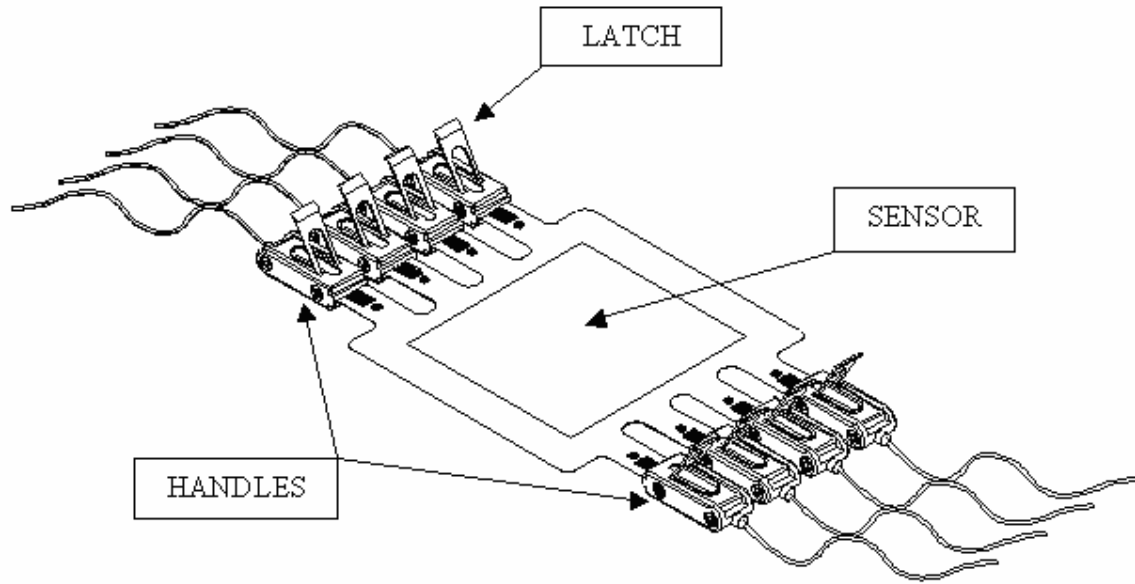
1. Lift the latch on the handle.
2. Slide the sensor tab with the **This Side UP** legend facing the raised handle latch. Slide the tab in until it reaches its mechanical stop. *Do not force the tab into the handle.*
3. Close the latch.



After following this procedure and starting the *TireScan* software, the message below the real-time window for the sensor should be '**Sensor OK**'. If the message is '**MISALIGNED**', re-insert the sensor using the above procedure. If the '**Sensor OK**' message still does not appear, consult the [Troubleshooting](#) section.

Caution! *Do not allow the handle to hang from the sensor. The handle may become damaged, resulting in a misaligned sensor.*

The following image displays the TireScan sensor setup with an 8-handle configuration:



Dual System Hardware Installation

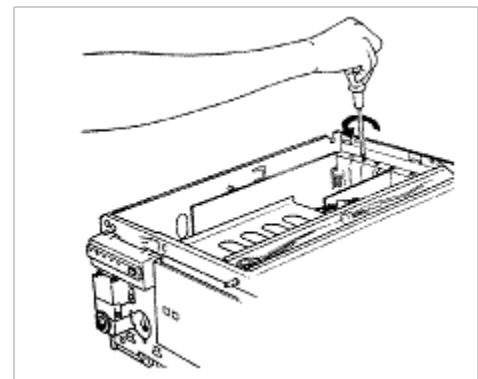
Note: Refer to the manual supplied with your computer for any specific details on installing a board in your computer's expansion slot.

Warnings:

1. Unplug your computer before beginning installation.
2. Touch the metal plating on the computer to ground yourself prior to handling the interface board to reduce the chance of static discharge.
3. Never force the interface board into a computer slot, and be careful not to twist or bend the board.

To install your *TireScan* interface board into a typical computer:

1. Disconnect power and remove the cover from your computer.
2. Remove the slot-cover plate from the available expansion slot with a screwdriver (shown at right).
3. **Carefully** insert the interface board into the slot. *Make sure to touch a metal object before handling the interface board.*
4. Fasten the interface board firmly to the computer using the screw from the slot-cover plate.
5. Test the system thoroughly to ensure it works correctly, before replacing the computer cover. Reinstall the computer cover.
6. Plug the *TireScan* handle cable into the receptacle on the *TireScan* interface board. The receptacle should be unobstructed and accessible at the back of your computer.
7. Plug your computer back in and turn the power on.



ACCURATE SCANNING ELECTRONICS

The following section provides information on the Accurate Scanning Electronics hardware, and procedures for installation and setup.

Accurate Component Identification

The following table provides a view of all the components that ship with your Accurate System:



PC Interface Board (SR-2)



Data Handle with Cable



Installation CD



User Manual

Also Included:

Tekscan I-Scan "family" Sensor(s) -- Depends upon application.
Sensor Carrying Case (*Part ZCAS-4850*)

PC Interface Board (SR-2)

The **PC Interface Board** is a data acquisition card that uses a 16-bit PCI expansion slot of your computer.



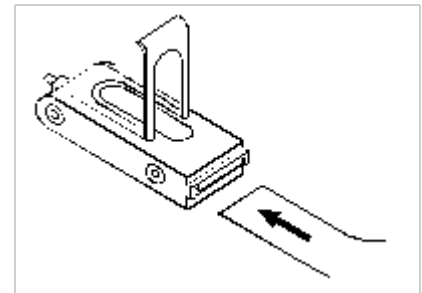
Accurate Handle

The **Accurate Handle** gathers the data from the sensor and processes it so that it can be sent easily to the computer. The sensor tab is placed into the handle. The handle connects to the PC Interface Board inside your computer.

The handle has a latch on its topside. In the 'Up' position, the latch retracts the contact pins inside the handle to allow insertion of the sensor tab.

To insert a sensor into the handle:

1. Lift the latch on the top of the handle.
2. Slide the sensor tab with the **This Side UP** legend facing the raised handle latch (see figure at right).
3. Slide the tab in until it reaches its mechanical stop. *Do not force the tab into the handle.*
4. Close the latch completely.



After starting the **Research** software, the message below the Real-time window should be 'Sensor OK'. If this message is 'MISALIGNED', remove and reinsert the sensor into the handle, ensuring that the sensor is right side up, and the handle latch is completely down. If the 'Sensor OK' message still does not appear, consult the [Troubleshooting](#) section.

When the hardware and software are properly installed, a sensor is properly inserted into the handle, and the correct *map* (sensor type) is selected, the *Real-time* window will display the pressure detected by the sensor.

Caution! *Do not allow the handle to hang from the sensor. The handle may become damaged, resulting in a misaligned sensor.*

Accurate Hardware Installation

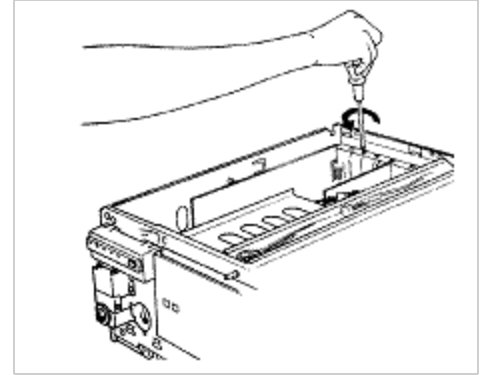
Note: *Refer to the manual supplied with your computer for any specific details on installing a board in your computer's expansion slot.*

Warnings:

1. Unplug your computer before beginning installation.
2. Touch the metal plating on the computer to ground yourself prior to handling the interface board to reduce the chance of static discharge.
3. Never force the interface board into a computer slot, and be careful not to twist or bend the board.

To install your *I-Scan* interface board into a typical computer:

1. Disconnect power and remove the cover from your computer.
2. Remove the slot-cover plate from the available expansion slot.
3. **Carefully** insert the interface board into the slot. *Make sure to touch a metal object before handling the interface board.*
4. Fasten the interface board firmly to the computer using the screw from the slot-cover plate.
5. Test the system thoroughly to ensure it works correctly, before replacing the computer cover. Reinstall the computer cover.
6. Plug the *I-Scan* handle cable into the receptacle on the *I-Scan* interface board. The receptacle should be unobstructed and accessible at the back of your computer.
7. Plug your computer back in and turn the power on.



SENSORS

The *I-Scan* sensor is an ultra-thin (.004", 0.10mm) flexible printed circuit. It has up to 2288 individual pressure sensing locations, which may also be referred to as 'sensing elements', 'sensels', or 'cells'. The sensels are arranged in rows and columns (Cartesian coordinates) on the sensor in most cases (not in 'Polar Sensors'). The sensor acts as a variable resistor in an electrical circuit. When the sensor is unloaded, its resistance is very high; when a force is applied to the sensor, the resistance decreases. This output resistance is then converted to a digital value (raw sum) in the range of 0 to 255.

Because of the nature of the pressure sensitive layer that is the heart of the *I-Scan* sensors, it is possible to produce sensors of varying sensitivity. Sensors with pressure ranges as low as 0 - 5 PSI (0 - 260 mmHg) and as high as 0 - 30,000 PSI (0 - 206 Mpa) have been produced. The active sensing area of the sensor is surrounded by substrate material that contains conductive leads. These leads connect the rows and columns of the active sensing area to the terminal section or tab of the sensor. The topside of the sensor is labeled **This Side UP** near the terminal section. The sensor tab must be inserted into the handle with the proper orientation.

Polar Sensors

Sensors with sensels arranged in rows and columns (Cartesian coordinates) are well suited to applications in which the load area is rectangular. In cases where the load area is circular or annular, sensors with sensels arranged in circles and radii (polar coordinates) may be used.

Polar sensors have sensels with different areas as a function of the sensel radial location. The ***I-Scan*** software accounts for these variations by breaking the polar sensor grid into a finer Cartesian grid. The displayed image is actually an image of these ‘virtual’ sensels.

Scan Rate

Scan Rate is defined as **Frequency**, or the number of frames that are recorded each second (frames/second). Scan rate is an important consideration when evaluating dynamic events. Faster scan rate gives greater visibility and insight into rapid dynamic events, and provides finer time resolution.

The following chart outlines the maximum scan rates possible with a variety of sensors for your ***I-Scan*** system. Scan rates can vary depending upon the sensor and number of handles used.

Note: This list is not a definitive list of sensors. This list provides Scan Rates for a portion of Tekscan Sensors.

<i>Sensor</i>	<i># of Handles</i>	<i>Connection Type</i>	<i>Sensor Scan Rate with VersaTek Hardware (Hz.)</i>
5026	2	CH-T	Up to 214
6010	2	CH-T	Up to 192
6020	2	CH-T	Up to 192
6030	2	CH-T	Up to 192
7501	2	CH-T	Up to 137

For further information about setting your Scan Rate or Frequency, refer to the [About Data Acquisition Parameters](#) section.

Sensor Maintenance and Care

- To properly clean the sensors, wipe them with a 70% Isopropyl Alcohol solution. To do this, slightly dampen a cloth with the alcohol solution, careful not to soak or saturate the cloth. Then wipe the sensors after each use.
- Sensors should always be stored flat in either the envelope in which they were shipped, or some other protective cover. Improper storage can adversely affect sensor life.
- Sensor Replacement/Disposal: Dispose of sensors in any waste container. Sensors are not biohazardous waste.

I-Scan Sensor Chart

Following is a list of Industrial "T" style and "CH-T" style sensors available for use with the *I-Scan* system. The "T" type sensor connection works with [VersaTek](#), [Evolution](#), [USB](#) or [Accurate](#) Handles, while the "CH-T" sensor connection works with [VersaTek](#) or [Dual](#) Handles.

Note: The VersaTek electronics works with both "T" and "CH-T" type sensor connections.

Note: Clicking on the sensor links below will open the internet catalog page for each sensor. You must be connected to the internet for the page to be viewable.

<i>Model (Map) Number</i>	<i>Sensor Connection</i>
3150	T
4000	T
4010N	T
4201	T
4205	T
5026	CH-T
5027	T
5033	T
5040	T
5040N	T
5051	T
5076	T
5101	T
5150N	T
5210N	T
5250	T
5260	T
5320	T
5510	T
5511	T
5515	T
5550	T
5555	T
5570	T
5612N	T
5617N	T
5620N	T
5630N	T
5632N	T
5660N	T
5800	T
5800N	T
6010N	CH-T
6020N	CH-T
6030N	CH-T

6077	T
6220	T
6230	T
6300	T
6500	T
6510	T
6700	T
6900	T
6911	T
7501	CH-T
8100	T
8110	T
8150	T
8155	T
9801	T
9830	T
9850	T
9851	T
9855N	T
9856N	T
9870	T

SOFTWARE

SOFTWARE INSTALLATION

Before installing the software, close all other applications. If you have any older versions of the software on your system, you do not need to uninstall the older version.

To install the software, place the installation CD in your CD-ROM drive. If the installation menu does not open automatically, click on **Start** at the bottom left of the screen, then click on **Run**. Type **D:\autoplay.exe** in the Command Line field of the Run dialog box (assuming "D" is the letter for your CD-Rom drive).

This will open the Installation Menu (shown at right).



Click the **Install** button to install the software on your computer. Use the other buttons to view or install other optional components of the software. It is highly recommended you install the documentation as well.

To Install the Software on a Network:

An additional licensing fee is associated with running the software on multiple computers linked to a network. Please contact **Tekscan** for more information at 1-(800) 248-3669 or via e-mail at marketing@tekscan.com.

Once the application has been installed, and the computer has been rebooted, the application can be run by double clicking on the icon on your desktop, or by clicking on the **Start** button at the bottom left of the screen, selecting **Programs**, then clicking the **Tekscan Software** icon (shown at right).

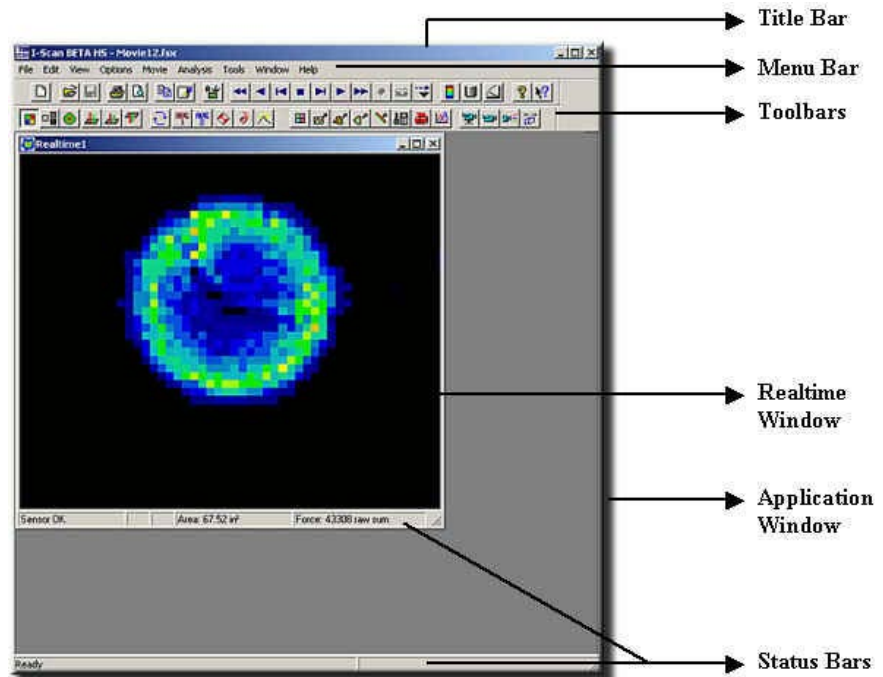


SOFTWARE COMPONENTS

I-Scan software is a 32-bit application. For software compatibility, see the [Hardware and Installation](#) section (specifically the [Computer Requirements](#) section). This section describes the many different components of the software, as well as the function of map files.

Main Window

When you initiate the **I-Scan** software program, the **Main Window** is displayed on your computer screen. The **Main Window** consists of the **Menu Bar**, **Toolbar**, and **Main Status Bar**, and may contain one or more **Real-time Window(s)**, **Movie**



Window(s), **Graph Window(s)**, and/or **Legend MENU BAR (s)**. The window that is selected is considered the active window. Only one window (Movie, Real-time, Graph, or Legend) can be active at one time, and the title bars of the other windows will be grayed out. The **Main Window**, and its components, are shown in the figure above.

Title Bars



Each window (*Main, Real-time, Movie, and Graph*) has a title bar along the top, which displays the name (title) of that window (e.g. 'Real-time 1' or 'Movie1.fsx'). If that window is paused, '(Paused)' is added to the title. If a recording is in progress, '(Rec)' is added to the title. Within brackets, the **Sensor Map** name is listed, then a colon, and then the **Sensitivity Adjustment** setting (found under **Tools -> Adjust Sensitivity**). Each title bar has the usual MS Windows control button at the left end and *minimize, maximize and close* buttons at the right end. Only one window may be active at one time; the active window's title bar will be blue, while the deactivated windows' title bars will be gray.

The following shows the various elements that make up the title bar: "Realtime1 (F:5051:Default)"

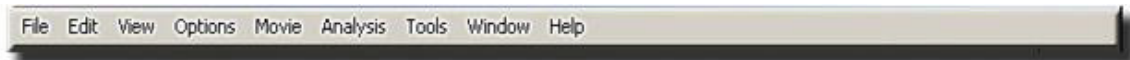
Realtime1 is the name of the movie window.

F:5051 is the name of the sensor map

Default is the Sensitivity Adjustment setting



Menu Bar



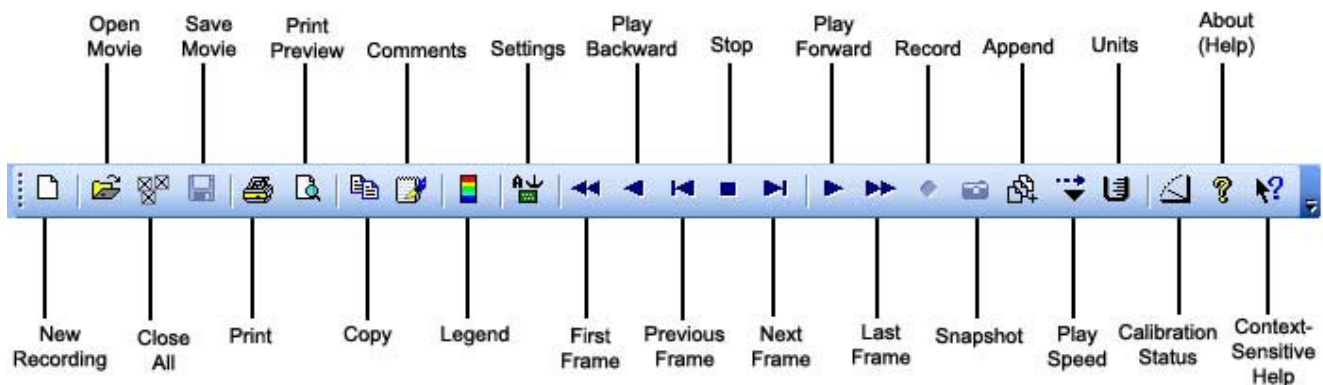
The **Menu Bar** provides the pull-down menus used to control the *I-Scan* program. Each *Menu Bar* option is described in detail in the 'Main Menu' section. When no Movie or Real-time windows are open, the *Menu Bar* contains only the *File, Options, and Help* menus. When a window is open, the other menus become available (see figure below). *Note that not all Menu Bar functions are available at all times. When a function is unavailable, it will be grayed out.*

Toolbars

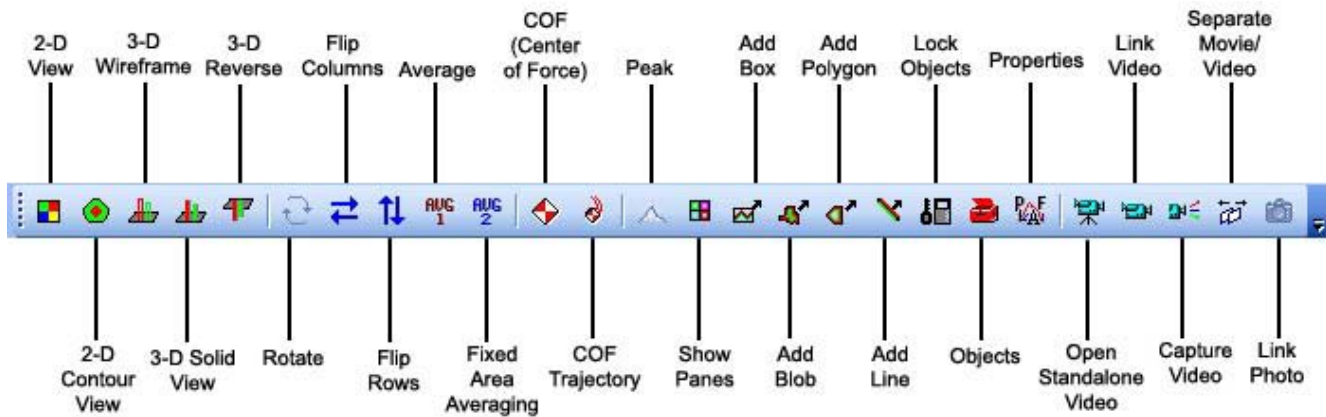
Most *I-Scan* functions can be performed using the **Toolbars**. The *Toolbars* provide the same options as the *Menu Bar*, but with icons replacing pull-down menus. Each icon on the Toolbars have a corresponding pull-down item in the Menu Bar; however, only the most frequently used items on the Menu Bar have an icon on the Toolbar.

Note that not all Toolbar functions are available at all times. When a function is unavailable, it will be grayed out. Some functions may also be optional software features.

The Upper (Main) Toolbar

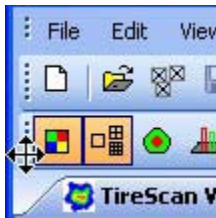


The Lower (Advanced) Toolbar

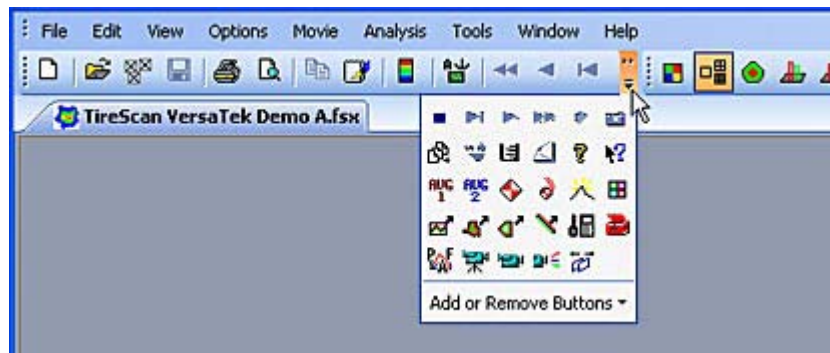


The upper (Main) Toolbar and lower (Advanced) Toolbar are “docked” to the top side of the application window above the [Tab Bar](#) and below the [Menu Bar](#) by default.

The Main and Advanced Toolbars can be moved independently of each other. For example, if you want to have a single toolbar, grab the bottom toolbar by the dotted line on the left side. Note that the cursor turns into a four-way arrow (shown below left). Then move it to the right of the toolbar above (shown below right).

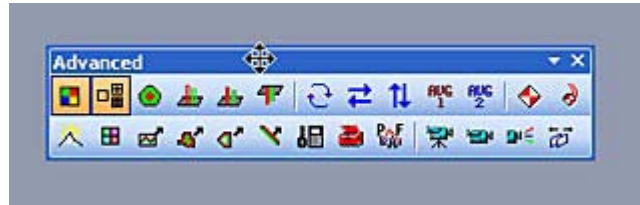


If, as in the image above left, you do not have enough space to display all the toolbar icons (due to the width of the application window being too narrow), a small drop-down arrow is displayed to the right side of the toolbar. Click this arrow and the remaining icons open on-screen (shown below). You can then select the icon of your choice.



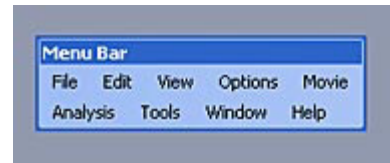
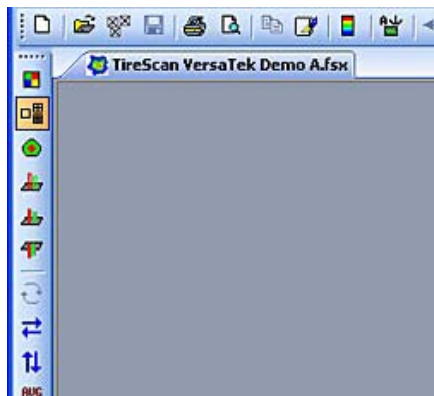
To “undock” a Toolbar, move it away from the top side of the application window towards the center of the application window. The Toolbar will gain a Title Bar, and can then be moved around the screen by grabbing this Title Bar. You can also turn the single row Toolbar into a double-row Toolbar. To do this, hover your mouse over the left or right side of the Toolbar.

When you see the cursor turn into a two-way arrow (shown below left), click and drag the side inward until it becomes two rows of icons (shown below right). To turn the Toolbar into a single row of icons again, reverse this process, pulling the side of the Toolbar outward until it becomes a single row.



To dock the Toolbar to a different side of the window, move it close to the side you want until the Toolbar “snaps” into place (shown below left).

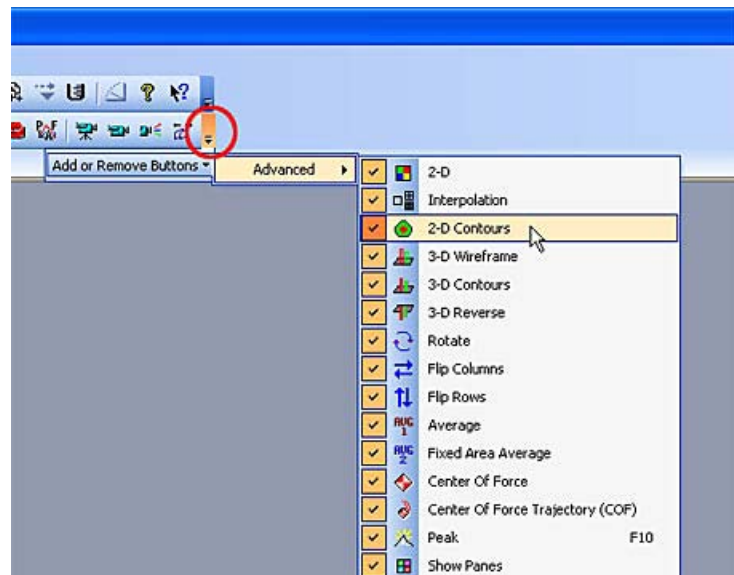
Note: The Menu Bar can also be moved, sized, docked, and undocked from the application in the same way as the Toolbars (shown below right).



Showing/Hiding Icons on the Toolbar

A new feature with the Toolbars is the ability to show/hide icons on the Toolbars. This gives you the freedom to tailor the Toolbar to your needs. If you never use an icon, you can simply hide it. If you always use a command or icon, you can keep it displayed.

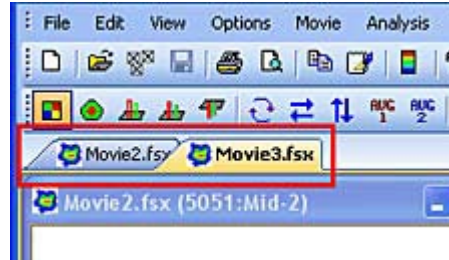
To show or hide an icon, click on the drop-down arrow located to the right of the Toolbar. Then hover and hold your mouse over the “Add or Remove Buttons” command. This opens the “Advanced” command. Slide your mouse over this command and all the icons located on the “Advanced” toolbar are displayed in a menu (shown at right). By default, they are all checked (displayed). Click on any icon in the menu in order to hide it (remove it from the Toolbar). Repeat this process to remove any further icons.



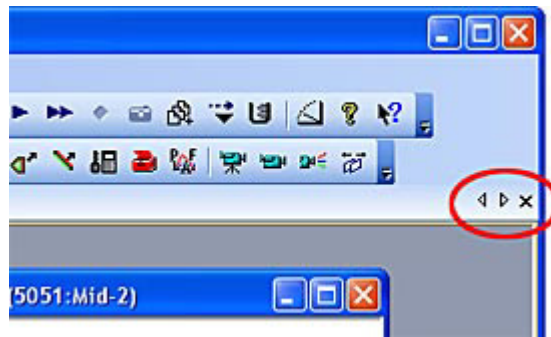
To add the icon back to the toolbar, go back into the Advanced menu of icons, and click the icon once again to check it. The icon is now displayed on the Toolbar.

Tab Bar

Directly under the Toolbars is the **Tab Bar**. Each separate window (Real-Time, Movie, or Graph) has its own associated tab. When a window has focus (its Title bar is highlighted), the tab moves to the front of the tab stack and the tab's font is boldened. You can make a window active (and give it focus) by clicking the Window's Tab. In addition, each tab will have its own color. Note that the **Farm Tread 7100.fsx** movie is located at the top of the layer stack in the image above and below.



On the right side of the Tab Bar, there are three icons. The left and right arrows are used in situations where the tabs cannot fit in the width of space provided along the Tab Bar. If you have more tabs than can be accommodated along this Tab Bar, use the arrows to move the Tab Bar left or right and scroll through the tabs. The "x" icon is used to close the currently active movie (the movie that is highlighted and has focus).



Main Status Bar

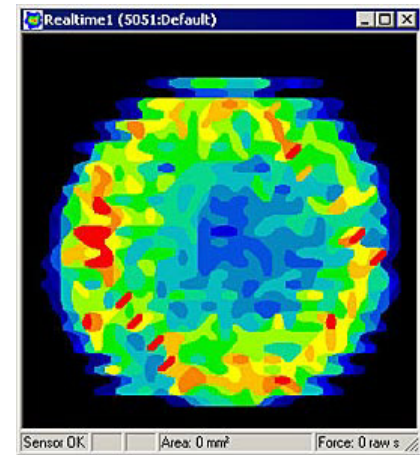
The **Main Status Bar** provides information about what is happening on your screen at any given moment. When the cursor is over a Toolbar icon,



the left side of the status bar will give a description of that icon; otherwise, it will say 'Ready'. When the cursor is over a Real-time or Movie window, the right side of the status bar will give the cursor position ('Row, Col:'), and the applied load ('Load') on the sensor at that point. Note that the cursor position corresponds to the *actual* row and column on the sensor (the origin, or zero point, is the top left corner of the window). You will see pressures in finer detail here in comparison to the ranges of pressure shown by the color bands on the display. The Load (pressure) shown may be affected by the display mode, such as *Averaging*, as well as the Legend's lower limit. Pressure values below this limit are disregarded, thereby changing the total Load value.

Real-time Window

The **Real-time window** displays the color-coded pressure information. This pressure data is displayed as it is happening, or in real-time. If no pressure is on the sensor, a blank *Real-time* window will appear. A *real-time* window will be opened when the software is initiated, and selecting File>>New Recording or New Session can open a different window. The number of *real-time* windows that may be opened is dependent on the number of handles connected to the system (one for each handle). The *I-Scan* software provides numerous options for displaying, recording, and analyzing the *real-time* data, which are described later.



Real-time Status Bar is located at the bottom of the real-time window, and displays the *sensor status* (or the *frame count* of a recording in progress), *Area*, and *Force*. The left side of this status bar displays the sensor status; either 'Sensor OK' or 'MISALIGNED!' If this message is 'MISALIGNED!' remove and reinsert the sensor into the handle, ensuring that the sensor is right side up, and the handle latch is completely down. When the real-time window data is being recorded, the left side of the status bar displays the 'Rec. Frame', or the *frame count* of the recording in progress.

Clicking the left-most section of this status bar is another way to **Pause** the screen (see the 'View' menu). When the indicator is popped out, it means that the screen is 'Paused' (see figure above). The center of this status bar will display a red and white icon to warn you if the Legend's lower limit has been changed (refer to 'Legend' section).



Area and **Force** are reported at the right side of the *Real-time Status Bar*. The Area shown is the calculated sensor loading area, and the **Force** is the total force on the sensor, as displayed in the window. The Area that is displayed here may be affected by changing the display mode, such as **Averaging**, or by raising the Legend's lower limit.

Note: *Pressure values below the Legend's lower limit are disregarded, thereby changing the displayed Area and Force values.*

Movie Window

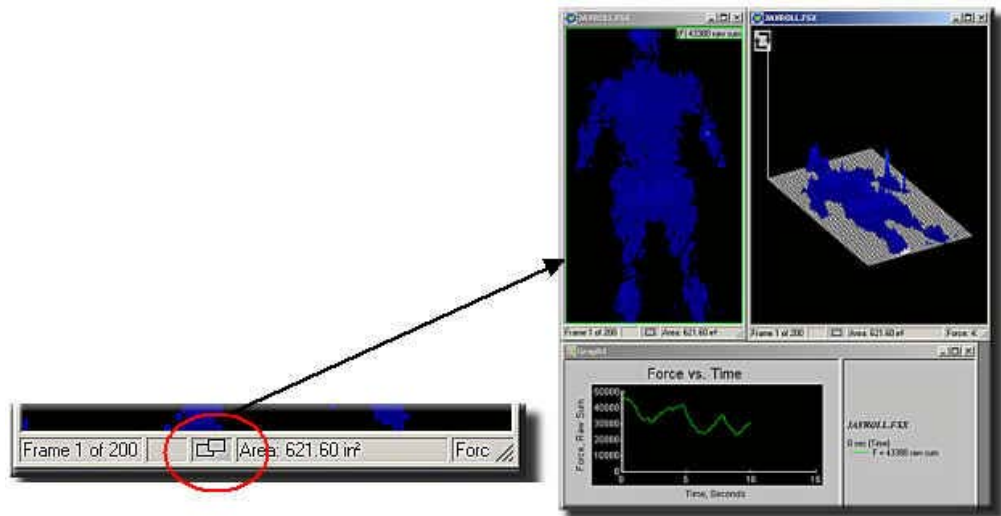
The **Movie Window** displays a previously recorded movie. Movie files can be opened by clicking the *Open Movie* icon in the Toolbar, or by selecting **File>>Open Movie**. Only *I-Scan* recordings, with the extension *.fsx, can be opened. The *I-Scan* software provides many options for displaying and analyzing recordings, which are described in later sections.

Movie Status Bar is located at the bottom of the *Movie Window*, and displays the *Frame count*, *Area*, and *Force*. The left side of the status bar displays the *frame count*, which is the current frame number vs. the total number in the movie. Clicking the left-most section of this status bar is another way to **Pause** the screen (see the 'View' menu). When the indicator is popped out, it means that the movie is 'Paused'. Right clicking this area and entering a frame number allows the user to jump to the desired frame number in the movie. The center of this status bar will display a red and white icon to warn you if the lower limit on the Legend has been changed (refer to 'Legend' section).

Area and **Force** are reported at the right side of the *Movie Status Bar*. The Area shown is the calculated sensor loading area, and the **Force** is the total force on the sensor, as displayed in the window. The Area that is displayed here may be affected by changing the display mode, such as **Averaging**, or by raising the Legend's lower limit.

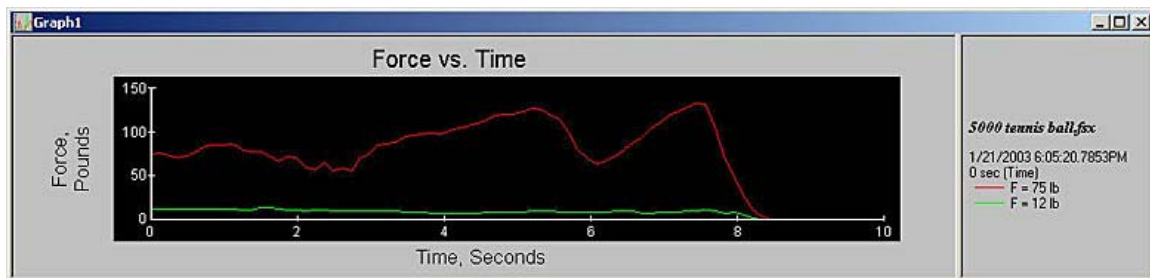
Note: Pressure values below the Legend's lower limit are disregarded, thereby changing the displayed Area and Force values.

In addition, there is a **Multiple Window View** button located in the middle of the status bar. Clicking this area opens 3 windows at once: a 2-D View, 3-D View, and Force vs. Time graph. Note that this is only available when working with a previously saved movie, and is not available within a real-time window.



Graph Window

The **Graph Window** displays the graph data from one or more Movie or Real-time windows. A *Graph* window is opened when an object (tiles, box, or line) is placed in a Movie or Real-time window. The color-coded graph trace for that object will be displayed in the corresponding **graph**. The display attributes of each Graph are defined in the **Analysis>> Properties** menu.

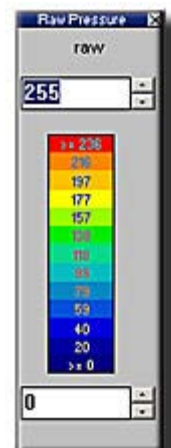


To the right of the *Graph* in the window, the names of the corresponding Movie or Real-time windows are displayed, followed by the actual data value totals. If the information on the right side of the Graph window is not fully visible, this area can be expanded by dragging (with the cursor) the line that separates it from the actual graph. Refer to the 'Analyzing Pressure Data' section for more information on the **I-Scan** graphing features.

Note: The Legend was referred to as the 'Pressure Scale' in the MS-DOS -based I-Scan software and documentation.

Legend

The Legend (shown at right) displays the pressure range that corresponds to each of the 13 possible colors in any open Movie or Real-time window. The number shown on each color in the Legend is the lowest value in that color's range. The values in the fields at the top and bottom of the legend are the upper and lower limits, respectively. The pressure units (or raw) are also displayed at the top of the legend. Selecting **Set Legend** from the "Options" pull-down menu, or clicking the corresponding Toolbar icon can place a legend in the Main Window.



A calibrated legend can be opened if the corresponding Movie or Real-time window has been calibrated; otherwise, only a raw legend may be opened. If it is a Raw legend, no pressure units will be associated with each range, and the default upper limit will be 255. All calibrated movies that are open simultaneously will be controlled by one calibrated legend. Likewise, all uncalibrated (raw) movies are controlled by one raw legend.

Increasing or decreasing the upper limit value can change the pressure range represented by each color in the legend. This limit can be changed (one increment at a time) by clicking the up or down arrow to the right of the number, or by typing in a new number. Decreasing this number will make the range of each color band smaller (finer), and increasing the number will make the range of each band larger. This upper limit does not, however, limit the high end of pressures displayed. All pressures above the maximum value will be displayed as red, but their values are still reported as Load in the Main Status Bar.

Note: Saturated sensels are always displayed as red, regardless of the Legend's upper limit setting. However, unsaturated cells may be displayed as red based on this setting.

The legend's lower limit can be changed in the same manner as the upper limit. Increasing this limit from 0 creates a threshold; sensed pressures below the value are disregarded, and will not be visible in the window as colors. In this way, this feature can be used to suppress the display of noise (usually caused by the presence of EMI or due to prolonged sensor use) that is indicated as low-level pressure.

Note that pressures that are not displayed in the window are also not included in the Area, Force, or Load values reported in the Main, Movie, and Real-time Status Bars.

If the Legend's lower limit is raised, the icon to the left is added to the status bar(s) of any affected Movie and/or real-time windows, to remind you that this limit has been changed.

Note: Changing the Legend's upper and lower limit values has no effect on the actual data; it merely alters the appearance of the displayed data.

Tips:

- The upper and lower limits for the last calibrated and raw legend are saved when exiting ***I-Scan***, and are restored when you restart the application.
- If the numbers in the colored boxes are interspersed with blank colored boxes, then the legend limits are adjusted too low. Increase the upper limit or decrease the lower limit until all boxes have numbers.
- If colored boxes with no numbers appear next to each other in the top of the legend (except the red box), then the upper limit is adjusted too high. Decrease the upper limit until a good color distribution is obtained.
- You can easily identify saturated cells by increasing the upper limit until there is a blank colored box below the red box. When there is at least one blank box below the red box, all red cells in the display are saturated, meaning they have reached their maximum output, and can no longer measure any changes in pressure (i.e. further increases in force on saturated sensels do not increase the reported value).
- If you click the right mouse button while the cursor is over a Legend, the following additional options become available:
 - **Move** - a four-way arrow appears, and the arrow keys (on your keyboard) can then be used to move the legend in the window. Simply selecting it with the cursor and dragging it to another location in the window can also move the legend.
 - **Close** - removes the legend from the window.
 - **Units** - displays the Units of Measure dialog box.
 - **Copy** - the legend will be copied to the MS Windows clipboard. From the clipboard, the legend can be pasted into other Windows applications as an image (bitmap) by selecting **Edit>>Paste** or **Paste Special**.

Map Files

The software uses a file called a map to convert the pressure detected by the hardware into the pressure data displayed in the real-time window. The map filename is the same as the sensor number (printed on actual sensor). Each sensor has a map associated with it. Without the map file, the software cannot understand information from the sensor. When the hardware and software are properly installed, a sensor is properly inserted into the handle, and the correct map file (with extension *.mp) is selected, the real-time window will display the pressure detected by the sensor.

Note that: A Real-time window will not open unless the hardware is properly connected.

Multi-Handle Map File Feature

When using a Tekscan system with multiple handles, multiple real-time windows or “virtual”, multi-sensor software maps, Tekscan software will retain the link between the sensor type, the map and calibration data for each handle. This greatly enhances the utility and ease of use of “Virtual” or Multi-sensor maps and multiple real-time windows. Each real-time and movie window is labeled with the handle letter and sensor map. Separate equilibration and calibration information will be maintained throughout the recording session for each handle.

I-Scan File Extensions

File extensions are placed at the end of a file name, to differentiate one type of file from another. **I-Scan** uses the following file extensions to designate various files: recordings (movies) sensor maps:

.fsx	recordings (movies)
.mp	sensor maps
.asc	ASCII data files (Center of Force)
.asf	ASCII data files (frames)
.asg	ASCII data files (graphs)
.fbx	object files
.cal	calibration files
.equ	equilibration files

Note: Each file type is described in more detail in the appropriate section.

Keyboard Shortcuts

Instead of using a mouse to control the software, certain keys on the keyboard may be used for a variety of "shortcut" command access. These keys are defined below:

Note: Depending on your software, some functions may be unavailable. As a result, their equivalent shortcut key will not be available.

<i>Shortcut Key</i>	<i>Function</i>	<i>Shortcut Key</i>	<i>Function</i>
F1	Help	F2 or Ctrl+R	Record
F3 or Ctrl+H	Snapshot	F4 or Ctrl+T	Play / Stop
F5 or Ctrl+F	Play Forward	F6	New Client
F7	Walk Mode	F8	Balance Mode
F9 or Ctrl+P	Print	F10	Enable / Disable Peak View
F11	Show / Hide Percentage Lines	F12	Start Over
Ctrl+B	Play Reverse	Ctrl+C	Copy
Ctrl+E	Edit Dialog	Ctrl+M	Edit Comments
Ctrl+N	New Recording	Ctrl+O	Open Movie

Ctrl+S	Save	Ctrl+V or Left Arrow	Play Previous Frame
Ctrl+X or Right Arrow	Play Next Frame	Ctrl+F8	SAM Mode
Ctrl+F9	Print Preview	Ctrl+F11	Show / Hide Object Lines
Ctrl+F12	Exit Program	Shift+F1	Context Help (Help On)
Up Arrow	Jump to First Frame	Down Arrow	Jump to Last Frame

When [Editing a movie](#), there are additional shortcut keys that are available. To access these keys, an edit box or edit cell must first be placed in the movie window and have focus (the edit box or edit cell must be selected). Then you can use the following shortcut keys to navigate the edit box or edit cell:

<i>Shortcut Key</i>	<i>Function</i>
Left Arrow	Moves the edit box or edit cell one cell to the left
Right Arrow	Moves the edit box or edit cell one cell to the right
Up Arrow	Moves the edit box or edit cell one cell upward
Down Arrow	Moves the edit box or edit cell one cell downward
Ctrl+Left Arrow	Shifts the right side of the edit box one cell to the left (making the box smaller)
Ctrl+Right Arrow	Shifts the right side of the edit box one cell to the right (making the box larger)
Ctrl+Up Arrow	Shifts the bottom side of the edit box one cell upward (making the box smaller)
Ctrl+Down Arrow	Shifts the bottom side of the edit box one cell downward (making the box larger)

The Main Menu

The **Main Menu** section is a comprehensive look at each option in the **Menu Bar**. The most frequently used items in the Main Menu also have an icon on the Toolbar. *Familiarity with MS Windows is assumed, so menu items that have the same function in standard Windows applications are not described in this manual.*

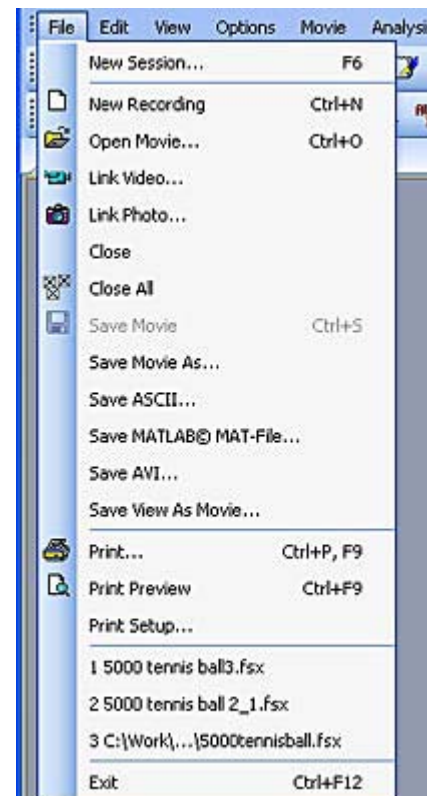
Note: Not all menu items are available all of the time. When an option is unavailable, it will appear grayed out.

File Menu

New Session: When beginning a new session, selecting this option will close all open windows (asking the user whether to save unsaved data), open a new real-time window, and reset all display options, as well as the sensitivity, calibration, and equilibration data.

Note: If you close the application, then restart the program and open a Real-time window, the display options (e.g. 3-D, COF, zoom %, rotation) and the sensitivity setting will be recalled.

New Recording: Opens a new real-time window, in addition to any other real-time or movie windows that is currently open. The new real-time window becomes the active window when it is opened. When New Recording is selected, the window size, display options (e.g. 3-D, COF, zoom %, rotation), sensitivity, and equilibration and calibration data, are recalled from the previous real-time window.



Open Movie: Displays the 'Open' dialog box, which allows you to open previously recorded *I-Scan* recordings or snapshots (with the *.fsx file extension). You may also open this dialog box by clicking on the Open Movie icon in the Main Toolbar. To open multiple files, hold down the <CTRL> key and click to highlight the desired files. All selected files will be opened when you click 'OK.'

Link Video (Optional): Opens a video and links the movie and video together. When 'Play' is selected, the movie and video will play simultaneously. These files are not synchronized.

Link Photo: The **Link Photo** feature allows you to open a photo file and link it to any frame of a Tekscan (*.fsx) movie. Since the photo is linked to a frame within the movie, it becomes possible to associate one or several photos to a single movie. The amount of photos is dependant only on the number of frames available for the movie. Conversely, you can only associate one photo per frame. You cannot have multiple photos linked to a single frame. The following image file formats are supported: *.bmp;*.jpg;*.gif;*.tif;*.pcx;*.tga. Refer to the [Linking a Photo to a Movie Frame](#) section for further information.

Close: Closes the currently active movie or real-time window.

Close All: Closes all open Movie, Real-time and Graph windows. You will be prompted to save any windows that have not been saved.

Save Movie: Saves the current recording as a file, with the specific file name and location of your choice. *I-Scan* movie files must have the extension *.fsx. Note that your system may limit file names to eight characters.

Save Movie As: Saves the current movie, with the specific file name and location of your choice. This menu item is also available when a movie window is active, and can be used to save a movie under more than one file name. *I-Scan* movie files must have the extension *.fsx. Note that your system may limit file names to eight characters.

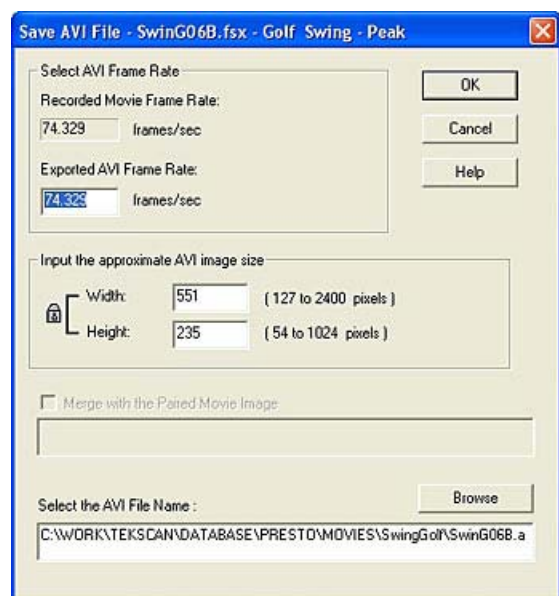
Warning! *If you open a DOS-version I-Scan movie file, then 'Save Movie As' the same file name, it will overwrite the older file. You will then be unable to open this movie using the DOS software.*

Save ASCII: Saves the current *I-Scan* movie or real-time window pressure (or center of force) data, or the current Graph data, as an ASCII (text) file, with the specific filename and location of your choice. The 'Objects' dialog box also contains a 'Save ASCII' button, which can be used to save graph data ASCII files. Frame data ASCII files must be saved with the extension *.asf, center of force (COF) ASCII files must be saved with the extension *.asc, and graph data must be saved with the extension *.asg. Refer to the 'Analyzing a Recording' section for more information.

Note: *Window (Movie, Real-time, or Graph) backgrounds are always printed out as white, regardless of whether or not View>>Background White is selected.*

Save MATLAB® Mat File: Saves the .fsx file to a MATLAB .mat file. For more information see the Help File "Appendix: Save MATLAB®" section.

Save AVI (optional feature): Allows you to export the currently selected movie into a digital video format (*.avi) that can be saved on your computer. When exporting the movie to a video, the following dialog opens (shown below left), where you can select the exported video frame rate and file name. By default, the movie is saved into the same folder where the Tekscan movie (*.fsx) is located. For more information, refer to the [Saving an AVI \(File Menu – Optional\)](#) section.



Save View as Movie: Allows you to save the current movie frame, or view, as a movie file (with extension *.fsx). This new movie file will consist of one frame, and will be saved exactly as it appeared when saved. If any View menu items are in effect when Save View as Movie is selected, they will be retained in the saved file. Note that COF Trajectory will not be displayed by the saved view, since it only affects movies with more than one frame.

Print: Allows you to print the currently active Real-time or Movie window to an MS Windows-compatible printer. The window will be printed exactly as it appears on the screen, and will include the options selected in Print Setup.

Print Preview: Shows you exactly how your printed page will appear. It is recommended that you preview all printed pages to ensure that your setup is correct. Refer to the 'Printing' section for more information.

Print Setup: Gives you a number of printing options that control what will be printed, and how this printed page will look. Refer to the 'Printing' section for more information.

Exit: Closes the *I-Scan* program.

Edit Menu



Copy: A powerful tool that allows you to analyze pressure data in a spreadsheet or word processing program, compare *I-Scan* movie frames, and create reports or documents using actual color movie frames. This command saves the currently active real-time window, movie frame, or Graph, to the MS Windows clipboard as both an image and as a text array of the actual pressure values.

Once copied to the clipboard, this data can be pasted into other Windows applications as either a graphic (bitmap) or text by selecting **Edit>>Paste Special**. If **Edit>>Paste** is selected, the window's sensor pressure data, as well as the comments, fields, and header information, will be automatically pasted in ASCII (text) format.

Clicking the right mouse button while the cursor is in a Real-time, Movie, or Graph window, and selecting Copy can also select the Copy option. Refer to the 'Analyzing Pressure Data' section for more information on the Copy menu item, as well as other available Copy options.

Note: For further information on how to export information from Tekscan software into other software applications, please refer to the [Copy & Export Options](#) section.

Comments: Opens up a dialog box that allows you to view, add, and edit the comments associated with a real-time or movie window. Real-time window comments are remembered until **File>>New Session** is selected or the software is restarted. When a real-time window is saved as a movie, the comments are included with the movie file. The comments for the movie can then only be changed by selecting Comments, changing the comments, and resaving the movie file.

If a real-time window is active, **Comments** also allows you to add and edit the comment field names associated with that window. If the field names are changed and saved in a newly recorded movie, any later recordings will automatically be updated to these field names.

The Comments option can also be selected by clicking the right mouse button while the cursor is in a real-time or movie window, and selecting **Comments**, or by accessing the 'Comments' tab in the Settings dialog box. Refer to the 'Including Comments with a Recording' section for more information.

Edit Mode: displays the 'Edit' dialog box, which enables you to modify a specific area of data within a movie or real-time window. **Edit Mode** allows you to place an edit box or edit cell in the active window, and then modify the sensels within this box. An edit box or cell can be used to remove data from one or more sensels, or to average sensel reading(s) with those of its neighbors, in either a movie or a real-time window. Refer to the 'Editing' section for more information on using and manipulating edit boxes and cells.

Cut Frames: opens the 'Cut Frames' dialog box, which enables you to discard unwanted frames from a recording using a number of methods. The three options in the 'Cut Method' pull-down field are 'Cut Frames', 'Keep Frames', and 'Filtered'. 'Cut Frames' allows you to discard specific frames, 'Keep Frames' allows you to keep specific frames, and 'Filtered' allows you to discard frames whose total applied force is less than a specified minimum value.

Load Edit File: enables you to load an existing edit file, with the extension *.fed, into the currently active movie or real-time window. An edit file is simply the saved edit boxes and/or cells from a previously edited window.

Save Edit File: enables you to save the edit box(s) and/or cell(s) from the currently active movie or real-time window as an edit file, with the extension *.fed. These edit boxes/cells can then be placed into a movie or real-time window by loading the edit file. Refer to the 'Editing' section for more information on the Edit Mode, Cut Frames, and Load/Save.

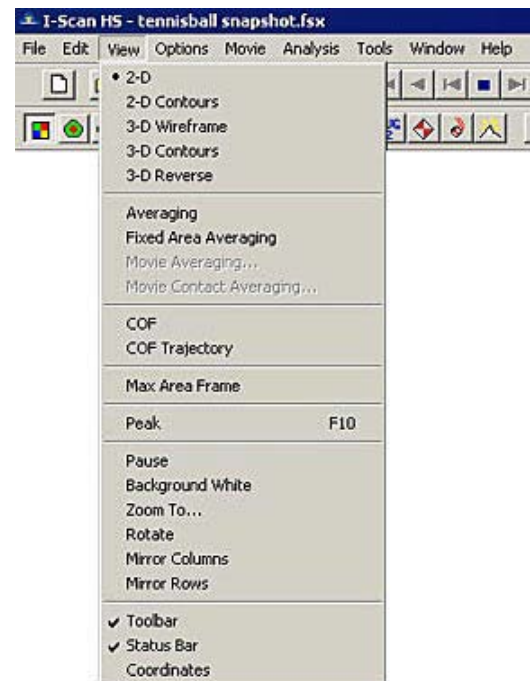
View Menu

The View menu items and corresponding Toolbar icons control the primary *I-Scan* display functions for the active movie or real-time window. When one of the 2-D or 3-D items is enabled, it will have a dot beside it in the pull-down menu, and its Toolbar icon will be depressed. Note that the sensor data does not change when changing modes; only the appearance of the display is altered.

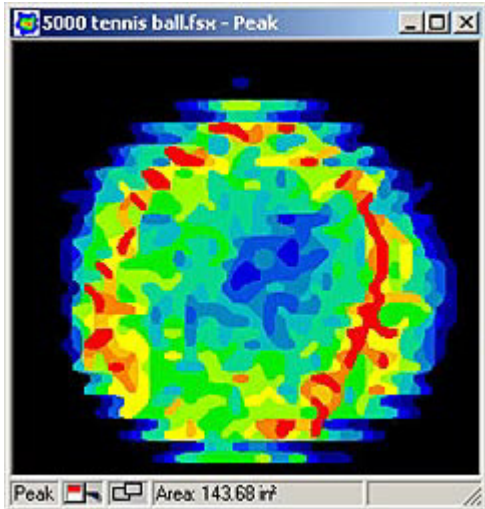
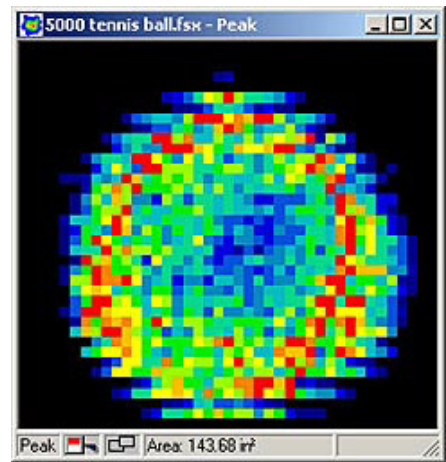
Notes:

1. When any of these View options is selected, it affects all open windows. To apply the selected option to only the active window, press the <CTRL> key and select the option simultaneously.

2. Any View menu items that are in effect are reset to the default when File>>New Session is selected. They are saved when the program is closed and reopened, or File>>New Recording is selected.



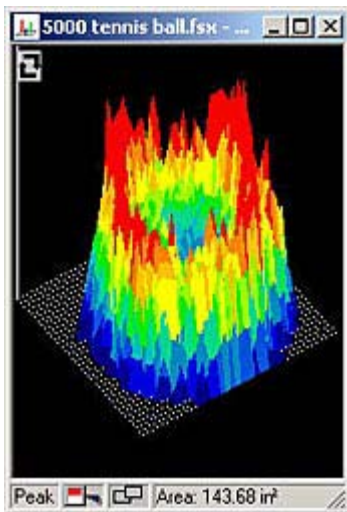
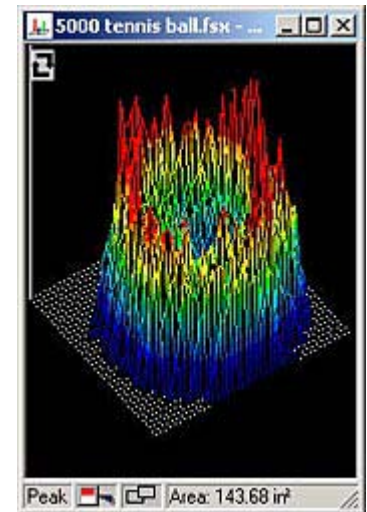
2-D: Displays the recording in two-dimensional form with the color of each sensel location representing the pressure sensed at that point (see image at right). This display looks the closest to the actual raw output of the sensor, and individual sensels can be identified. **2-D** is the default view.



2-D Contours: Displays the recording as a two-dimensional, contoured image. The sharp corners of the sensor output are smoothed, making the pressure boundaries easier to discern (see image at left).

3-D Wireframe: Displays the recording as a three-dimensional image, in which higher pressures are shown as peaks (shown at right). The relative peaks

provide visual insight into the gradients within each color band. 'Wireframe' means that the image is not a solid image, and the image is represented by a series of interconnected lines. This display can often provide a better view of complex 3-D data.

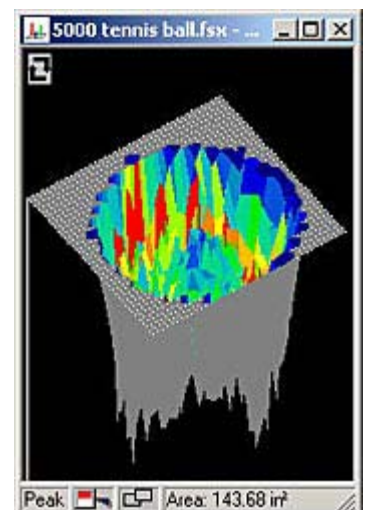


3-D Contours: Displays the recording as a three-dimensional image, in which higher pressures are shown as solid peaks (shown at left). The 3-D Contour image will look similar to the **3-D Wireframe** image, except that the image will be represented as a solid object. This display can provide a view of complex data that is easier to comprehend than the **3-D Wireframe** image.

3-D Reverse: Displays the recording as a three-dimensional image, in which higher pressures are shown as inverse peaks (shown at right). The **3-D**

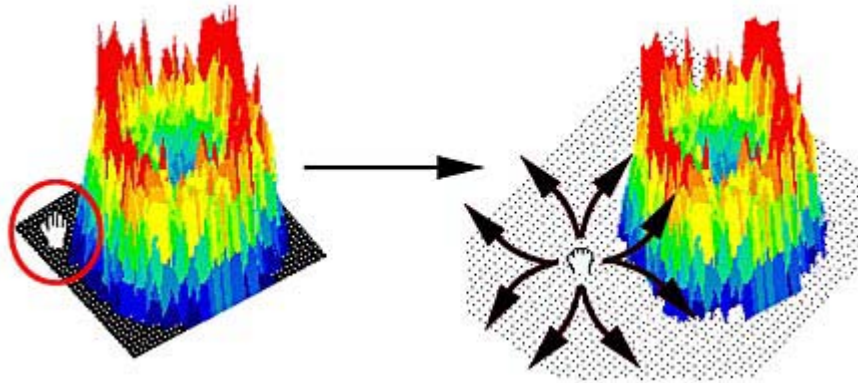
Reverse image is simply an inverted, or reversed, **3-D Contours image**. **3-D Reverse** is especially useful for viewing the exact footprint of the applied force.

The 3-D displays are especially useful in comparing pressure areas that are above the maximum pressure setting on the Legend. In the 2-D displays, all pressures above the maximum pressure setting are shown as red, and cannot be distinguished from each other; whereas in 3-D Wireframe, Contour, and Reverse modes, pressures are displayed as peaks, and relative pressure differences are readily apparent.



When you are in a "3-D" view, a separate "rotate" function is available. Moving your cursor over the screen, you will notice the cursor changes to a "hand" icon. Left-clicking your mouse will grab the 3-D image and you can then rotate the display in the direction of your choice by navigating with your mouse.

Note that the bottom plane of the image is not available, and rotating the image past this point will flip the image to show the next available plane. The arrow keys on the keyboard can also be used to shift the image incrementally in any direction.



Averaging (Avg1): Averaging displays the image with each cell's pressure value modified to reflect the value of its neighbors. It does include neighboring cells that have zero load. This results in a smoother image, with an increase in total loading area since certain cells with a previous zero value now have a value. When **Averaging** is in effect, it has a check mark beside it in the pull-down menu, and its Toolbar icon is depressed. As an example, in the group of nine cells shown below, the averaged pressure value of X is calculated using the following equation:

A	B	C
D	X	E
F	G	H

$$X_{avg} := \frac{A + C + F + H}{2} + B + D + E + G + X$$

Where the load recorded in each cell (A-G) is used to calculate the averaged pressure in cell X. Note that if cell X is at the edge of the sensor, the values of the neighboring cells that are not loaded are not used in the calculation.

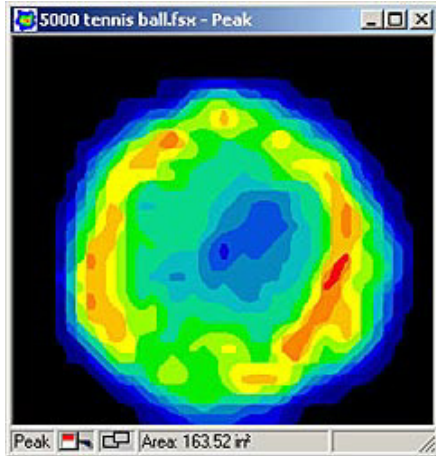
Fixed Area Averaging (Avg2): This option displays each cell's pressure value modified to reflect the value of its neighbors. It does not include neighboring cells that have a zero load value. There is no increase in the total contact area in the display. The result is a smoother image. When **Fixed Area Averaging** is in effect, it will have a check mark beside it in the pull-down menu, and its Tool bar icon will be depressed. As an example, in the group of nine cells shown below, the averaged pressure value of X is calculated using the following equation:

A	B	C
D	X	E
F	G	H

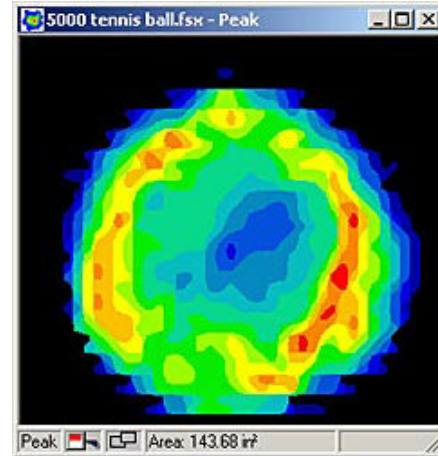
$$X_{avg} := \frac{A + C + F + H}{2} + B + D + E + G + X + X$$

The denominator is reduced to reflect the "weight" of the cells with zero load in the equation. For example, if cell A is zero, the denominator is reduced by half. If D is zero, the denominator is reduced by one. If X is zero, the denominator is reduced by two.

Averaging

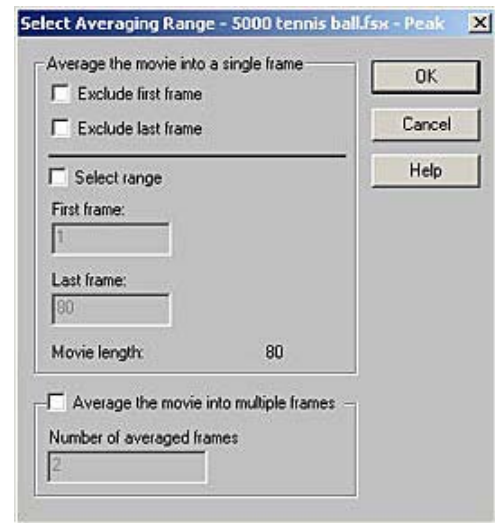


Fixed Area Averaging:



Movie Averaging: Displays the averaged pressure value of each cell for the entire recording, or for a selected range of frames, in one or more composite frame(s). When a movie has been averaged into a single frame, the Movie Status Bar will read 'Averaged Frame' (followed by the range of frames that were averaged); when a movie has been averaged into multiple frames, the Movie Status Bar will reflect the updated number of frames. This menu item is not available in real-time mode, and will not affect a snapshot, since a snapshot contains only one frame.

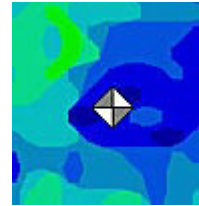
When Movie Averaging is in effect, it will have a check mark beside it in the pull-down menu and the movie playback options will be unavailable. Refer to the 'Analyzing Pressure Data' section for more detailed information. Movie Averaging applies only to the active window, and is not affected by the <CTRL> key.



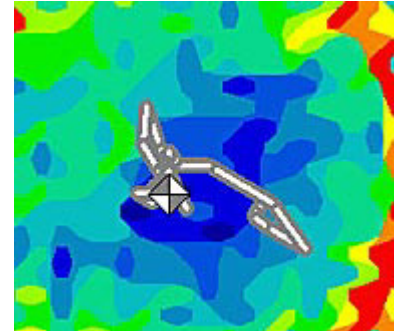
Movie Contact Averaging: Displays the averaged pressure value of each cell for the entire recording, or for a selected range of frames, in one or more composite frame(s). This menu item is very similar to the Movie Averaging option, except that it calculates the cell's average pressure value in a slightly different manner. The difference between these two types of movie averaging is that cells with zero load are disregarded in this option, whereas they are factored into the formula with a zero value in the Movie Averaging option.

When a movie has been averaged into a single frame, the Movie Status Bar will read 'Averaged Frame' (followed by the range of frames that were averaged); when a movie has been averaged into multiple frames, the Movie Status Bar will reflect the updated number of frames. This menu item is not available in real-time mode, and will not affect a snapshot, since a snapshot contains only one frame. When **Movie Contact Averaging** is in effect, it will have a check mark beside it in the pull-down menu and the movie playback options will be unavailable. Refer to the 'Analyzing Pressure Data' section for more detailed information. Movie Contact Averaging applies only to the active window, and is not affected by the <CTRL> key.

COF (Center of Force): Shows the center of all of the forces in the Movie or Real-time window. This feature is used to show how the forces are 'balanced' on the sensor. The center point is represented on the screen by a gray version of the red and white icon that represents **COF** on the Toolbar. You can view the **COF** for each separate box and tile in the Movie or Real-time window, in addition to the **COF** for the entire window. **COF** is only available in the 2-D display modes. **COF** is 'toggled' on and off; when it is enabled, it will have a check mark beside it in the pull-down menu, and its Toolbar icon will be depressed. Refer to the 'Analyzing Pressure Data' section for more information.



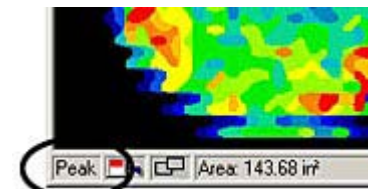
COF (Center of Force) Trajectory: Displays the movement of the Center of Force for the duration of the recorded movie. The movement of the COF can be tracked by playing a movie one frame at a time, with COF Trajectory selected. The trajectory is represented on the screen by a gray and white line that 'trails' the COF marker. In a real-time window, **COF Trajectory** simply shows the center of force. If **Peak** is also selected, the trajectory of the COF for the entire movie will be shown, and the playback controls will be unavailable.



As an optional feature, you can view the COF Trajectory for each separate box and tile in a Movie window, in addition to the trajectory for the entire window.

Max Area Frame: Advances the current movie to the frame that has the largest contact area. The contact area is displayed in the center of the Movie Status Bar. Note that only pressures above the legend's lower limit (threshold) are displayed, and accounted for in the contact area. Refer to the 'Analyzing Pressure Data' section for more details.

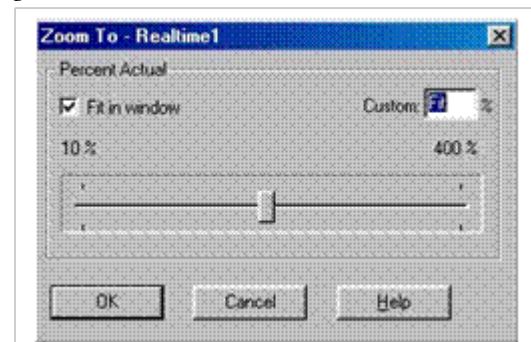
Peak: Displays the maximum pressure value that each cell reached during the recording, in one composite frame. The movie playback options are not available when this item is selected. **Peak** is available in all 2-D and 3-D display modes. **Peak** will not affect a snapshot since a snapshot contains only one frame. If **Peak** is in effect while COF Trajectory is enabled, the trajectory of the COF for the entire range of frames is displayed. **Peak** is toggled on and off; when it is enabled, it will have a check mark beside it in the pull-down menu, its Toolbar icon will be depressed, and the Movie Status Bar will say '**Peak**'.



Pause: Causes the active movie or real-time window to freeze, or to ignore all movie playback or recording commands. Pausing a real-time window causes the '**Sensor OK**' Status Bar pane to pop out, and pausing a Movie window causes the Frame Count indicator Status Bar pane to pop out. Clicking on the '**Frame Count**' or '**Sensor OK**' pane will toggle Pause on and off. When a movie or real-time window is paused, a check mark is placed beside it in the pull-down menu, and '**(Paused)**' is added to the title bar at the top of the window. Pause applies only to the active window, and is not affected by the <CTRL> key.

Background White: Changes the background areas (where no pressure is sensed) of the active Movie, Real-time, or Graph window from black to white. Changing the background color to black is useful when viewing lighter colors, such as yellow, in the display. This item can also be selected by clicking the right mouse button while the cursor is over a window, and selecting **Background White**. When **Background White** is in effect, it will have a check mark beside it in the pull-down menu.

Zoom To: Allows you to resize the display inside the active movie or real-time window. Percentages refer to the actual size of the sensor. Zoom percentages are available up to 400%, and any size up to this value can be entered in the 'Custom Fit' area. The default size (when 2-D or 2-D Contours is in effect) is **Fit in Window**. **Zoom To** applies to only the active window, and is not affected by the <CTRL> key.



Note: *The displayed image size is dependent on a number of factors, so when the image is zoomed to 100%, it may not be exactly the same size as the actual sensor.*

When you are in one of the 3-D views, an additional zooming feature becomes available. **I-Scan** allows you to change the vertical size of the 3-D view by adjusting the 'Percent Vertical' percentage between 10 and 100%, or by entering a custom value.

Rotate: Turns the entire view clockwise in 90 degrees increments from its current orientation. All 2-D and 3-D views are rotated clockwise. When a window is rotated, the zero point (which is initially in the upper left corner of each window) is also rotated. If you get confused at any point, put the cursor over a sensel and look at the coordinates displayed in the status bar to get the true row and column numbers.

When in 3-D Wireframe, Contour, or Reverse modes, a separate '**Rotate**' function is also available. Click anywhere on the screen with your cursor to rotate the display 5 degrees counterclockwise. The cursor (arrow) keys on the keyboard have the same effect as clicking the mouse cursor. Hold down the <SHIFT> key while clicking to get top-forward vertical rotation. Hold down the <CTRL> key with either of the previous functions to get the reverse effect.

Mirror Columns: mirrors the entire sensor across the rows.

Mirror Rows: mirrors the entire sensor across the columns.

Toolbar: Used to display or hide the Main Toolbar. The **Toolbar** can be toggled on and off by clicking this menu item. When the **Toolbar** option is in effect, it will have a check mark beside it in the pull-down menu. When the **Toolbar** is in effect, its two levels can be relocated (separately) in the Main Window by placing the cursor anywhere on one of them (not on an icon) and dragging. If you place the cursor on an edge of the Toolbar, a two- or four- way arrow will appear, and its shape and size can be changed. The <CTRL> key does not affect the **Toolbar** option.

Status Bar: Used to display or hide the Main, Movie, and Real-time Status Bars. These status bars can be toggled on and off by clicking this menu item. If a Graph is the active window, this menu item affects the Main Status Bar. If a Movie window is active, this menu item affects the Movie Status Bar. If a real-time window is active, this menu item affects the real-time Status Bar. When the **Status Bar** option is in effect, it will have a check mark beside it in the pull-down menu. The <CTRL> key does not affect the **Status Bar**.

Coordinates: Displays the X and Y coordinates in any open movie or real-time window. If the window is in a 3-D view mode, the sensor origin (0,0) point is marked by a vertical black or white line, and the X and Y-axes are not displayed.

For more information about how the View menu commands operate, see "Display Options" under the "Analyzing Pressure Data" section.

Options Menu



Set Legend: Places a Raw or Calibrated **Legend** in the Main Window. The **Legend** shows the pressure range that corresponds to each of the 13 possible colors in any open movie or real-time window.

A *calibrated* legend will be available if the current movie or real-time window has been calibrated; otherwise, only a raw legend will be available. All *calibrated* windows that are open simultaneously will be controlled by one calibrated legend. Likewise, all uncalibrated (raw) movies are controlled by one raw legend. Refer to the 'System Overview' section for more detailed information on using the Legend.

Note: *The Legend was referred to as the 'Pressure Scale' in the MS-DOS -based I-Scan software and documentation.*

Hardware Information: Displays a dialog box that shows which Tekscan hardware is attached to the system and how it is configured. This dialog box tells you which **Receiver** (*VersaTek, Evolution, USB, Parallel adapter, or Super Receiver*) is connected between the handle and the computer, and the number of **Handles** that are connected. The other information (**Address, IRQ Line, and DMA Channel**) describes the computer's internal setup, and is intended primarily for service personnel.

Acquisition Parameters: Displays the 'Data Acquisition Parameters' dialog box, which enables you to select the movie recording parameters. You may set the number of frames to record, and either the period (elapsed time between frames) or frequency (frames per second), but not both. The movie duration (in seconds) is calculated by the *I-Scan* software, based on the other values, and displayed below the frequency. If you would like the system to wait for a specific amount of time before recording, enter a value (in seconds) in the 'Delay Recording' field (under Start Options).

Note: *Allowable ranges for recording parameters are shown in parentheses.*

At the bottom of the dialog box is the '**Noise Threshold**' field, followed by its available range. This threshold is used to filter out unwanted low-level force readings (noise). The software sets measured forces equal to or less than the threshold value to zero.

To use the standard Internal Triggering feature to start recording when the sensor reaches specified pressure or area values, click the check box. The Triggering button to the right of the dialog box become available. Clicking this button displays the 'Triggering' dialog box, which allows you to 'trigger' the software to start and stop recording pressure data based on specified internal events. By default, the 'Enable Triggering' option is not checked, and triggering is not in effect. Refer to the 'Taking a Recording' section for more information on setting the '**Data Acquisition Parameters.**'

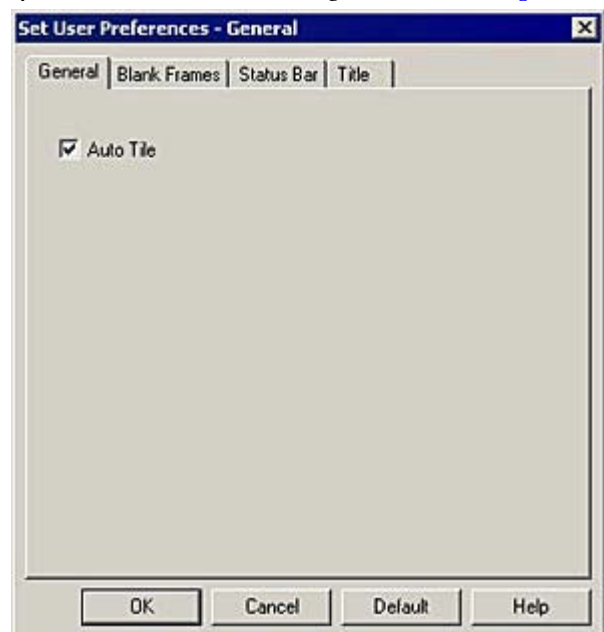
Tekscan offers an optional External Triggering/Synchronizing software package. This package allows the software to receive an external triggering event, as well as to generate a signal synchronized to the recording. Refer to the "[Optional Accessories](#)" section for more information on the External Trigger.

Set User Preferences: Allows you to customize certain operational features of the system according to your preferences.

You may enable/disable the following options:

General Tab:

Auto Tile: When Auto Tile is checked, and one or two view (real-time or movie) windows are open, the system will automatically tile all open windows (including graphs) when placing a graph. If more than two view windows (or graphs) are open, Auto Tiling has no effect.



Blank Frames Tab:

Auto Remove Blank Frames: If you select "From Beginning of Movie", any blank frames that are in the beginning of your recordings (ie: no force is measured) will be removed automatically. If you select "From End of Movie", any blank frames at the end of your movie will be removed. A frame is considered "blank" by the system if all of the forces on the sensor are below the "Noise Threshold" setting in Acquisition Parameters (Options menu). The frame count in the Movie Status Bar will reflect the change in the number of frames.

Status Bar Tab:

The options under this tab enable you to decide which information will be displayed in the Movie Status Bar. If "Frame Count" is selected, the standard frame count (e.g., "Frame 2 of 5") will be shown. If "Relative Time (Seconds)" is selected, the elapsed time of the movie will be shown (e.g. "Time 2.5 of 4.0 sec"). If "Absolute Date/Time" is selected, the date and time that the movie was recorded is displayed.

Title Tab:

Show in Title: The "Handle" information, "Sensor/Map" and "Sensitivity" can be displayed in the title bar of any Real-time or Movie windows. By default the title bars will display this information. Uncheck these boxes if you do not want this information displayed. By default, only the Handle option is enabled.

Measurement Units: Displays a dialog box, which allows you to choose the units that will be displayed for length, force, and pressure, and the number of decimal places that will be displayed for force and pressure. The default units are inches for Units of Length, pounds for Force, and PSI for Pressure. The default number of decimal places depends on the units that are selected.

When the cursor is over a Movie or Real-time window, the Main Status Bar displays the sensor row and column numbers that the cursor is currently above. The status bar also displays these coordinates in standard engineering units, such as inches or centimeters. This feature is disabled if 'cells' is selected as the Unit of Length in the dialog box. The units and decimal place settings are applied to all Movie and Real-time windows, and Graphs, regardless of which view is currently active. Once you change these settings, the software will remember them when you start a new session or close the application.

In the "Measurement Units" dialog box, you can also select the increment by which the upper and lower limits of the legend will be increased and/or decreased. The allowable increments are 1, 10, 100, 1000, and 10000. Note that if you select an increment other than 1, you may still manually enter any upper or lower limit into the legend limits.

Select Sensor: Displays the "Select Sensor" dialog box, which enables you to designate the sensor and map that are currently installed on your *I-Scan* system. The system needs this information in order to correctly display the data in the real-time window. The dialog box lists the available maps, the number of handles required for that specific map, the system's available handles, and the type of receiver (*VersaTek*, *Evolution*, *USB*, or *Super Receiver*) used with each specific handle. You may select any one of the maps that are listed, but make sure that it is the correct one for the sensor you are using (the map number is the same as the sensor number printed on the actual sensor).

Note: *Only one sensor map may be selected for each real-time window, but some maps require multiple handles. Refer to the 'System Overview' section for a description of map files.*

Select the correct handle(s) for the sensor you are using (a check mark will be placed next to your selection). Once the sensor map and handle(s) have been selected once, the software will set them as defaults, and the 'Select Sensor' dialog box won't be displayed at startup. If you change the type of sensor or the handle that you are using, make the appropriate changes using the **Select Sensor** option.

Settings: Displays the 'Settings' dialog box for the currently active movie or real-time window. This dialog box allows you to view the window's general information, comments, calibration data, and sensitivity setting, all in one place. You may also select 'Settings' by clicking the right mouse button with your cursor over a window, and clicking on 'Settings'. To access the desired information for the window, click on one of the following tabs in the dialog box.

General: The general information includes the sensor data, such as the sensor type and number of rows and columns, and the movie data, such as the noise threshold and the selected units.

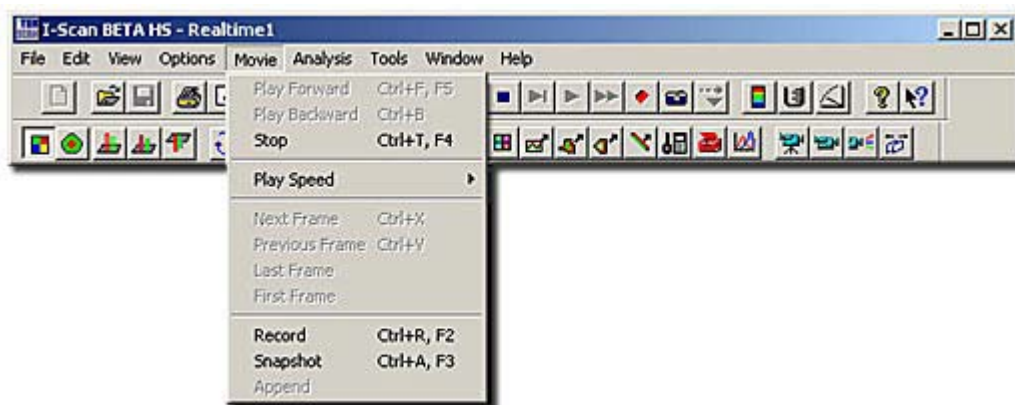
Comments: This page displays the comments associated with the current window. If you want to make any changes to these comments, click the 'Change Comments' button - the 'Comments' dialog box will be displayed. Any changes will be displayed immediately. Refer to the 'Including Comments with a Recording' section for more information.

Calibration: This page displays the calibration status and data for the current window. If the window is calibrated, the calibration data, such as the scale factor and saturation pressure, will be displayed. Depending on the calibration status of the window, you may perform a number of calibration functions using the four buttons at the bottom of this dialog box.

The 'Calibrate' button enables you to perform a calibration. The 'UnCalibrate' button allows you to remove the calibration data from the window. The 'Load' button allows you to load a calibration file into the window, and the 'Save' button allows you to save the window's calibration data as a file. These buttons will be grayed out if they are not available, based on the calibration status of the window. Refer to the 'Calibration and Equilibration' section for more information on calibration.

Sensitivity: This page displays the sensitivity setting of the current movie or real-time window. If you want to make a change to the sensitivity setting in a real-time window, click the 'Sensitivity' button - the 'Adjust Sensitivity' dialog box will be displayed. The 'Sensitivity' button will be unavailable (grayed out) when a Movie window is active. Any changes will be displayed immediately. Refer to the 'Calibration and Equilibration' section for more information on adjusting the sensitivity.

Movie Menu



The **Movie** menu items and corresponding Toolbar icons control the movie playback functions. The playback menu items and icons control all open movies in unison, regardless of which movie is active. Press the <CTRL> key simultaneously if you want the selected playback option to affect only the active window. Refer to the 'Taking a Recording' section for more information on these options.

Play Forward: Plays the recording forward, starting at the current frame.

Play Backward: Plays the recording in reverse, starting at the current frame. To enable '**Continuous Play**', press the <SHIFT> key, and simultaneously select **Play Forward** or **Play Backward**. The movie will play in the desired direction indefinitely, looping back to the first or last frame each time.

Stop: Stops either playback or recording. If a recording is in progress, Stop will cause the software to stop recording the active Real-time window, even if the selected number of movie frames has not yet been recorded. You can also stop recording by pressing the <F4> key. The <CTRL> key does not affect the **Stop** option.

Next Frame: Plays the recording forward one frame each time it is clicked.

Previous Frame: Plays the recording in reverse one frame each time it is clicked.

Last Frame: Positions the recording at the end (final frame).

First Frame: Positions the recording at the beginning (first frame).

Record: Used to make a recording (movie). When **Record** is selected, the software begins recording frames of sensor data as it appears in the current real-time window, using the **Acquisition Parameters** selected in the **Options** pull-down menu. You can also begin recording by pressing the <F2> key.

If Triggering is in effect, and **Record** is selected, recording will not begin until the start trigger event occurs. When data is being recorded, '(Rec)' is added to the real-time window title bar, and the window's status bar displays each frame number as it is recorded. Movies created by the **I-Scan** system are stored as*.fsx files.

Snapshot: Used to make a single-frame recording (movie). When **Snapshot** is selected, the software records a single frame of sensor data as it appears in the current real-time window. You can also take a snapshot by pressing the <F3> key. When data is being recorded, '(Rec)' is added to the real-time window title bar. The only **Acquisition Parameters** that apply to a Snapshot are '**Triggering**' and '**Noise Threshold**'. Movies created by the **I-Scan** system are stored as*.fsx files.

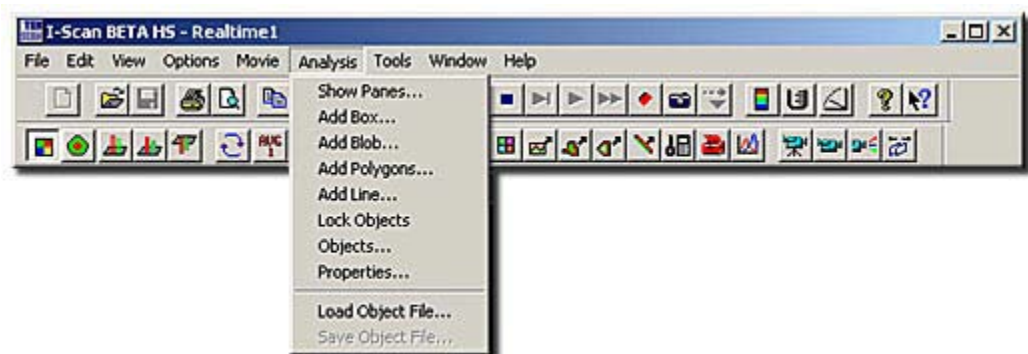
Append: Allows you to record additional frames of data at the end of an existing movie. The **Append** menu item is only available when a Movie window (recording) is open, and will be unavailable (grayed out) if a real-time window cannot be opened. The original recording may be one that was closed and reopened, or one that has just been recorded.

When you select **Append** (and choose your desired handle), a real-time window is opened (the title bar will say "**Append to ...**"). The frames of data you record in this real-time window will automatically be added to the end of the previous movie. Refer to the '**Taking a Recording**' section for more information on using Append mode.

Analysis Menu

The **Analysis** menu option provides all of the tools necessary to analyze the data displayed in the Playback (movie) and Real-time windows. Package A, an optional software add-on package, includes Add

Blob, Add Polygons and the Tracking box features. If you have the optional features enabled in your software, corresponding buttons will appear on your Toolbar. Refer to the '**Analyzing a Recording**' section for more information on each of these menu items.



Show Panes: Places a box, or 'pane', in each open Real-time or Movie window. To place a pane in only the active window, press the <CTRL> key while selecting the **Show Panes** option. Press the <SHIFT> key while selecting this option to add four panes (quadrants) to each window. The data from each of these four panes can then be displayed, analyzed, and graphed separately from the rest of the window's pressure data.

Add Box: Allows you to place a box on the active Playback or Real-time window. The data inside that box can then be displayed, graphed, copied and analyzed separately from the rest of the window's data. When **Add Box** is selected, a box symbol is added to the cursor, and clicking the cursor at the desired point in a movie or real-time window may then place a box. You can also add a box by pressing the <CTRL> key and clicking the cursor in an open window, or by clicking the right mouse button with the cursor over an active window, and selecting **Add Box**.

Add Line: Allows you to place a line on the active Playback or Real-time window, which can be used to measure the distance or pressure difference between two points. This line data can then be graphed separately from the rest of the window's data. When **Add Line** is selected, a line symbol is added to the cursor, and clicking the cursor at the desired point in a movie or real-time window may then place a line. You can also add a line by pressing the <SHIFT> key and clicking the cursor in an open window, or by clicking the right mouse button with the cursor over an active window, and selecting **Add Line**.

Add Blob (Optional): This feature enables you to study a group of loaded sensels separately from the rest of the window. See the "[Optional Accessories](#)" section for more information.

Add Polygons (Optional): This feature allows you to add a custom-shaped box to a real-time or movie window. See the "[Optional Accessories](#)" section for more information.

Tracking Box (Optional): This feature allows you to add a box to the movie or real-time window that will 'track,' or follow, the loaded sensels in the window. See the "[Optional Accessories](#)" section for more information.

Lock Objects: Locks the relative position of all graphing objects (lines, boxes, blobs or polygons), which have been placed in the movie window. When a window contains multiple graph objects, and the Lock Objects button is depressed, moving on object will also move all other objects in the window so that they retain the same position to the object that was moved.

Objects: Displays the '**Objects**' dialog box, which can be used to control each tile, box and line that is placed in a Real-time or Movie window. In this dialog box, objects (tiles, boxes or lines) can be added, deleted, changed, saved as an ASCII (text) data file, or saved/loaded as an '**Object file**'. You can also open this dialog box by clicking the right mouse button with the cursor over an active window or graph, and selecting **Objects**.

Properties: Opens the '**Properties**' dialog box, which controls the display attributes of all objects (boxes and lines) in an active real-time or movie window, or the X- and Y-axes display attributes of an active graph in the Main Window. You can also open this dialog box by clicking the right mouse button with the cursor over an active window or graph, and selecting **Properties**.

If a real-time or movie window is active when you select this option, tile and box options will be displayed in the upper ('Boxes') section of the dialog box, and line options will be displayed in the lower ('Lines') section. The selected data values are displayed in a small box in the top right corner of each box, or at the midpoint of each line.

If a Graph window is active when you select this option, the graph's Y-axis display options will be displayed in the upper ('Y-axis' and 'Y-mode') sections of the dialog box, and the X-Axis options will be displayed in the lower sections. **Load Object File** allows you to load previously saved objects (tiles, boxes, and lines) into the current movie or real-time window.

Load Object File: Can also be accessed by clicking the Load Object File button in the 'Objects' dialog box. When **Load Object File** is selected, an 'Open' dialog box is displayed, and 'Object Files' (with the extension *.fbx only) can be opened. When an object file is loaded into the current window, the objects will be placed at their saved coordinates. If the current window is not the same size as the original window, the software will still attempt to place the objects at the correct coordinates.

Save Object File: Allows you to save the objects (panes, boxes, and lines) from an active real-time or movie window as a file. Clicking the **Save Object File** button in the 'Objects' dialog box can also access this file. This option is necessary because, when a real-time window is saved as a movie file, the objects in the window are not saved as part of the file. When **Save Object File** is selected, a 'Save As' dialog box will be displayed, and the file can be saved as an 'Object File', with an *.fbx extension. Objects are saved exactly as they appear in the window, with the same properties and coordinates.

Tools Menu

The Tools menu items control the sensor calibration and equilibration functions, as well as the handle manual sensitivity adjustment. The Tools menu items do not have corresponding Toolbar icons. Refer to the 'Calibration and Equilibration' section for more specific information on each menu item.



Calibration: Opens the 'Calibration' dialog box, which enables you to calibrate the sensor using either a linear or a 2-point power law calibration. You can also access this dialog box by clicking on the 'Calibration' tab in the Settings dialog box. A calibration can be performed either before or after a recording has been taken (known as a 'Movie (Frame) Calibration'). *I-Scan* also allows you to calibrate a "virtual" or "multi-region" sensor, either as a single window or as separate 'tiles'. You can perform both linear and 2-point power law calibrations, for a real-time or movie window, for this virtual sensor.

From the 'Calibration' dialog box, you can add, edit, or delete calibration points, as well as manually adjust the handle sensitivity and change the system's current measurement units. Once a calibration is performed, the dialog box displays a graphic representation of the calculated calibration curve, based on the points you have entered. The resulting calibration can be saved as a file by selecting **Save Calibration**.

Dynamic Calibration I-Scan: Enables you to perform a dynamic calibration, which is a calibration performed while recording real-time data. Select Dynamic calibration from the **Tools** pull-down menu. This will open the **Dynamic Calibration** dialog box.

You may enter one or two force values, depending on whether you would like to perform a 1- or 2- point dynamic calibration. Then click the Record button in the dialog box to begin recording data. As the system is recording, you must load the sensor, and then click the 'Force 1' button at the appropriate time. A calibration point will be added using the current frame data. If you wish to add a second calibration point, repeat these steps and click the 'Force 2' button. Once the dynamic calibration is complete, a message will be displayed to tell you which frame(s) were used for the calibration.

UnCalibrate: Simply removes the sensor's calibration data. When this item is selected, a dialog box appears, giving the option of canceling the UnCalibrate operation. UnCalibrate cannot be used to uncalibrate a Movie window, and will only be available when a calibrated Real-time window is open. If you want to recalibrate the sensor during the same testing session, you may either use UnCalibrate to reset the calibration of the current window, or select **File>>New Session** to get a new Real-time window.

Load Calibration File: Allows you to use the calibration data from a previously saved file for your current Movie or Real-time window. When this item is selected, an 'Open' dialog box appears, and you can specify the file name and destination (path) of the calibration file you want to load. Calibration data from calibration files (with the extension *.cal) or from *I-Scan* movies (with the extension *.fsx) may be loaded.

Save Calibration File: Allows you to save your current calibration data (in a movie or real-time window) as a file for future use. When this item is selected, a 'Save As' dialog box appears, and the calibration file can be saved with the specific file name and destination (path) of your choice. *I-Scan* calibration files must be saved with the extension *.cal.

When a recording is saved, the calibration data used to make the recording is saved with the movie file. If you calibrate a sensor and want to save the data separately for later use, however, you must explicitly save it as a calibration file.

Note: The calibration file saves only the calibration data associated with the sensor, and does not include the equilibration data (if an equilibration was performed).

Equilibration: Opens the 'Equilibration' dialog box, which enables you to equilibrate the sensor, and to save or load equilibration data as a file. *I-Scan* also allows you to equilibrate a "virtual" or "multi-region" sensor, either as a single window or as separate 'tiles'. You can create multiple (up to 10) equilibration points for a sensor (or for a 'tile'), which may be necessary when you want to equilibrate the sensor at a number of different pressures.

Once an equilibration has been performed, a graphic representation of the pressures in the real-time window (in grayscale) is displayed in the dialog box. This is intended to show the relative high and low spots of the sensor, and the low levels of compensation necessary to ensure uniform results. The resulting equilibration can be saved as a file by selecting **Save Equilibration**. If you wish to remove the equilibration data during the same testing session, you may either select **File>>New Session**, or select **UnEquilibrate** to reset the equilibration of the current real-time window.

UnEquilibrate: discards the current real-time window's equilibration data. When this item is selected, a dialog box appears, giving the option of canceling the UnEquilibrate operation. This option may be used if you make a mistake in performing your initial equilibration, or if you want to perform another equilibration without exiting the *I-Scan* program.

Load Equilibration File: allows you to use the equilibration data from a previously saved file for your current real-time window. When this item is selected, an 'Open' dialog box appears, and you can specify the file name and destination (path) of the equilibration file you want to load. You can also access this option by clicking on the 'Load Eq. File' button in the 'Equilibration' dialog box. Only *I-Scan* equilibration files (with the extension *.equ) may be loaded.

Save Equilibration File: Allows you to save your current equilibration data (in a Real-time window) as a file for future use. When this item is selected, a 'Save As' dialog box appears, and the equilibration file can be saved with the specific file name and destination (path) of your choice. You can also access this option by clicking on the 'Save Eq. File' button in the 'Equilibration' dialog box. *I-Scan* equilibration files must be saved with the extension *.equ.

When a recording is saved, the equilibration data used to make the recording is saved with the movie file. If you equilibrate a sensor and want to save the data for later use, however, you must explicitly save it as an equilibration file.

Note: The equilibration file saves only the equilibration data associated with the sensor, and does not include the calibration data (if a calibration was performed).

Adjust Sensitivity: opens the dialog box, which enables you to select the handle sensitivity from 11 possible settings, ranging from Low-1 to Hi-4. If you have the option, you can “customize” the sensitivity setting by manually entering the gain values (Vref and Vtest). The sensitivity (also referred to as ‘adjustable gain’) adjusts the data output range from the computer (0 to 255 raw) to the actual force range output from the system handle. This dialog box may also be accessed by clicking the Sensitivity button in the ‘**Calibration**’ dialog box, or by clicking the ‘**Sensitivity**’ tab in the Settings dialog box.

Once you have made your selection, the real-time window will show the updated pressure data. When a recording (*.fsx) or calibration (*.cal) is saved as a file, the sensitivity setting is saved with the file.

Window Menu



Cascade: Use this command to arrange multiple opened windows in an overlapped fashion.

Tile (Vertically or Horizontally): Use this command to arrange multiple opened windows in a non-overlapped fashion, either vertically (side by side) or horizontally.

Arrange Icons: Use this command to arrange the icons for minimized windows at the bottom of the Main Window. The names of the currently open Real-time, Movie, and Graph windows are listed beneath the Arrange Icons option. A check mark appears next to the name of the active window.

Help Menu



Contents: Displays the ‘**Table of Contents**’ for the Help information.

Search: Displays an index of the available Help topics.

About *I-Scan*: Displays the About *I-Scan* dialog box, which gives some basic information about the Tekscan software.

Technical Support: provides information on how to contact Tekscan for technical assistance.

CALIBRATION AND EQUILIBRATION

I-Scan is a resistive-based technology. The application of a normal force to an active sensor results in changes in the resistance of each sensing element (sensel) in inverse proportion to the force applied. Calibration is the method by which this digital output is converted to an actual engineering unit, such as PSI or g/cm². Calibration enables us to compare the output of the same sensor in various environments and allows us to compare calibrated outputs of various sensors.

Note: For further guidelines and information about Equilibration and Calibration, please refer to the “Equilibration and Calibration Practical Suggestions” addendum found in your system software Help File under the “Help” menu.

Some variation exists between individual elements on any given sensor. This variation is acceptable in the majority of applications. The **I-Scan** system provides a method, called Equilibration, by which this source of error can be minimized, if desired. Equilibration is accomplished by applying a highly uniform pressure across the individual sensing elements. Each element within the sensor should produce a uniform output. When this is not the case, the software determines a unique scale factor for that sensel to compensate for the slight variation.

SENSOR PERFORMANCE CHARACTERISTICS

Before effective use can be made of the Equilibration and Calibration features, an understanding of **I-Scan** sensors is important. While the equilibration and calibration functions of the software address the uniformity and average non-linearity of a sensor, many other sensor characteristics can influence the accuracy of the sensor measurements. In addition, the conditions under which the sensor is used can significantly affect its performance.

This section will discuss some of these characteristics. Where applicable, methods for generically addressing these sensor characteristics from a calibration or equilibration standpoint are presented.

Repeatability

Repeatability is the ability of a device to respond in the same way to a repeatedly applied stimulus. It is necessary to ‘exercise’ a measurement device several times before calibrating it or using it for measurement. The same is true with **I-Scan** sensors. This is done to reduce the amount of change in the sensor response due to repeated loading and unloading.

New sensors, and sensors that have not been used for a length of time, must be loaded three to five times before they are used (called ‘conditioning’ the sensor). Refer to the “Conditioning Sensors” section for more information.

Important! It is essential that a sensor be properly ‘conditioned’ prior to use and calibration.

Linearity

One of the characteristics that the calibration feature addresses is the non-linearity inherent in the sensor’s (and system’s) response to an applied force or pressure. As with many of the characteristics of the sensor, the linearity is a function of the particular sensor in question. The design of the sensor, the sensitivity range, the interface material, and the loading method all influence the degree of non-linearity.

Uniformity

A typical **I-Scan** sensor can have more than 2200 individual sensing elements, or ‘sensels’. For many reasons, the individual response of these independent sensels may vary slightly. This uniformity can be measured by applying a known uniform pressure to the entire sensing area of the sensor, or a large portion of the sensing area. The software’s equilibration function can be used to compensate for the variations in individual sensel output.

Hysteresis

As with many measurement devices, ***I-Scan*** sensors experience hysteresis. Hysteresis is the difference in the sensor output response during loading and unloading, at the same applied force. For static forces and applications in which the force is only increased, and not decreased, the effects of hysteresis are minimal. If an application includes load decreases, as well as increases, there may be error introduced by hysteresis that is not accounted for by calibration and equilibration.

Drift

Drift is the change in sensor (and system) output when a constant force is applied over a period of time. Among other things, the sensor design, the sensor sensitivity, the interface material, the applied load, and environmental conditions may influence the drift. It is important to take drift into account when calibrating the sensor, so that its effects can be minimized. The simplest way to accomplish this is to perform the sensor calibration in a time frame similar to that which will be used in the application.

Temperature Sensitivity

The output of an ***I-Scan*** sensor will vary with temperature. Please refer to the data sheets for the actual temperature coefficient for your sensor type. To account for this thermal coefficient of the sensors, calibrate the sensor at the temperature at which it will be used in the application. If the sensor is being used at different temperatures, a calibration should be performed at each of the temperatures. The calibration associated with each temperature should then be saved as a calibration file (*.cal), and then loaded when using the sensor at that temperature (only).

Sensor Life / Durability

The durability of ***I-Scan*** sensors is generally very good, and in most cases, they are reusable. The actual life of a particular sensor depends on the application in which it is used. Under severe conditions, such as against hard surfaces, sharp edges, non-flat surfaces, sliding surfaces, etc., a sensor may have a very limited life. Rough handling of a sensor will also shorten its useful life. For example, a sensor that is repeatedly installed in a flanged joint will have a shorter life than a sensor installed in the same joint once and used to monitor loads over a prolonged period.

After each installation, visually inspect your sensors for physical damage. Damage to individual cells, rows, or columns do not affect the output of intact elements, and may be accounted for using the software's Editing feature. Malfunctioning sensor elements will induce biases during the calibration steps and should not be recalibrated.

Practical Sensor Loading Considerations

When calibrating and equilibrating a sensor, you should consider two main points: how the forces or pressures are applied to the sensor, and whether or not the load or pressure is being applied uniformly over the contact area of the sensor. These considerations are very important because the calibration of the sensor will only be as good as the known conditions that were used to calibrate it.

Many applications will require minimal attention to these considerations. However, you should be aware of them when designing your applications for the ***I-Scan*** system. Incorporate any considerations that apply to your specific application.

IMPORTANT! Please read the 'Sensor Performance Characteristics' section for information on how to get accurate results from your sensor readings.

Shimming Sensors

If the thickness of a sensor will change the load path in an application, a shim should be installed so that the load path is not altered. Typically, this is the situation when you want to measure the contact pressure between two flat, parallel rigid members whose mating surface is larger in extent than the active area of the sensor.

One aspect of this problem is that some sensors are thicker outside the sensing area than within it (many of the sensors with green dielectric have this trait). This can cause 'bridging' of the rigid member across the outer areas of the sensor. A shim of the same size and shape as the sensing area may be used to build up the sensing area to the same thickness as the surrounding sensor area. The other aspect of the problem is when the presence of the sensor alters the load path between the two rigid load members.

For example, suppose that you wish to measure the pressure between two flat, parallel rigid members. The applied force is 100 lbf and the contact area is 8 sq. in. The theoretical pressure is 100 divided by 8 = 12.5 PSI. Now place a sensor with an active sensing area of 1.5 sq. in. between these same two members. Since they are rigid and parallel, the 100 lb load is now carried entirely by the sensor. The sensed pressure is 100 divided by 1.5 = 66.7 PSI. The presence of the sensor has radically changed the load path and also the contact pressure.

The solution to this problem is to cover the entire load member area, except for the part covered by the sensor itself, with a shim of the same thickness and stiffness as the sensor. This effectively restores the member contact area to 8 sq. in.

Constructing Sensor Shims

It is best to use shim material of the same thickness as the sensor (If necessary, the sensor itself should also be shimmed to uniform thickness.). The standard *I-Scan* sensor thickness is 0.003 inches [0.076 mm]. Custom sensors may be thicker or thinner, depending on their design. Preferably, the shim material should also have stiffness similar to that of the sensor material (typically polyester or polyimide film). This assures that the sensor-shim layer has a relatively uniform stiffness.

To construct a shim, start by cutting a piece of shim material slightly larger than the entire load member contact surface. This is readily done using a razor knife or scissors. Then shape the shim to fit around the sensor by cutting away the area of the shim material into which the sensor will be placed. Also, cut out any through-holes necessary for assembly of the load members.

Now place the sensor in the shim cutout to form a uniform thickness sensor-shim layer that may be placed between the load members. Adhesive tape may be used to hold the shim and sensor together. Place the tape along the edge of the shim outside the contact area; otherwise, the tape thickness will affect the contact loading. Often an application requires using more than one sensor. In this case, all of the sensors may be 'spliced' into a single shim.

Conditioning Sensors

Sensors must be exercised by loading them three to five times before they are used (called '**conditioning**' the sensor). Conditioning helps to lessen the effects of Drift and Hysteresis, and is required for new sensors, and for sensors that have not been used for a length of time.

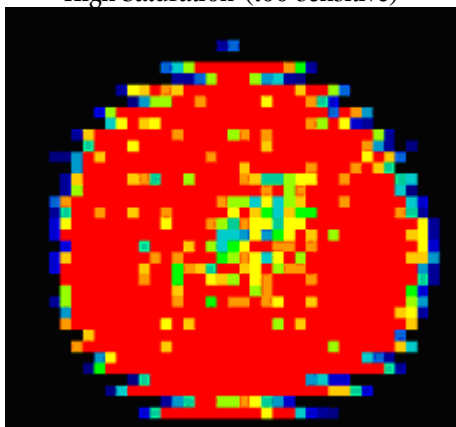
For best results, the sensor should be conditioned by loading it to pressures approximately 20% greater than expected during the actual application. The method of loading during this conditioning period should employ material of similar compliance to the actual material that will be used in the application.

Important! It is essential that a sensor be properly 'conditioned' prior to calibration, equilibration and use.

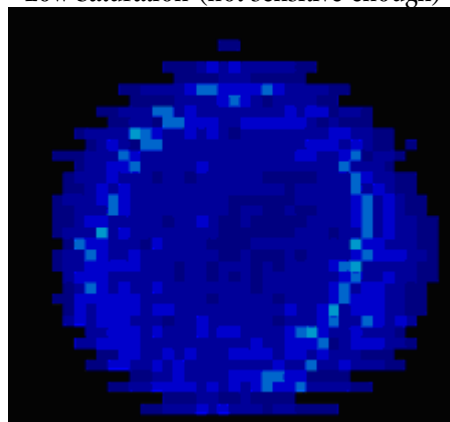
Saturation

The saturation pressure is the point at which the sensor output no longer varies with applied pressure. A sensel is saturated when its raw digital output is 255 or greater. These sensels are always displayed as red, regardless of the Legend's upper limit setting. If a significant number of sensels become saturated during calibration, equilibration, or testing, the calculated average digital output will be incorrect because increases in pressure on the saturated sensels yield no increase in the digital output of those cells.

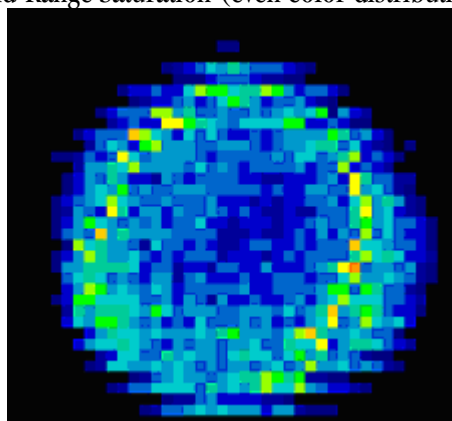
High Saturation (too sensitive)



Low Saturation (not sensitive enough)



Mid-Range Saturation (even color distribution)



If the sensor saturates at a lower pressure than desired, the sensitivity can be adjusted to increase the sensor's effective range. The saturation value is directly affected by the sensitivity setting; as the sensitivity is increased, the data output range decreases. Refer to the '**Sensitivity (Gain) Adjustment**' section for instructions.

Note: The sensitivity cannot be adjusted if you have the PC Interface Board (SR-2) hardware configuration.

Compliance

Compliant materials tend to "cushion" high points of contact, and "fill in" low points of contact. It acts as a special filter. In general, the effect of inserting compliant material (such as a rubber sheet) between two hard surfaces (such as two metal plates) is to lower the peak pressure areas, increase pressure in low pressure areas, and increase the contact area.

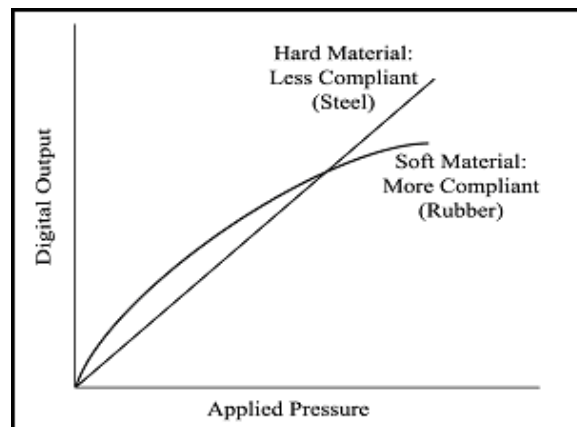
At a “micro level” there is a big difference between two hard materials coming together and a hard material bearing on a soft one, or soft on soft. Two soft materials give more contact area within the active area. Two hard materials give less contact area within the active area. When two hard surfaces touch, there often are three tiny regions with very high contact pressures, and large regions with no contact. The statement from high school geometry can be seen in action: Three points determine a plane. Imagine the deflection of a steel surface compared to the deflection of foam rubber with the same load. You can experiment with different materials and the same load to quantify the difference in contact area.

To elaborate on this issue, compare a bathroom scale to an array sensor for determining a person’s weight. First, standing on a bathroom scale, the person weighs 200 pounds. The bathroom scale gives the same result whether the person is bare foot, in hiking boots, or hiking boots with gravel in the treads. The scale does not respond to pressure distribution.

Standing bare foot on a sensor array loads it with soft flesh that squirms outward with low pressure as weight is applied, increasing the contact area. There is moderate pressure under bones, but no high pressure points. With hiking boots, there is high pressure under the contact point of the lugs, and no contact pressure between the lugs. With gravel caught in the lugs, if the point of the gravel strikes a non-responsive area between sensels there might be no output. If the point of the gravel loads a sensel, it may saturate it. Saturated sensels contribute their limit to the total load, but no more, even when loaded two or ten times their limit.

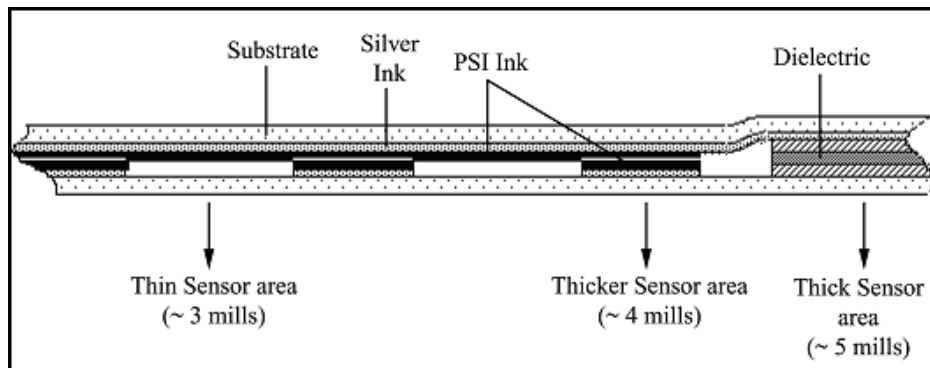
Another example of material compliance is placing clay on a window screen. The clay is soft, and therefore, very compliant. The points at which the screen grid meet and cross relates to the sensel grid, or sensing area. If you place the clay evenly on the screen with a low pressure, the clay will rest flat along the grid, and the crossing points support the load. The more pressure that is applied to the clay, the more the clay will seep through the screen (through the “empty” areas). Places where the clay extrudes through the screen relate to the non-sensing areas of the sensor. The empty areas are bearing some of the load. The relative change in output becomes smaller and smaller as more load is applied. The change in output diminishes with increasing load.

The difference in output between hard (less compliant) and soft (more compliant) materials:



The image below shows a greatly exaggerated view of a hypothetical cross section of a sensor through both the sensing area and the non-sensing area. The left portion of the sensor is the sensing area and the thicker portion to the right is outside the sensing area. The black squares in the middle area are the sensels, while the white areas in the middle are non-sensing areas.

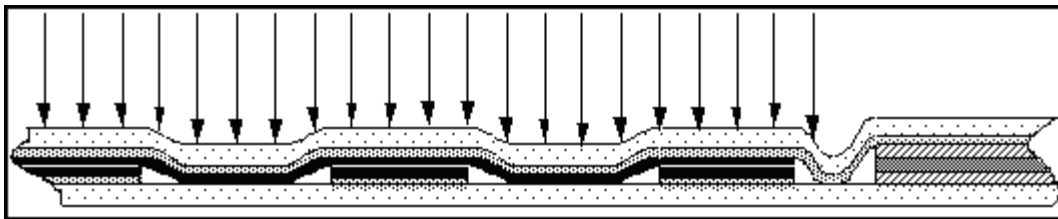
Edge View of Sensing Area:



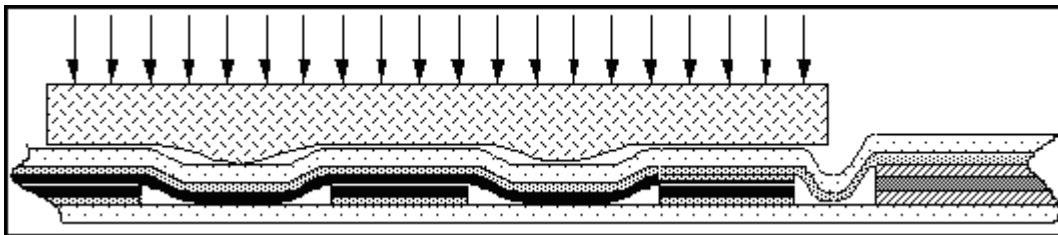
In general, for the section shown above, there are three basic stacking arrangements, in order of thickness. The thickest portion is the section that is outside the sensing area. The next thickest is the intersection points between rows and columns (where the sensels are located). The thinnest is the area between the intersection points in the sensing area (non-sensing area).

Note: some sensors' Dielectric Areas are thinner, which reduces the change in thickness along the sensor as a whole.

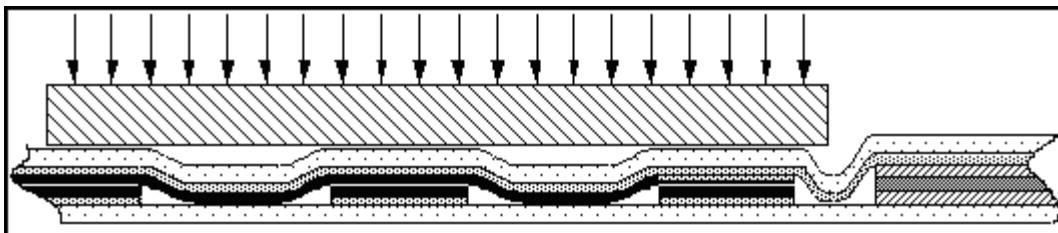
Sensor Loaded With Infinitely Compliant Hydrostatic Pressure:



Sensing Area Loaded With Moderately Compliant Material:



Sensor Loaded With Non-Compliant Material:



When loading the sensor, careful consideration needs to be given to the materials touching the sensor:

- When loading two hard, flat surfaces (e.g. two plates of steel), it is recommended to insert a sheet of paper, or thin urethane sheet between the two plates.
- When loading a part with high coefficient of friction and high squirm, such as rubber, onto the sensor, a thin sheet of Teflon paper works well to minimize shear forces, which can cause an output but cannot be quantified. Sensors are intended to measure compressive forces.

Cleaning the Sensor

Clean the sensor after each use by wiping it down with a damp rag or cloth with some alcohol. Store the sensors safely and flat (horizontal for large sensor pads) at room temperature in a protective case. Sensors that become wrinkled during storage may show artifacts during measurements. If the tab becomes wrinkled, continuity between handle pins and sensor pads may be lost.

In addition, it is important that the surfaces coming into contact with the sensor are cleaned thoroughly. Deposits and debris from these surfaces can cause damage or inaccurate readings from the sensor.

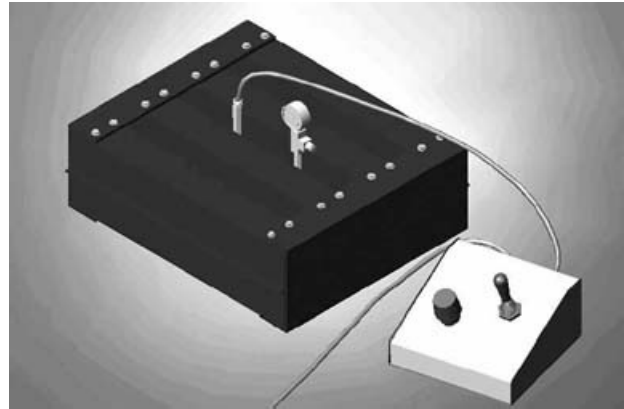
LOAD APPLICATION GUIDELINES

The following specific guidelines will help you obtain the best equilibration and calibration results possible, and are strongly recommended:

- Apply a calibration load that approximates the load to be applied during system use, using one of the following three types of devices:
 - **Dead weights** - Impractical for higher loads
 - **Comprehensive testing device** (such as an MTSTM or Instron™) - Essential for higher loads
 - **Uniform pressure applicator** - Suitable for lower pressures. The most common of these devices is an equilibration/calibration device, which inflates a rubber diaphragm against the sensor surface. Tekscan has developed equilibration/calibration devices with pressure ranges up to 500 PSI.

Note: A uniform pressure applicator is the only acceptable way to load the sensor for equilibration.

The image at right is an example of a Tekscan equilibrator (which contains an equilibration/calibration device):



- A material/sensor interface similar to that being tested should be employed during calibration. The influence of the surface compliance of mating surfaces has an effect on the response of the sensor.
- When applying the load for calibration or equilibration, cover as much of the sensor area as possible (the minimum loaded area is 25%). The software does not consider the sensor loaded (and will not start calibrating/ equilibrating) unless at least 1% (3% for equilibration) of the sensels are loaded.
- When applying the load for equilibration, cover the entire sensor area, if possible. The software assumes that the same pressure is applied to each loaded sensel, and any unloaded sensels will not be equilibrated.
- If you exceed the target load for calibration, you must remove and reapply the load.
- Avoid loading a large area of the sensor to near saturation when calibrating or equilibrating. If the sensor saturates at a lower pressure than desired, adjust the sensitivity.

- Make certain that no sensels become saturated (raw digital output of 255) during calibration or equilibration. If there are a significant number of saturated sensels, the calculated average digital output will be incorrect; increases in pressure on the saturated sensels yields no increase in the digital output of those cells.
- To avoid saturating sensels, it is important to clean the surfaces that are in contact with the sensor during the loading process. For example, scale deposits, corrosion or large pieces of dirt will cause the sensels in those areas to be loaded more highly than other areas. In many cases, these point loads will carry a significant portion of the applied force and will saturate at very low applied forces.

Method of Loading During Calibration and Equilibration

There are two types of loading that may be employed when calibrating and equilibrating a sensor: “Uniform Pressure Loading” and “Non-Uniform Applied Force Loading.” Uniform pressure loading is the only acceptable way to load a sensor for equilibration, and is the preferred method for calibration. However, this method of loading is limited by the user’s ability to apply a uniform load, and is often not practicable. In these cases, non-uniform loading must be utilized.

Note: Regardless of which method of loading is used, best results are achieved by properly conditioning the sensor prior to calibration/equilibration, and accounting for Hysteresis, Drift and Temperature Sensitivity.

Non-Uniform Applied Force Loading

This loading method does not require the user to apply an absolutely uniform force distribution. This method merely requires that a known force be applied to the sensor in as even a manner as possible. For calibration, the software calculates an average applied pressure based on the area of loaded sensels and the force value entered by the user. This means that care should be taken to ensure the loaded area of the sensor remains constant for all calibration points. It is also important that at least 25% of the sensels be loaded when using this loading method, to ensure that the average applied force calculations are representative of an average sensel.

Uniform Pressure Loading

This loading method requires that the user apply a known uniform pressure to the sensor. This is most easily done (for lower pressures) using an inflatable equilibration/calibration device, which consists of a thin, flexible membrane that is pressurized against the sensor (Tekscan has developed equilibration/calibration devices with pressure ranges up to 500 PSI). For higher pressures, you may have no means of safely and practically applying a known uniform pressure to the sensor. In these cases, reasonably flat and parallel platens can be used to load the sensor to ensure a reasonably even distribution of force over an area. When using an equilibration/calibration device tester to apply a uniform pressure, there are many things to consider in loading the sensor properly. Some of these are listed below:

- **Equilibration/calibration device Material Selection:** The type of material used for an equilibration/calibration device will have an impact on how the sensor is loaded. Most equilibration/calibration device materials are thin, flexible and compliant. If the application the sensor is to be used for has mating surfaces that are less compliant, non-uniform applied force loading is the preferred method for calibration. Some equilibration/calibration device-type materials can possess some structural stiffness, which will influence the actual pressure being applied to the sensor, especially at lower pressures.
- **Entrapped Air:** If the entire sensing area of the sensor is loaded using a equilibration/calibration device-type fixture, the entrapped air between the two layers of the sensor will act as an air bearing. This means that the entrapped air is supporting the applied load and the sensels are not being loaded properly.

There are two methods that can be used to avoid the entrapped air problem. One method is to not load the entire sensing area of the sensor with the equilibration/calibration device. By leaving a sizable area of the sensor unloaded by the equilibration/calibration device, the entrapped air has a place to go during the loading process. The second method involves cutting slits in the sensing area of the sensor (between rows and columns so no leads are touched) so the entrapped air can escape during loading of the sensor. To do this, follow the steps below:

1. Place the sensor on a flat table, and with a razor, cut slightly into each end of the sensor. A small slit, 1-2 mm. in length, is sufficient.
2. Place a very small amount of talcum on the sensor to prevent it from sticking to the equilibration/calibration device fixture.
3. Remove any excess, as it could manifest itself as a peak load area.
4. Run your finger along the sensor to force the air out of the slits you have made.

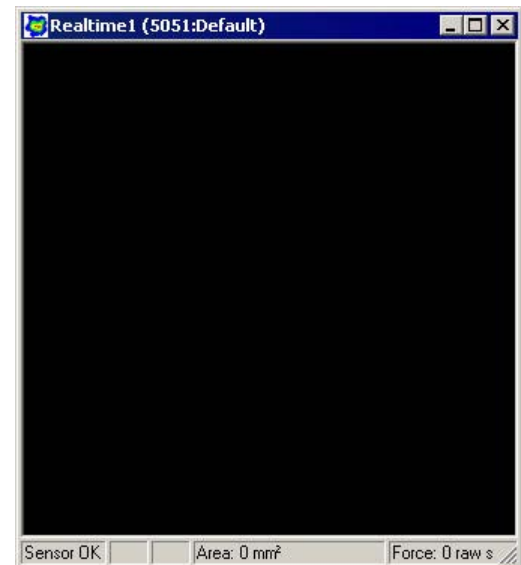
Sensitivity (Gain) Adjustment

Sensitivity (also referred to as 'Adjustable Gain') adjusts the data output range from the computer (0 to 255 raw) to the actual force range output from the system handle. Adjusting the system sensitivity can alter the sensor's effective force range. The sensitivity should be adjusted whenever necessary to keep the real-time display in a usable range (i.e. both the high and low pressures are visible), or if the sensor saturates at a lower pressure than desired. The saturation value is directly affected by the sensitivity setting; as the sensitivity is increased, the data output range decreases.

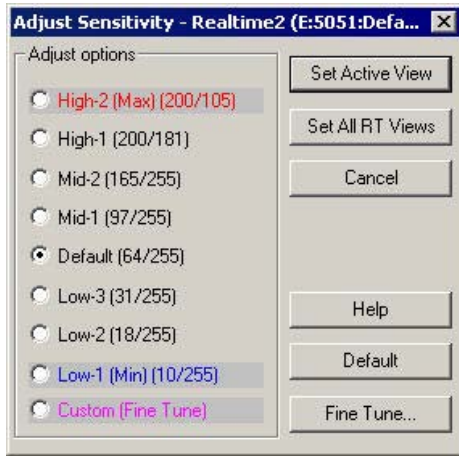
IMPORTANT! Sensitivity adjustments should be performed before calibrating or equilibrating. Otherwise, the calibration data will be lost, and the equilibration data may not be as accurate as possible.

To adjust the handle sensitivity:

1. Open a real-time window, and select it to make it active.
2. Load the sensor with the equilibration/calibration test weight, following the '**Loading Guidelines**' in this section.
3. Select **Adjust Sensitivity** from the **Tools** pull-down menu, or click on the '**Sensitivity**' button in the '**Calibration**' dialog box. The '**Adjust Sensitivity**' dialog box will be displayed (shown at right). You can also access the '**Adjust Sensitivity**' dialog box by selecting **Settings** from the **Options** pull-down menu, clicking on the '**Sensitivity**' tab, then clicking on the '**Sensitivity**' button.



Selecting **Tools -> Adjust Sensitivity** opens the following dialog directly:

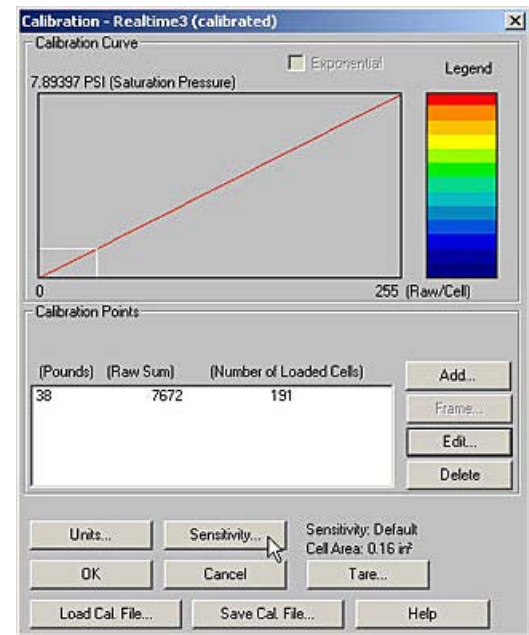


There are two options, which determine how the sensitivity adjustment is applied:

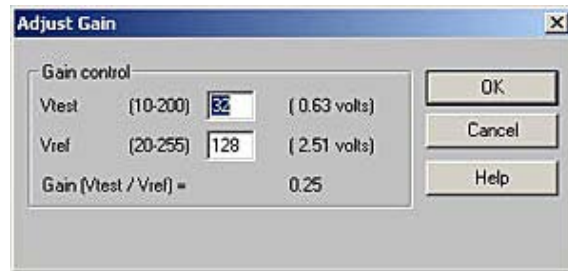
Set Active View: Adjusts the sensitivity of the currently selected Real-time window only.

Set All RT Views: Adjusts the sensitivity of all open Real-time windows (This option is only available if more than one handle is connected to your computer).

Selecting **Tools -> Calibration** opens the following dialog, from which you can select the Sensitivity button to open the "Adjust Sensitivity" dialog:



4. In the 'Adjust Sensitivity' dialog box, there are 8 possible sensitivity settings, ranging from **Low-1** to **High-2** and including a **Custom (Fine Tune)** option. Click on the desired sensitivity setting to select it, or click the Default button for the default setting. If you have the 'Custom (Fine Tune)' option, the gain values (**Vref** and **Vtest**, respectively) are displayed for each of these possible settings, and the **Fine Tune** button will be available in the dialog box to the right.
5. The optional 'Fine Tune' feature allows you to enter a custom sensitivity (gain) setting, which may be necessary if you need a more exact setting than is offered by the choices in the 'Adjust Sensitivity' dialog box. To do this, click the **Fine Tune** button, or select the 'Custom' setting, to open the 'Adjust Gain' dialog box (see figure below).



6. In this dialog, type in the desired test voltage (**Vtest**) and reference voltage (**Vref**); these two values will be used in the equation below to calculate the new gain:

$$V_{test} / V_{ref} = \text{Gain}$$

7. Click <OK> to accept the new values. This new gain (sensitivity) value will override the previous sensitivity setting, and 'Custom' will be selected in the 'Adjust Sensitivity' dialog box, followed by the Vref and Vtest values that you have entered.

8. When you are satisfied with your choice, click the <OK> button to close the 'Adjust Sensitivity' dialog box.
9. Once you have made your selection, the real-time window will show the updated pressure data. Sensitivity settings are recalled when the application is closed and re-opened, or File>>New Recording is selected, but not when **File>>New Session** is selected. When a recording (*.fsx) or calibration (*.cal) is saved as a file, the sensitivity setting is saved with the file.

EQUILIBRATION

Each sensor is somewhat unique - the process for creating sensors results in some variation. In particular, the distribution of pressure-sensitive ink throughout a sensor is not precisely uniform. In addition, as a sensor is used, certain areas may become less responsive than others. The purpose of the equilibration process is to compensate for these slight differences.

IMPORTANT! Equilibration is optional, but if it is necessary, it must be performed before calibration.

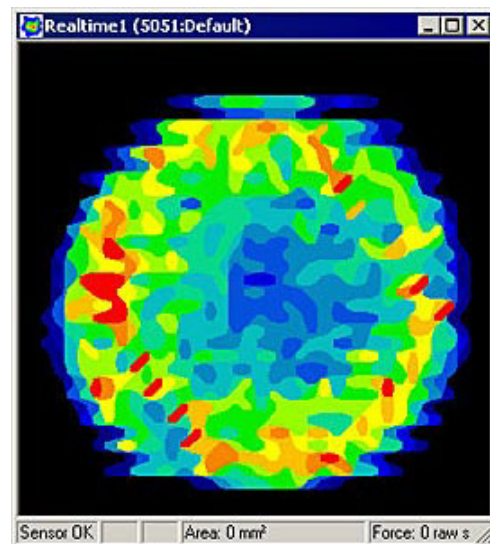
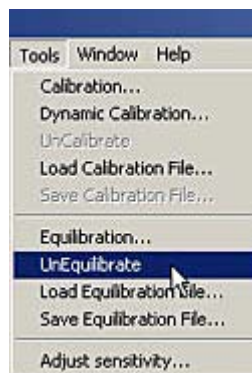
Switchable Equilibration

The software stores equilibration data and raw sensor data separately. Equilibration can therefore be done before or after a movie has been made, and added to the fsx after the movie is recorded.

You can save and load equilibration data while in a real-time window and from within a movie that has an equilibration file. You can also turn the loaded equilibration file on or off. This gives the user the ability to see the effect equilibration has on a movie by choosing to display the fsx data in an unequilibrated or equilibrated mode. One can also load equilibration files into previously recorded movies and save equilibration data as a separate file that can be loaded into real-time windows at any time.

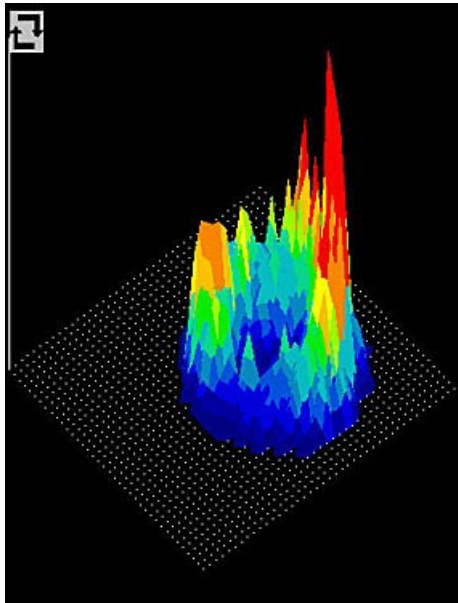
To equilibrate a sensor:

1. Open a real-time window, and select it to make it active.
2. Remove any previous equilibration data by selecting **UnEquilibrate** from the **Tools** pull-down menu.

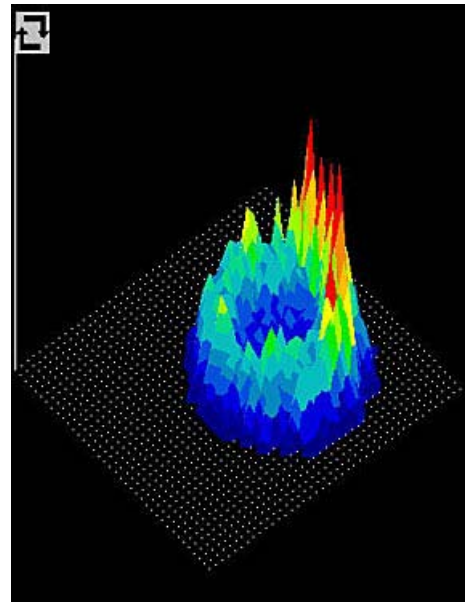


Hint: Before starting the equilibration process, change the Real-time display to a 3-D view. Note the irregularities in the detected pressure. After equilibration, the 3-D display will be much smoother.

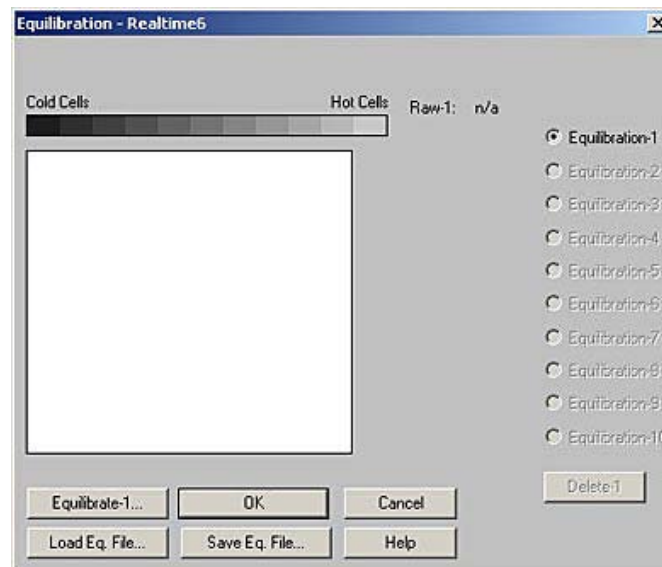
Unequilibrated Movie file in 3-D Contours View



Equilibrated Movie file in 3-D Contours View

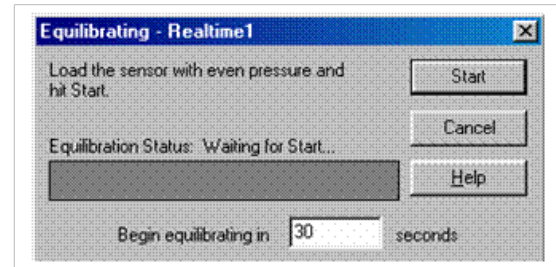


3. Select **Equilibration** from the **Tools** pull-down menu. The '**Equilibration**' dialog box will appear.

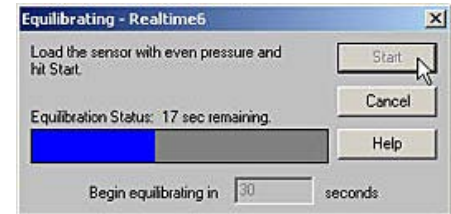


4. Apply a uniform pressure to the entire sensor, following the '**Loading Guidelines**' in this section. In particular, remember the following notes:
 - Pressure must be applied using the Uniform Pressure Loading method.
 - You must be able to apply an even load to perform an equilibration (uniform pressure applicator or compressive device). Therefore, you cannot use dead weights to perform an equilibration.
 - When applying the load, cover the entire sensor area, if possible. The software assumes that the same pressure is applied to each loaded sensel, and any unloaded sensels will not be equilibrated.
 - If only a certain area of the sensor is equilibrated, all actual tests must be done using the equilibrated area of the sensor to ensure accuracy.

5. Click on the **Equilibrate-1** button. When the 'Equilibrating' dialog box appears (see figure at right), enter the amount of time (in seconds) you want to allow the load to stabilize before the equilibration is performed (enter zero to cancel this delay). The software will remember this 'wait' time and save it as the default.



6. Click **Start**. A timer will count down the time remaining (in seconds) before the equilibration is performed (see image at right). Once this time has elapsed, the software will begin collecting sensel data and equilibrating the sensor. During this process, a unique scale factor is applied to each sensel's value to make them uniform.

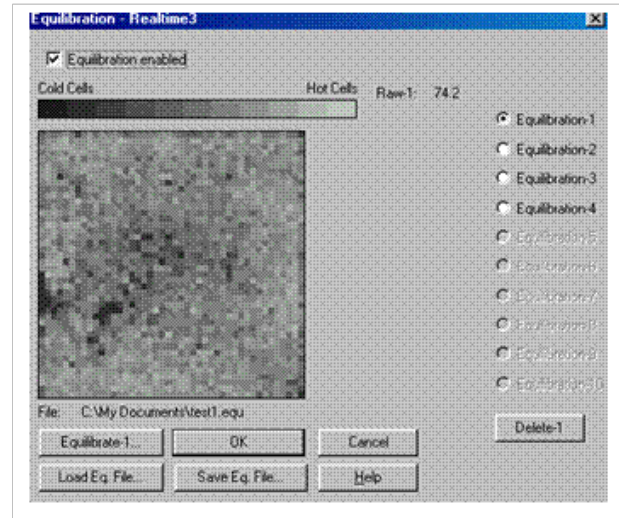


Note: If no load (less than 3% of sensels loaded) is on the sensor, the equilibration process won't be performed and the message 'Insufficient Area Loaded' will be displayed. However, it will automatically begin when a load is sensed.

Once the equilibration process is complete, a graphic representation of pressures in the real-time window (grayscale) is displayed in the dialog box. A grayscale legend appears above, and shows relative pressure ranges from 'cold' cells (dark) to 'hot' cells (light).

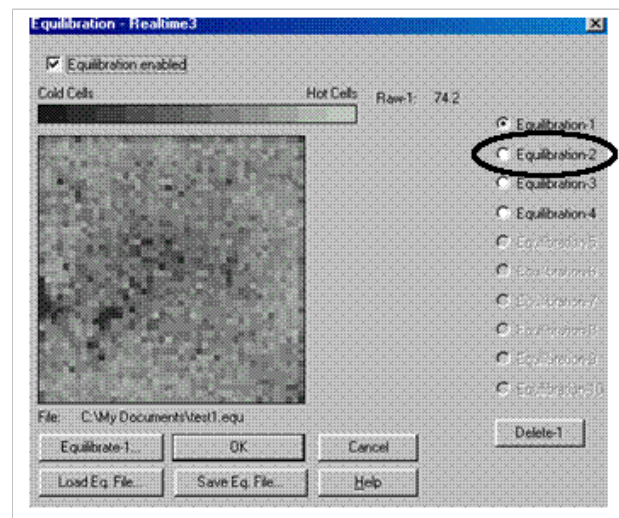
The image at right shows the relative high and low spots of the sensor, and the levels of compensation necessary to ensure uniform results.

7. If you would like to add a second equilibration point for the sensor, click the **'Equilibration-2'** radio button (on the right side of the dialog box), then click the **Equilibrate-2** button at the bottom of the dialog box to start the equilibration process. Repeat these steps to add up to 10 equilibration points. Multiple points may be necessary when you want to equilibrate the sensor at a number of different pressures.



Once you have created one or more equilibration points, you can display the data for a specific point by selecting its radio button to the right. The selected point's raw value will be displayed (e.g. "Raw-1: 56") and its graphic equilibration data will be displayed in the window to the left.

8. Click **<OK>** to close the **'Equilibration'** dialog box. This pressure values should appear uniform in the real-time window.



Multi-Tile Equilibration

I-Scan allows you to equilibrate a "virtual" or "multi-region" sensor, either as a single window or as separate 'tiles'. A multi-tile equilibration is performed as described in the 'Equilibration' section, with the following differences:

1. When you open the **'Equilibration'** dialog box (**Tools->Equilibration**), any tiles are shown as color-coded boxes in the display window. This graphic representation of the sensor shows each tile's size and position in the overall virtual sensor.

Note: The multi-tile options will only be available when a "virtual" or "multi-region" Map is selected (in Options->Select Sensor).

2. You can choose to perform an equilibration on a single tile, or on the entire virtual sensor. To equilibrate the entire virtual sensor, click (to de-select) the **'Tiles'** check box. The individual 'tile' buttons to the right will become unavailable. Click the **Equilibrate-1** button to open the **'Equilibrating'** dialog box and start the equilibration process.
3. To equilibrate a single tile, place a check mark in the **'Tiles'** check box, and select one of the color-coded 'tile' buttons that become available to the right. The word 'tile' next to each of these buttons will be the same color as its accompanying tile. Click the **Equilibrate-1** button - the 'Equilibrating' dialog box will open, with instructional text that is the same color as the tile being equilibrated.

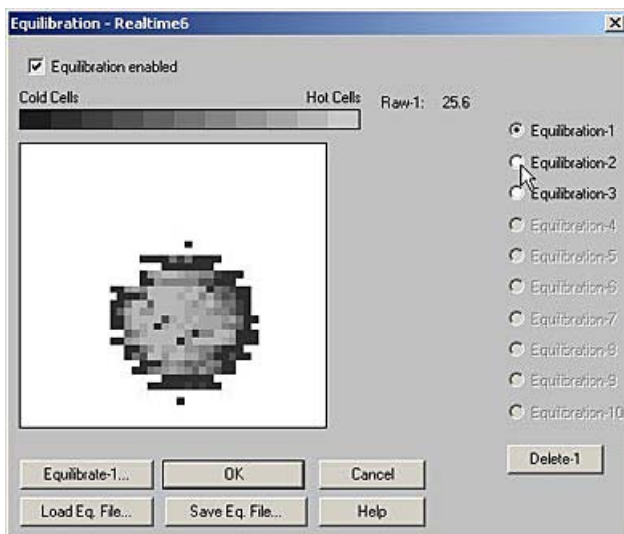
Important! When performing a multi-tile equilibration using multiple sensors, ensure that they are all in the same force range.

4. Once an equilibration has been performed, the equilibration data (**'Levels'** and **'Raw'**) for whichever tile is selected, or for the entire virtual sensor, will be displayed in the **'Equilibration'** dialog box. If you performed an equilibration on separate tiles, this text ('Levels' and 'Raw') will be the same color as its accompanying tile. If you created multiple equilibration points for a single tile or for the entire virtual sensor, you can view the data for a specific point by selecting it to the right (e.g. "Equilibration-2").

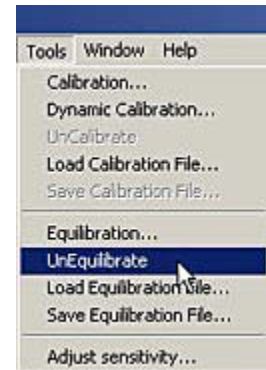
Once a tile has been equilibrated, the word 'tile' next to that tile's button will be displayed entirely in capital letters; before equilibration, it will be entirely in lowercase letters.

UnEquilibrating

If for any reason you wish to discard the current equilibration data (e.g. to perform another equilibration without exiting the **I-Scan** program), simply select **UnEquilibrate** from the **Tools** pull-down menu.

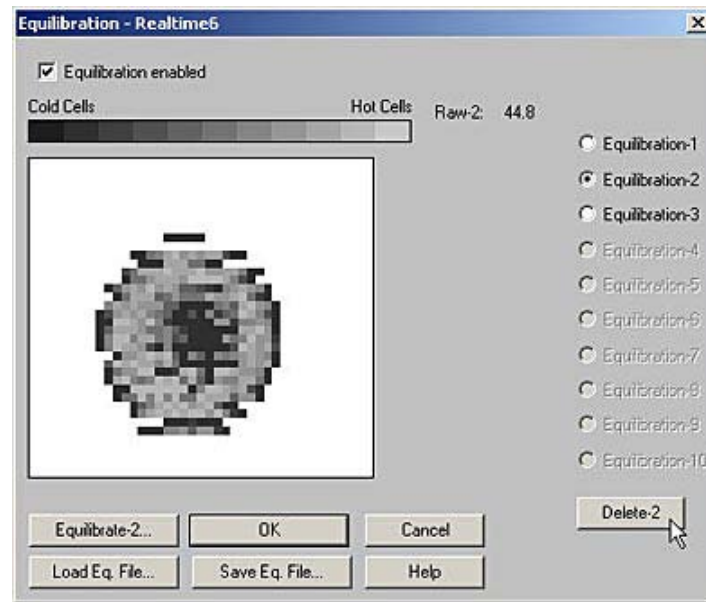


If you are in the process of equilibrating a sensor, and wish to reload an equilibration point, do the following:

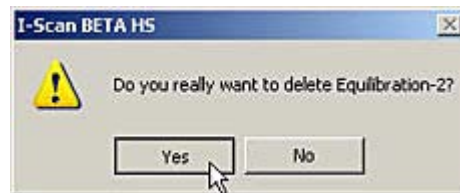


1. Assuming you are currently in the **Tools -> Equilibration** dialog, and have already inserted an equilibration point(s), select the equilibration point to be deleted, by clicking the radio button (shown at left).

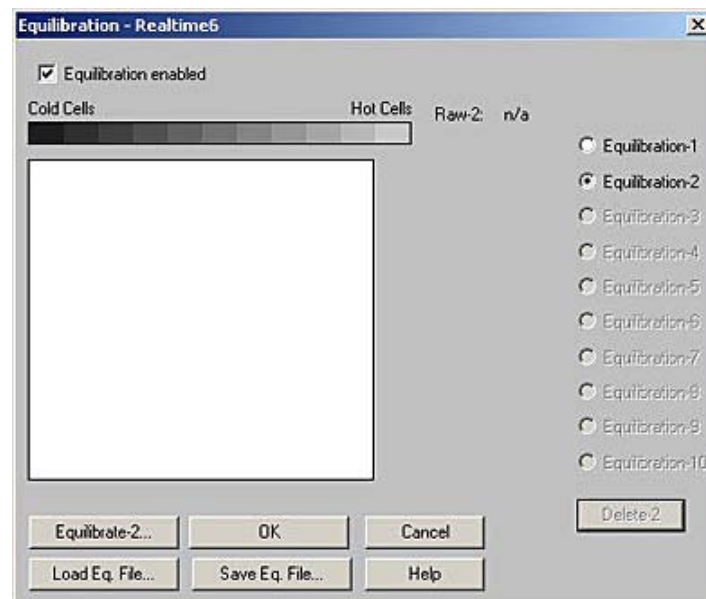
- Next, Click the Delete button below the Equilibration points within the dialog (in the bottom right corner).



- A dialog prompt opens asking if you really want to delete the specified Equilibration point. Click the Yes button.



Your Equilibration point is deleted, and you can enter a new point, if needed.

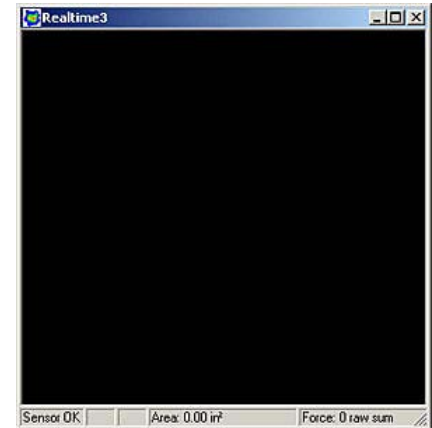


Loading/Saving Equilibration Files

When a recording is saved, the equilibration data used to make the recording is saved with the movie file. If you equilibrate a sensor and want to save the data for later use, however, you must explicitly save it as an equilibration file. The equilibration file saves only the equilibration data associated with the sensor, and does not include the calibration data (if a calibration was performed). Equilibration data is recalled when **File>>New Recording** is selected, but not if the application is closed or **File>>New Session** is selected. Note that each sensor must be equilibrated individually; so one equilibration file may not be used with multiple sensors.

To save an equilibration file:

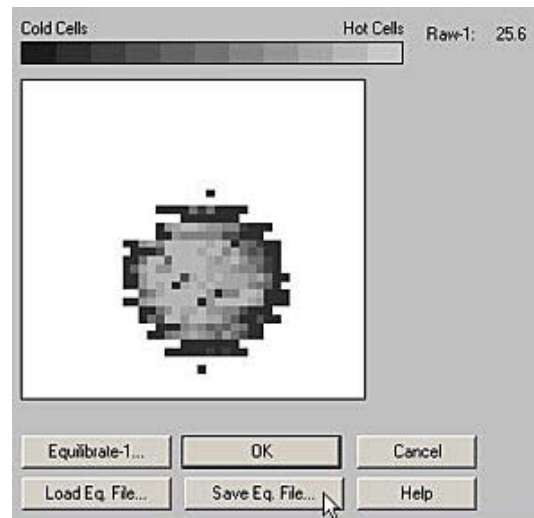
1. Open a real-time window, and perform an equilibration.
2. Select **Save Equilibration File** from the **Tools** pull-down menu. Clicking on the **Save Eq. File** button in the 'Equilibration' dialog box can also access this option.



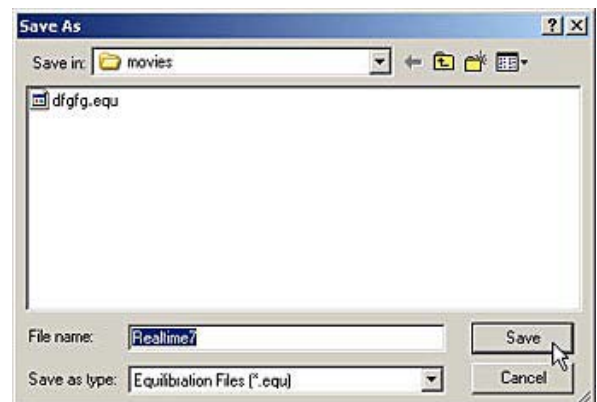
Selecting from the Tools menu



Selecting from the Equilibration dialog.



3. When the 'Save As' dialog box appears, specify the file name and destination (path), then click **<OK>**. *I-Scan* equilibration files must be saved with the extension *.equ.



To load an equilibration file:

1. Open a new real-time window.
2. Select **Load Equilibration File** from the **Tools** pull-down menu. Clicking on the **Load Eq. File** button in the 'Equilibration' dialog box can also access this option.

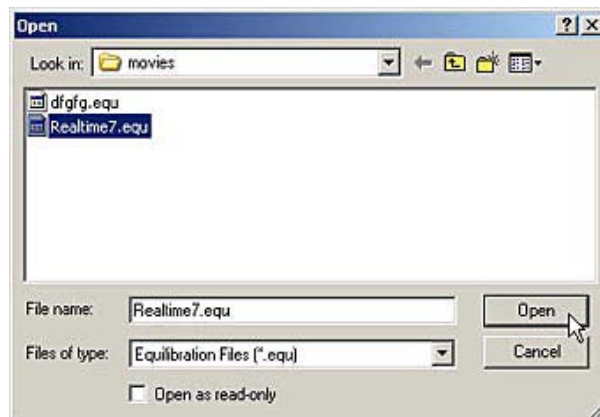
Selecting from the Tools menu



Selecting from the Equilibration dialog.



3. When the 'Open' dialog box appears, specify the file name and destination (path) of the desired equilibration file, then click <OK>. Only **I-Scan** equilibration files (with the extension *.equ) may be loaded.



Note: The software will check the loaded equilibration file to ensure it was saved using the same map file as the current real-time window.

CALIBRATION

The **I-Scan** system enables you to perform two different types of calibration -- linear and 2-point power law -- either before or after a recording has been taken. This section describes how to perform a calibration before recording data; refer to the 'Movie (Frame) Calibration' section for instructions on calibrating an existing movie.

When you perform a linear calibration, you apply a known load to the sensor. The **I-Scan** software then performs a linear interpolation between zero load and the known calibration load. A linear calibration is the most simple to perform, and is suitable for tests in which the load range is limited. Calibration data is stored with movie (*.fsx) files, or explicitly in calibration (*.cal) files, and is displayed with the header information.

When you perform a 2-point power law calibration, you apply two different known loads to the sensor. The software then performs a power law interpolation based on zero load and the two known calibration loads, using the equation $y=ax^b$. A 2-point power law calibration is preferable if measurement loads vary considerably during testing. As a rule of thumb, the applied calibration loads should be approximately 20% and 80% of the expected maximum test load.

When calibrating a sensor, the software calculates an average applied pressure based on the area of loaded sensels and the force value entered by the user. This means that care should be taken to ensure the loaded area of the sensor remains relatively constant for all calibration points. It is also important that at least 25% of the sensels be loaded when using this loading method, to ensure that the average applied force calculations are representative of an average sensel.

IMPORTANT! Sensors must be properly conditioned and equilibrated (if necessary) prior to calibration and use.

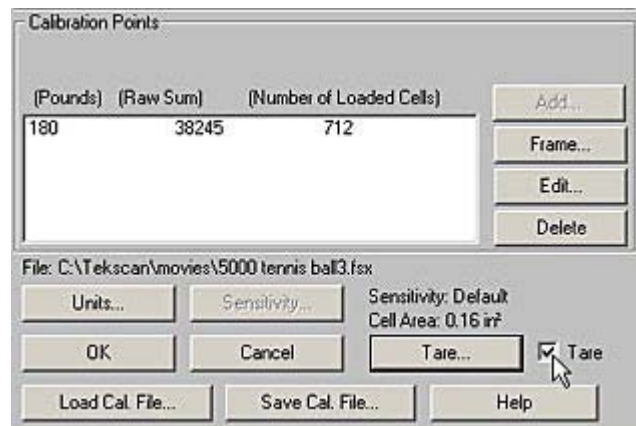
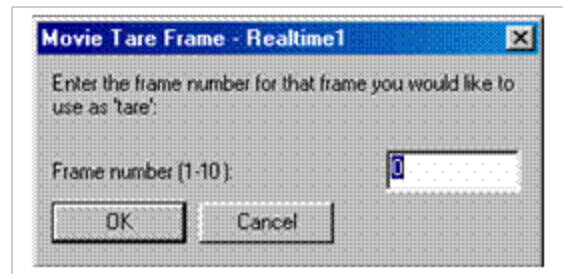
Tare

This feature modifies the calibrated data to correct for residual pressure (referred to as the 'Offset Load') on the sensor at zero load. This operation is useful if the sensor is under some load for the entire duration of the recording, or if the sensor must be secured in place by hardware which would cause load to be detected by the sensor. The raw data is not affected by the Tare function, but an extra calculation is performed when converting the raw data to calibrated data. Any cells reading raw pressure are set at zero raw before the calibration scale factor is applied. **Tare** is recommended only for single point calibration and may increase error when using a multipoint calibration.

Consider the following example:

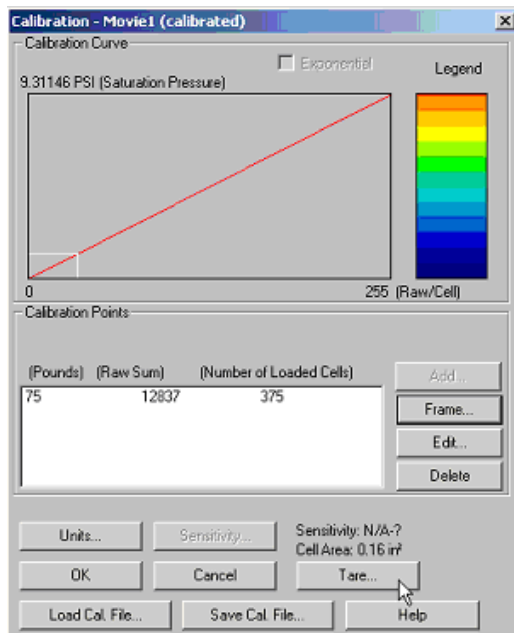
If a sensor with saturation pressure under 50 PSI is wrapped around a cylinder, the wrapping will induce some load, because the inner and outer skins of the sensor will rub against each other. After calibrating, go into the software and click on Tools -> Calibration. Then click Tare. I-Scan allows a consistent output that is to be ignored (to be "tared out").

Note: I-Scan can be toggled back and forth between gross weight and net weight by adding or removing the check mark in the Tools -> Calibration dialog.



Note: Tare must be set before a sensor is calibrated

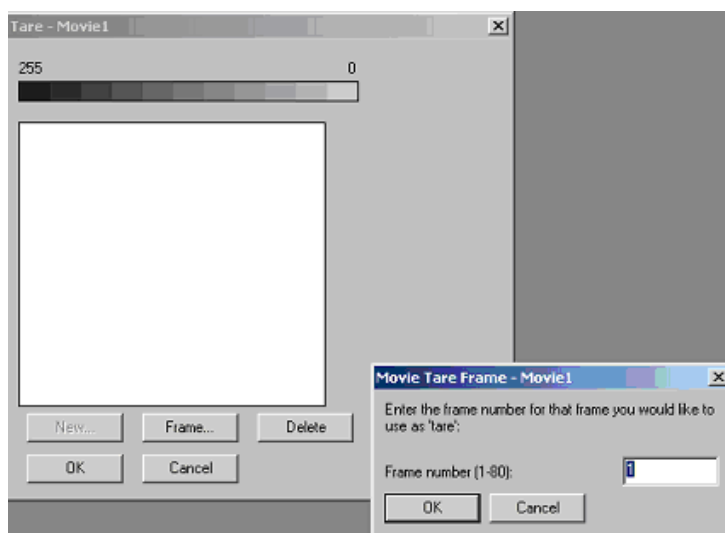
1. Open a new real-time window, and select it to make it active.
2. Select **Calibration** from the **Tools** pull-down menu. The 'Calibration' dialog box will appear. You can also access the 'Calibration' dialog box by selecting Settings from the Options pull-down menu, clicking on the 'Calibration' tab, then clicking on the Calibrate button.



3. Click the 'Tare' button to bring up the 'Tare' dialog box (shown at left).



4. Select either 'New' for a real-time window, or 'Frame' (see image below) for a previously recorded movie.



- **New:** Enter the number of seconds you wish to wait before the software records the Tare. Apply the offset load and click 'Start.' The Tare will be applied, and the 'Tare' dialog box will display the residual forces that will be corrected by the Tare function.
 - **Frame:** If you have previously recorded a movie and wish to use Tare on the forces visible in a specific frame, click 'Frame.' Enter the frame number you wish to use to set the Tare, and click <OK>. The Tare will be applied, and the 'Tare' dialog box will display the residual forces that will be corrected by the Tare function.
5. When you are satisfied with your selections, click <OK> to return to the 'Calibration' dialog box. When Tare is in effect, the Status Bar will read, "Net Force". If you wish to save your Tare information separately from a Calibration file, perform the Tare as described above. Before performing calibration functions, click the 'Save Cal File' button. Your Tare will be saved as a *.cal file for use later.

- To load a saved Tare file (*.cal), click the "Load Cal File" button in the Calibration dialog box.

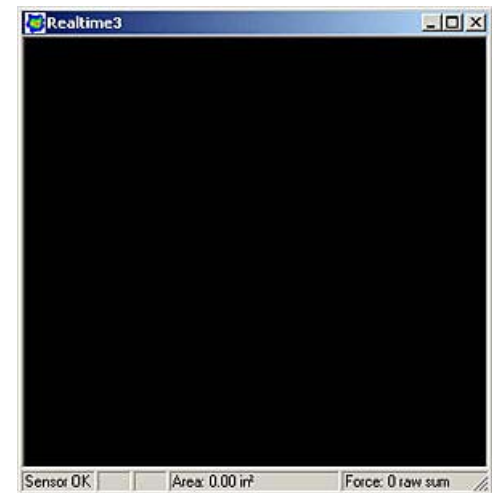
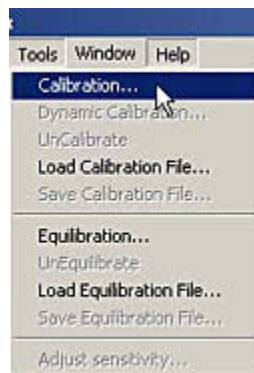
Notes:

- 1. If an Offset Load is used for Tare, it must be applied during the Calibration process.*
- 2. Removal of the Offset Load during Calibration or Movie recording will cause inaccurate data to be recorded.*

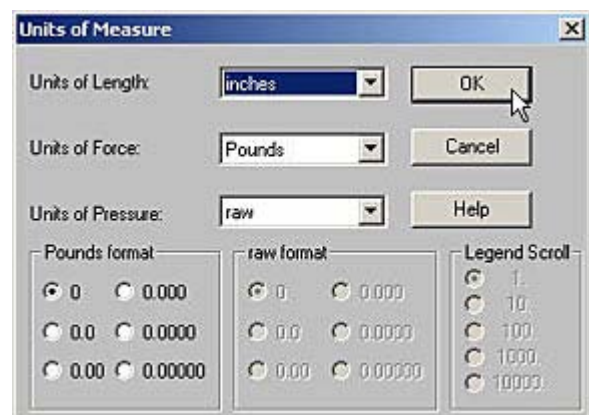
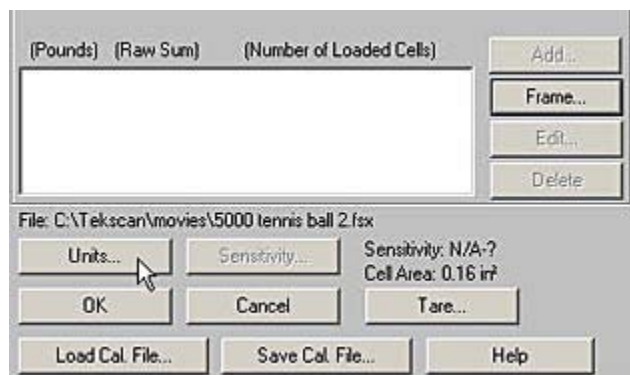


Linear Calibration

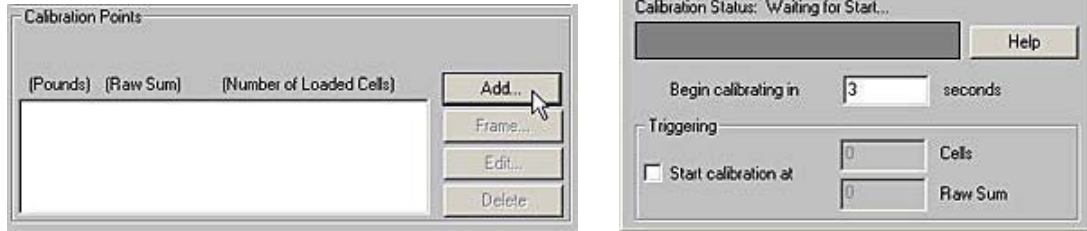
- Open a new Real-time window, and select it to make it active.
- Select **Calibration** from the **Tools** pull-down menu. The 'Calibration' dialog box will appear. You can also access the 'Calibration' dialog box by selecting **Settings** from the **Options** pull-down menu, clicking on the 'Calibration' tab, then clicking on the Calibrate button.



- Select the force units you want to use in the calibration. The default force unit is pounds; if you want to use another unit, click on the Units button and make your selection in the 'Units of Measure' dialog box.



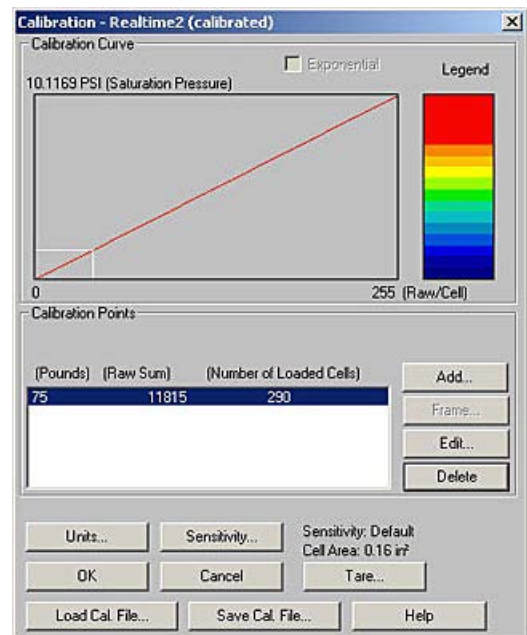
4. Click the **Add** button in the '**Calibration**' dialog box. This will bring up the Calibration Point dialog box.



5. In the '**Applied Force**' field, enter the known force value in the appropriate units.
6. In the '**Begin calibrating in ____ seconds**' field, enter the amount of time (in seconds) you want to allow the load to stabilize before the calibration is performed. This value is set to 30 seconds by default; however, **I-Scan®** will save any value that you enter as the default (you can enter a zero to cancel the delay).
7. If you want to trigger the start of the calibration process, place a check mark in the '**Start Calibration at**' check box (under '**Triggering**' at the bottom of the dialog box), and enter values into the force and area fields (in Raw Sum and Cells, respectively). The calibration process will not start until both the force and area values are reached; however, you can enter a zero in either field to disable it (both values are zero by default). If calibration triggering is selected, the calibration timer (in Step 6) will not start counting down until after the trigger event.
8. Apply the calibration load to the sensor, following the guidelines and recommendations of this section. As a general rule, make sure that the calibration load is relatively stable and approximates the load range that will be applied during testing.
9. Click **Start**. The system will wait for any calibration triggering conditions to be met, and then a timer will count down the time remaining (in seconds) before the calibration is performed. Once this time has elapsed, the software will begin collecting sensel data for the calibration point. During this process, the software performs a linear interpolation between zero load and the calibration load.

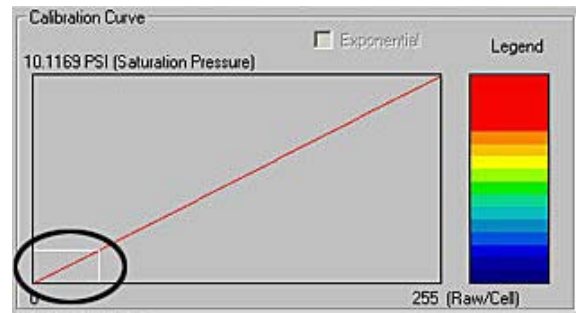
Note: If no load (less than 1% of sensels loaded) is on the sensor, the calibration process won't be performed and the message 'Insufficient Area Loaded' will be displayed. It will automatically begin when a load is sensed.

Once the calibration process is complete, a graphic representation of the calculated linear calibration curve is displayed in the '**Calibration**' dialog box. To the right of this calibration curve is an accompanying legend that shows the relative pressure levels of all points (0-255 raw) on the curve. The calculated saturation pressure is also displayed in this dialog box (refer to the 'Saturation' section for more information).



The white line(s) in the calibration dialog window graph represent(s) the average raw output(s) of the calibration point(s).

10. Review the calibration point data (weight entered, Raw Sum, and Number of Loaded Cells). If any of this data is not satisfactory, highlight the point, and click **Edit** (see image below) to change the entered weight, or Delete to remove the calibration point and start over. If the point is acceptable, click <OK>.



(Pounds)	(Raw Sum)	(Number of Loaded Cells)
75	11815	290

Buttons: Add..., Frame..., Edit..., Delete

Edit Calibration - Realtime2

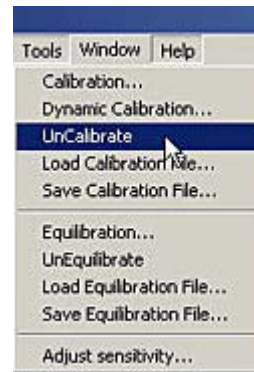
Force (Pounds): 75

Raw Sum: 11815

Number of Loaded Cells: 290

OK Cancel

11. Once you have completed the calibration, check the results by placing the cursor over a loaded sensel, reading the '**Load**' reported for that sensel in the Main Status Bar, then making sure that the value falls into the correct range in the Calibrated Legend (with the Legend's lower limit set to zero).
12. If you wish to recalibrate the sensor during the same testing session, you may either select **File>>New Session** to get a new Real-time window, or select **UnCalibrate** to delete the calibration point in the current window.

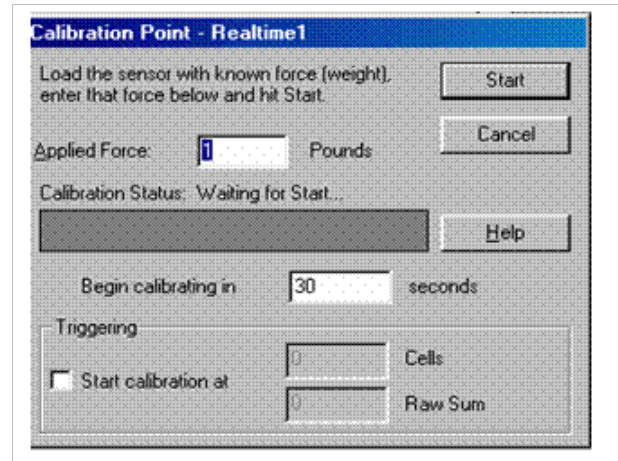


13. You can view the calibration status and data for an active window at any time by selecting Settings from the Options pull-down menu, and clicking on the '**Calibration**' tab. If the window has been calibrated, the calibration data, such as the scale factor and saturation pressure, will be displayed. Depending on the calibration status of the window, you can also calibrate, uncalibrate, or load or save a calibration file from this page. Refer to the description of "Settings" in the 'Main Menu' section for more information.

Note: Refer to the "Loading / Saving a Calibration File" section for details on saving Real-time or movie window calibration data as a file.

2-Point Power Law Calibration

1. Open a new real-time window, and select it to make it active.
2. Select **Calibration** from the **Tools** pull-down menu. The '**Calibration**' dialog box will appear. You can also access the 'Calibration' dialog box by selecting Settings from the Options pull-down menu, clicking on the 'Calibration' tab, then clicking on the Calibrate button.
3. Select the force units you want to use in the calibration. The default force unit is pounds; if you want to use other units, click on the Units button and make your selection in the '**Units of Measure**' dialog box.
4. Click the **Add** button in the '**Calibration**' dialog box. You will see the dialog box shown at right.
5. In the '**Applied Force**' field, enter the first known force value in the appropriate units.
6. In the '**Begin calibrating in ____ seconds**' field, enter the amount of time (in seconds) you want to allow the load to stabilize before the calibration point data is taken. This value is set to 30 seconds by default; however, *I-Scan* will save any value that you enter as the default (you can enter a zero to cancel the delay).
7. If you want to trigger the start of the calibration process, place a check mark in the '**Start Calibration at**' check box (under 'Triggering' at the bottom of the dialog box), and enter values into the force and area fields (in Raw Sum and Cells, respectively).



The calibration process will not start until both the force and area values are reached; however, you can enter a zero in either field to disable it (both values are zero by default). If calibration triggering is selected, the calibration timer (in Step 6) will not start counting down until after the trigger event.

8. Apply the first calibration load to the sensor, following the guidelines and recommendations of this section. As a general rule, make sure that the calibration loads are relatively stable and approximate the load range that will be applied during testing.
9. Click **Start**. The system will wait for any calibration triggering conditions to be met, and then a timer will count down the time remaining (in seconds) before the calibration is performed. Once this time has elapsed, the software will begin collecting sensel data for the calibration point.

Note: If no load (less than 1% of sensels loaded) is on the sensor, the calibration process won't be performed and the message 'Insufficient Area Loaded' will be displayed. It will automatically begin when a load is sensed.

10. Click the **Add** button in the '**Calibration**' dialog box to add a second point. In the '**Applied Force**' field, enter the second known force value in the appropriate units.
11. In the '**Begin calibrating in ____ seconds**' field, enter the amount of time (in seconds) you want to allow the load to stabilize before the calibration point data is taken (you can enter a zero to cancel the delay).
12. If you want to trigger the start of the calibration process, place a check mark in the '**Start Calibration at**' check box (under 'Triggering' at the bottom of the dialog box), and enter values into the force and area fields (in Raw Sum and Cells, respectively). You can enter a zero in either field to disable calibration triggering.

13. Apply the second calibration load to the sensor, following the guidelines and recommendations of this section.

Note: *It is recommended that you allow the sensor to 'relax' for 3-5 times the total time the sensor has been loaded, before adding the second calibration point.*

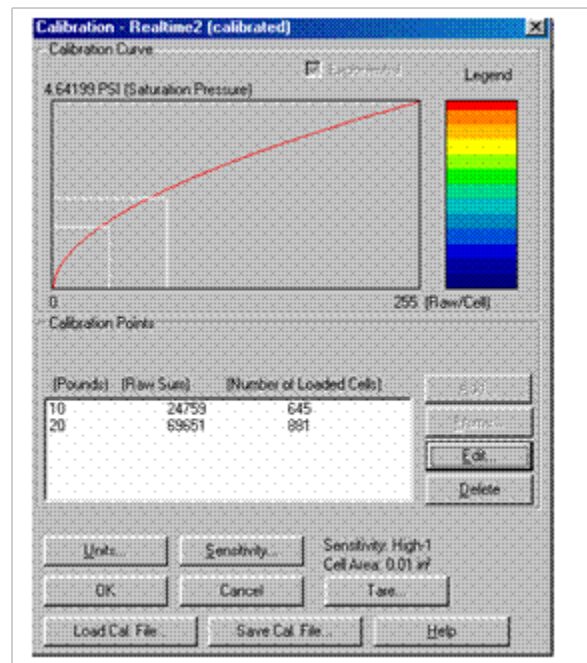
14. Click **Start**. The system will wait for any calibration triggering conditions to be met, and then a timer will count down the time remaining (in seconds) before the calibration is performed. Once this time has elapsed, the software will begin collecting sensel data for the second calibration point. The software automatically performs a 2-point power law calibration when a second calibration point is added (a check mark is placed next to the 'Exponential' option at the top of the dialog box, to show that this method was used).

If the calibration points are not valid for some reason, the error message '**2-Point Exponential Curve Calculation Failed**', followed by a more descriptive explanation, will be displayed. If you receive any of these error messages, you should read and follow the entire sensor loading guidelines and considerations in this section, then perform another calibration, ensuring the following:

- The loaded area is constant for the two-calibration points.
- The entire load path is through the sensing area of the sensor.
- There are not a significant number of saturated sensels during calibration.
- The sensor's average digital output is sufficiently different for the two-calibration points (i.e. enough difference between the two applied forces).

Note: *Following these guidelines should ensure a valid calibration.*

Once the calibration process is complete, a graphic representation of the calculated calibration curve is displayed in the '**Calibration**' dialog box (see figure below). To the right of this calibration curve is an accompanying legend that shows the relative pressure levels of all points (0-255 raw) on the curve. The calculated saturation pressure is also displayed in this dialog box (refer to the 'Saturation' section for more information).



15. Review the calibration point data (weight entered, Raw Sum, and Number of Loaded Cells) for the two points. If any of this data is not satisfactory, highlight the point, and click Edit to change the entered weight, or Delete to remove the calibration point. If both points are acceptable, click <OK>.
16. Once you have completed the calibration, check the results by placing the cursor over a loaded sensel, reading the 'Load' reported for that sensel in the Main Status Bar, then making sure that the value falls into the
17. If you wish to recalibrate the sensor during the same testing session, you may either select **File>>New Session** to get a new Real-time window, or select UnCalibrate to delete all of the calibration points in the current window.

18. You can view the calibration status and data for an active window at any time by selecting Settings from the Options pull-down menu, and clicking on the **'Calibration'** tab. If the window has been calibrated, the calibration data, such as the scale factor and saturation pressure, will be displayed. Depending on the calibration status of the window, you can also calibrate, uncalibrate, or load or save a calibration file from this page. Refer to the description of Settings in the Main Menu section for more information.

Note: Refer to the 'Loading/Saving a Calibration File' section for details on saving Real-time or Movie window calibration data as a file.

Multi-Tile Calibration

I-Scan allows you to calibrate a "virtual" or "multi-region" sensor, either as a single window or as separate 'tiles'. You can perform both linear and 2-point power law calibrations, for a Real-time or Movie window, for this virtual sensor.

A multi-tile calibration is performed as described in the **'Calibration'** section, with the following differences:

1. When you open the **'Calibration'** dialog box (**Tools>>Calibration**), you have the option of selecting either **'Show Curves'** or **'Show Panes'** in the upper (**'Calibration Curve'** section).

Note: The multi-tile options will only be available when a "virtual" or "multi-region" map is selected (in Options>>Select Sensor).

If you select **'Show Panes'**, a graphic representation of the total sensor is displayed, with each tile a different color to show its size and position in the overall virtual sensor. If you select **'Show Curves'**, the color-coded calibration curve for each tile, or for the entire 'virtual' sensor, is displayed in the upper section (after the calibration is performed).

2. In the lower (**'Calibration Points'**) section, you may choose to perform a calibration on a single tile, or on the entire virtual sensor. To calibrate a single tile, place a check mark in the **'Tiles'** check box, and select one of the 'tile' buttons that become available to the right. The word 'tile' next to each of these buttons will be the same color as its accompanying tile.

To calibrate an entire virtual sensor, remove the check mark from the **'Tiles'** check box. The 'tile' buttons to the right will become unavailable.

3. Click the **'Add'** button. The **'Calibration Point'** dialog box will open, and you will be able to perform a calibration, as described in the previous section. If you are adding a calibration point for a single tile, the **'Applied Force'** and **'Calibration Status'** text in this dialog box will be the same color as the tile.

The following application guidelines will help you obtain the best multi-tile calibration results possible, and are strongly recommended. If performing a calibration for multiple tiles:

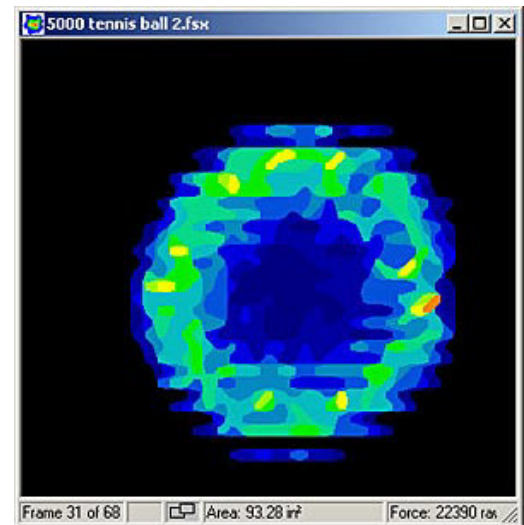
- When using multiple sensors, ensure that they are all in the same force range.
 - Calibrate all tiles using the same method (Linear or 2-Point Power Law).
 - After a 2-point calibration, review the curves and ensure that they all appear similar. If they are drastically different, recalibrate or replace that specific sensor.
4. Once a calibration has been performed, the calibration data values (**'Saturation Pressure'**, **'Force Unit'**, **'Raw Sum'**, and **Number of Loaded Cells**) for whichever tile is selected, or for the entire virtual sensor, will be displayed in the lower (**'Calibration Points'**) section of the dialog box. If you added a calibration point for a single tile, this text will be the same color as its accompanying tile. Once a tile has been calibrated, the word **'tile'** next to that tile's button is entirely in capital letters; before calibration, it is entirely in lowercase letters.

5. To display the calibration curve and data for the entire virtual sensor, make sure there is no check mark in the 'Tiles' check box. Once a calibration has been performed, for either a single tile or the entire sensor, selecting or de-selecting 'Tiles' will remove the sensor's calibration data.
6. Once a calibration has been performed, you can manipulate any of the calibration point(s). If the tiles were calibrated separately, you can change the calibration point(s) for a single tile, separately from the other tiles; including adding, deleting, and editing points, as well as calibrating a movie.
7. A Calibration file can be saved before all tiles have been calibrated. The file must be saved before it is applied to the real-time window. This is useful for virtual maps where multiple sensors are included as one map and each tile represents a separate sensor. Since partial equilibration is also available, this feature allows the user to equilibrate, calibrate, and check each tile/sensor before proceeding to the next tile/sensor, and before any recordings are taken.

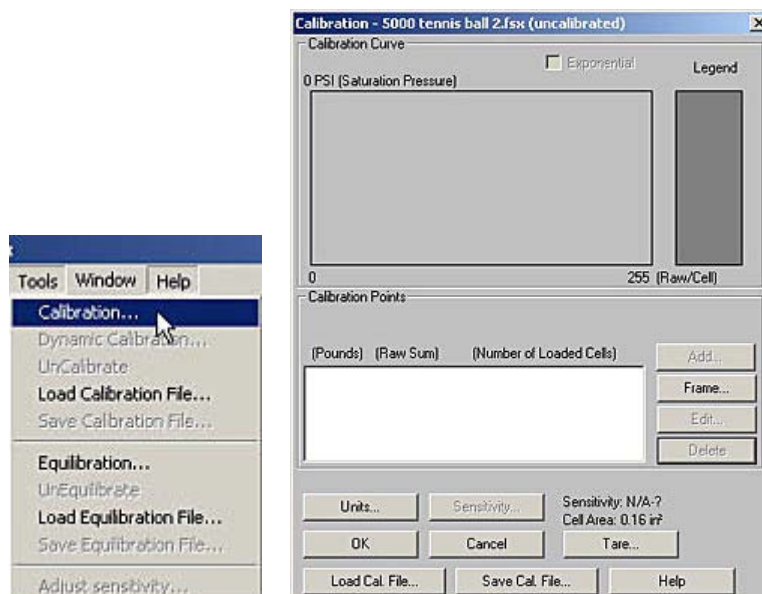
Movie (Frame) Calibration

The *I-Scan* system enables you to perform two different types of calibration; linear and 2-point power law, on an existing Movie window. This section describes how to calibrate a recording; refer to the 'Calibration' section earlier in this chapter for instructions on how to calibrate a Real-time window.

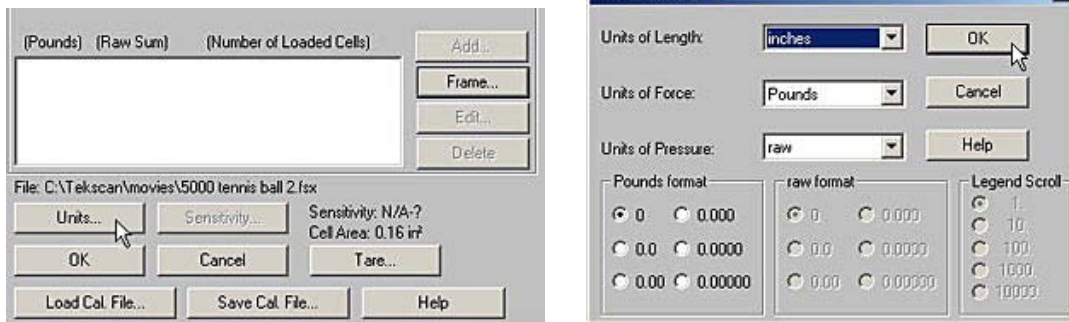
To perform a movie (frame) calibration, enter the total force applied to the sensor in a specific frame of the movie. The *I-Scan* software then performs a linear interpolation between zero load and this known calibration load. To perform a 2-point power law calibration, enter the total force applied for a different frame of the movie. The software then performs a power law interpolation based on zero load and the two known calibration loads, using the equation $y=ax^b$.



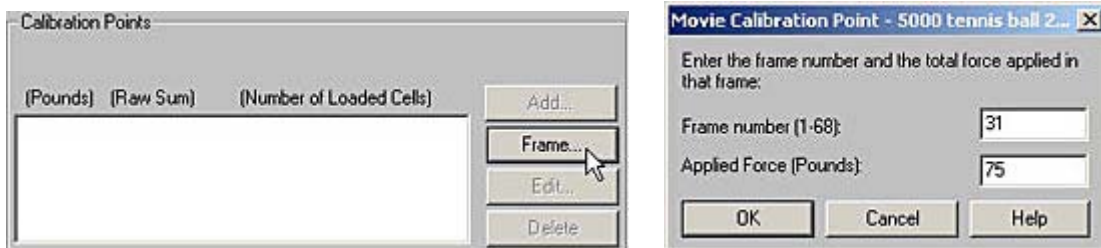
1. Take a recording, or open an existing movie file.
2. With the Movie window active, select **Calibration** from the **Tools** pull-down menu. The 'Calibration' dialog box will appear. You can also access the 'Calibration' dialog box by selecting Settings from the Options pull down menu, clicking on the 'Calibration' tab, then clicking on the Calibrate button.



3. Select the force units you want to use in the calibration. The default force unit is pounds; if you want to use another unit, click on the Units button and make your selection in the 'Units of Measure' dialog box.



4. If there are any existing calibration point(s), highlight that point, then click the 'Delete' button to remove it. Movie calibration may only be performed if the movie is uncalibrated. An error message will be displayed if you attempt to add a point when there exist calibration point(s).
5. Click the Frame button to open the 'Movie Calibration Point' dialog box (shown below).



6. In the 'Frame Number' field, enter the movie frame number you wish to use for the calibration. The movie's frame range is displayed in brackets (shown above).
7. In the 'Applied Force' field (shown above), enter the total force applied to the sensor in the movie frame. The selected force units are displayed in brackets. The software will remember this 'applied force' and save it as the default.
8. Click <OK>. The system will use this point to perform a linear calibration for the entire movie. During this process, the software performs a linear interpolation between zero load and the calibration load.

Note: If no load (less than 1% of sensels loaded) is on the sensor, the calibration process won't be performed and the message 'Insufficient area loaded in this frame' will be displayed. An example of this is shown below:



9. If you want to add a second calibration point, click the Frame button again, to open the **'Movie Calibration Point'** dialog box. In this dialog box, enter the frame number and the total force applied to the sensor in that frame (in the **'Frame Number'** and **'Applied Force'** fields, respectively) for the second calibration point.

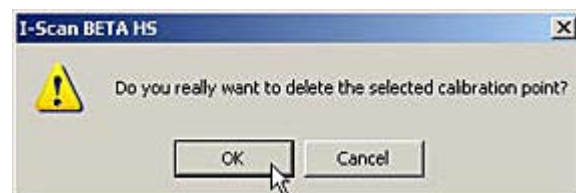
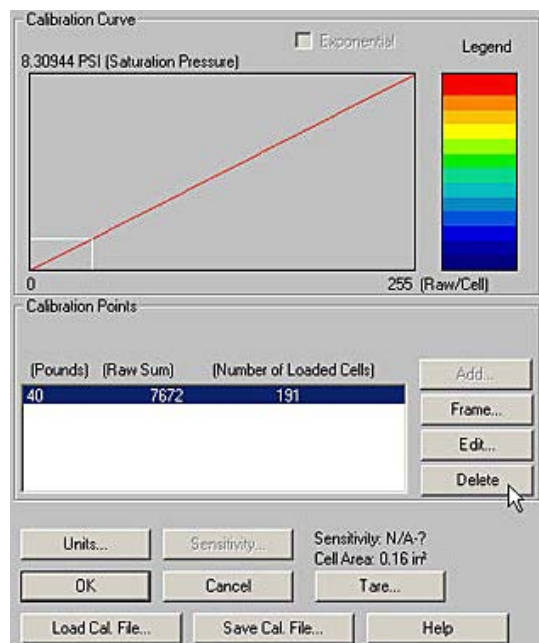
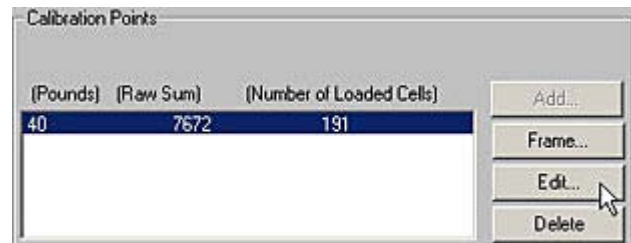
Note: You must use different movie frames for the two calibration points.

10. Click **<OK>** to add the second calibration point. The system will automatically perform a 2-point power law calibration when this second point is added. During this process, the software performs a power law interpolation based on zero load and the two known calibration loads, using the equation $y=ax^b$.

If the calibration points are not valid for some reason, the error message **'2-Point Exponential Curve Calculation Failed:'** followed by a more descriptive explanation, will be displayed. If you receive any of these error messages, you should review and follow the entire sensor loading guidelines and considerations described earlier in this section.

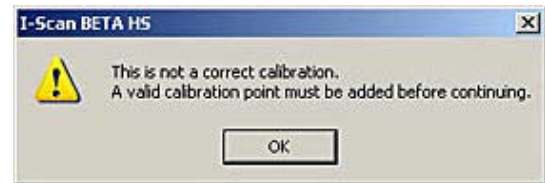
Once the calibration process is complete, a graphic representation of the calculated linear or 2-point calibration curve is displayed in the **'Calibration'** dialog box. To the right of this calibration curve is an accompanying legend that shows the relative pressure levels of all points (0-255 raw) on the curve. The calculated saturation pressure is also displayed in this dialog box (refer to the **'Saturation'** section for more information).

11. Review the calibration point data (applied force, Raw Sum, and Number of Loaded Cells) for the point(s). If any of this data is not satisfactory, highlight the point, and click Edit to enter a different applied force (the **'Edit Calibration'** dialog box is shown below), or Delete to remove the calibration point. If the point(s) is acceptable, click **<OK>**.

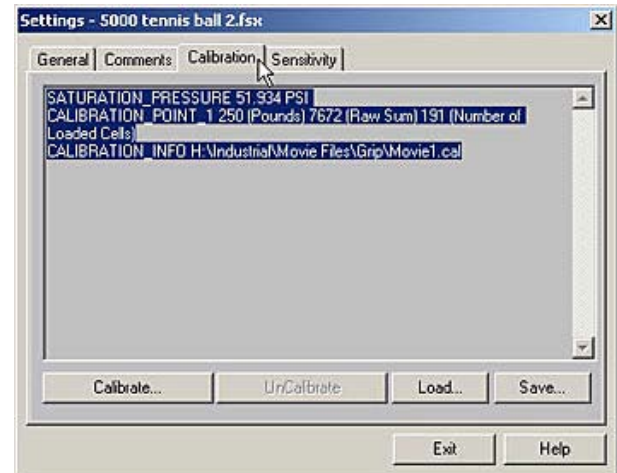


12. Once you have completed the calibration, check the results by placing the cursor over a loaded sensel, reading the 'Load' reported for that sensel in the Main Status Bar, then making sure that the value falls into the correct range in the Calibrated Legend (with the Legend's lower limit set to zero).

13. If you want to recalibrate a Movie window at any time, you must delete any existing calibration points, and add new calibration point(s), in the 'Calibration' dialog box. The system will not allow you to exit the calibration dialog unless you add new points first.



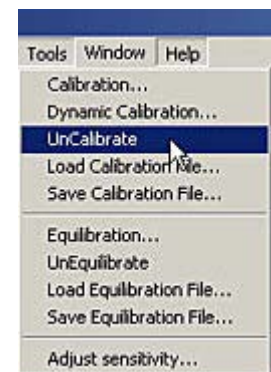
14. You can view the calibration status and data for an active Movie window at any time by selecting **Settings** from the **Options** pull-down menu, and clicking on the '**Calibration**' tab. If the window has been calibrated, the calibration data, such as the scale factor and saturation pressure, will be displayed. Depending on the calibration status of the window, you can also calibrate, or load or save a calibration file from this page. Refer to the description of "Settings" in the '**Main Menu**' section for more information.



***Note:** Refer to the 'Loading/Saving a Calibration File' section for details on saving Movie window calibration data as a file.*

UnCalibrating

If for any reason you wish to discard the current calibration data (e.g. to recalibrate without exiting the *I-Scan* program), simply select **UnCalibrate** from the **Tools** pull-down menu. You can also uncalibrate a window by selecting **Settings** from the **Options** pull-down menu, clicking on the '**Calibration**' tab, then clicking on the '**UnCalibrate**' button.

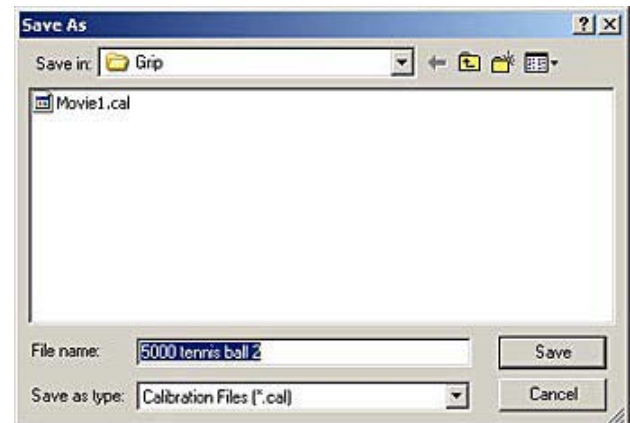
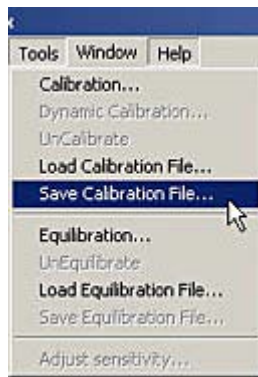
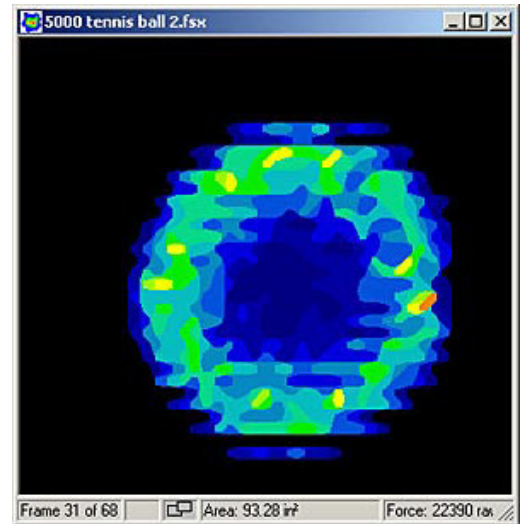


Loading/Saving Calibration Files

When a recording is saved, the calibration data used to make the recording is saved with the movie file. If you calibrate a sensor and want to save the data for later use, however, you must explicitly save it as a calibration file. The calibration file saves only the calibration data associated with the sensor, and does not include the equilibration data (if an equilibration was performed). Calibration data is recalled when **File>>New Recording** is selected, but not if the application is closed or if **File>>New Session** is selected. Note that each sensor must be calibrated individually, so calibration files may not be used with multiple sensors.

To save a calibration file:

1. Open an un-calibrated movie file, or open a new real-time window, and perform a calibration.
2. Select **Save Calibration File** from the **Tools** pull-down menu. You can also access this option by selecting **Settings** from the **Options** pull-down menu, clicking on the 'r' tab, then clicking on the 'Save Calibration' button.

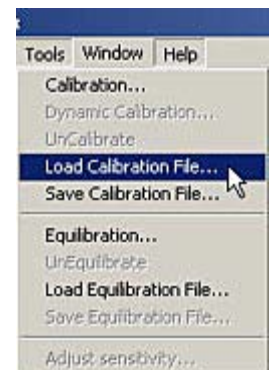


3. When the 'Save As' dialog box appears, specify the file name and destination (path), then click <OK>. **I-Scan** calibration files must be saved with the extension *.cal.

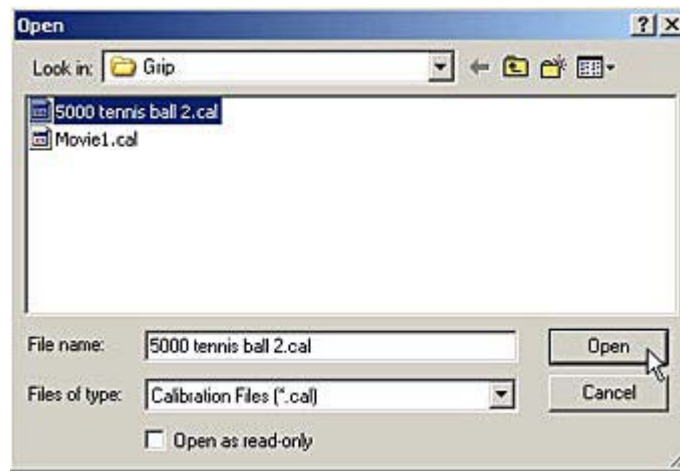
To load a calibration file:

1. Open a movie file, or a new real-time window.
2. Select **Load Calibration File** from the **Tools** pull-down menu. You can also access this option by selecting **Settings** from the **Options** pull-down menu, clicking on the 'Calibration' tab, then clicking on the 'Load Calibration' button.

Note: If you load a calibration file into a calibrated Movie or Real-time window, the file's data will replace the window's previous data.



3. When the 'Open' dialog box appears, specify the file name and destination (path) of the desired calibration file, then click <OK>. Only ***I-Scan*** calibration files (with the extension *.cal) may be loaded.



Note: The software will check the loaded calibration file to ensure it was saved using the same map file as the current Real-time window.

TAKING A RECORDING

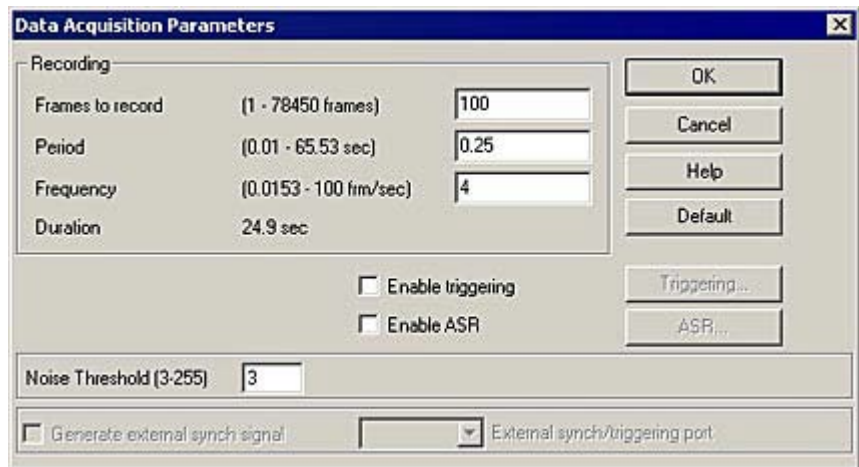
I-Scan allows you to capture sensor pressure measurements over a period of time through a process called ‘recording’. An *I-Scan* recording is similar to a video recording, since you can rewind your recording, play it and replay it as many times as you wish. With an *I-Scan* recording, you can also view the recorded information in a variety of graphical representations. *I-Scan* recordings are also referred to as ‘movies.’

Note: Unlike video recordings, I-Scan movies must be explicitly saved in order to be available for review in future sessions.

ABOUT DATA ACQUISITION PARAMETERS

Before you take a recording, you should set the recording parameters to values appropriate for your test dynamics and your computer system capacity. Later in this section you will be given step-by-step instructions for changing recording parameters. First we introduce you to some general information about recording parameters.

When a recording is made, the recorded information is stored in your computer’s memory (RAM) until it is written to disk. It is not written to disk until you explicitly save the recording. The size of your recording is limited by the amount of available RAM. The *I-Scan* software checks how much RAM is available and lets you know the maximum size (in frames) available for your recording when you are setting recording parameters. There are a number of ways to free up more RAM for recording, if necessary, such as closing all other applications.



The recording parameters available to you are:

- **Frames to Record:** the total number of frames to be recorded
- **Frequency:** the number of frames that will be recorded each second (frames/second)
- **Period:** the amount of time between frames (seconds/frame). Period is related to Frequency by the following relationship: **Period = 1 / Frequency**
- **Duration:** the length of time (seconds) that a recording will take to complete (the first frame is recorded at time=0). The user cannot enter the duration; the software, based on the other three parameters, calculates it.

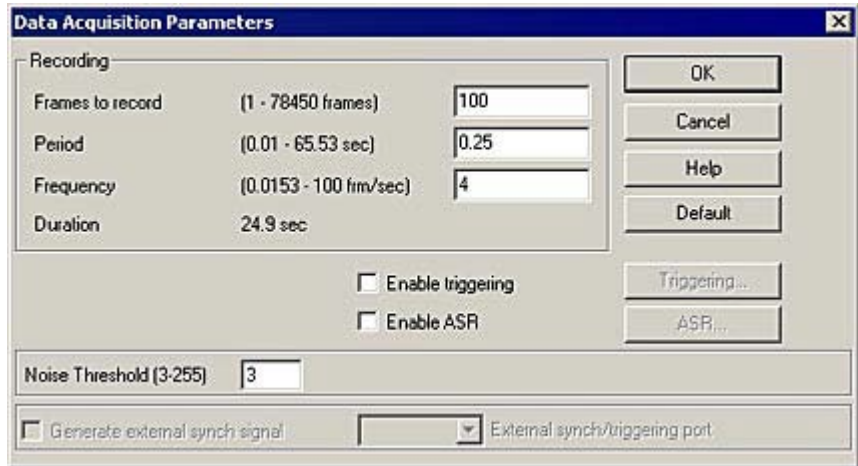
Three of these parameters have the following mathematical relationship: **(Frequency * Duration) + 1 = Frames to Record**

When setting the movie recording (data acquisition) parameters, you will want to set frequency and duration based upon the period of time of interest and the rate at which pressures change. You may have to make a trade-off between total movie duration and frame rate based on memory availability.

Note: The maximum allowable recording frequency is dependent on your hardware. For the Evolution System the maximum is 100 frames per second. For the USB System, the maximum is 9.84 frames per second. For information on the maximum Frequency attainable with your Scanning Electronics, refer to the [Scan Rate](#) section. For High Speed I-Scan systems, refer to the [Scan Speed](#) section.

SETTING THE RECORDING (DATA ACQUISITION) PARAMETERS

1. Click on **Options** in the **Menu Bar**. The Options pull-down menu will be displayed.
2. Click on **Acquisition Parameters** in the pull-down menu. The '**Data Acquisition Parameters**' dialog box will be displayed, as shown below.
3. Enter the number of frames that you wish to record in the **Frames to Record** field.



- Enter either the **Period** or the **Frequency** in the appropriate field. When you change either one of these parameters, the other parameter will automatically change (after you exit that field) to maintain the mathematical relationship. The Duration (in seconds) is automatically re-calculated by the software each time to Frame to Record, Period, or Frequency is changed.
- Set the appropriate **Start Options** for your recording. If you would like the system to wait for a specific amount of time before recording, enter a value (in seconds) in the '**Delay Recording**' field. By default, this value will be zero.

If you want to trigger the recording to start or stop based on an event(s), refer to the "Triggering a Recording" section for instructions. The time delay will be counted down before the program considers any triggering options.

4. Review the values in each field and adjust as necessary. When you are satisfied with the values, click **<OK>**. These recording parameter settings remain in effect until explicitly changed (even if you close the application), or until the 'Default' button is clicked. Clicking '**Default**' will set all the Data Acquisition Parameters to their default values.

More about Data Acquisition Parameters

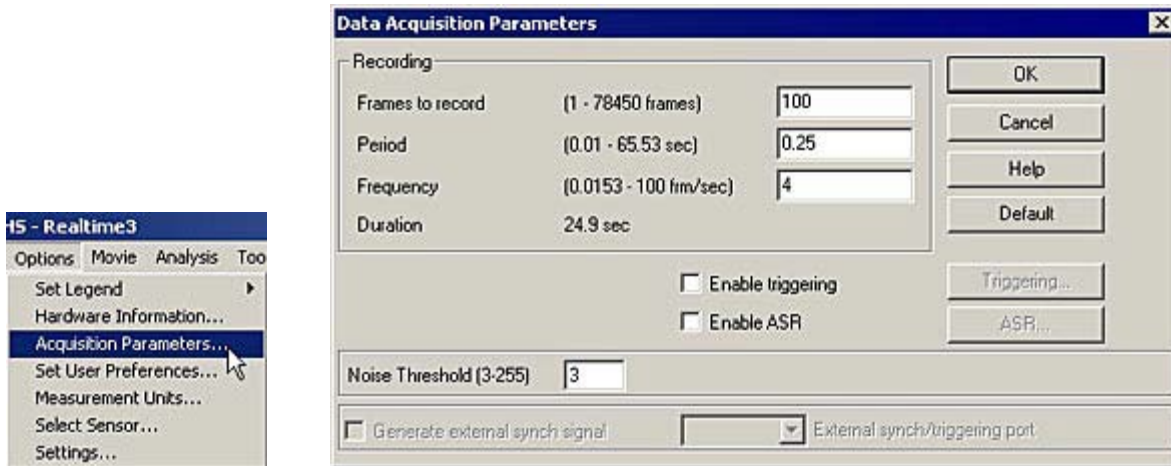
In parentheses next to each field in the Data Acquisition Parameters dialog box is the range of acceptable values for that field. The allowable ranges for the Period and Frequency are constant, but the range for Frames to Record will change. This range will resemble 1 - NMAX (where NMAX is a number). NMAX is the maximum number of frames that can be recorded based upon the computer system's resources (especially the available RAM) and the hardware attached to the system (e.g. number of handles). You can increase the available N, which generally means closing other applications that are running. If you enter a value that exceeds NMAX, an error message will appear, prompting you to enter an integer in the specified range. You will not be allowed to leave the field until a valid value is entered. The dialog can only be closed with invalid numbers in a field by clicking on **<Cancel>**.

NOISE REDUCTION

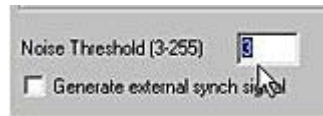
After a period of use or in the presence of EMI, a sensor may indicate noise as low-level pressure. To reduce the effect of such noise on your recordings, you can set a noise threshold. Digital output values that are equal to or below this threshold are set to zero by the software, thereby filtering out unwanted low level force readings (noise). See *Data Acquisition Parameters* under ‘Setting the Recording Parameters’.

To set up the system for noise reduction:

1. Select **Acquisition Parameters** from the **Options** pull-down menu.

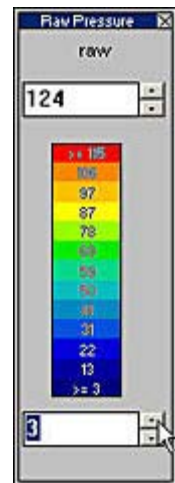


2. Enter a value between 0 and 255 (raw units) in the **Noise Threshold** field at the bottom of the ‘**Data Acquisition Parameters**’ dialog box. The default value is 3.



3. When you are satisfied with the value you have entered, click <OK>.

Note: An alternative way to mask noise that appears in the display is to raise the minimum value in the Legend. This does not eliminate the noise from the recording; however, it only keeps it from being displayed.



TRIGGERING A RECORDING

The *I-Scan* software provides great flexibility in ‘triggering’ when to start and stop recording pressure data. The triggering option enables the user to start and stop recording based on a number different events. These events include sensor load, sensor contact area, or a pre-defined number of frames. With the optional external trigger/synch package, the software can also be set up to start and stop recording based on an external signal. See “[Optional Accessories](#)” for more information about triggering. and Video Features.

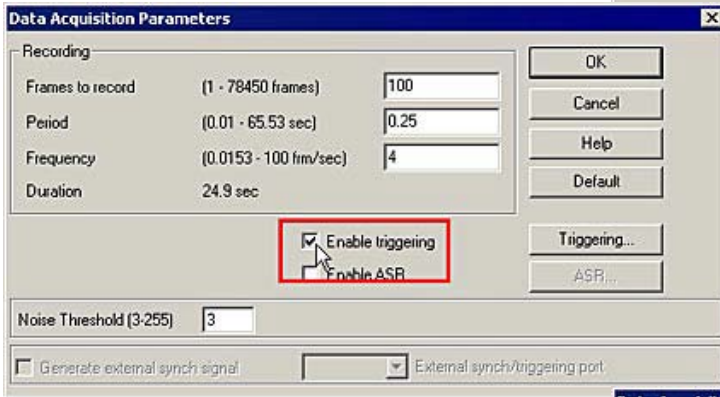
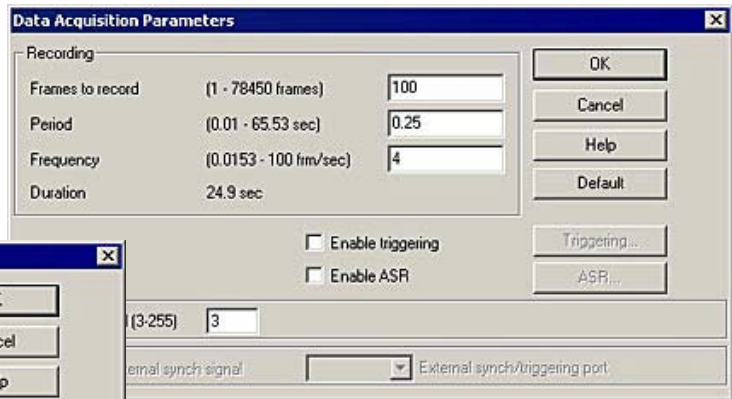
The software also provides the ability to record groups of frames (as a single movie) using multiple start and stop events, or to retrieve a specified number of frames previous to the start trigger event. These options are called “Group Recordings” and “Pre-Triggering,” respectively.

Triggering is disabled by default, and data recording will begin when Record is selected or the Start button is pressed on the sensor handle, and will end when Stop is selected, or the specified number of frames have been collected.

Note: If more than one Real-time window is open, the force and area values set for triggering are the sum of the total values for all windows.

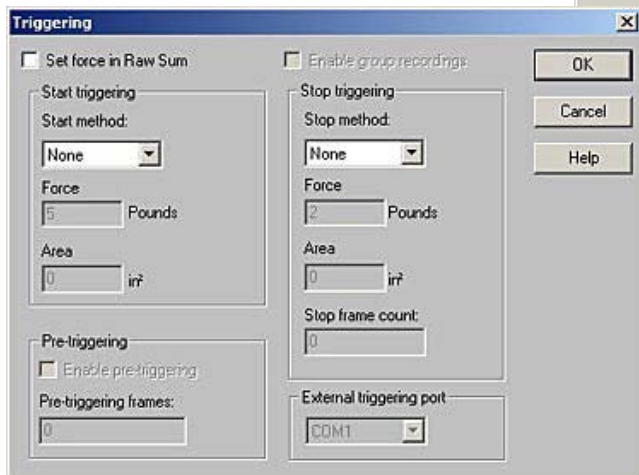
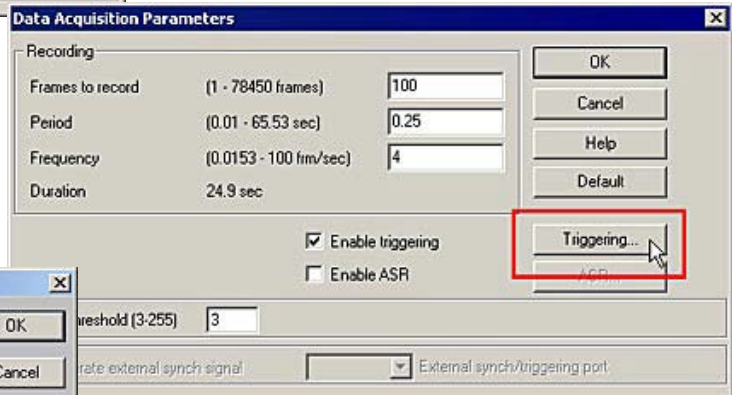
To access the internal triggering option:

1. Select **Acquisition Parameters** from the **Options** pull-down menu (shown at right).



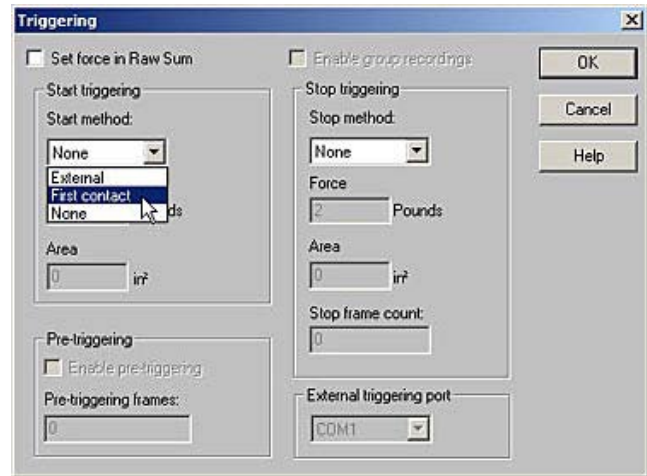
2. Click in the **'Enable Triggering'** box on the left side of the dialog box (shown at left). The **Triggering** button, on the right side of the dialog box, will become available.

3. Click on the **Triggering** button (shown at right) to open the **'Triggering'** dialog box (shown below).



To set the 'start triggering' events:

1. In the 'Start Triggering' area, select '**First Contact**' as the Start Method. Selecting 'None' will disable triggering. The '**Force**' and '**Area**' fields become available.
2. Enter the force and contact area on the sensor at which you want recording to begin. When the load and contact area on the sensor both reach or exceed these thresholds, the software will start recording. You should enter small values in these fields (e.g. 1 lb. and 0.25 in²), to ensure that an actual event, and not noise, actually triggers the recording. To disable triggering for either force or area, set its value to 0.




Note: The triggering force and area are not affected by the Legend's lower limit setting. Therefore, triggering may occur even though no pressures are visible in the Real-time window.

3. If a calibration has been performed on the real-time window, you have the option of setting the force units to raw digital units instead of using the calibration units.



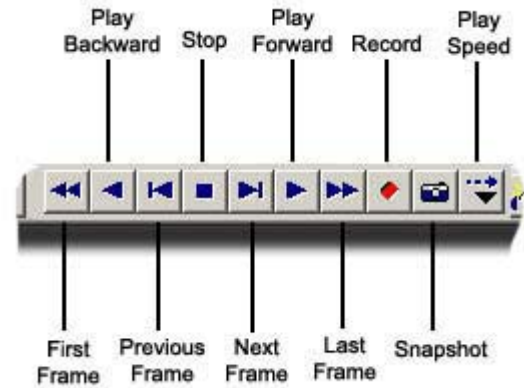
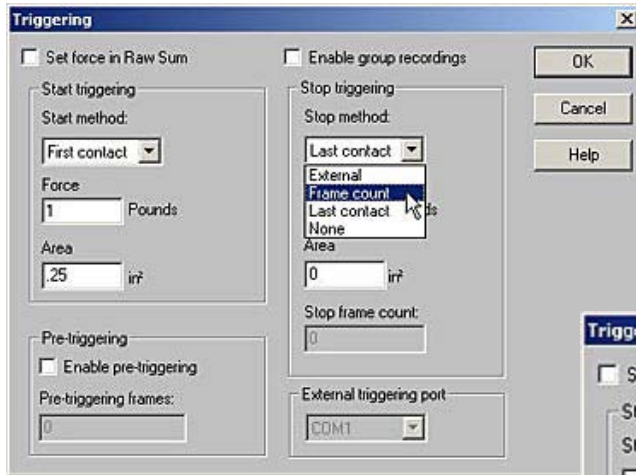
To set the units to raw, click in the '**Set Force in Raw Sum**' check box above the '**Start Triggering**' area. If a calibration has not been performed, this check box will be grayed out, and raw units must be used.

4. Once recording is initiated by the user (by selecting **Record** or **Snapshot** from the Main Menu or Toolbar), pressure data collection will not start until the start trigger event occurs. If no frames of data have been collected (i.e. no trigger event was received), and recording is stopped, the system will revert to real-time mode.

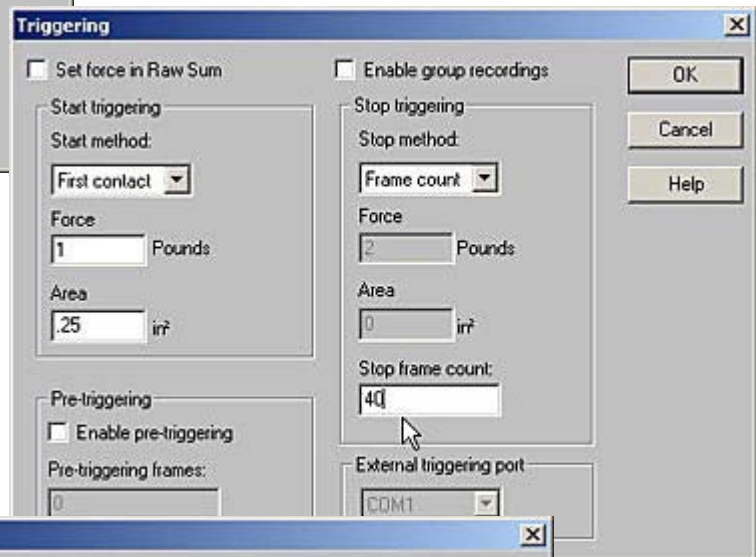
- The recording will stop when specified by the '**Stop Triggering**' event selections. If you wish to stop recording manually, select **Stop** from the **Movie** menu, click on the Stop icon in the Toolbar, or press the <F4> key on your keyboard.

To set the 'stop triggering' events:

- In the '**Stop Triggering**' area, select '**Frame count**' as the Stop Method.

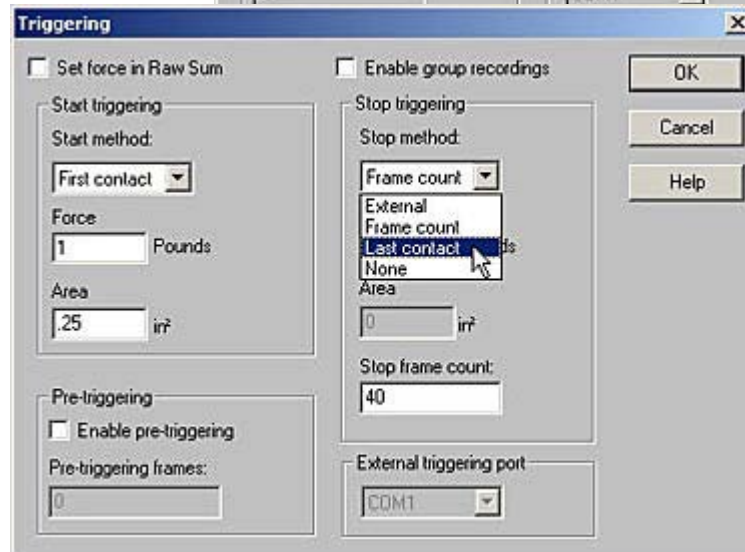


1. If you select Frame count as the Stop Method, the '**Stop frame count**' field will become available. Enter the number of frames you want to record (in this field); the software will stop recording when this number of frames has been collected.



Note: The number of frames that you enter as the 'Stop frame count' must be less than the 'Frames to Record' that are selected in the Acquisition Parameters (Options menu).

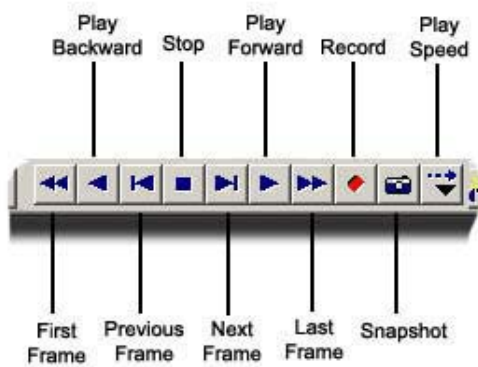
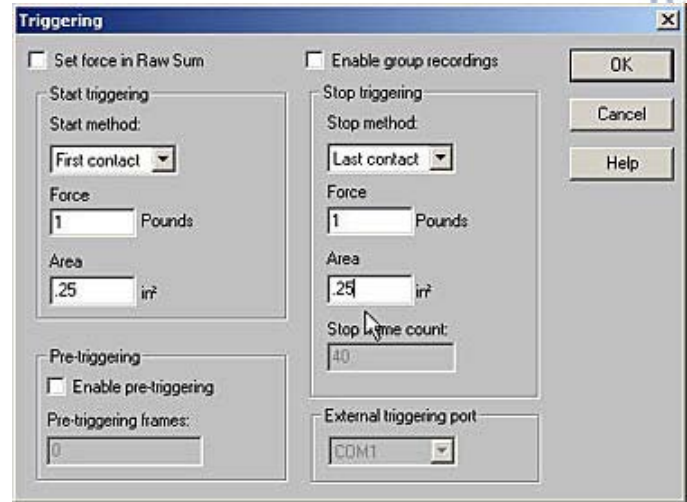
- If you select **Last Contact** as the Stop Method, the '**Force**' and '**Area**' fields become available.



4. Enter the force and contact area on the sensor at which you want recording to end. When the load and contact area on the sensor both reach or drop below these thresholds, the software will stop recording.

To disable 'stop triggering' for either force or area, set its value to 0. The units for the area and force settings will be the same as those used to start triggering.

4. If you wish to stop recording manually, before the stop trigger event occurs, select **Stop** from the **Movie** pull-down menu, click on the Stop icon in the Toolbar, or press the <F4> key on your keyboard.



Note: The number of frames that you enter as the 'Stop Frame Count' is limited by the 'Frames to Record' that are selected in the Acquisition Parameters.

Note: Your recording may include fewer frames than expected if either of the 'Remove Blank Frames' options, in 'Set User Preferences' (Options menu), are selected.

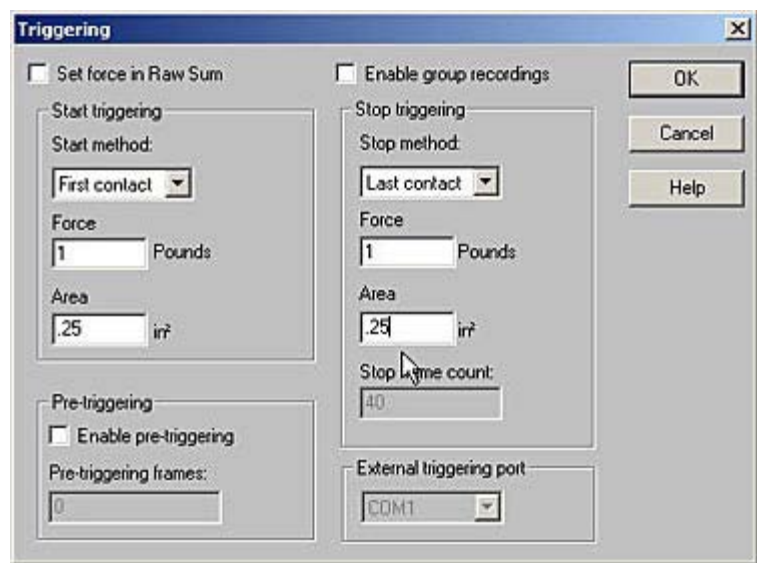
GROUP RECORDINGS

It is possible to record groups of frames as a single movie (called a 'Group Recording') using multiple start and stop events. When this is done, the software will continue to start and stop recording based on the selected triggering events until either the number of frames specified in the 'Frames to Record' field of the Acquisition Parameters (Options menu) have been recorded, or the user manually stops the recording.

Note: 'Pre-Triggering' and 'Enable group recordings' cannot be in effect at the same time. One option will be grayed out when the other is selected.

To set up a group recording:

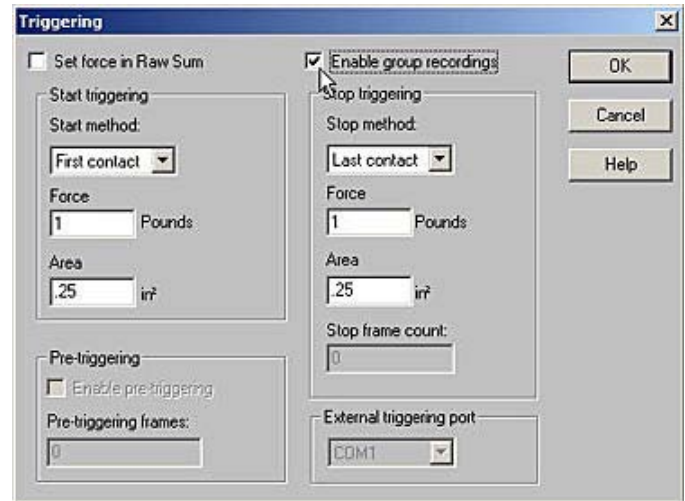
1. Set the Stop and Start Triggering options as desired.



2. Place a check mark in the 'Enable group recordings' box above the 'Stop Triggering' area in the dialog box, then click <OK>.

Note: All frames within a group are recorded at the same frequency, which is defined in the Acquisition Parameters dialog box.

Note: When you view a movie that has multiple groups of frames, the time axis in the graph display will not show the correct times. It treats the frames as if they all occurred in sequence; however, this does not include the time in which the software was waiting for a trigger event.



PRE-TRIGGERING

The Pre-Triggering option enables you to retrieve frames of pressure data that precede the start trigger event. These frames are then included at the beginning of the recording. This option is often used in high-speed applications, in which the trigger events occur very rapidly, and 'pre-triggering' ensures that no pressure data is missed.

To set up pre-triggering:

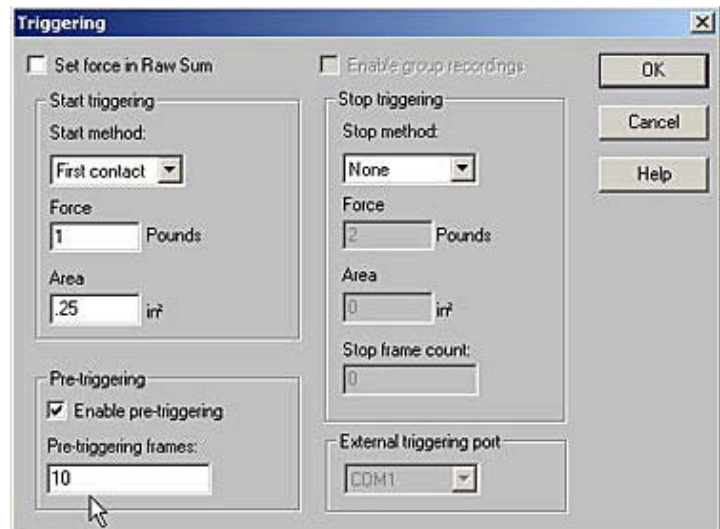
1. Access the Triggering options, as described in the "Triggering a Recording" section. Set up your "Start Triggering" options. In the 'Pre-Triggering' area, place a check mark in the 'Enable pre-triggering' box to select this option.

Note: 'Pre-Triggering' and 'Enable group recordings' cannot be in effect at the same time. One option will be grayed out when the other is selected.

2. In the 'Pre-triggering frames' field, enter the number of frames (preceding the start trigger event) that you want to retrieve. The number of frames entered is limited only by the 'Frames to Record' that are selected in the Acquisition Parameters (Options menu).

Note: Your recording may include fewer frames than expected if either of the 'Remove Blank Frames' options, in 'Set User Preferences' (Options menu), are selected.

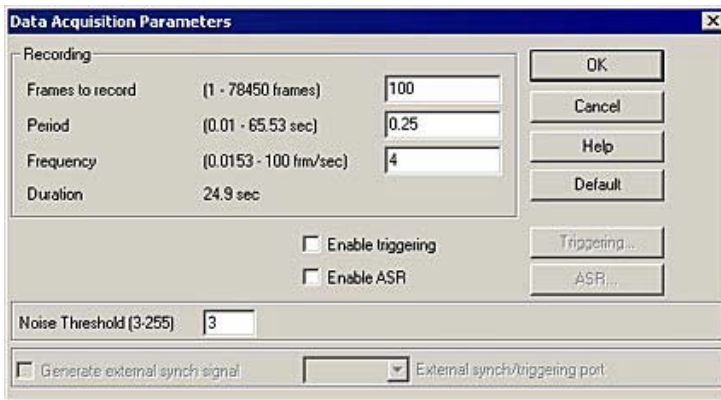
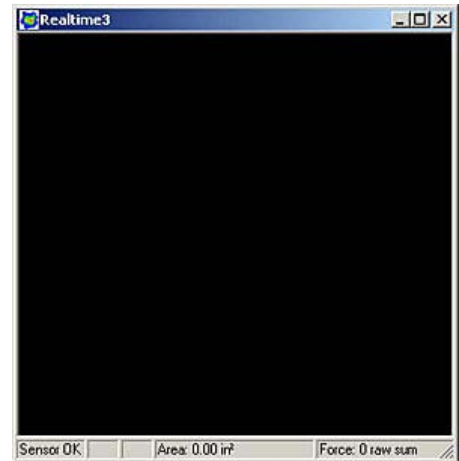
3. Click <OK> to apply your selection.



TAKING A RECORDING

To initiate a recording, do the following:

1. Open a new real-time window (see image at right). Refer to the “Calibration and Equilibration” section, and perform a sensitivity adjustment, equilibration, and/or calibration, if necessary.
2. Select **Acquisition Parameters** from the **Options** pull-down menu, and ensure that the recording parameters are set correctly for your application (see below). Also ensure any triggering options are set correctly.

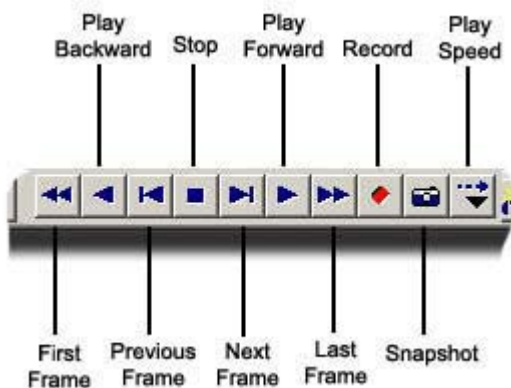
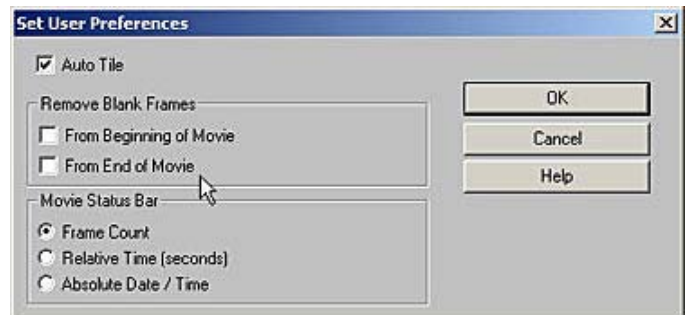


3. If you wish to remove blank frames at either the beginning or end of the recording, select **Set User Preferences** from the **Options** pull-down menu, and place a check mark next to ‘**From Beginning of Movie**’ or ‘**From End of Movie**’, respectively (refer to the images below).



Note: A frame is considered blank by the system if all of the forces on the sensor are below the Noise Threshold setting in Acquisition Parameters (Options menu).

4. Load the sensor exactly as specified in the “Calibration and Equilibration” section.
5. Select **Record** or **Snapshot** from the **Movie** pull-down menu, or click the corresponding icon on the Toolbar (see image below), to start recording pressure data. You can also begin recording by pressing the <F2> key (or <F3> to take a snapshot) on your keyboard.



Note: The Snapshot option records a single-frame movie.

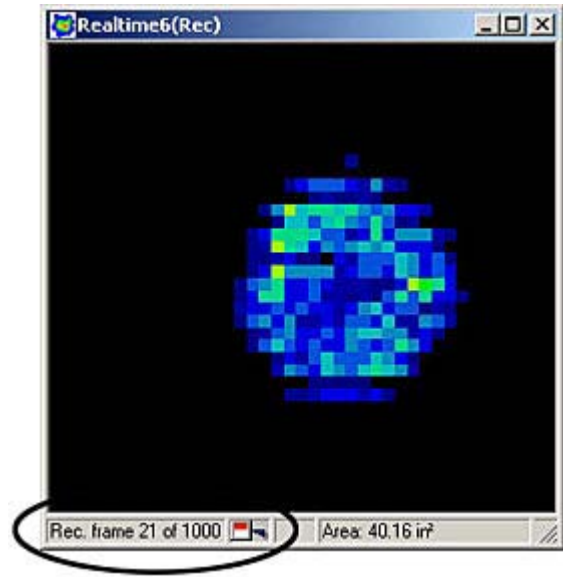
Another way to start a recording is by pressing the long blue button on the top of your sensor handle (referred to as '**Remote Recording**').

***Note:** Remote recording control is an option in the I-Scan system. It may not be available on your system.*

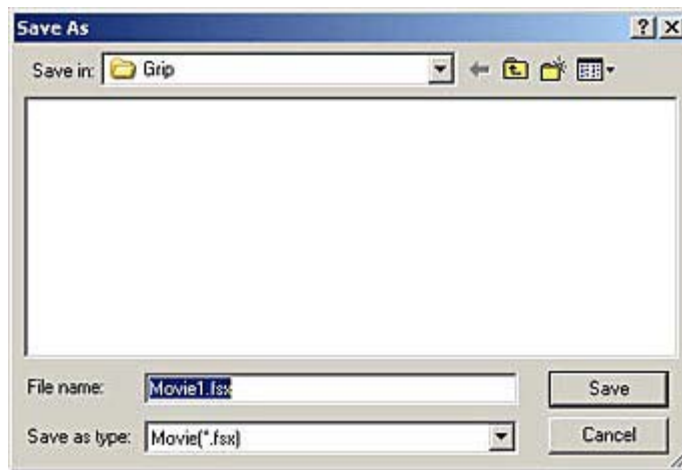
When recording is in progress, '**(Rec)**' is added to the real-time window's title bar, and the Real-time Status Bar displays 'Rec. Frame X of Y' (where X is the current frame number and Y is the total number of frames to be recorded – see image at right).

6. When the desired number of frames has been collected (if triggering is not in effect), select **Stop** from the **Movie** pull-down menu or click the **Stop** icon in the Toolbar, to stop recording data.

As soon as recording is completed, the real-time window becomes a Movie window. Now you have the recording available, and are ready to review the recorded data. Refer to the "Reviewing a Movie" section.



7. Once you have completed recording, you can select **Append** from the **Movie** menu to add additional frames of data to the end of the movie (refer to "Apply Mode" for more information).
8. If you want to save the movie as a file, select **Save Movie** from the **File** pull-down menu, and save the movie under the filename and directory of your choice. ***I-Scan*** movie files must have the extension *.fsx.



***Note:** The View settings (e.g. window position, zoom %, rotation, averaging, COF), as well as the sensitivity setting and calibration data, are all saved with the movie file.*

APPEND MODE

I-Scan allows you to record additional frames of data at the end of an existing movie, using the Append feature. The Append option is only available when a Movie window (recording) is open. The original recording may be one that was closed and reopened, or one that has just been recorded.

To append data to an existing movie:

1. Take a recording, or open a previously recorded movie.
2. Select **Append** from the **Movie** pull-down menu. The 'Select Sensor' dialog will be displayed. Under "Available Handles", you may select which handle to use for the appended real-time window. The "Available Maps" will be grayed out, because the appended data must use the same map as the initial movie.

Once you select a handle, a real-time window will automatically be opened (the title bar will say "Append to...").

The Append menu item will be unavailable (grayed out) if a real-time window cannot be opened.

3. Take a recording. As soon as you finish recording, the frames of data you have recorded will automatically be added to the end of the existing movie. The frame count displayed in the movie's status bar will reflect the additional frames.

Calibration Notes:

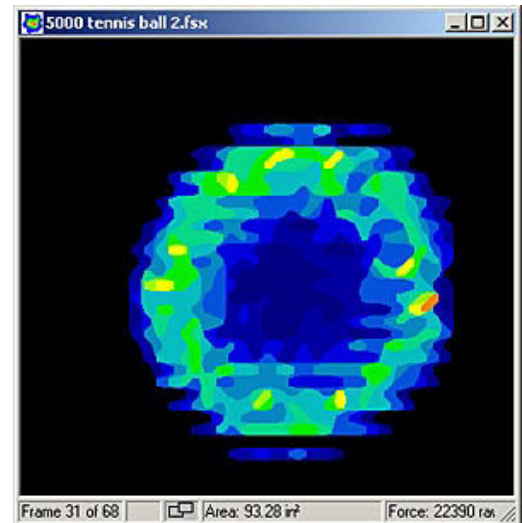
- The 'Append' real-time window will use the same calibration data as the original movie window (if it has been calibrated). The appended frames cannot be calibrated separately from the original recording.
- When an 'Append' real-time window is open, the original movie window cannot be calibrated (the Calibration and Load Calibration File menu items are grayed out).

Equilibration Notes:

- The 'Append' Real-time window can be equilibrated separately from the original movie window; however, this is not recommended, since the final (appended) movie will use two sets of equilibration data.
- If a movie has been recorded, and has not been closed and reopened, any appended frames will use the equilibration data from that movie window. If the movie has been closed and reopened, any appended frames will use the equilibration data from the last open Real-time window, if there was one.

INCLUDING COMMENTS WITH A RECORDING

The Comments option allows you to view, add, and edit the contents of the comment fields associated with the current active Movie or Real-time window. The comment fields can be filled in before a recording is taken (i.e. in a real-time window), between recording and saving, or after saving. These comment fields can be useful for recording information about a particular series of tests, such as the test date and the technician.



The '**Comments**' dialog box (shown below) is accessed by selecting **Comments** from the **Edit** pull-down menu, or clicking on the Comments icon in the Toolbar. You can also view the comments for an active window by selecting Settings from the Options pull-down menu, and clicking on the '**Comments**' tab. Click on the '**Change Comments**' button on to display the '**Comments**' dialog box.

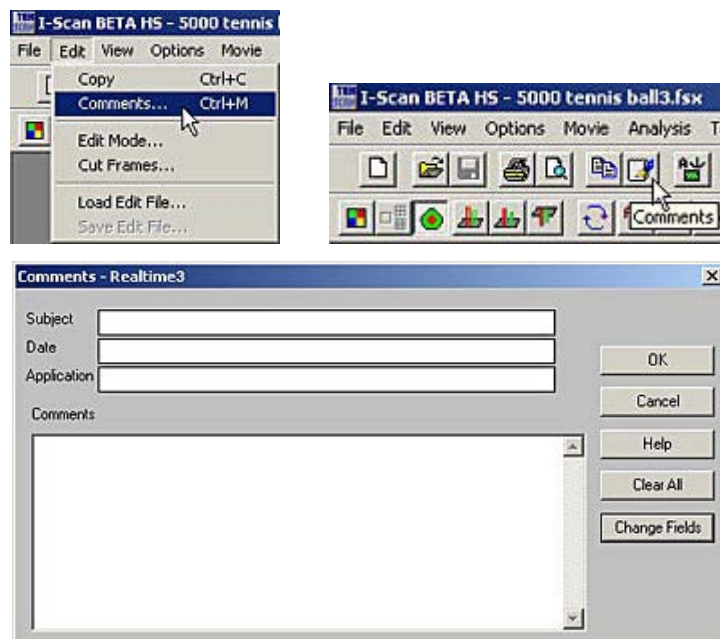
***Note:** The title of each of the spaces in the Comments dialog box is referred to as a 'field name', and the entered information is referred to as a 'field entry'.*

If a Movie window is active, you may enter information in each field, but cannot change field names. Once comments are added or altered in a Movie, the recording must be saved again in order to save these comment changes. If the field entries are changed in a newly recorded movie, all other newly recorded movies will automatically be updated with the information entered in these fields (information under the 'comments' field is not transferred between movies).

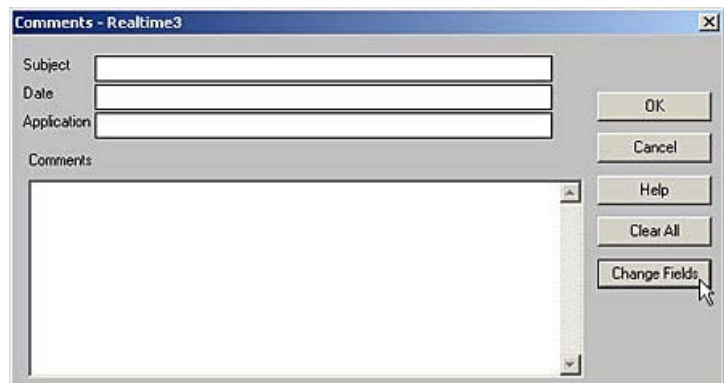
If a real-time window is active, you may enter information in each field, and also add, delete, and change the field names. Once comments are changed in a real-time window, they will be remembered until New Session is selected from the File menu, or the application is closed. If this real-time window is recorded and saved as a movie, the window's comments will also be saved.

To edit a field name:

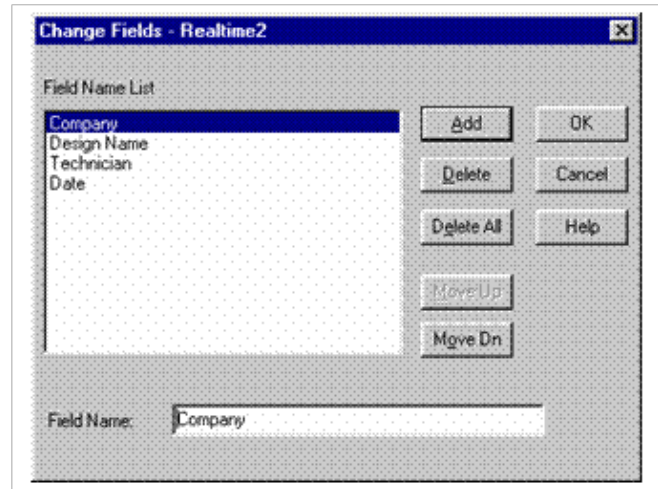
1. With a real-time window as the active view, select **Comments** from the **Edit** pull-down menu, or click the Comments icon in the Toolbar. The '**Comments**' dialog is opened.



2. Click on the **Change Fields** button to access the '**Change Fields**' dialog box (shown at right), which allows you to add, delete and change field names.
3. In the '**Field Name List**', select (highlight) the field name you want to edit (only one may be selected at a time). The currently highlighted field will be displayed in the edit field (under '**Field Name**') at the bottom of the dialog box.



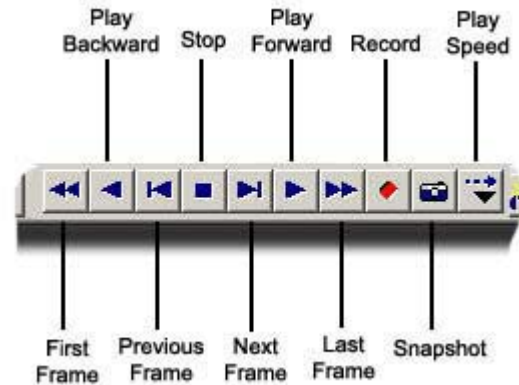
4. Use the edit buttons to edit the currently highlighted field:
 - **Add:** adds a new field name. Type in the new field name under 'Field Name'.
 - **Delete:** removes the highlighted field name
 - **Delete All:** removes all field names
 - **Move Up:** moves the currently highlighted field up one row.
 - **Move Dn:** moves the currently highlighted field down one row.
5. When you are satisfied with the changes, click <OK>. The changes will be remembered across sessions.



REVIEWING A RECORDING

Once you have made a recording (movie), it is available for your review. Your movie consists of a certain number of frames of pressure data that were captured during the recording period. This section will review the basics of movie playback.

The playback options control all open movie windows (and corresponding graphs) in unison, regardless of which movie is active. Press <CTRL> if you want the selected playback option to affect only the active window. If a movie window is 'Paused', it will not respond to the playback commands.



The next section ('Analyzing Pressure Data') describes how to interpret the pressure information in a recording or Real-time window.

To play back a recording:

Either record a current Real-time window, or load an *I-Scan* movie file (with extension *.fsx). Play the movie using the Movie pull-down menu items, or the corresponding Toolbar icons:

- **First Frame:** positions the recording at the beginning (first frame).
- **Play Backward:** plays the recording in reverse, starting at the current frame.
- **Previous Frame:** plays the recording in reverse one frame each time it is clicked.
- **Stop:** stops playing the recording
- **Next Frame:** plays the recording forward one frame each time it is clicked.
- **Play Forward:** plays the recording forward, starting at the current frame.
- **Last Frame:** positions the recording at the end (final frame).
- **Record:** Starts the recording.
- **Snapshot:** Takes a single frame recording.
- **Play Speed:** adjusts the speed at which the recording is played back. You can choose from one of the following speeds: Slowest, Medium Slow, Normal (default), Medium Fast, and Fastest. The play speed can be adjusted either before or during movie playback.
- **Continuous Play:** To enable Continuous Play, press the <SHIFT> key, and simultaneously select Play Forward or Play Backward. The movie will play in the desired direction indefinitely, looping back to the first or last frame each time.

Note that the Play Speed option has a Toolbar icon (and no pull-down menu item), and that the <CTRL> key does not affect the Play Speed option.

LINKING A PHOTO TO A MOVIE FRAME

The **Link Photo** feature allows you to open a photo file and link it to any frame of a Tekscan (*.fsx) movie. Since the photo is linked to a frame within the movie, it becomes possible to associate one or several photos to a single movie. The amount of photos is dependant only on the number of frames available for the movie. Conversely, you can only associate one photo per frame. You cannot have multiple photos linked to a single frame. The following image file formats are supported: *.bmp;*.jpg;*.gif;*.tif;*.pcx;*.tga.

Note: The “Photo Link” option is only available if a movie is open within the Tekscan software. If there is no open movie, the “Link Photo” option is grayed out and unavailable.

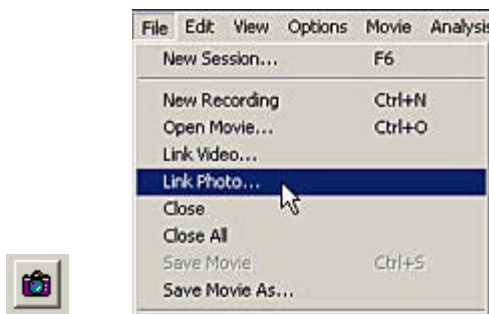
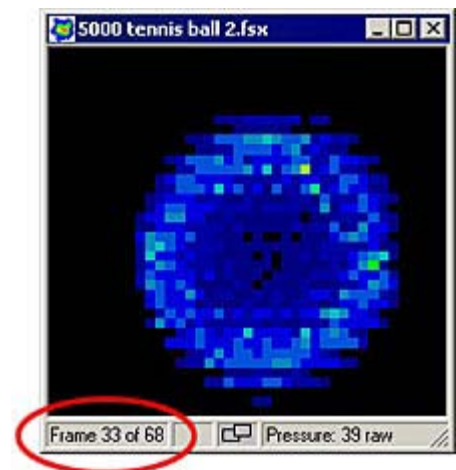
The following highlights some benefits from using this feature.

- Procedure documentation is enhanced for all or any part of a movie. You can link several photos to a single movie, thereby documenting each step of a given procedure.
- Create a visual catalog for your movies. Once a photo is associated to a movie frame, you can instantly recognize when the movie was recorded by viewing the linked photo. In addition, you can [include comments](#) to the movie to explain why the photo is linked to the movie.
- If you have several different applications for your Tekscan system, this feature can help you associate an application to a movie.

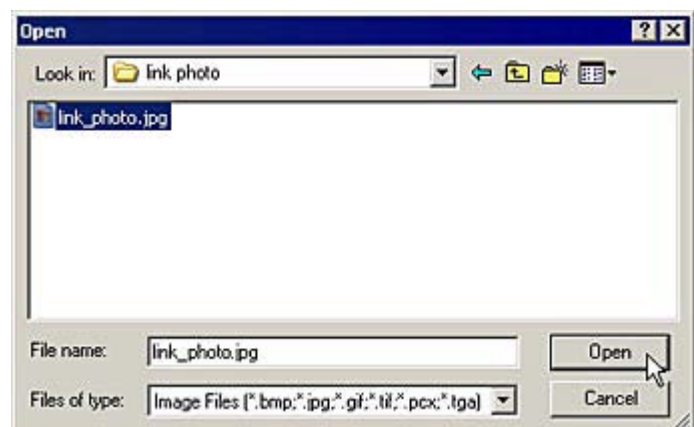
Associating a Photo to a Movie Frame

Follow the steps below to associate a photo to a movie frame:

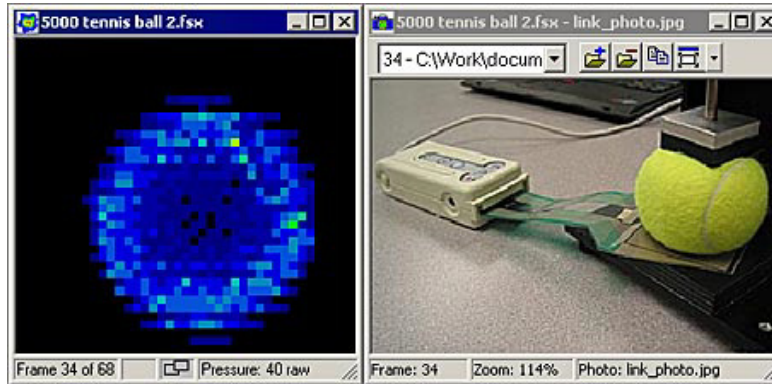
1. Open the movie for which you would like the photo associated.
2. Advance the movie to the frame location for which you would like the photo associated (see image at right). You must select a frame for your movie. If you do not, the photo is automatically associated with frame 1.
3. Click the Photo Link icon on the toolbar, or go into File -> Link Photo (shown below).



4. The “Open” dialog displays (shown at right), where you can select a photo file from your hard drive. Locate the appropriate file, highlight it, and click **Open**.



5. The photo file opens and snaps to the right edge of the movie window.



Only one photo can be linked to each frame of the movie. If you attempt to link a photo to a frame which already has a linked photo, the message shown at right opens on-screen asking if you would like to overwrite the existing photo with the new photo:

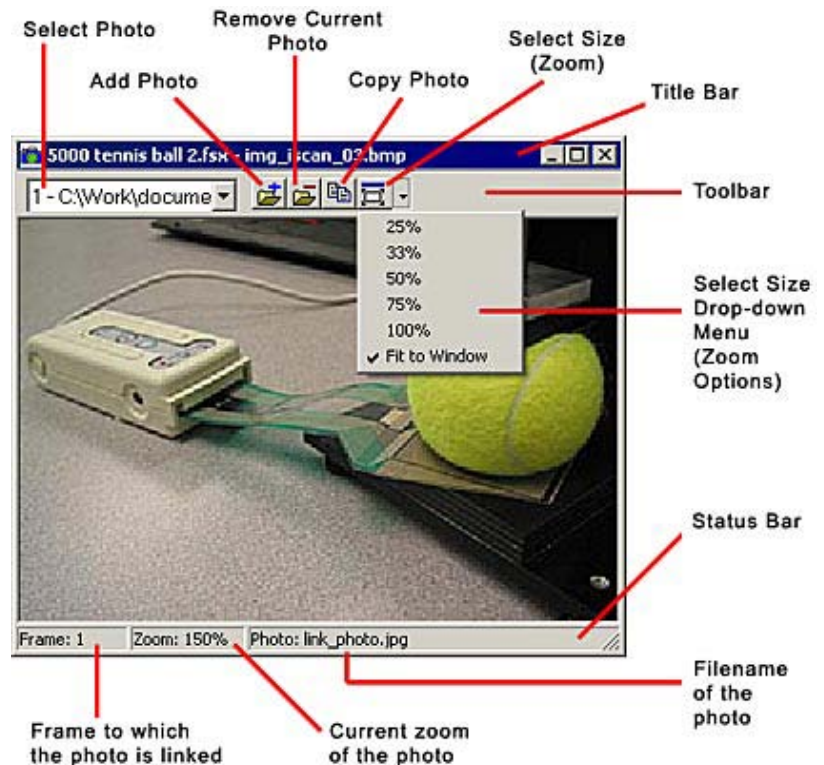


When there is only one photo linked to the frame of a movie, removing it causes the warning message shown at left to appear on-screen

Toolbar and Status Bar Options

When a photo is linked to the frame of a movie, the following toolbar and status bar options are available from the top and bottom of the photo window, respectively:

- **Select Photo:** This option lists all photos that are linked to the current movie. Clicking on the drop-down arrow allows you to select from any of these linked photos. When a photo is selected, the movie automatically jumps to the corresponding linked frame.
- **Add Photo:** Allows you to add a photo and link it to a movie frame.
- **Remove Current Photo:** Allows you to remove the currently displayed photo, thereby removing its link to the movie's frame.
- **Copy Photo:** This command copies the photo to the windows clipboard. You can then paste the photo to any other application that supports photo files, such as Microsoft Word or Excel.



- **Select Size (Zoom):** Allows you to quickly select from a number of preset zoom factors. Options are 25%, 33%, 50%, 75%, 100% and Fit to Window. These scale factors are relative to the original photo size. You can also drag from the bottom-right edge of the window to expand or contract the photo and photo window, while maintaining the photo's aspect ratio. *Note: the photo's current scale is also displayed as a percentage within the middle of the status bar.*

Movie Playback

During movie playback, when one photo is linked to a movie, the photo is continually displayed for the entire duration of the movie.

If more than one photo is linked to the movie, when the movie is played, the photo switches to the next photo at the movie frame for which the photo is linked. For example, if you associate image1 to frame 1 and image2 to frame 10, when the movie plays from the beginning, image1 is displayed until the movie reaches frame 10. When the movie reaches frame 10, image2 is displayed until it reaches another frame for which a photo is associated, or until the end of the movie if no other photos are linked to the movie.

Smart Docking

By default, the photo is docked to the right side of the movie window (called "Smart Docking"). When you drag the movie around the screen (by clicking on the title bar of the movie and dragging with your mouse), the photo window moves with the movie. This helps you identify photo windows in situations where you have several movie and photo windows open on-screen at once.

Conversely, you can move the photo window independently of the movie window by clicking and dragging on the photo window's title bar. The reason for this is so that you can dock the photo window to a different side of the movie window. If, for example, you want the photo to dock to the left side of the movie window instead of the right, you can click and drag the photo window's title bar and move the photo window to the right side of the movie window. When you then move the movie window around the screen, the photo window automatically docks to the left side of the movie window.

If you prefer to unhinge the photo window entirely so that the movie window moves independently from the photo window, right click and select **Disable smart docking** from the context menu (see image below). You can now move both windows independently of one another.



Copying Linked Photos

Using the Copy Photo command located as an icon on the toolbar, you can copy any photo to the windows clipboard. The copy can then be pasted to any other application that supports photo files, such as Microsoft Word, PowerPoint or Excel.

When you copy and paste the photo in this manner, the photo is always displayed at its original photo size. If you want to keep the photo's size the same as what you see on-screen, right-click over the photo window. From the context menu, select **Copy Photo (Current View Size)**. When the photo is pasted into other programs, the size will remain consistent with the photo's current view displayed within the Tekscan software.



Missing Photo Links

When a photo is linked to a movie frame, there is a connection that is made between the photo file location and the movie. If the photo is moved, this connection is broken. When this happens, your photo is no longer displayed when the movie is opened (see image at right).

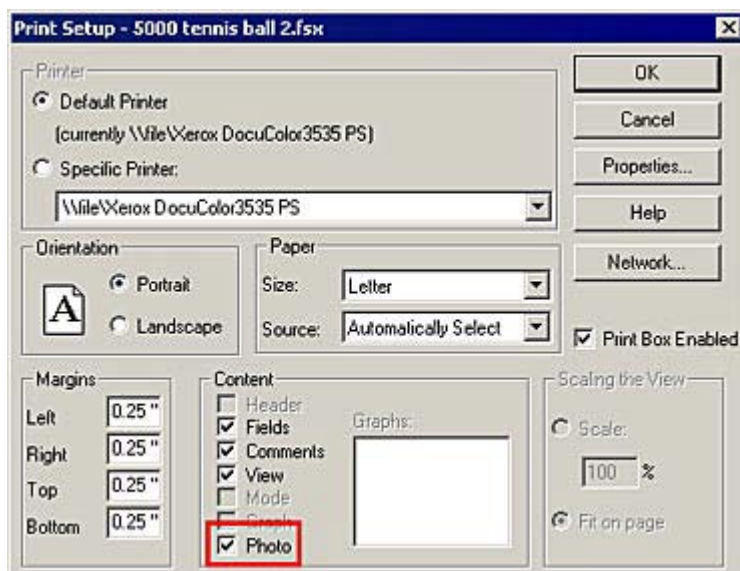
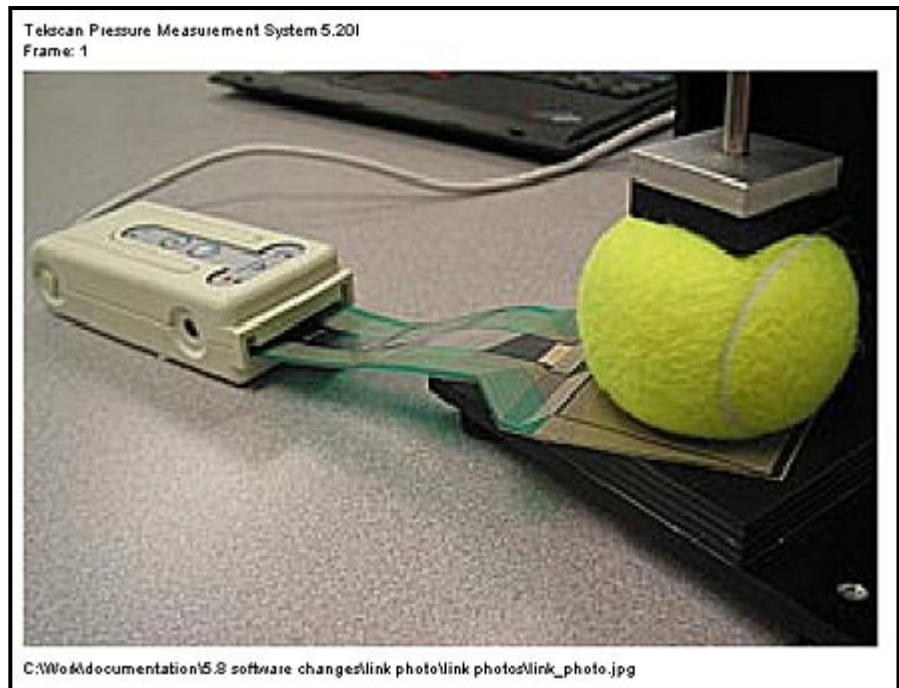
To correct this situation, first remove the file using the **Remove Current Photo** icon on the toolbar, and then link the photo once again by clicking the **Add Photo** icon on the toolbar. When the "Open" dialog is displayed, select the new location for the photo; where the photo was moved. The photo is visible once again.



Printing the Movie Frame and Photo

To print the photo without the movie, first select the photo window by clicking the photo window's Title Bar. It will turn blue. Then go into File -> Print. The Photo prints out along with the Tekscan software version, Frame number to which the photo is associated, and full path to the file's location on your hard drive (see image at right).

If you would like to print both the movie and photo window at the same time, first select the movie window by clicking the movie window's Title Bar. It will turn blue. Then go into File -> Print Setup. Ensure the "Photo" checkbox has a checkmark located in it (see image below). This is checked by default.



Click **OK** to close this dialog box. Then go to **File -> Print**. Both the movie frame and photo are printed at the same time.

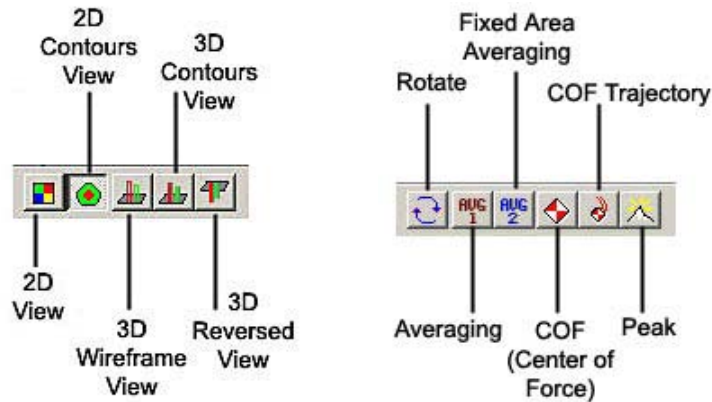
ANALYZING PRESSURE DATA

As you review real-time or recorded (Movie) pressure data, you will want to make use of the many analysis tools that *I-Scan* provides. The four main analysis options are: "Display", "Objects", "Graphs", and "Save ASCII." Each of these choices provides you with a number of ways of presenting the data, each of which allows you to focus on specific aspects of the contact pressure. The following sections describe how to best utilize these analysis options for your application.

DISPLAY OPTIONS

The following *I-Scan* display options are essential in analyzing pressure data: 2-D and 3-D Views, Averaging, Fixed Area Averaging, Movie Averaging, Movie Contact Averaging, COF (Center of Force), COF Trajectory, Max Area Frame, and Peak. *Note that Movie Averaging, Movie Contact Averaging, Max Area Frame and Peak are not available in Real-time mode.*

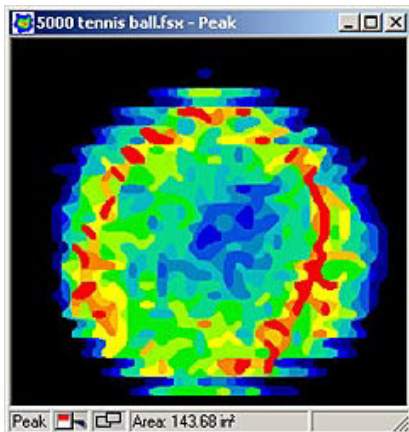
The following displays the main "Views" sections of the lower toolbar:



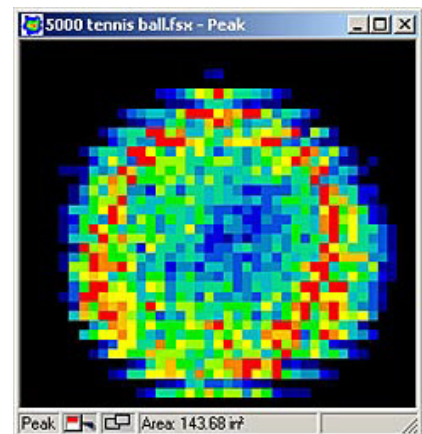
Note: When any of these View options is selected, it affects all open windows. To apply the selected option to only the active window, press the <CTRL> key and select the option simultaneously.

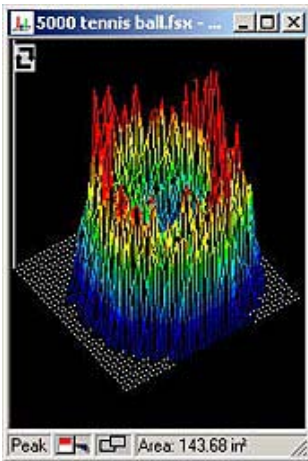
Note: Any View menu items that are in effect are reset to the default when File>>New Session is selected. They are saved when the program is closed and reopened, or File>>New Recording is selected.

2-D displays the recording in two-dimensional form with the color of each sensel location representing the pressure sensed at that point on the sensor. This display looks the closest to the actual raw output of the sensor, and individual sensels can be identified. **2-D** is the default view (see image at right).



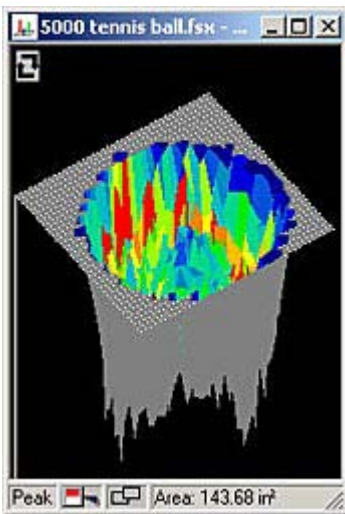
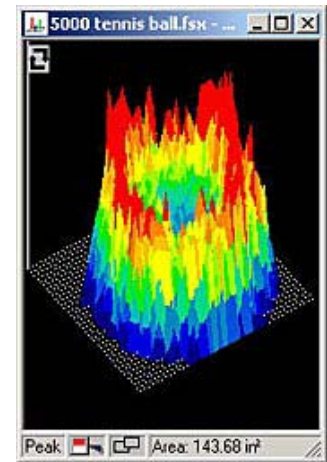
2-D Contours displays the recording as a two-dimensional, contoured image. The sharp corners of the sensor output are smoothed, making the pressure boundaries easier to discern (see image at left).





3-D Wireframe displays the recording as a three-dimensional image, in which higher pressures are shown as peaks (see image at left). The relative peaks provide visual insight into the gradients within each color band. **'Wireframe'** means that the image is not a solid image, and is represented by a series of interconnected lines.

3-D Contours displays the recording as a three-dimensional image, in which higher pressures are shown as solid (contoured) peaks (see image at right). The **3-D Contours** image looks similar to the **3-D Wireframe** image, except that the image is represented as a solid object. This display can often provide a view of complex pressure data that is easier to comprehend than the Wireframe image.

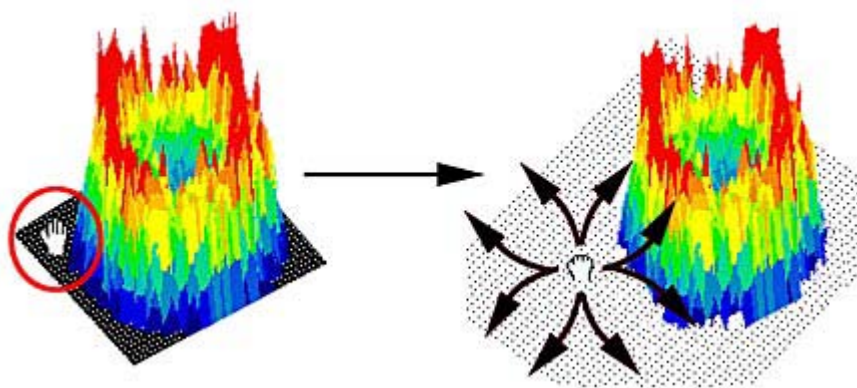


3-D Reverse displays the recording as a three-dimensional image, in which higher pressures are shown as inverse peaks (see image at left). The **3-D Reverse** image is simply an inverted, or reversed, 3-D Contour image. 3-D Reverse is especially useful for viewing the exact footprint of the applied force.

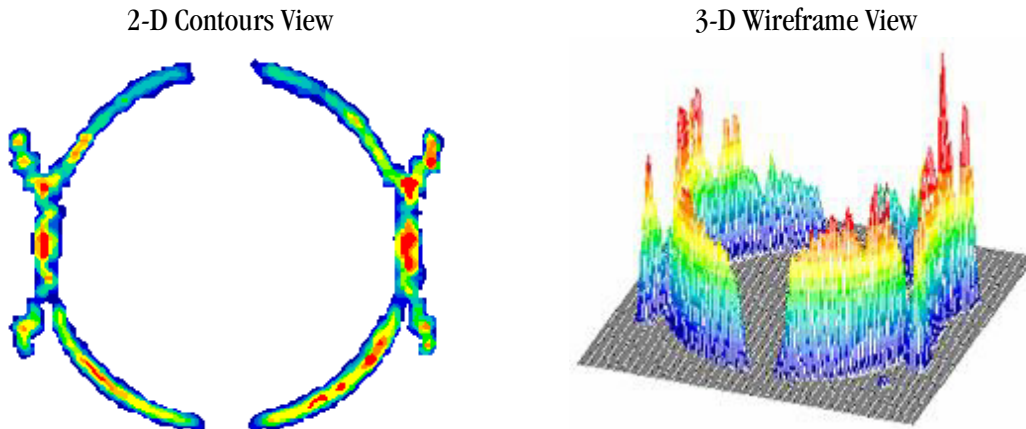
The 3-D displays are especially useful in comparing pressure areas that are above the maximum pressure setting on the Legend. In the 2-D displays, all pressures above the maximum pressure setting are shown as red, and cannot be distinguished from each other; in **3-D Wireframe**, **Contour**, and **Reverse** modes, pressures are displayed as peaks, and relative pressure differences are readily apparent. As you can see in the two figures on the following page, the **2-D** displays show where there are areas of high pressure, but the **3-D** displays give a much clearer view of exactly how high the pressures are at those points.

When you are in a "3-D" view, a separate "rotate" function is available. Moving your cursor over the screen, you will notice the cursor changes to a "hand" icon. Left-clicking your mouse will grab the 3-D image and you can then rotate the display in the direction of your choice by navigating with your mouse.

Note that the bottom plane of the image is not available, and rotating the image past this point will flip the image to show the next available plane. The arrow keys on the keyboard can also be used to shift the image incrementally in any direction.



The following shows our sensor in a ring configuration with 2 of the 5 available views:



Averaging (Avg1): Averaging displays the image with each cell's pressure value modified to reflect the value of its neighbors. It does include neighboring cells that have zero load. This results in a smoother image, with an increase in total loading area since certain cells with a previous zero value now have a value. When **Averaging** is in effect, it has a check mark beside it in the pull-down menu, and its Toolbar icon is depressed. As an example, in the group of nine cells shown below, the averaged pressure value of X is calculated using the following equation:

A	B	C
D	X	E
F	G	H

$$X_{avg} := \frac{\frac{A + C + F + H}{2} + B + D + E + G + X}{7}$$

Where the load recorded in each cell (A-G) is used to calculate the averaged pressure in cell X. Note that if cell X is at the edge of the sensor, the values of the neighboring cells that are not loaded are not used in the calculation.

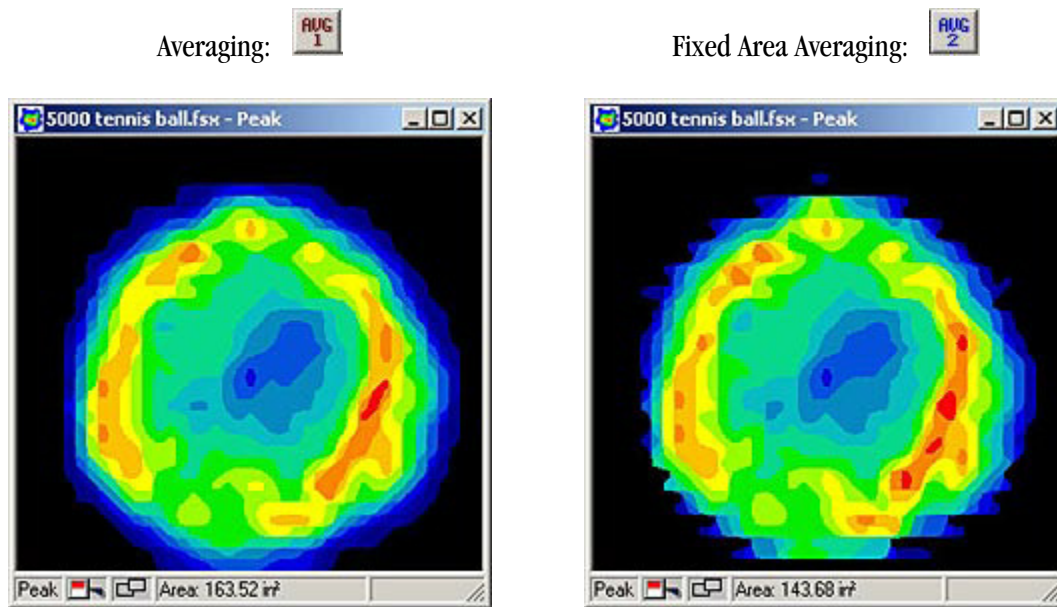
Fixed Area Averaging (Avg2): This option displays each cell's pressure value modified to reflect the value of its neighbors. It does not include neighboring cells that have a zero load value. There is no increase in the total contact area in the display. The result is a smoother image. When **Fixed Area Averaging** is in effect, it will have a check mark beside it in the pull-down menu, and its Tool bar icon will be depressed. As an example, in the group of nine cells shown below, the averaged pressure value of X is calculated using the following equation:

A	B	C
D	X	E
F	G	H

$$X_{avg} := \frac{\frac{A + C + F + H}{2} + B + D + E + G + X + X}{8}$$

The denominator is reduced to reflect the "weight" of the cells with zero load in the equation. For example, if cell A is zero, the denominator is reduced by half. If D is zero, the denominator is reduced by one. If X is zero, the denominator is reduced by two.

The following images show the difference between Averaging (Avg1) and Fixed Area Averaging (Avg2):



Movie Averaging opens the 'Select Averaging Range' dialog box, which allows you to display the averaged pressure value of each cell for the entire recording, or for a selected range of frames, in one or more composite frame(s). First go to **View -> Movie Averaging**.

This opens the Movie Averaging dialog below:

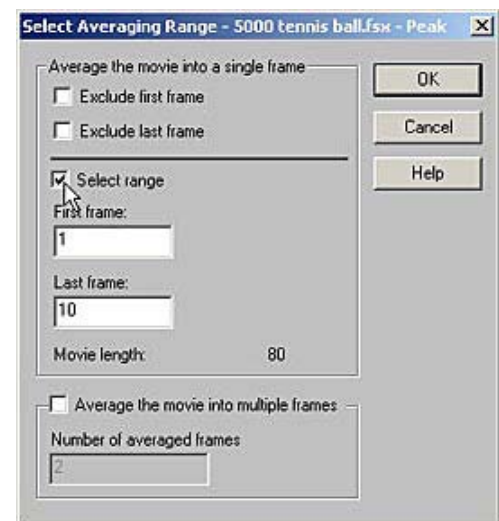


To average the movie into a single or multiple frames, do the following:

- **Average the movie into a single frame**

To exclude the first or last frame from the averaged frame, click in the 'Exclude first frame' or 'Exclude last frame' box. To include only a specific range of frames in the averaged frame calculation, click in the 'Select range' box, and then define the range by

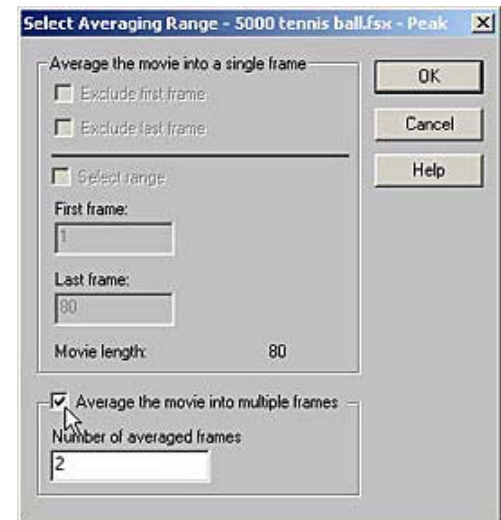
entering frame numbers into the 'First frame' and 'Last frame' fields (shown at right).



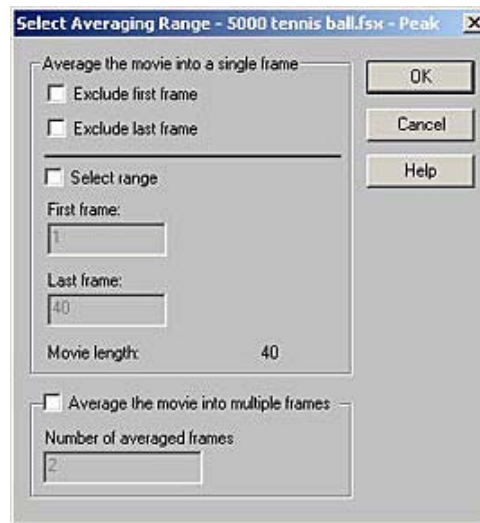
- **Average the movie into multiple frames**

To average the movie into multiple averaged frames, place a check mark in the 'Average the movie into multiple frames' check box (shown at right), and enter a value in the 'Number of averaged frames' field.

The movie will be broken into groups of frames, each of which will contain the number of frames you entered. Each of these groups of frames will then be averaged into a single frame. For example, if your movie has 50 frames, and you enter 10 in this field, your averaged movie will have 5 frames. The first frame would have been created by averaging the first 10 frames, the second frames from averaging the second 10 frames, and so on.

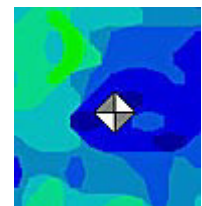


Movie Contact Averaging opens the 'Select Averaging Range' dialog box, which allows you to display the averaged (contact) pressure value of each cell for the entire recording, or for a selected range of frames, in one or more composite frame(s). Refer to the description of 'Movie Averaging' (above) for instructions.



Movie Contact Averaging is the same as the Movie Averaging option, except that it calculates the cell's average pressure value in a slightly different manner. The difference between these two types of movie averaging is that cells with zero load are disregarded in this option, whereas they are factored into the formula with a zero value in the Movie Averaging option.

COF (Center of Force) shows the center of all of the forces in the Movie or Real-time window. This feature is useful for showing how the forces are 'balanced' on the sensor. The center point is represented on the screen by a gray version of the red and white icon that represents **COF** on the Toolbar.

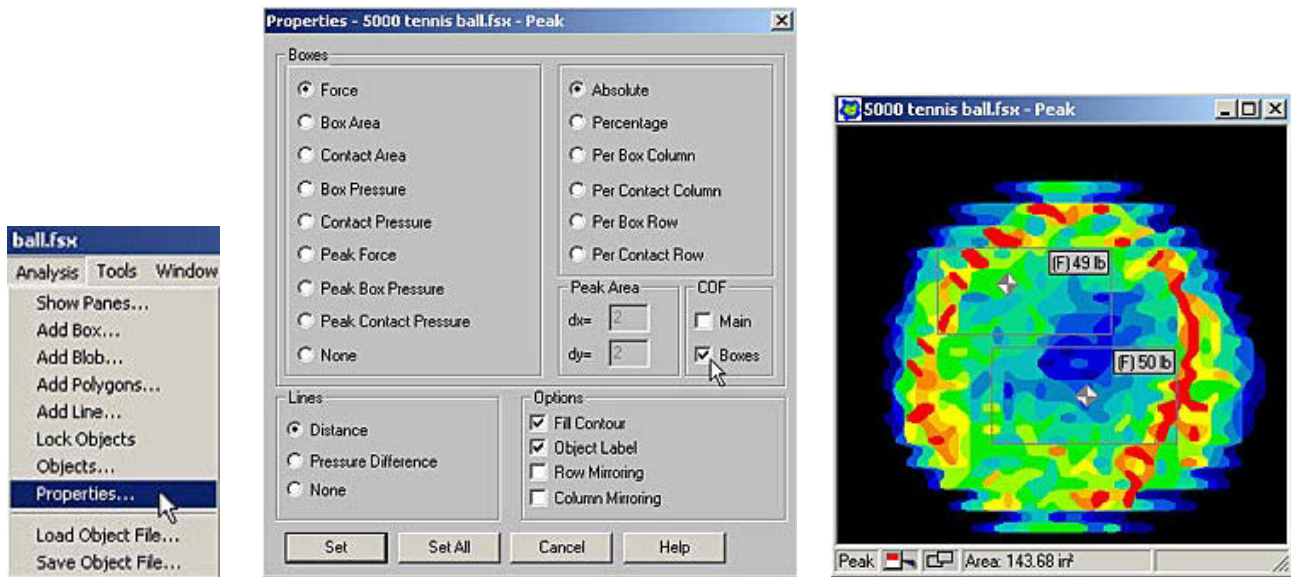


The COF coordinates (in X, Y) are determined using the following two equations:

$$X_{cof} = \frac{\sum_{i=0}^{Cols-1} \sum_{j=0}^{Rows-1} (i * F_{ij})}{\sum_{i=0}^{Cols-1} \sum_{j=0}^{Rows-1} F_{ij}} \quad Y_{cof} = \frac{\sum_{i=0}^{Cols-1} \sum_{j=0}^{Rows-1} (j * F_{ij})}{\sum_{i=0}^{Cols-1} \sum_{j=0}^{Rows-1} F_{ij}}$$

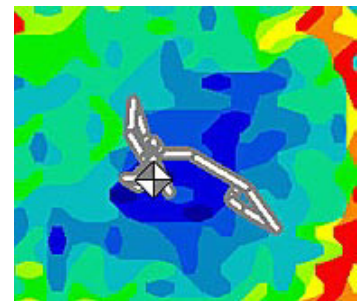
Where F is the force at each sensel, the COF calculation takes all forces on the sensor into account, regardless of the Legend's lower limit.

As an optional feature, you can view the COF for each separate box and tile in a Movie or Real-time window, in addition to the COF for the entire window. Each COF marker will be the same color as its corresponding box or tile. You can disable one or both of these COF features by selecting or de-selecting "Main" or "Boxes", under "COF" in the 'Properties' dialog box. The COF boxes are disabled by default.



COF (Center of Force) Trajectory displays the movement of the Center of Force for the duration of the recorded movie. The movement of the COF can be tracked by playing a movie one frame at a time, with COF Trajectory selected. The trajectory is represented on the screen by a gray and white line that trails the COF marker.

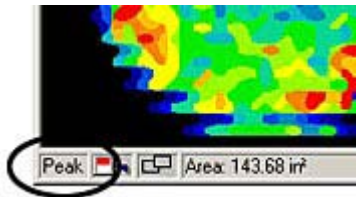
In a Real-time window, COF Trajectory simply shows the center of force. If Peak is also selected, the trajectory of the COF for the entire movie will be shown, and the playback controls will be unavailable.



As an optional feature, you can view the COF Trajectory for each separate box and tile in a Movie window, in addition to the trajectory for the entire window. Each COF Trajectory marker will be the same color as its corresponding box or tile. You can disable one or both of these features by selecting or de-selecting "Main" or "Boxes", under "COF" in the 'Properties' dialog box. The COF Trajectory boxes are disabled by default.

Max (Maximum) Area Frame advances the current movie to the frame with the largest contact area. The actual contact area value is displayed in the center of the Movie Status Bar. Only pressures above the legend's lower limit (threshold) are displayed and accounted for in the contact area. Note that **Max Area Frame** is not available in real-time mode.

Peak/Phase displays the maximum pressure value that each sensel reached during the recording, in one composite frame. This allows you to view, simultaneously, the highest pressures experienced by each part of the sensor during the recording. Peak is available in all 2-D and 3-D display modes. Peak will not affect a snapshot, since a snapshot contains only one frame.



When this item is enabled, the Movie Status Bar will say 'Peak' (see image at left), and the movie playback options will be unavailable. If Peak is in effect while COF Trajectory is enabled, the trajectory of the COF for the entire range of frames is displayed.



Save View As Movie (File Menu) is not a display option, but can be essential to the analysis of recorded pressure data. **Save View As Movie** allows you to save the current movie frame, or view, as a movie file (with extension *.fsx). This new movie file will consist of one frame, and will be saved exactly as it appeared when saved (i.e., any View menu items that were in effect when **Save View as Movie** was selected will be retained in the saved file.).

Note that COF Trajectory will not be displayed by the saved view, since it only affects movies with more than one frame.



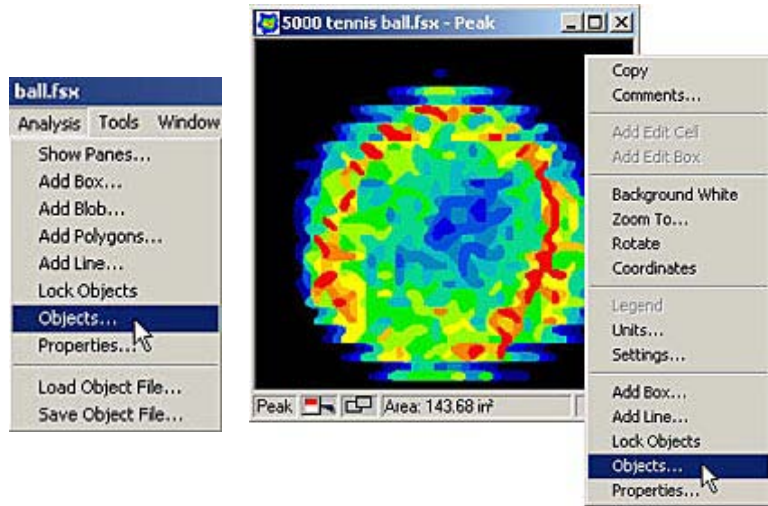
OBJECTS

Objects (Tiles, Boxes, and Lines) can be placed in a Movie or Real-time window, and used to display data characteristics for specific areas of the sensor. Tiles enclose the entire window in one box, or divide the window into four equal boxes. The data inside a Box can be displayed and analyzed separately from the rest of the window's data. Lines can be used to measure the distance or pressure difference between two points on the sensor. Three additional analysis features (Polygons, Blob, and Tracking Box) are also available as an optional package. See "[Optional Accessories](#)" for more information about these features.

Note: Each window has 12 colors available between the boxes, lines, and tiles; after this, all added objects will be gray, and the associated data will not be displayed in a graph.

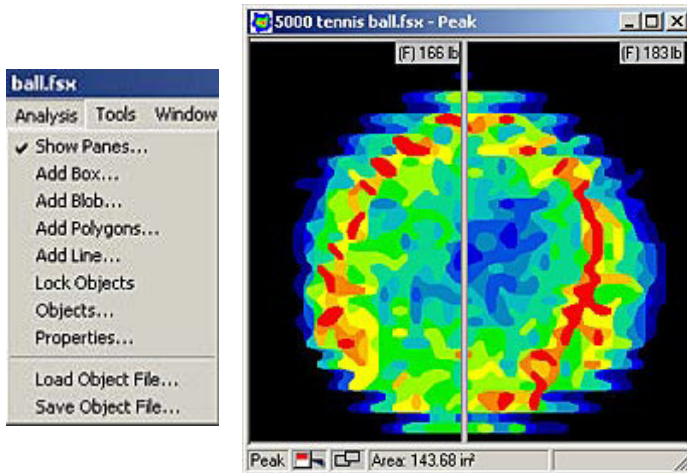
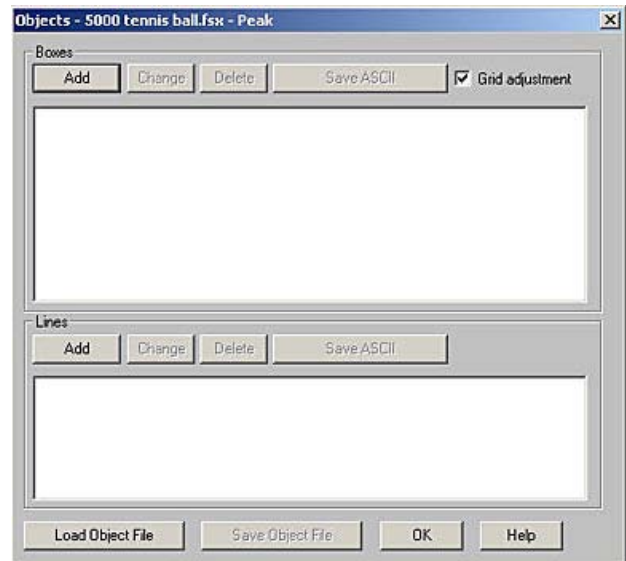
To add Tiles to a Movie or Real-time window:

1. Select **Show Panes** from the **Analysis** pull-down menu (or click the corresponding Toolbar icon), or click on the **'Add'** button in the **'Objects'** dialog box (and select **'Tiles'**). The **'Objects'** dialog box can be accessed by selecting **Objects** from the **Analysis** menu (or clicking the corresponding Toolbar icon) or by clicking the right mouse button while the cursor is over a window, and selecting **'Objects'**.



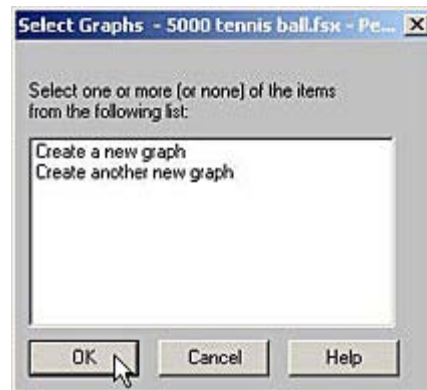
The **Objects** dialog box is opened (shown at right).

When the **Show Panes** option is enabled, the icon is depressed, and a check mark appears next to the option in the **Analysis** menu. Clicking the icon again will deselect this option and remove the tiles (shown below).



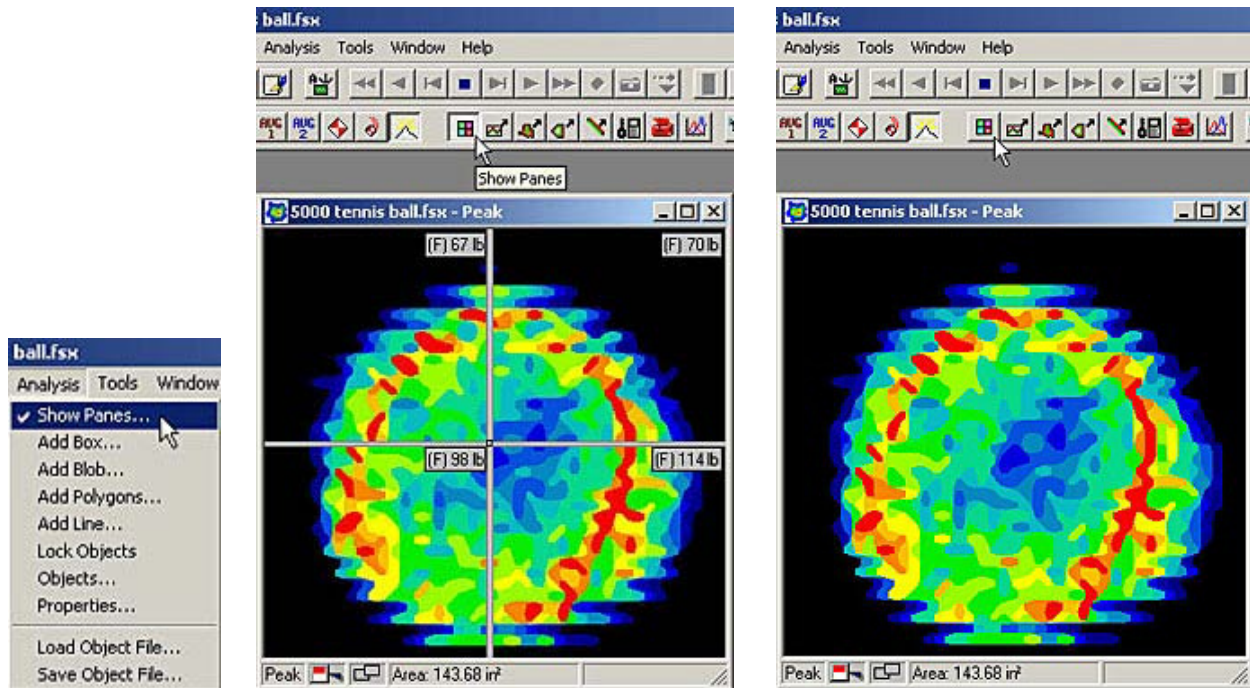
2. The 'Select Graphs' dialog box (shown at right) will appear, and prompt you to select 'Create a new graph' or 'Create another new graph'. You may select one of these, both of these, or neither of these.

If you want to place a single tile (box) in the window, without an accompanying graph, click on 'Create a new graph' (to deselect it). The tile will be gray, and no accompanying graph or graph trace will appear. If you wish to graph the data in the tile box(es), refer to the 'Graphing Section'.

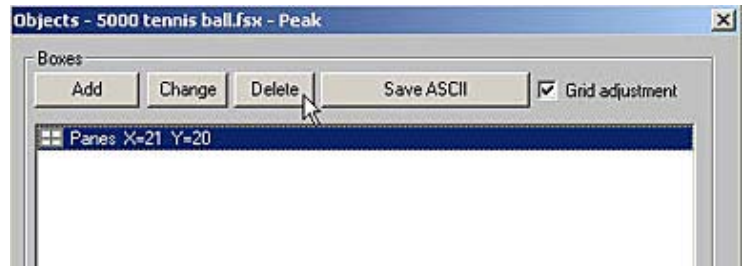


3. Click <OK>. A single tile (pane) will be placed in each open window, enclosing the entire window. If you press the <SHIFT> key and select Show Panes, four (4) tiles will be placed in each window. If you press the <CTRL> key and select this option, the tiles will only be added to the active window.
4. Once placed into a Real-time or Movie window, the tiles can be moved and/or resized so that they enclose the desired areas of the sensor. Refer to the "Moving and Sizing Objects" section for more information.
5. Select which values (or none) you want to display in the small box at the top right corner of each tile. By default, the displayed value will be the total force applied to the sensor in each individual tile. Refer to the "Changing Object Display Data" section for information.

- If you want to remove the tiles, either click on **Show Panes** in the **Analysis** menu (or on the Toolbar icon); right mouse click on one of the lines that create the tiles, then select **Delete**; or select the tiles and press **Delete** on your keyboard.



It is also possible to remove tiles by accessing the **Objects** dialog box, selecting the object(s) you want to delete, then clicking on the **Delete** button.



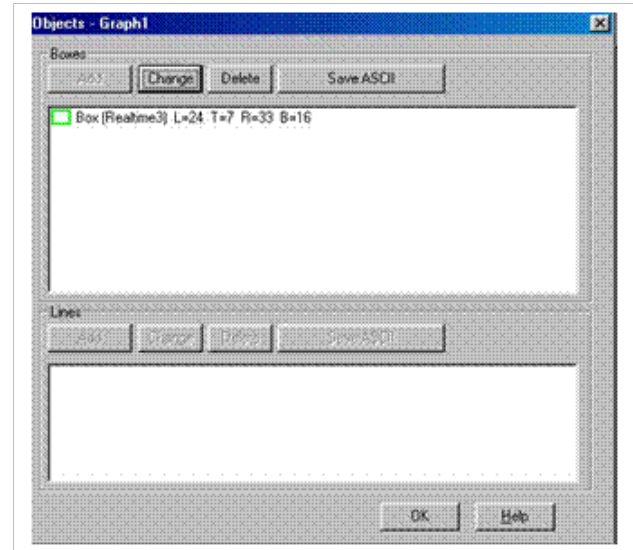
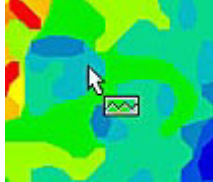
To add a Box or Line to an active Movie or Real-time window:

- Select **Add Box** or **Add Line** from the **Analysis** pull-down menu (or click the corresponding Toolbar icon), or click the right mouse button while the cursor is over a window, and select **Add Box** or **Add Line**.



It is also possible to add a box or line by clicking the Add button in the 'Objects' dialog box. The 'Objects' dialog box can be accessed by selecting **Objects** from the **Analysis** menu (or clicking the corresponding Toolbar icon), or by clicking the right mouse button while the cursor is over a window, and selecting **Objects** (see image at right).

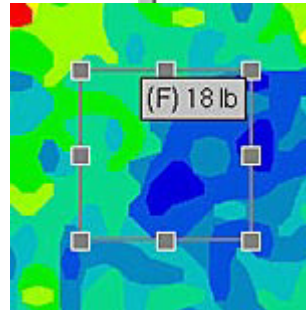
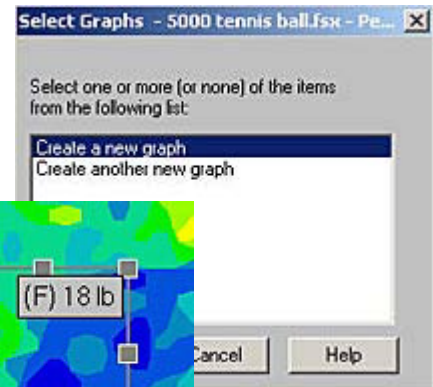
2. Place the cursor over the active Real-time or Movie window, and the 'Add Box' or 'Add Line' symbol will be added to the cursor. Click the left mouse button to place the box or line at that point in the window.



Shortcut: You can also add a box or line to a Movie or Real-time window using the <CTRL> and <SHIFT> keys on your keyboard. Press the <CTRL> key and click the mouse cursor in an open window to add a box at that point, or press the <SHIFT> key and click in an open window to add a line.

3. When you place a box or line in a window, the 'Select Graphs' dialog box will appear (shown below), and prompt you to select 'Create a new graph' or 'Create another new graph'. You may select one of these, both of these, or neither of these.

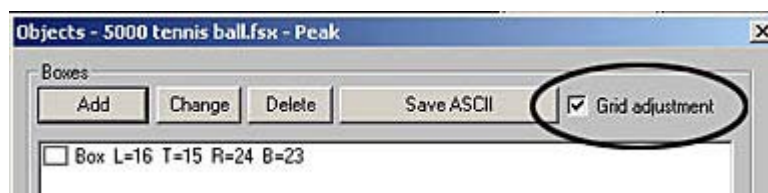
To place a box or line in the window, without an accompanying graph, click on 'Create a new graph' (to deselect it), then click <OK>. The box or line will be gray, and no accompanying graph or graph trace will appear. If you wish to graph the data for the box or line, refer to the "Graphing Object Data" section. Your Box (see images at right) or Line will be placed inside the movie window.



4. Once placed into a Real-time or Movie window, boxes and lines can be moved and/or resized so that they enclose the desired areas of the sensor. Refer to the "Moving and Sizing Objects" section for more information.

Grid Adjustment

The Grid Adjustment affects box placement. If the 'Grid Adjustment' box is selected (checked) in the 'Boxes' section of the 'Objects' dialog box, all four sides of each box will be aligned with the edges of sensor rows and columns. Each side of the box will be moved to the nearest row or column edge, even if this changes the box's size. If this option is not selected, the boxes may be positioned anywhere inside the window.

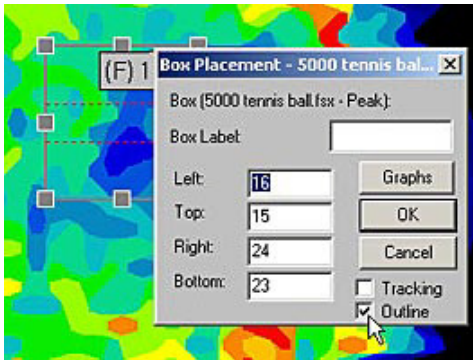
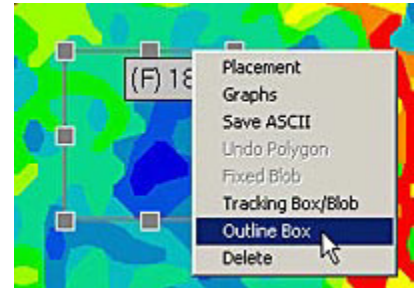


If a box is already in place, and Grid Adjustment is checked, the box will move to the nearest sensor row and column edge. Tile edges are always aligned with sensor rows and columns, and the '**Grid Adjustment**' option does not affect their placement.

1. Select which values (or none) you want to display in the top right corner of each box, or on the midpoint of each line. By default, this value will be the total force applied to the sensor for each box, and the distance between the endpoints for each line. Refer to the "Changing Object Display Data" section for information.

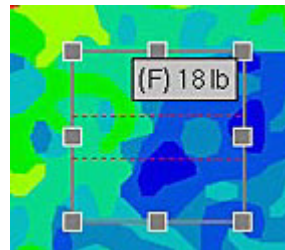
If you want to remove a box or line, either right mouse click on the object, then select '**Delete**', or select the object and press '**Delete**' on your keyboard. It is also possible to remove a box or line by accessing the '**Objects**' dialog box, selecting the object(s) you want to delete, then clicking on the 'Delete' button. You can select multiple objects in the '**Objects**' dialog box by holding down the **<CTRL>** key and clicking on any objects you wish to select. To select a range of objects, hold down the **<SHIFT>** key and click on the first and last objects in the range.

3. If you have the optional **Gross Contact Box** feature enabled, the software will draw a line around the perimeter of contact area within a box. To use this feature, add a box to the movie window. Highlight the box and right mouse click on the edge of the box. Select the "**Outline Box**" option from the **Menu** list (see image at right).

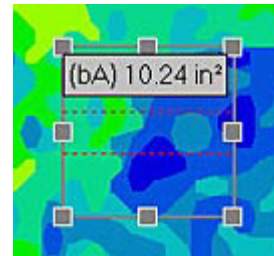
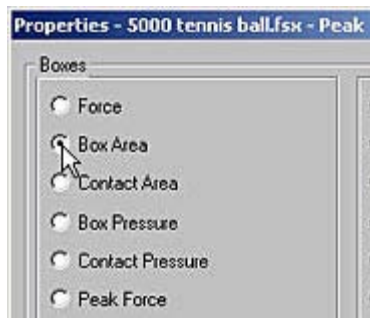


You can also enable this feature by placing a check mark in the "**Outline Box**" checkbox in the '**Box Placement**' dialog box (see image at right).

Once the outline option is on, the box will be drawn by a dotted line, and a solid line will be automatically drawn that outlines the contact perimeter (see image below).



The forces and pressures data and graphs displayed for this box will reflect only the cells surrounded by this perimeter line. Selecting **Object Area** from the **Properties** menu will accurately reflect the area enclosed by the outlining perimeter.



MOVING AND SIZING OBJECTS

Once placed into a Real-time or Movie window, objects (tiles, boxes, or lines) can be moved and/or resized. The size or location of an object can be changed using one of the following three methods:

Method 1: Direct Selection

Method 2: Numerical Placement

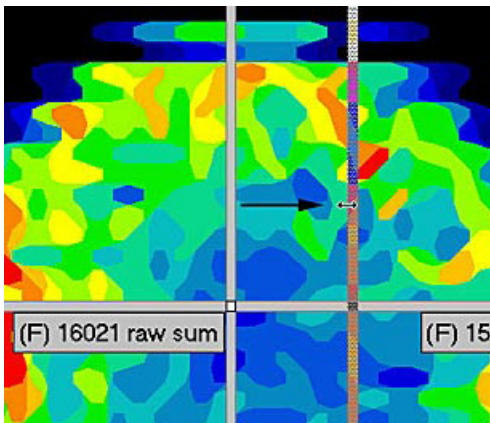
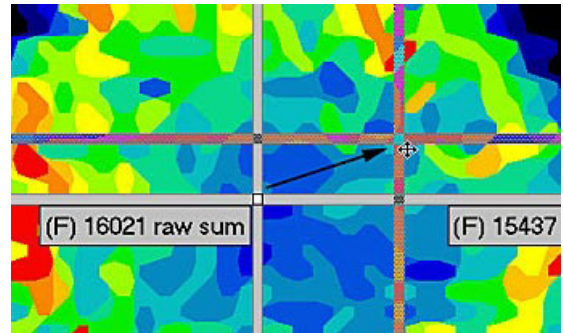
Method 3: Through the Objects Dialog Box

Note: *Box placement is affected by the Grid Adjustment setting. Refer to the "Objects" section for information on this setting.*

Method 1 - Direct Selection

To move/resize (four) tiles:

1. To move the center-point: Place the cursor at the intersection point of the two lines that create the tiles; a four-way arrow will appear, and that 'node' (center-point) can be dragged in any direction. Hold down the left mouse button, and drag the node to the desired position (see image at right).
2. To move the lines: Place the cursor on one of the lines that create the tiles; a two-way arrow will appear, and that line can be dragged in either direction. Hold down the left mouse button, and drag that line to the desired position (see image below left).



Note: *Dragging a line all the way to one side will eliminate the tiles created by that line, but will not eliminate the line.*

To move a box or line:

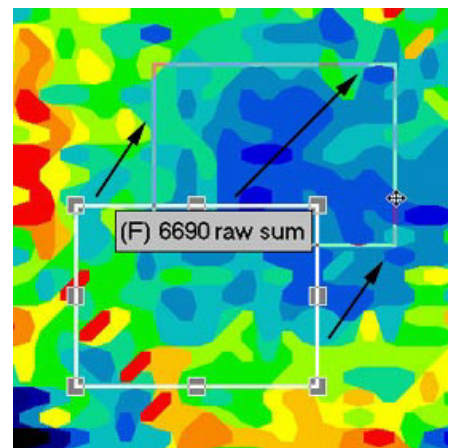
1. Click on the box or line to make it active. Place the cursor over an edge of the box or line, and a four-way arrow will appear.

Note: *When a box or line is active, its nodes (small squares around the perimeter) become visible*

2. Hold down the left mouse button, and

drag the box or line to the desired position inside the window (see image at right).

The entire box or line can be moved anywhere inside the window, without changing its shape. However, the object cannot be dragged out of a window; its size will be reduced if this is attempted.

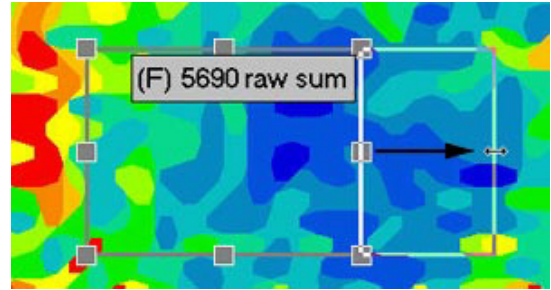


To resize a box or line:

1. Click on a box or line to make it active. Place the cursor over one of the eight nodes (one at each corner, and one at the midpoint of each side) for a box, or over one of the two nodes (one at each endpoint) for a line.

- When the cursor is placed over a box node, a two-way arrow appears. When the cursor is placed over a line node, a four-way arrow appears. Hold down the left mouse button, and drag the node until the box or line is the desired size and shape.

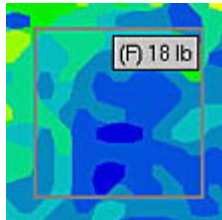
Note: If you want the edge of the box to encompass the edge of the sensor, drag the box beyond the border of the window and release it. It will be truncated to the size of the window.



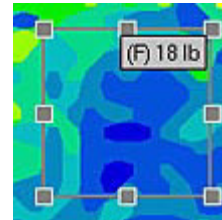
Method 2 - Numerical Placement

- Click on an object (pane, box, line) to make it active. An object is active when its nodes are visible (the nodes for tiles are at the intersection of any lines).

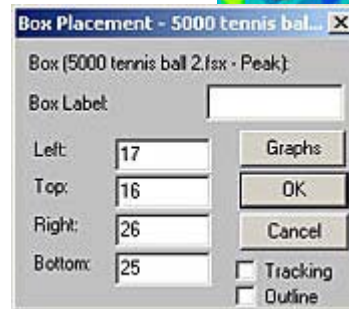
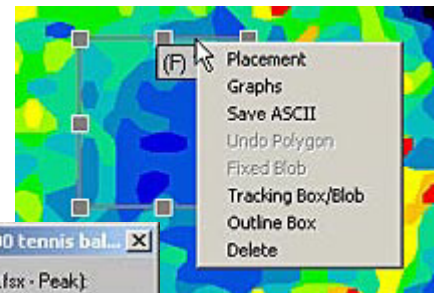
An example of a Box that is not selected



The same Box, selected.



- Click the right mouse button, and you will get the following options: 'Placement', 'Graphs', or 'Delete' (see image at right).
- Select 'Placement'. This will bring up the 'Placement' dialog box for that object (shown below right). At the top of the 'Placement' dialog box will be either the Movie name (*.fsx) or the Real-time window number (e.g. 'Real-time 2'), followed by the name(s) of any graph(s) in which the object data is displayed.
- Enter the desired coordinates in the fields to move and/or resize the object, then click <OK>. The coordinates that may be entered for the objects are: Tiles (center-point - X, Y), Box (Left, Top, Right, Bottom), Line (endpoints - X1, Y1 and X2, Y2).



Note: The Movie or Real-time window is a graphical representation of the actual sensor, so the coordinates in the window correspond to actual rows and columns on the sensor. All coordinates are given from the top left corner of the window. The zero point will also rotate if the window is rotated.

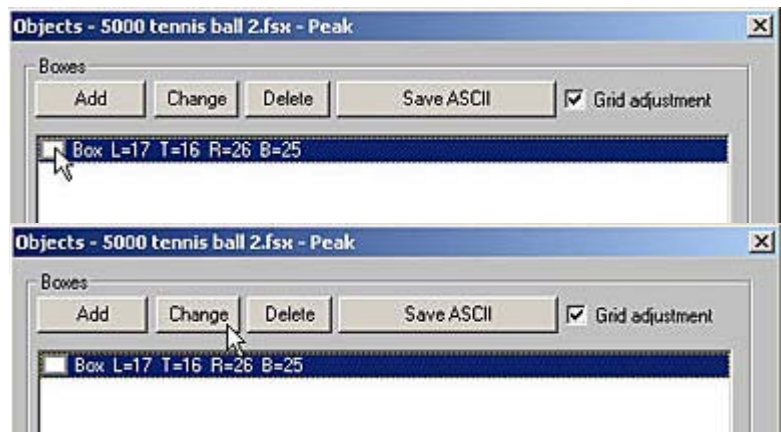
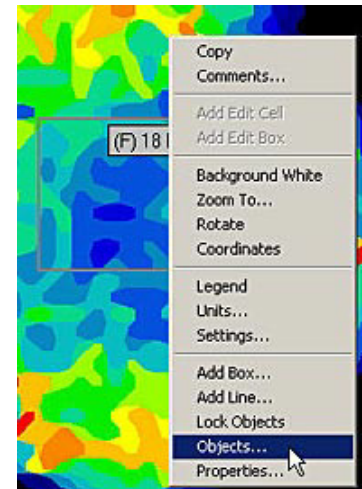
Method 3 - Objects Dialog Box

- Open the 'Objects' dialog box (below). This dialog box can be accessed by selecting **Objects** from the **Analysis** pull-down menu (or clicking the corresponding Toolbar icon), or by clicking the right mouse button while the cursor is over a window, and selecting **Objects**.

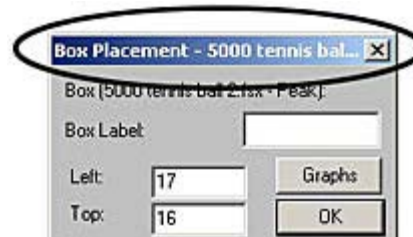
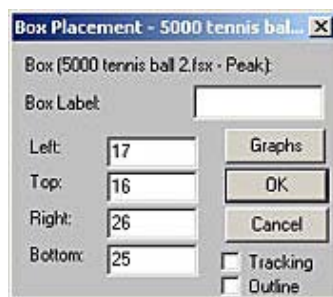
Note: With this right-click method, you must click from anywhere outside the lines of the object, and have the object deselected.

The 'Objects' dialog box displays information about current object placement coordinates, size and color-coding (see images at right). Each object is shown graphically, followed by the name of each graph that displays that object's data, and the object's position (coordinates) in the real-time or movie window.

2. Select (highlight) the object you wish to move or resize. Click the 'Change' button to bring up the 'Placement' dialog box for that object.
- 3.
4. Click on the 'Change' button in the top ('Boxes') section of the dialog box to change tiles or boxes, and in the lower ('Lines') section to change lines (see both images at right).



At the top of the 'Placement' dialog box will be either the movie name (*.fsx) or the Real-time window number (e.g. 'Real-time 2'), followed by the name(s) of any graph(s) in which the object data is shown (see images below).



4. Enter (type in) the desired coordinates in the fields to move and/or resize the object, then click <OK>. The coordinates that may be entered for the objects are: Tiles (center point - X, Y), Box (Left, Top, Right, Bottom), Line (endpoints - X1, Y1 and X2, Y2).

CHANGING OBJECT DISPLAY DATA

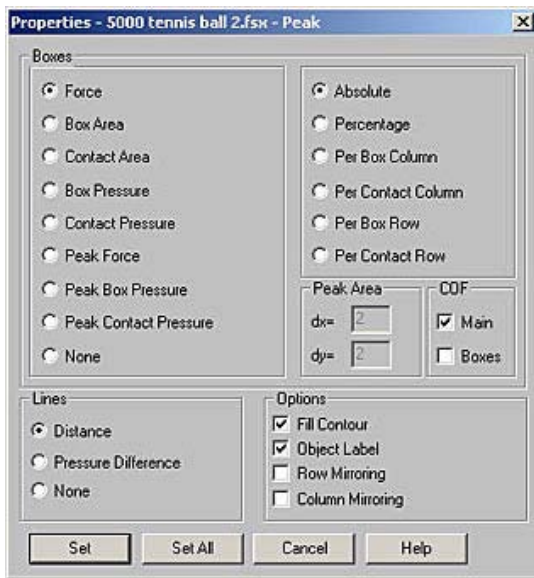
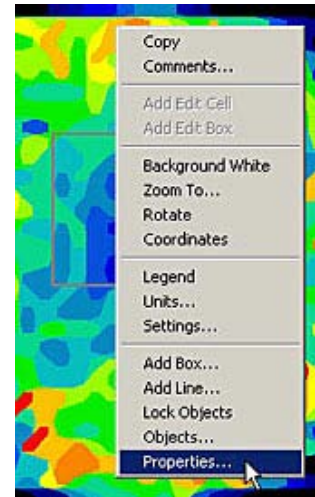
I-Scan allows you to select which values (or none) you want to display with each object (tiles, box, or line). The **Properties** option controls the display options of each object that is placed in a Real-time or Movie window.

Note: The units that are selected for length, force, and pressure in the 'Units of Measure' dialog box will be displayed for all values. If a calibration has not been performed, force and pressure will be displayed in 'Raw' units.

To change the data displayed with an object:

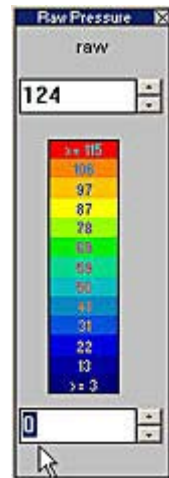
1. Open the **'Properties'** dialog box. **Properties** can be accessed by selecting **Properties** from the **Analysis** pull-down menu (or clicking the corresponding icon in the Toolbar), or by clicking the right mouse button while the cursor is in a real-time or movie window, and selecting **Properties**.

The **'Properties'** dialog box contains the data display options for all of the objects in the active window (see image below). Tiles and box options are displayed in the upper (**'Boxes'**) section of the dialog box, and line options are displayed in the lower (**'Lines'**) section. Tiles are simply treated as separate boxes when defining display options.



Note: The Legend's lower limit should always be set to zero when displaying object data. Pressure values below this limit are disregarded, thereby changing the displayed 'Area', 'Force', and 'Pressure' (see image at right).

2. Select which values (or none) you want to display in the small box at the top right corner of each box or tile. Any changes will affect all boxes and tiles in the active window (only).



Each value is based solely on that area of the sensor that lies inside that box or tile. You may choose from the following options:

- **Force (default):** Total force on the sensor area within the box.
- **Box Area:** Actual sensor area inside the box.
- **Box Pressure:** Total pressure on those sensels inside the box. This is calculated by dividing the total force applied in the box by the total area of the box.
- **None:** No value is displayed.



There are two display options that include the word **'Contact'**. In these cases, the **'Contact'** values include only those sensels that have some (greater than zero) load applied to them.

- **Contact Area:** Area of only the loaded (or 'contact') sensels inside the box.
- **Contact Pressure:** Pressure on the loaded sensels, which is calculated by dividing the force by the 'contact'

There are a number of display options that include the word '**Peak**'. In these cases, '**Peak**' means the small area (default size is 2 sensels by 2 sensels) inside the box or tile that contains the highest values of the selected property. This peak area will be indicated by a small white or black box, whose size is determined by the 'dx' and 'dy' values in the '**Peak Area**' section of the dialog box.

- **Peak Force:** Highest area of force inside the box or tile.
- **Peak Box Pressure:** Highest-pressure area inside the box or tile, calculated as the force inside the peak box divided by the peak box area.
- **Peak Contact:** Same as the 'Peak Box Pressure', except the pressure is calculated using only the contact area, instead of the entire box area. The contact area is the area within the peak box that has greater than zero contact pressure

3. If '**Peak Force**', '**Peak Box Pressure**', or '**Peak Contact Pressure**' has been selected, you may change the size of the peak area. This peak area is displayed graphically as a white or black box (depending on the background color).

You may define the size of the peak box by entering the following values in the '**Peak Area**' section at the right of the dialog box:



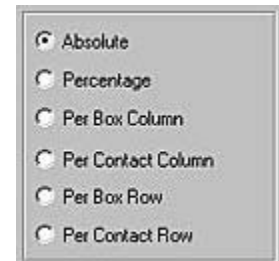
dx the length (number of sensels) of the X (horizontal) side of the peak box.

dy the length (number of sensels) of the Y (vertical) side of the peak box.

Note: Each of these values ('dx' and 'dy') must be between 1 and the size of the box; if larger than the box, the peak area will become the entire box's width (or height).

On the right side of the '**Properties**' dialog box, you can further define the box attributes that have been selected for the box(s). Note that any unavailable options will be grayed out. You may choose from the following options:

- **Absolute (default):** Force, area, or pressure as an absolute value.
- **Relative:** Force, area, or pressure inside the box as a percentage of the total force, area, or pressure for the entire sensor.
- **Per Box Column:** Force, area, or pressure for only those sensor columns that fall inside the box.
- **Per Box Row:** Force, area, or pressure for only those sensor rows that fall inside the box.
- **Per Contact Column:** Force, area, or pressure for only the loaded (or 'contact') sensor columns that fall inside the box.
- **Per Contact Row:** Force, area, or pressure for only the loaded sensor rows that fall inside the box.



On the right side of the dialog box, you can choose the Center of Force options (under '**COF**'). Select one, both, or neither of the following options:

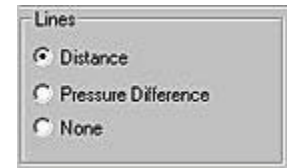
- **Main:** The gray COF or COF Trajectory marker will be visible in any open Movie or Real-time windows (when the View>>COF or COF Trajectory option is in effect). The 'Main' item is enabled by default.
- **Boxes:** The color-coded COF marker for each box and tile will be displayed inside that object in the Movie or Real-time window (when the View>>COF option is in effect). This option also applies to COF Trajectory, which can only be displayed in a Movie window. Each COF or COF Trajectory marker will be the same color as its corresponding box or tile, while the COF marker for the entire window will be gray. The "Boxes" item is disabled by default.



4. Select which values (or none) you want to display in the small box on the midpoint of each line. By default, this value will be the distance between the endpoints of the line (this value corresponds to an actual distance on the sensor). Any changes will affect all lines in the active window only.

At the bottom of the '**Properties**' dialog box, under '**Lines**', is a list of the attributes that may be displayed. You may choose from the following options:

- **Distance (default):** Length of the line
- **Pressure Difference:** Difference in pressure between the sensels beneath the two endpoints of the line.
- **None:** No value will be displayed.



5. In the lower left corner of the dialog box, select which other display options you would like to use:

- **Fill Contour:** If this box is checked, the 2-D Contours view shows each contour filled completely with the color corresponding to the sensed pressure range. If this box is not checked, pressure boundaries are represented by colored '**topographic**' lines only.
- **Object Label:** Check this box if you would like to display the label for each object in the Real-time or movie window. If you prefer to view objects without their labels, uncheck this box.
- **Row Mirroring:** If this box is checked, the entire sensor will be mirrored across the rows.
- **Column Mirroring:** To mirror the sensor across columns, check this box.



GRAPHING OBJECT DATA

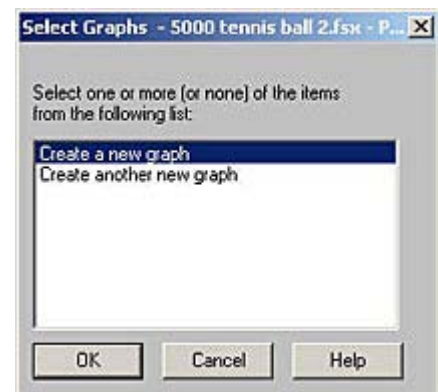
I-Scan enables you to represent specific area(s) of Real-time or Movie window data in a Graph. The data associated with each object (tiles, box, line) in a window is depicted in the Graph as a color-coded '**trace**'.

To create a graph:

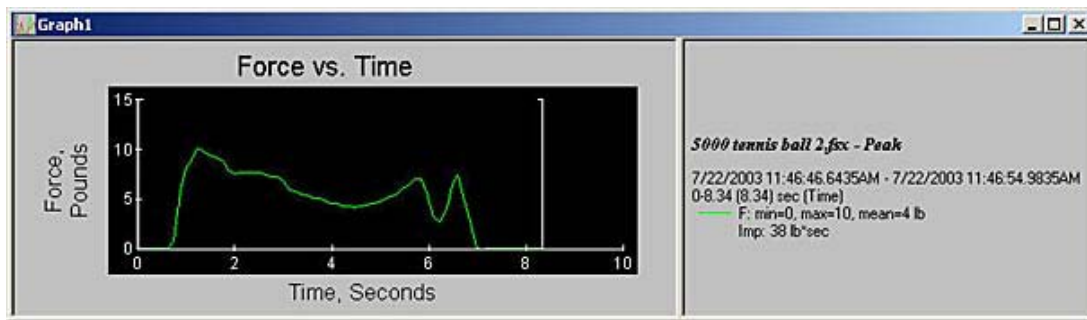
1. Add an object to the active Movie or Real-time window. Refer to the "Objects" section for instructions. When you add an object, the '**Select Graphs**' dialog box is displayed, with the following options: '**Create a new graph**' or '**Create another new graph**'. By default, '**Create a new graph**' will be highlighted (see image at right).

Note: Each option that is selected will be highlighted (multiple selections are possible), and clicking on an item will select or deselect it.

2. Make sure that '**Create a new graph**' is selected in the '**Select Graphs**' dialog box, then click <OK>. A graph (Graph 1) will be placed in the bottom portion of the screen, presenting the object's data as a color-coded graph trace. The graph trace will be the same color as the object, and the actual data values will be displayed to the right of the graph.



The default graph is '**Force vs. Time**' for Movie windows ('**Force vs. Distance**' for Real-time), but many different variables can be displayed.



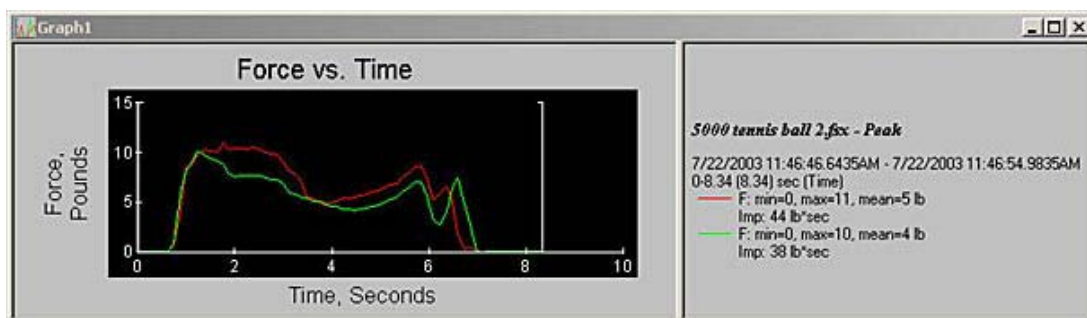
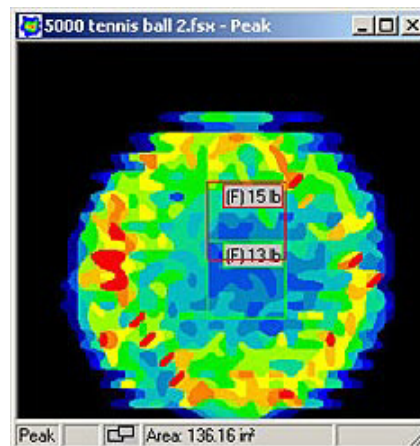
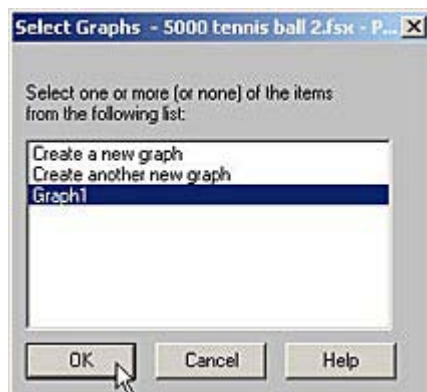
Note: The graph will only include traces for objects in which non-zero pressures were detected during the recording.

If '**Create another new graph**' is also selected, a second graph (Graph 2) will be opened, and will contain the same object's color-coded information. If '**Create another new graph**' is selected, but 'Create a new graph' is not, only one new graph (Graph 1) will be opened.

Note: Each window has 12 colors available between all objects; after this, each added object will be gray and the associated data will not be displayed in a graph.

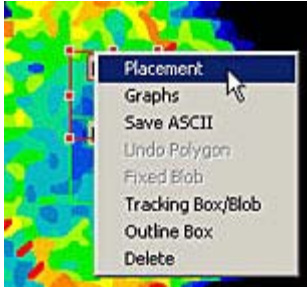
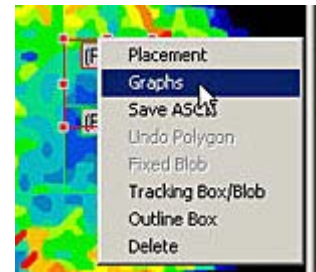
To add another object to the same graph:

Add another object to the active Movie or Real-time window. When the '**Select Graphs**' dialog box (shown below) is displayed, select an existing graph (such as '**Graph 1**') and click <OK>. The object's color-coded trace will be displayed in Graph 1. Any graphs that are currently open will also be listed as an option in the '**Select Graphs**' dialog box.



To change which graph displays an object:

1. Select an existing object, to make it active. Click the right mouse button on one of the lines of the object, and then select either '**Graphs**' or '**Placement**'. If you select Graphs, the '**Select Graphs**' dialog box will be displayed, and you can make your changes (see image at right).



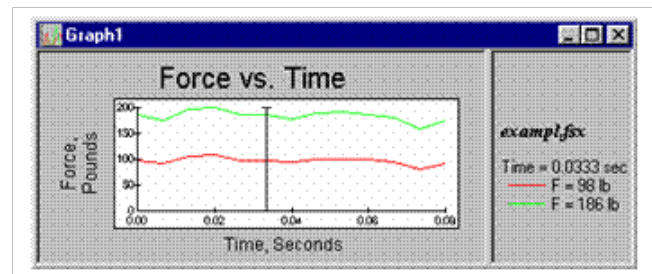
If you select Placement, the 'Placement' dialog box will appear (see image at left). Click the **Graphs** button in this dialog box to display the '**Select Graphs**' dialog box. Clicking on the Change button in the 'Objects' dialog box can also open the 'Placement' dialog box.

2. Modify the graph format, if necessary. Refer to "Changing Graph Display Options" for further instructions.

Note: The display options for objects and graphs are selected separately, so the data displayed in the graph can be different from that displayed in the actual object.

UNDERSTANDING THE GRAPH DATA

Review the data to the right of the graph. This information includes the name of the object's corresponding movie or real-time window, followed by the actual (color-coded) Y-axis data values and the X-axis value. For each colored graph trace, a value is displayed that indicates the force, pressure, or area (Y-axis value) for that object at that point on the X-axis. These values correspond to the intersection of the white vertical 'time line' with the colored line graph.

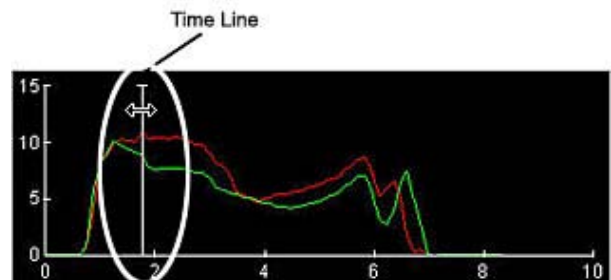


If this information is not fully visible, you may view the data by using the scroll bars, or you may expand this area by dragging (with your cursor) the divider that separates it from the actual graph. Refer to "Changing Graph Display Options" for information on setting the graph's X- and Y-axis parameters.

Using the graph functions:

The white vertical time line marks the point on the X-axis for which the Y-axis reading is being displayed. The data values displayed to the right correspond to the intersection of the vertical time line with the colored line graph. If data from more than one Movie window is displayed in the graph, the small number 1 or 2 above the time line indicates which window that particular line is tracking.

If none of the graph's object data is from a Real-time window, the X-axis can be set to 'Time' or 'Frame'. In this case, the time line displays the elapsed time or frame number at that point in a movie. You can observe both the time line and the data values to the right change in unison as you play back a recording. When the mouse cursor is over the time line, a two-way arrow appears, and the line can be dragged to any position on the graph. Dragging the time line will speed up the playback of a recording. Click the mouse cursor anywhere in the display area of the graph and the time line will instantly jump to that point.



Note: A 'Paused' movie will not respond to movements of the time line.

If you place your cursor over a Graph window, and click the right mouse button, some additional options become available.



- **Copy:** Will copy the active Graph window to the Windows clipboard, from which it can be pasted into other applications as an image (bitmap) or as ASCII (text).
- **Background White:** Will toggle the graph's (only) background color between black and white.
- **Zoom In / Zoom Out:** options allow you to zoom in on a specific area of the graph; the scale of the X and Y-axis will change to reflect the smaller or larger view area. Note that the zoom options are not available when the graph has less than 16 frames, or relates to a real-time window. Refer to

the 'Main Menu' section for a description of the other options ('Units', 'Objects', and 'Properties').

Click on the 'X' in the top right corner of the Graph window (standard Windows function). The graph will automatically close if the corresponding object(s) is deleted.

CHANGING GRAPH DISPLAY OPTIONS

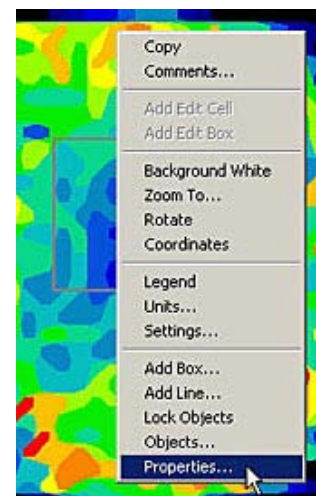
I-Scan enables you to control how each graph will be presented on the screen, independently of the corresponding Movie and/or Real-time window(s). If Properties is selected when a graph is the active window, the 'Properties' dialog box will contain the data display options for that graph. Changing the attributes of a graph will not change those of its corresponding object, and vice versa.

The graph display options are the same as those available for a box in a movie or real-time window, except the word 'object' replaces the word 'box', since the data may correspond to a box or line.

Note: The units that are selected for length, force, and pressure in the 'Units of Measure' dialog box will be displayed for all values. If a calibration has not been performed, force and pressure will be displayed in 'Raw' units.

To change the data displayed in the Graph:

1. Select the Graph window to make it active. Open the 'Properties' dialog box. It can be accessed by selecting **Properties** from the **Analysis** pull-down menu (or clicking the corresponding icon in the Toolbar), or by clicking the right mouse button while the cursor is on the Graph, and selecting Properties.

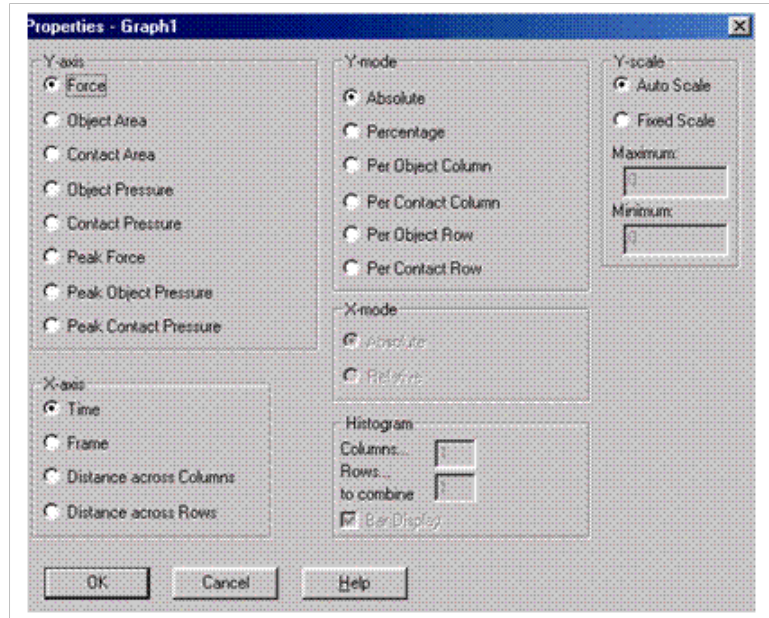


The 'Properties' dialog box contains the data display options for the X and Y-axes of the active graph. 'Y-Axis' options are displayed in the upper section of the dialog box, and 'X-Axis' options are displayed in the lower section.

- Choose the display options for the Y-axis. On the left side of the 'Properties' dialog box, under 'Y-axis', is a list of the attributes that may be displayed on the vertical axis of the graph. Any changes will affect only the active graph.

The Legend's lower limit should always be set to zero when displaying object data.

Pressure values below this limit are disregarded, thereby changing the displayed 'Area', 'Force', and 'Pressure.'

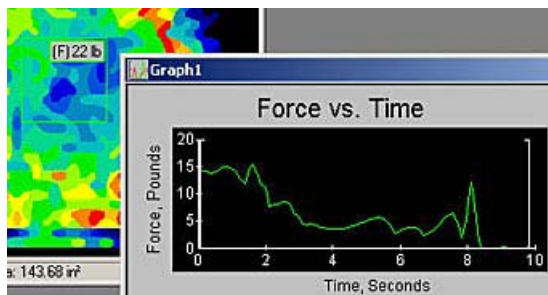


The values for each option below are only for the area (sensels) either inside the box, or beneath the line.

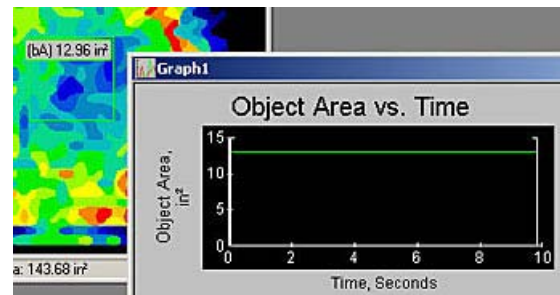
Attributes (force, area, pressure) associated with a line are simply those attributes of the sensels that are in contact with the line.

- Force (default):** Total force for the object.
- Object Area:** Actual sensor area of the object.
- Object Pressure:** Total pressure of the object. This is calculated by dividing the total force applied by the total area.

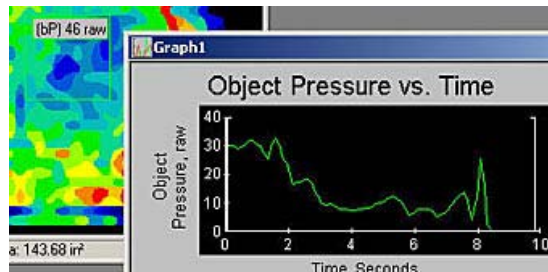
Force Vs. Time (Default)



Object Area Vs. Time



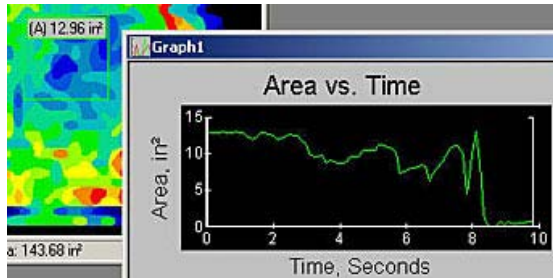
Object Pressure Vs. Time



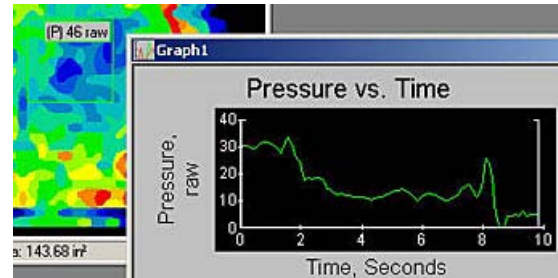
There are two display options that include the word 'Contact'. In these cases, the 'Contact' values include only those sensels that have some (greater than zero) load applied to them.

- **Contact Area:** Area of only the loaded, or 'contact' sensels.
- **Contact Pressure:** Pressure on the loaded sensels, which is calculated by dividing the force by the contact area

Contact Area Vs. Time



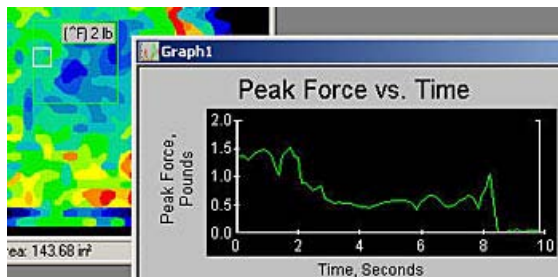
Contact Pressure Vs. Time



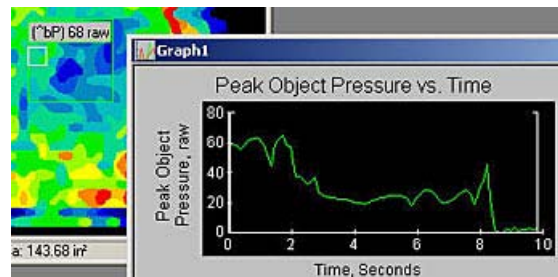
There are a number of display options that include the word 'Peak'. In these cases, 'Peak' means the small area (default size is 2 sensels by 2 sensels) of the object that contains the highest values of the selected property.

- **Peak Force:** Highest area of force in the object.
- **Peak Object Pressure:** Highest-pressure area in the object, calculated as the force inside the peak box divided by the peak box area.
- **Peak Contact Pressure:** Same as the 'Peak Object Pressure', except the pressure is calculated using the force and contact area.

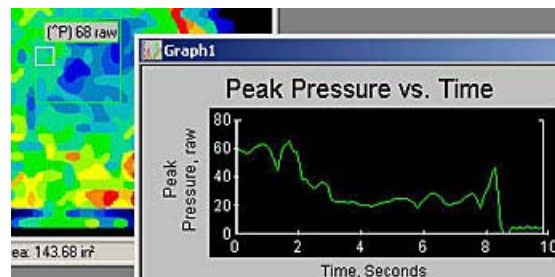
Peak Force Vs. Time (Default)



Peak Object Pressure Vs. Time

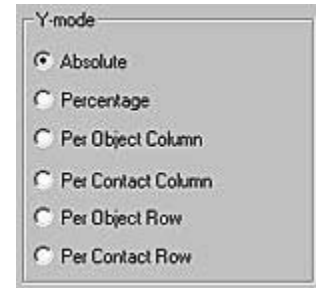


Peak Contact Pressure Vs. Time



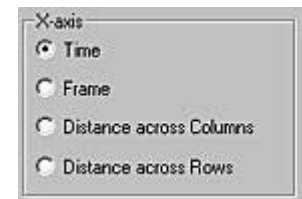
3. On the right side of the **'Properties'** dialog box (under **'Y-Mode'**), you may further define the Y-axis attributes that have been selected for the graph. Note that any unavailable options will be grayed out. You may choose from the following options:

- **Absolute (default):** Force, area, or pressure as an absolute value.
- **Percentage:** Force, area, or pressure for the object as a percentage of the total force, area, or pressure for the entire sensor.
- **Per Object Column:** Force, area, or pressure for only those sensor columns under the object.
- **Per Object Row:** Force, area, or pressure for only those sensor rows under the object.
- **Per Contact Column:** Force, area, or pressure for only those loaded, or contact, columns under the object.
- **Per Contact Row:** Force, area, or pressure for only those loaded, or contact, rows under the object.



4. Choose the display options for the X-axis. At the bottom of the **'Properties'** dialog box, under **'X-axis'**, is a list of the attributes that may be displayed on the horizontal axis of the graph. Any changes will affect only the active graph. You may choose from the following options:

- **Time (movies only):** Elapsed time
- **Frames (movies only):** Movie frame number
- **Distance Across Columns:** Width of the sensor in the X-axis
- **Distance Across Rows:** Length of the sensor on the X-axis.



When **'Time'** or **'Frame'** is selected, the data is displayed as a line graph; and when **'Distance Across Rows'** or **'Distance Across Columns'** is selected, the data is displayed as a bar graph.

5. When **'Distance Across Columns'** or **'Distance Across Rows'** has been selected, the following two options become available in the **'X-mode'** section on the right side of the dialog box:

- **Absolute:** The zero point of the window will be used as the zero point for the X-axis.
- **Relative:** The edge of the specific object will be used as the zero point for the X-axis.

6. When **'Distance Across Columns'** or **'Distance Across Rows'** is selected, the following options become available on the lower right side of the dialog box, under **'Histogram'**:

- **Columns:** The number of bars (grouped columns) that will be displayed in the histogram.
- **Rows:** The number of bars (grouped rows) that will be displayed in the histogram.

These two options allow you to change the number of bars displayed in the graph. You must enter an integer between 1 and the maximum number of sensor columns or rows (44 is most common).

The data from the columns or rows will be combined into the number of groups chosen, and the properties (force, area, or pressure) of these grouped columns or rows will be summed and displayed as one bar. For example, if 'Columns' is set to 4, and the sensor being used has 44 columns, the X-axis of the graph will show 11 bars, with each bar displaying the total force, area, or pressure on four sensor columns.

- **Bar Display:** If selected, the data in the histogram will simply be displayed as separate bars (grouped columns or rows). If this item is not selected, the midpoint of each of the separate bars is connected to make a smoother graph trace.

7. Select your options for Y-Scale:

- **Auto Scale:** The software automatically applies a scale to the Y-axis. This is the default scale selection.
- **Fixed Scale:** To enter your own scale to be displayed on the Y-axis, click to select this option. Enter values in the Maximum and Minimum fields. These numbers will be used as the maximum and minimum values for the Y-axis of the graph.

SAVING/LOADING OBJECT FILES

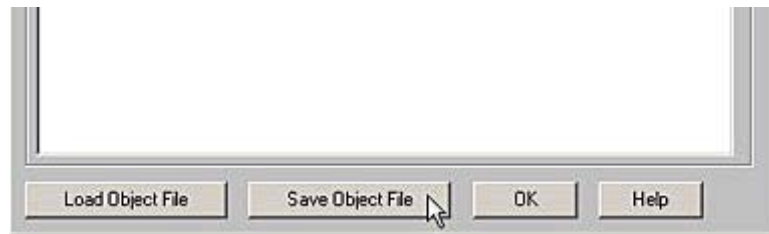
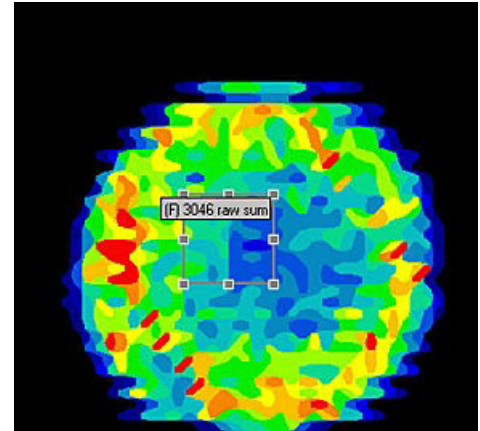
I-Scan allows you to save the objects (tiles, boxes, and lines) from an active Real-time or Playback window as a retrievable file. This option is necessary because, when a current window is saved as a movie file, the objects in the window are not saved as part of the file.

***Note:** If you change any object placements and don't save them, the software WILL NOT ask you if you wish to save them when you exit.*

To save an Object file:

When you are satisfied with the size and placement of the objects in your Movie or Real-time window, select **Save Object File** from the **Analysis** pull-down menu.

Clicking the **Save Object File** button in the 'Objects' dialog box can also access this option.



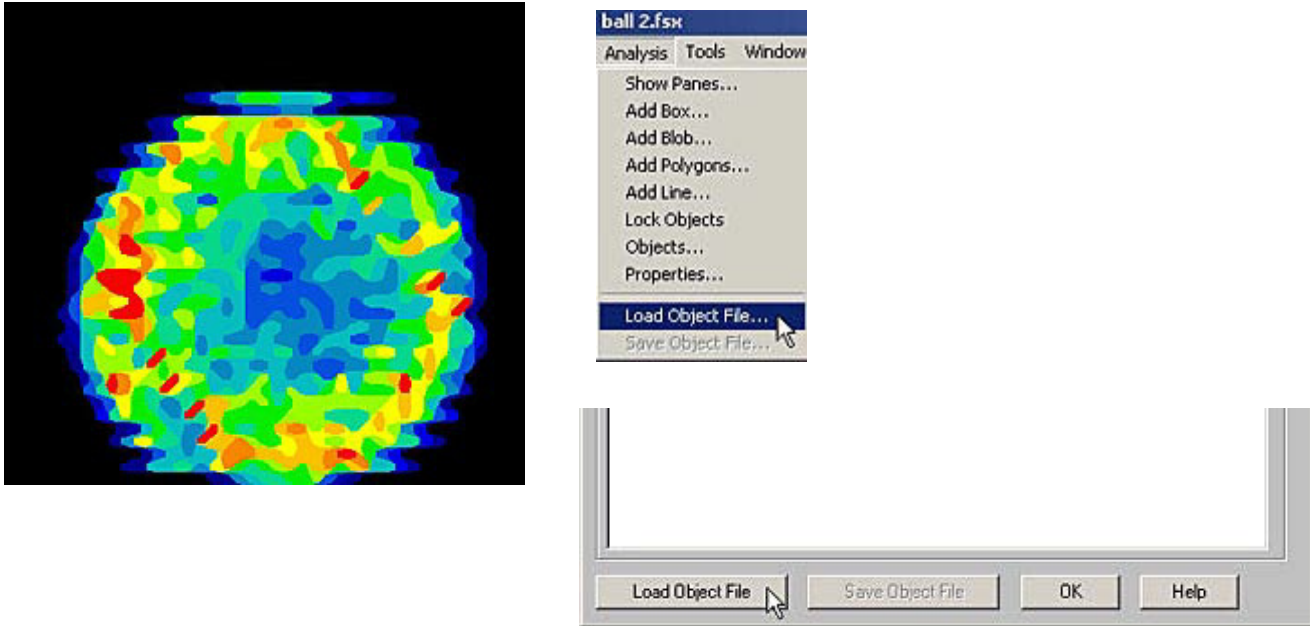
When **Save Object File** is selected, a '**Save As**' dialog box will be displayed, and the file can be saved as an object file (with an *.fbx extension). The default filename will be the name of the active window (e.g. "Movie1.fbx" or "Real-time2.fbx"). Objects are saved exactly as they appear in the window, with the same properties and coordinates.

To load an Object file:

Open a Movie or Real-time window, and then select **Load Object File** from the **Analysis** pull-down menu.

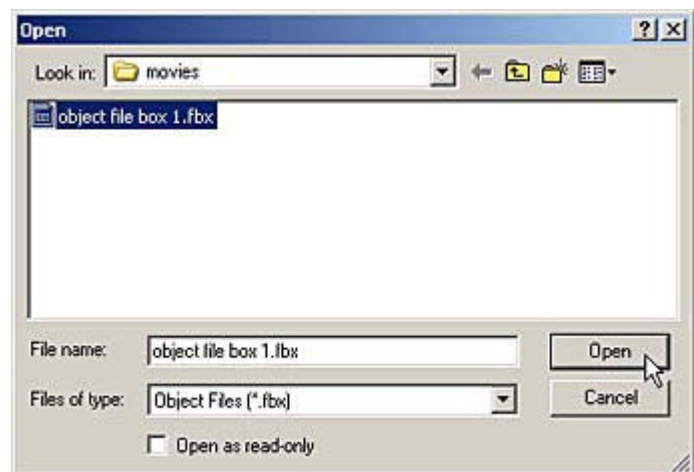


Clicking the **Load Object File** button in the 'Objects' dialog box can also access this file.



When Load Object File is selected, an 'Open' dialog box is displayed, and object files (with the extension *.fbx only) can be opened. When an object file is loaded into the current window, the objects will be placed at their saved coordinates. If the current window is not the same size as the original window, the software will still attempt to place the objects at the correct coordinates.

Note: Objects that are loaded from an object file may not be the same color as when they were saved, since the software assigns colors to objects in the order they are placed.



ASCII DATA

This section explains how to Save and Read movies, frames, and objects as ASCII data, so they can be exported into a spreadsheet program like MS Excel.

Saving ASCII Data

I-Scan enables you to save the current Movie, Real-time or Graph window data as an ASCII (text) file, with the filename and location of your choice. Data can be saved for an entire window, or for a single object. Specifically, you can save frame sensel pressure data, center of force (COF) data, or graph data.

Since these files are in ASCII format, they can be imported into a spreadsheet program (e.g. MS Excel), or they can be opened in a text editor or word processing program (e.g. MS Word) in order to review and manipulate the data.



Note: The orientation of ASCII data within a file is independent of the rotation of the display window on the screen.

To create an ASCII file for an entire window:

1. Click on the Movie or Real-time window of interest to make it active.
2. Select Save ASCII from the **File** pull-down menu (shown at right).
3. When the 'Save ASCII' dialog box (shown below) is opened, select the desired 'Data Type' - either 'Frame data' or 'Center of force (COF)'.

Note: Real-time windows are simply treated as single-frame movies.

If you select 'COF', the coordinates of the center of force for each frame is saved. The output ASCII file contains a list of COF row/column coordinates for the selected range of frames. If there is no data in a frame, "(-1, -1)" will be output to the file for that frame.

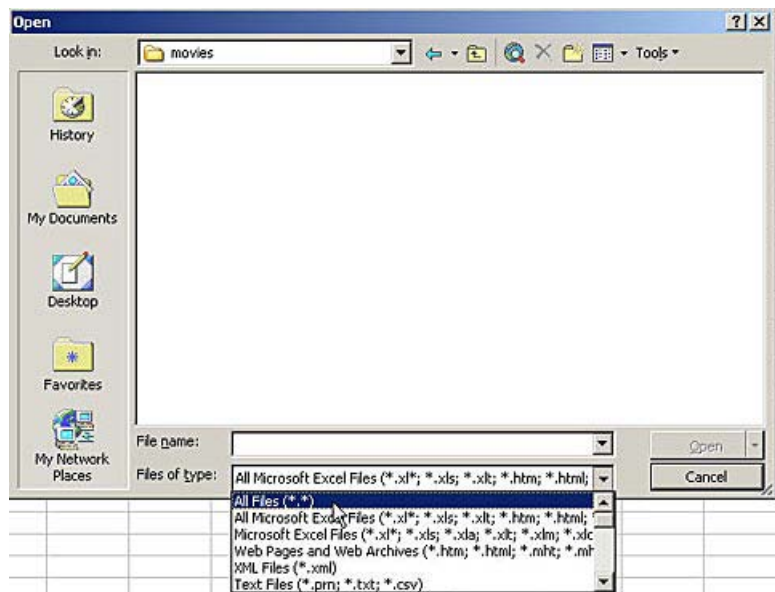
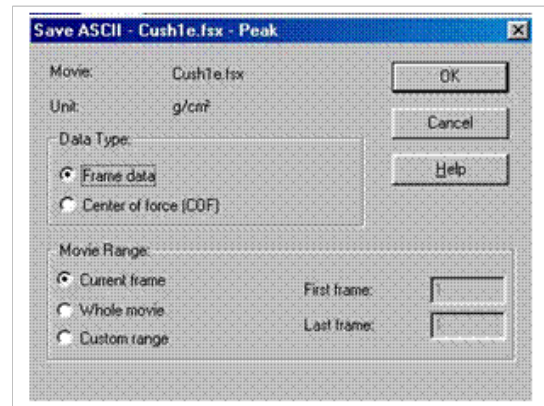
If you select 'Frame data', sensel pressures are saved for the selected range of frames. The output ASCII file contains a table of sensed pressures in sensor row/column format.

4. Select the desired 'Movie Range'. You may save data for the current frame, the whole movie or a custom range of frames. If you select a custom range of frames, you must specify the start and end frame of the data to be saved. The start frame number must be less than or equal to the end frame number, but greater than zero. The end frame number must be greater than or equal to the start frame number, but less than or equal to the number of frames in the movie.

If a Real-time window is active, the current information will be saved, regardless of which option is selected. For ASCII information, Real-time windows are simply treated as single-frame movies.

5. When you have specified the type of data and the data range, click <OK>. The 'Save As' dialog box will appear, and you must specify the file attributes - drive, directory, filename, file extension - or use the defaults. The Frame data ASCII files must be saved with the extension *.asf, and center of force (COF) ASCII files must be saved with the extension *.asc. Note that your system may limit file names to eight characters.

When one of these files is opened, the movie and sensor information (i.e. sensor type, number of rows and columns, etc.) is displayed, followed by the data. The following image shows the data for a single frame (the peak frame) within MS Excel. Note that when opening the file in MS Excel you must change the file type drop-down to "All Files" and then select the appropriate file.



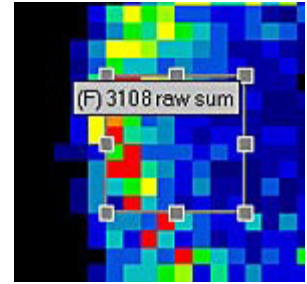
1. *Journal of the American Medical Association*, 1997; 278: 1039-1044.

[illegible]

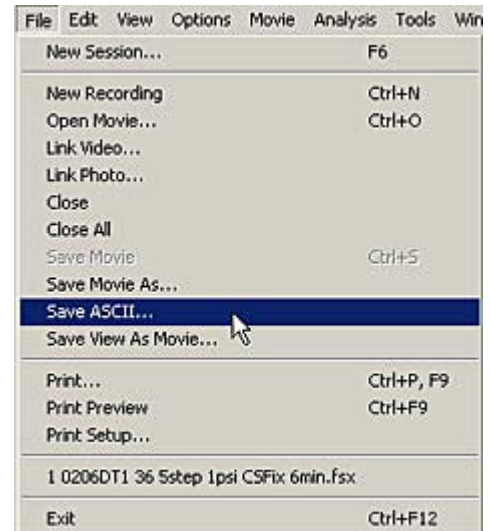
Saving an ASCII file for an Object

The following outlines how to save an ASCII file for an object, such as a Box or a Line.

1. Click on the Movie, Real-time, or Graph window of interest to make it active. In this example, a box object was placed on the movie (shown at right).

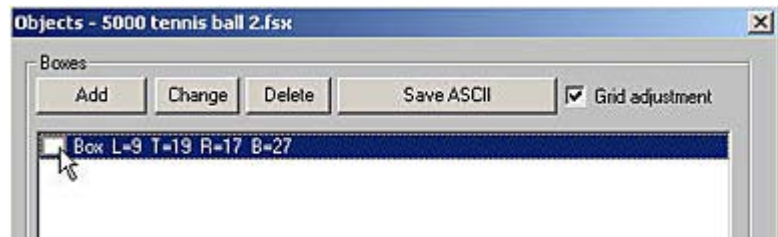


2. Open the 'Objects' dialog box. This dialog box can be accessed by selecting **Objects** from the **Analysis** pull-down menu (or by clicking the corresponding Toolbar icon), or by clicking the right mouse button while the cursor is over the window, and selecting 'Objects'.

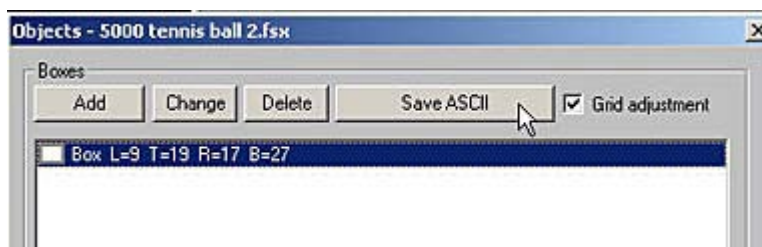


If the active window is a graph window, you can also open the 'Objects' dialog box by selecting **Save ASCII** from the **File** menu (see image at right).

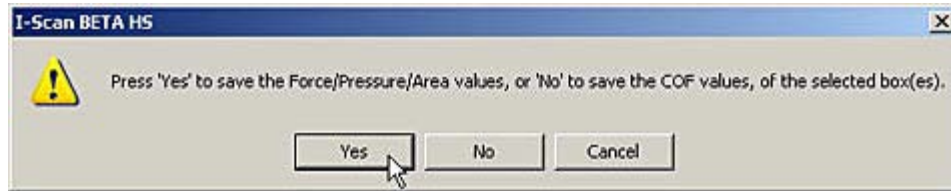
3. In the 'Objects' dialog box, select (highlight) the object(s) you wish to save as an ASCII file. You may select a single object, or multiple objects, in both the 'Boxes' (upper) and 'Lines' (lower) sections. To select more than one object simultaneously, hold down the **<CTRL>** key and click on any object you wish to select. To select a range of objects, hold down the **<SHIFT>** key and click on the first and last objects in the range.



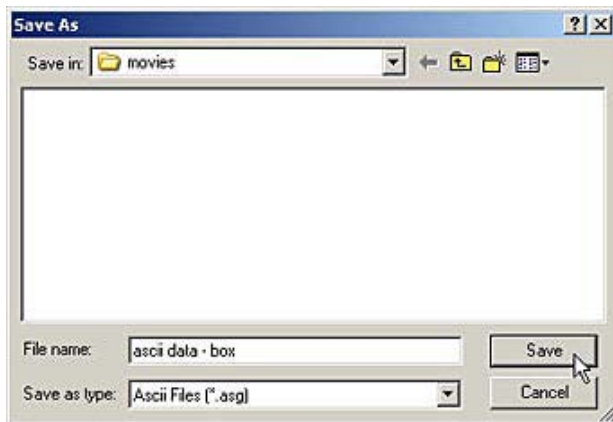
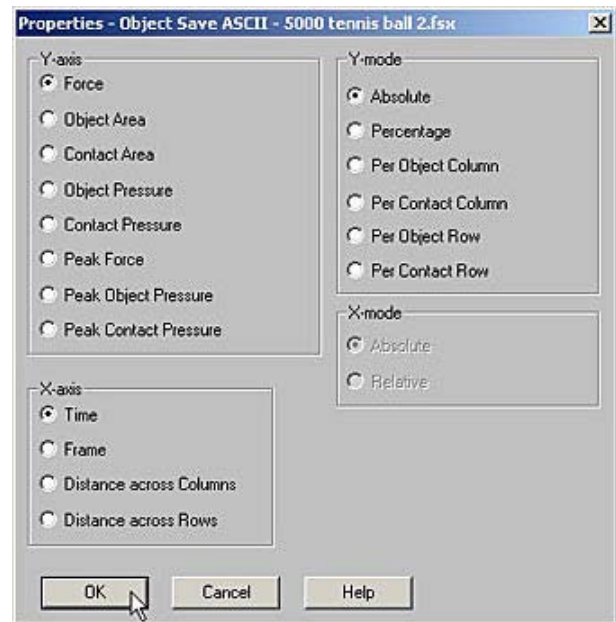
4. Once you've highlighted the object(s), click the 'Save ASCII' button in the appropriate ('Boxes' or 'Lines') section.



- A message will open asking if you want to save the Force/Pressure/Area values or save the COF values. Select your choice.



- The **'Properties'** (graph) dialog box will be displayed, and you may choose which values to save for the X- and Y-axis in the ASCII file, without changing the graph display. Click <OK> once you've made your selections.
- The **'Save As'** dialog box appears (shown below), and you must specify the file attributes - drive, directory, filename, file extension - or use the defaults. The data for the selected object(s) will be saved into the specified file. Object ASCII files must be saved with the extension *.asg.



The output file will contain the elapsed time ('Time'), frame number ('Frame'), the Y-axis value (force, area, pressure) that was selected in Properties (default is force), and the object's color (e.g. 'Y: Yellow'), for each object that was selected in the 'Objects' dialog box.

	A	B	C	D	E	F
1	DATA_TYPE	Graph				
2	VERSION	Tekscan Pressure Measurement System 5.201				
3	FILENAME	C:\Tekscan\movies\5000 tennis ball 2.fsx				
4	SENSOR_TYPE	5350				
5	ROWS	38				
6	COLS	41				
7	ROW_SPACING	0.4 inches				
8	COL_SPACING	0.4 inches				
9	SENSEL_AREA	0.16 in ²				
10	NOISE_THRESHOLD	3				
11	SECONDS_PER_FRAME	9.3e-005				
12	MICRO_SECOND	1				
13	TIME	Tuesday, July 22, 2003 11:46:46 AM				
14	SENSITIVITY	N/A-?				
15	START_FRAME	1				
16	END_FRAME	68				
17	UNITS	Raw Sum				
18	ASCII_DATA	@@				
19	Frames(X), Time(X), Force(Y)					
20	1, 0, 0					
21	2, 0.125, 0					
22	3, 0.249, 0					
23	4, 0.374, 0					
24	5, 0.498, 0					
25	6, 0.623, 0					
26	7, 0.748, 0					
27	8, 0.875, 0					
28	9, 1.001, 0					

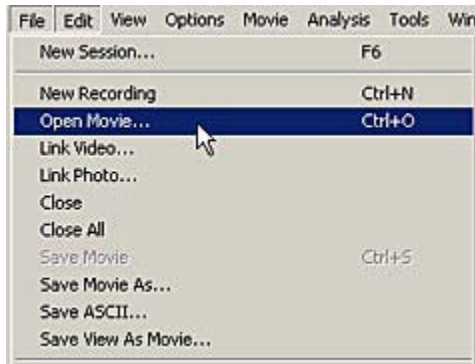
Note: For ASCII information, Real-time windows are simply treated as single-frame movies.

When one of these files is opened, the movie and sensor information (i.e. sensor type, number of rows and columns, etc.) is displayed, followed by the data. The image at right shows an example of the ASCII file opened in MS Excel:

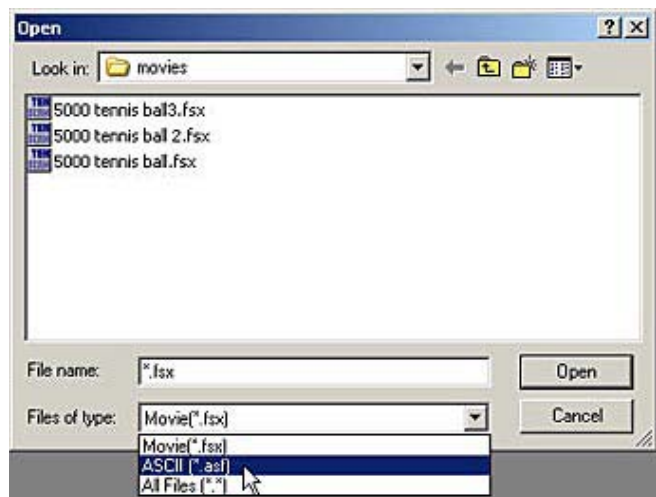
Reading ASCII Data

I-Scan software has the ability to open ASCII files and view them as pressure movies. Only files with the *.asf file extension may be opened.

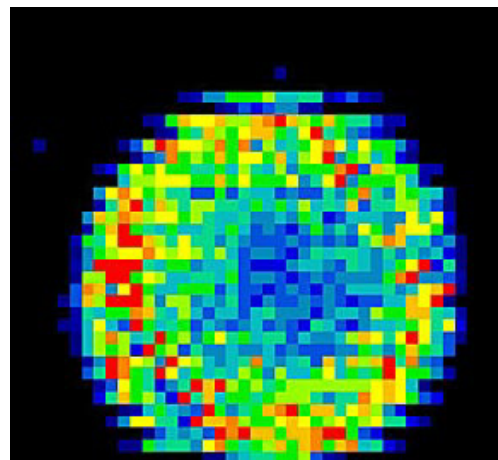
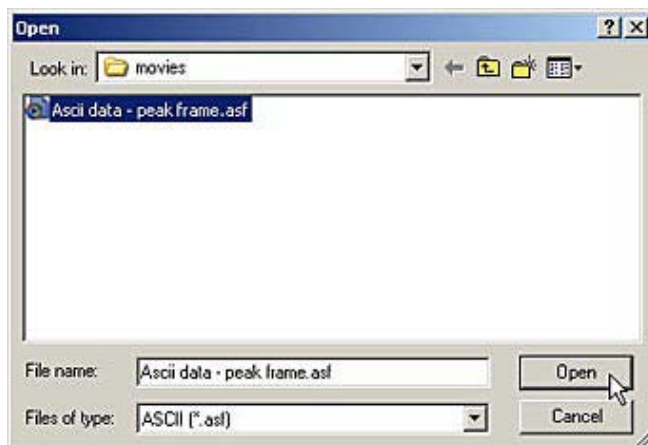
To read an ASCII file:



1. Select **Open Movie** from the **File** pull-down menu (see image at left). This will bring up the 'Open' dialog box.



2. Click the down arrow in the 'files of type' field (see image at right). Select **ASCII (*.asf)**.
3. All available files will be listed. Select the desired movie and click **Open** (see images below).



COPY & EXPORT OPTIONS

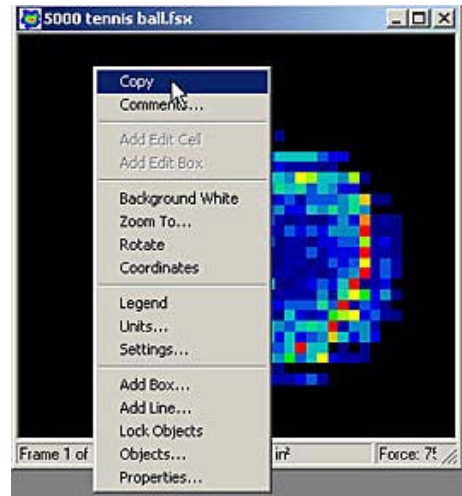
COPY OPTIONS

I-Scan provides a number of copy options that allow you to analyze pressure data in a spreadsheet or word processing program, compare *I-Scan* movie frames, and create reports or documents using actual color movie frames.

When this pressure data is copied, it is saved to the MS Windows clipboard as an image (bitmap), an ASCII (text) file of the actual pressure values, or both. This data can then be pasted into other Windows applications by selecting **Edit>> Paste Special** in that program and selecting 'bitmap' or 'text'. If the pressure data is pasted as text, some general information, such as the window's name, header information, comments, date/time, sensor information, and calibration data, is included.

You may copy *I-Scan* pressure data using one of the following methods:

- To copy an entire Movie, Real-time, or Graph window: Click on the desired window to make it active. Select **Copy** from the **Edit** pull-down menu (or click the corresponding Toolbar icon), or click the right mouse button while the cursor is above the window and select **Copy**. The entire active window's pressure data will be copied to the MS Windows clipboard as both an image (bitmap) and as a text file of the actual pressure values.



The image at right shows the results of Pasting the copied information from a real-time window into MS Excel.

	A	B	C	D	E	F	G	H	I
1	<<[TITLE								
2	5000 tennis ball.fsx								
3	FILENAME C:\Tekscan\movies\5000 tennis ball.fsx								
4	TIME 1/21/2003 6:05:20.8PM								
5	>>[TITLE								
6	<<[HEADER								
7	SENSOR_TYPE 5350								
8	ROWS 38								
9	COLS 41								
10	ROW_SPACING 0.4 inches								
11	COL_SPACING 0.4 inches								
12	SENSEL_AREA 0.16 in²								
13	NOISE_THRESHOLD 3								
14	SECONDS_PER_FRAME 9.3e-005								
15	MICRO_SECOND 1								
16	UNITS raw								
17	TIME Tuesday, January 21, 2003 6:05:20 PM								
18	SATURATION_PRESSURE 9.31145 PSI								
19	CALIBRATION_POINT 1 75 (Pounds) 12837 (Raw Sum) 375 (Number of Loaded Cells)								
20	CALIBRATION_INFO C:\Tekscan\movies\5000 tennis ball.fsx								
21	SENSITIVITY N/A.?								
22	>>[HEADER								
23	Frame 1 of 80 (1/21/2003 6:05:20.7853PM)								
24	START_FRAME 1								
25	START_ROW 1								
26	END_ROW 38								
27	START_COL 1								
28	END_COL 41								
29	ASCII_DATA BB								
30	0	0	0	0	0	0	0	0	0
31	0	0	0	0	0	0	0	0	0
32	0	0	0	0	0	0	0	0	0

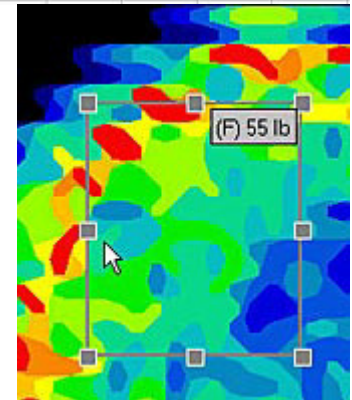
To copy only a specific area of a Movie or Real-time window: Add a box (refer to the **'Adding an Object'** section) to the Movie or Real-time window, enclosing the area of interest. Select **Copy** from the **Edit** pull-down menu (or click the corresponding Toolbar icon), or click the right mouse button while the cursor is over the window and select **Copy**. The pressure data from the portion of the window that is inside the box will be copied to the clipboard, as both an image (bitmap) and as a text file of the actual pressure values (see images at right).

You can be sure the box is the only object selected, when you see all 8 nodes in place (the square corner and side markers surrounding the object (in this case, a box above). Notice the mouse pointer is pointing to a node in the image at right.

Once the box is selected, you can use **Edit -> Copy** to copy only the contents of the box in an ASCII format.

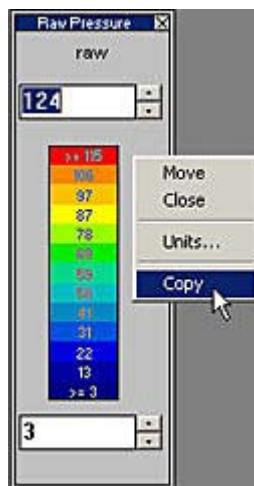
- To copy the Legend: Right mouse click anywhere on the **Legend** (color scale), and select **Copy**. The legend will be copied to the Windows clipboard as an image (bitmap) only. The Copy command (Edit pull-down menu) cannot be used to copy the legend to the clipboard.

	A	B	C	D	E	F	G	H	I	J	K
13	NOISE_THRESHOLD 3										
14	SECONDS_PER_FRAME 9.3e-005										
15	MICRO_SECOND 1										
16	UNITS raw										
17	TIME Tuesday, January 21, 2003 6:05:20 PM										
18	SATURATION_PRESSURE 9.31146 PSI										
19	CALIBRATION_POINT_1 75 (Pounds) 12837 (Raw Sum) 375 (Number of Loaded Cells)										
20	CALIBRATION_INFO C:\Tekscan\movies\5000 tennis ball fsx										
21	SENSITIVITY N/A-?										
22	>>))HEADER										
23	Peak										
24	START_FRAME 1										
25	START_ROW 11										
26	END_ROW 23										
27	START_COL 12										
28	END_COL 22										
29	ASCII_DATA BB										
30	62	46	41	157	166	119	105	89	70	59	97
31	176	137	88	69	48	87	61	89	67	36	50
32	40	131	151	82	83	81	64	65	77	78	82
33	100	61	73	78	76	79	67	47	64	64	53
34	135	75	76	74	81	86	85	57	80	60	43
35	43	49	27	78	60	37	66	59	54	61	50
36	53	74	58	56	72	69	81	62	57	59	55
37	68	56	57	77	63	59	53	70	47	37	36
38	44	81	72	47	74	66	55	74	63	40	56
39	71	60	74	41	58	63	88	49	36	26	48
40	84	82	74	80	65	60	57	61	24	23	21
41	61	29	88	61	59	45	60	34	22	32	39
42	85	60	49	61	47	55	62	56	31	34	47
43	BB										



Example how to copy the Legend

Example of the Legend Pasted into MS Excel



- To copy the entire screen: Press the **'Print Screen'** key on your computer's keyboard. A picture of the entire screen, including any open Movie, Real-time, or Graph windows, as well as the Toolbar and Main Menu, will be placed into the Windows clipboard as an image (bitmap) only.

EXPORT OPTIONS

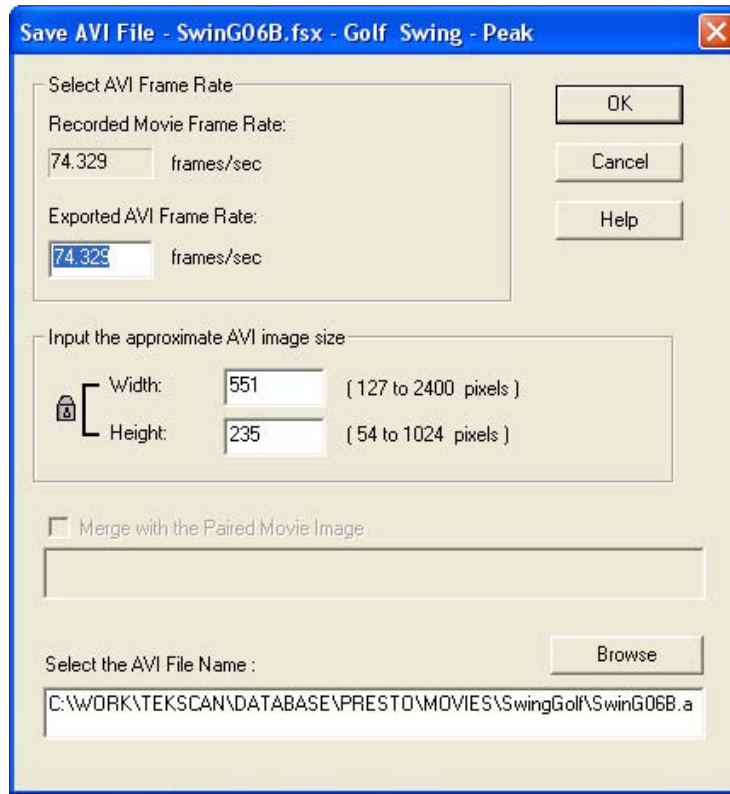
Note: This chart provides an overview of what can be exported from Tekscan into other programs and how this information can be exported. For a full explanation of each option, please refer to the "Copy & Export Options" section found in your system software Help File under the "Help" menu.

		Where do you want to use the exported information?				
		Word Processor (MS Word) Presentation (MS PowerPoint) Spreadsheet (MS Excel)	Image File (Static - .BMP)	Video File (Dynamic - .AVI)	Printout (Hard Copy)	Report File (PDF)
		Method Used to export data				
What do you want to export from Tekscan?	Movie Frame (2-D or 3-D Image)	Copy & Paste Special	Copy & Paste	N/A	Print	Acrobat or PDF995
	Legend (Image)	Copy & Paste	N/A	N/A	Print	Acrobat or PDF995
	Graph (Image)	Copy & Paste Special	Copy & Paste	N/A	Copy & Paste Special	Copy & Paste Special
	Whole Movie or Whole Movie with synchronized video (Video)	SnagIt or Hypercam	N/A	SnagIt or Hypercam	N/A	N/A
	Entire Application Window (Image)	Alt + Print Screen	Alt + Print Screen	N/A	Alt + Print Screen	Alt + Print Screen
	Comments (Text)	Copy & Paste	N/A	N/A	Print	Acrobat or PDF995
	Movie Frame (Array Data)	Copy & Paste or ASCII data	N/A	N/A	Copy & Paste	Copy & Paste

SAVING AN AVI (FILE MENU - OPTIONAL)

The **Save AVI** is an optional feature which allows you to export the currently selected Tekscan movie (*.fsx file) into a digital video format (*.avi file) that can be saved on your computer. The following instructions explain how to use this feature. An .avi file can be:

- Emailed to colleagues for viewing on their computer.
 - Imported into a MS PowerPoint presentation to highlight animated pressure profiles.
 - Used in a video analysis system to synchronize pressure data with visual images of a subject in motion.
1. Select **File>Save AVI**. The following dialog opens (shown below), where you can select the exported video frame rate, video dimensions (in pixels), and file name.



Note: Selecting a lower or higher frame rate affects the size and quality of the video. A lower frame rate means smaller .avi file size and lower quality video. A higher frame rate means a larger .avi file size, but does not necessarily provide a better video quality. If you are going to send the resulting video via email, use a frame rate between 15 and 25. Otherwise, the default setting (same frame rate as the .fsx file) is recommended.

Note: What you see on-screen in the Movie is what will be saved as an .avi video. This means that any view options (objects, panes, COF Trajectory, View Mode -- 2-D or 3-D) will be included in the .avi.

2. If you wish, adjust the Exported AVI Frame Rate (frames/sec). This can be any number between 1 and 1000.

3. Enter the dimensions of the AVI video in the Width and Height input fields. The default width and height listed is the .fsx dimensions that were on-screen before you entered the "Save AVI" dialog. If you resize the movie window, then go into the "Save AVI" dialog, these numbers will update. In addition, the width and height are constrained proportionally (their aspect ratio remains constant). This means that entering a value in the Width field will update the Height field relatively. This is done to ensure the video aspect ratio (Length x Width) remains the same when resized. Finally, the Width and Height values are approximate. This is due to the difference between sensel and pixel sizes. They are accurate +/- 5% approximately.
4. By default, the movie is saved into the same folder where the Tekscan movie (*.fsx) is located -- in the [Client File Management System](#) (database). If you wish to save the movie to a new location, click the **Browse** button. The Browse button opens the "Save As" dialog, where you can specify a folder on your computer where the file should be saved, as well as the file name. When finished selecting the proper folder and name for the file, click the **Save** button.

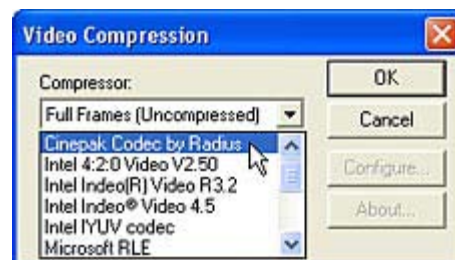
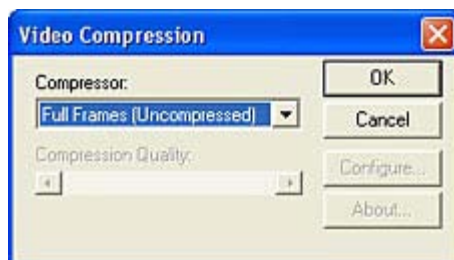


Note: *Movie files can only be saved as an avi file type.*

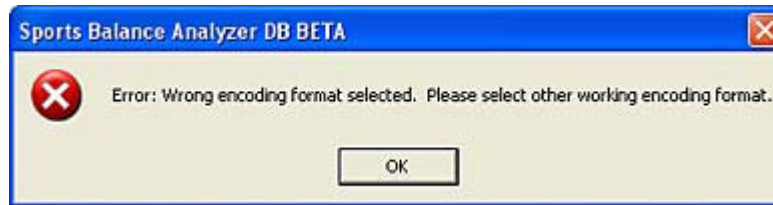
5. Back in the "Save AVI File" dialog, click **OK**. The "Video Compression" dialog opens, where you can select the compressor, compression quality, and configure compression options. By default, the .avi file is encoded with **Full Frames (Uncompressed)**, meaning that no compression is applied to the video. This could potentially result in a very large file size. To reduce size, you can apply a compressor to the video. If you wish to do so, click the arrow to the right of the "Compressor" drop-down box. A list of all codecs installed on your computer are displayed.

Note: *The choices available in your list may be different from what is displayed below, depending which compressors are installed on your computer.*

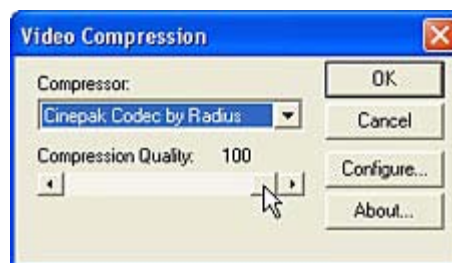
Not all codecs installed on your computer will work with the **Save AVI** feature. Tekscan recommends the **Cinepak Codec by Radius** codec. This is a fairly generic codec that is available on most computers and this codec has been tested to work with the **Save AVI** feature.



In the event a codec cannot be used, you will receive the following warning message below. If this happens, try using the **Cinepak Codec by Radius**.

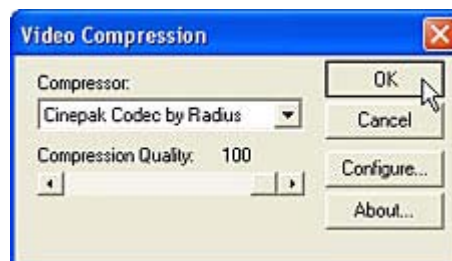


6. Once a compressor is selected from the list, you may be able to select the Compression Quality on the "Video Compression" dialog (shown below). Since Tekscan movies are small to begin with, lowering the Compression Quality is not necessary, and will not result in a smaller file size or poor image quality. Therefore, it is recommended you keep the default Compression Quality as is (100).



7. Click the **OK** button on the "Video Compression" dialog. The software converts the *.fsx file to an *.avi file and saves it in the designated location on your computer. Depending on the length of the movie and your compression settings, this may take some time. A blue status bar indicates the progress of video rendering. Once finished, you can use any video playback software (such as Microsoft Media Player or Apple Quick Time) to play the video file.

Note: Codec is short for "video encoder and video decoder." It allows video to be encoded when created, and decoded when played back. This means that the codec used to encode the video must be installed on any computer which is playing back the video. If it is not, then the video will not play and an error is generated. This should not be an issue if you are playing back the video on the same computer from which the video was created. However, this can be a problem if the video is transferred to other computers. If transferring videos from one computer to another, ensure the destination computer also has the appropriate codec installed.

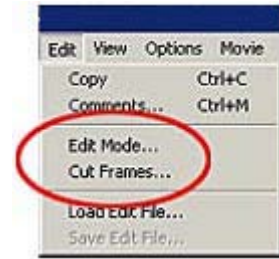


EDITING

The **I-Scan** software allows you to edit data using two methods: by removing unwanted frames from a movie (Cut Frames), or by modifying specific areas of data in a Movie or Real-time window (Edit Mode).

The **Edit Mode** functions permit you to remove data from one or more sensels, or to average sensel reading(s) with those of its neighbors, in either a Movie or a Real-time window.

The **Cut Frames** feature allows you to cut specific unwanted frames from a movie.

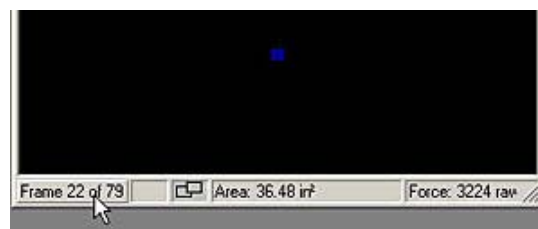
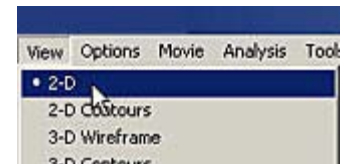


REAL-TIME EDITING

Damage may occur to a sensor due to use (excessive wrinkling, folding, etc.) or trimming. Individual sensels or even rows and columns may develop shorts or opens. Shorts may appear as red sensels in the real-time window, even when no pressure is applied to the sensor. Opens will show up in the real-time window as missing rows, columns, or sensels, even though there is a load being applied to those areas of the sensor. With Real-time Editing, you can define a "mask" for these sensels, which replaces their readings with an average of those of their neighbors in the display, or you can remove specific sensels altogether.

IMPORTANT! 'Real-time Editing' is meant to be used to 'repair' one or two bad rows / columns or a few hot spots on a sensor. Do not use it to fix large areas of the sensor or adjacent rows or columns, since the data you obtain with such a sensor will be poor. If this is necessary, the sensor should be replaced.

1. Click on the real-time window to make it the active window. Click on the **View** menu, and select **2-D**. In 2-D mode, it is possible to see each individual sensel, making editing much easier.
2. When **Edit Mode** is selected, the real-time window will automatically be paused. It will return to normal when the dialog box is closed. You can also manually **'Pause'** (freeze) the real-time window in which you want to edit sensels, by selecting **View>>Pause**, or by clicking on **'Sensor OK'** in the Real-time status bar. The title bar should say **'(Paused)'**. This frozen frame should clearly show the bad row, column or cell to be fixed. It will be very difficult to correctly edit sensels if the bad cells are not shown.



3. Click on the **Edit** menu, and select Edit Mode. The Edit dialog box that appears (at right) will apply only to the currently active Movie or Real-time window (the name of the active window will be shown in the Edit box title).



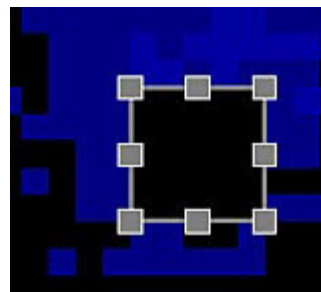
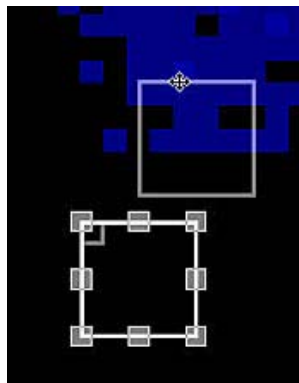
4. Select the 'Add Edit Cell' or 'Add Edit Box' buttons on the left of the dialog box. 'Add Edit Cell' is used when you want to edit an individual sensel. 'Add Edit Box' is used when you want to edit a group of sensels.



Note: The edit options can also be selected by placing the cursor over the active window, clicking the right mouse button, and selecting the desired option, as shown at right.

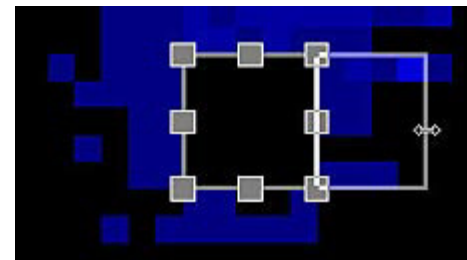
Note: You can use keyboard shortcuts to perform certain functions once an edit box has been placed in the window. Refer to [Keyboard Shortcuts](#) for more information.

5. Place the cursor at the desired point in the real-time window (a 'pencil' symbol will be added to the cursor) and click the left mouse button. A gray edit box or edit cell will be placed at that point.
6. Once an edit box/cell has been placed, it can be moved in the window, just as an object can be moved. Click on any edge of the edit box/cell with the left mouse button, and a four-way arrow will appear. Hold down the left mouse button and drag the cursor to move the edit box/cell until it is in the sensor area that you wish to edit (see images below).



The dimensions of an edit box may also be changed. Click on one of the eight nodes on the box's outline, and a two-way arrow will appear. Hold down the left mouse button and drag the node until the edit box outlines the group of sensels that you wish to edit (see image at right).

Note: The software automatically aligns the edges of edit boxes and edit cells with the edges of sensor rows and columns.



When you place any edit cells/boxes on a real-time window and then record the movie, the next real-time window you open will already have the edit cells/boxes placed in the movie window in the same position; provided you are still within the same session. Clicking on the New Session button removes this inheritance, and any further real-time windows you open will not contain any edit cells/boxes.

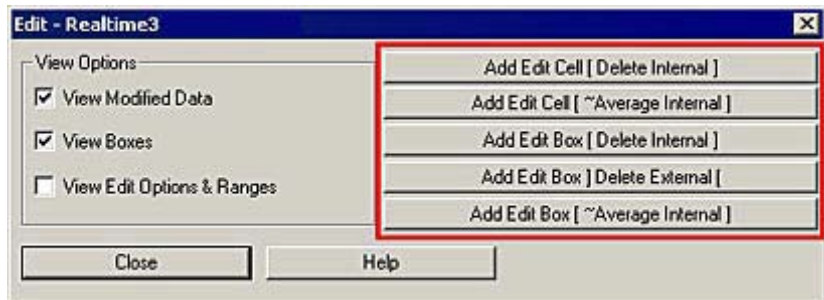
With a real-time window, when you place edit cells/boxes, and then close the window or application, you will receive the following message warning you that your edit cells/boxes will be lost if you do not save them.



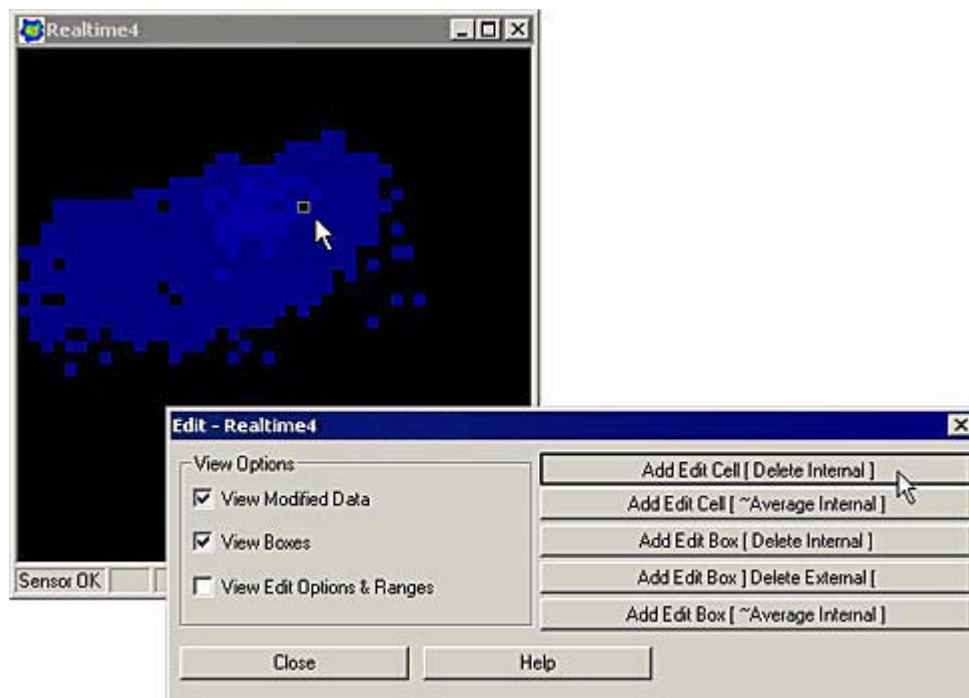
Be sure that you save any edit cells/boxes within an "edit file." This procedure is discussed in the "Saving / Loading Edit Files" section.

Edit Options:

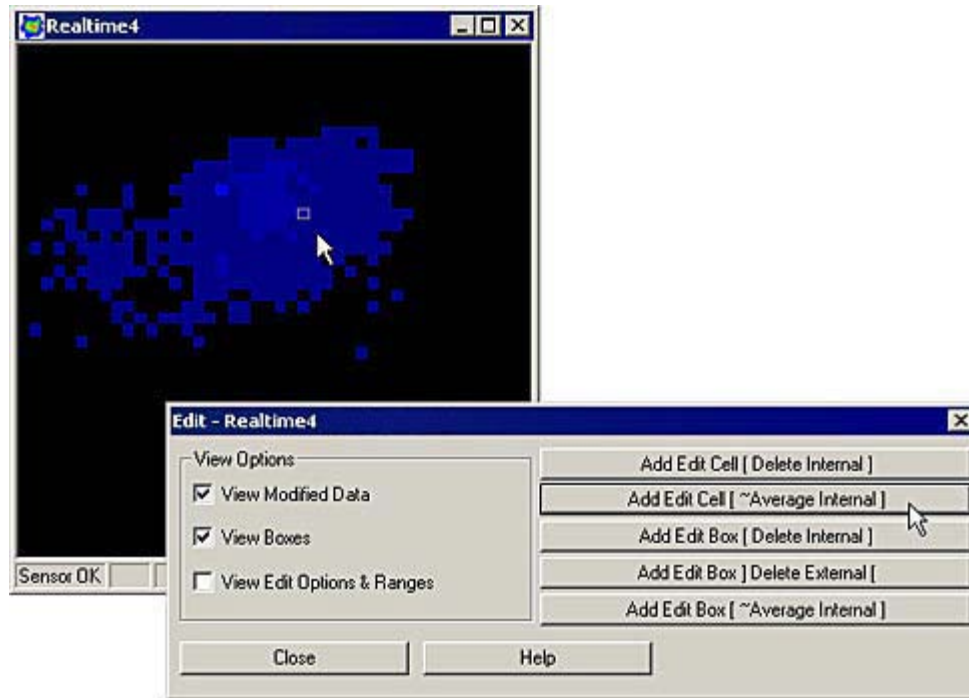
Once you have placed the edit box/cell around the damaged sensel(s), the selected Edit option will be applied to each of the sensor cell values inside or outside the Edit box/cell, depending. The possible "Edit Options" are as follows:



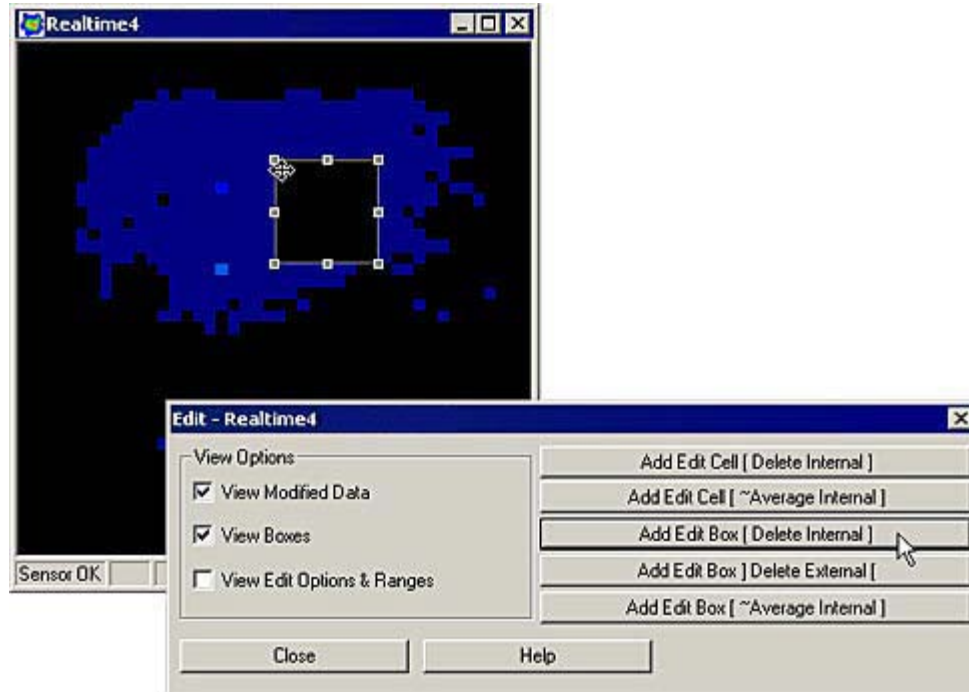
- **Add Edit Cell [Delete Internal]:** Sets the pressure reading of the edit cell to a value of 0 (no pressure).



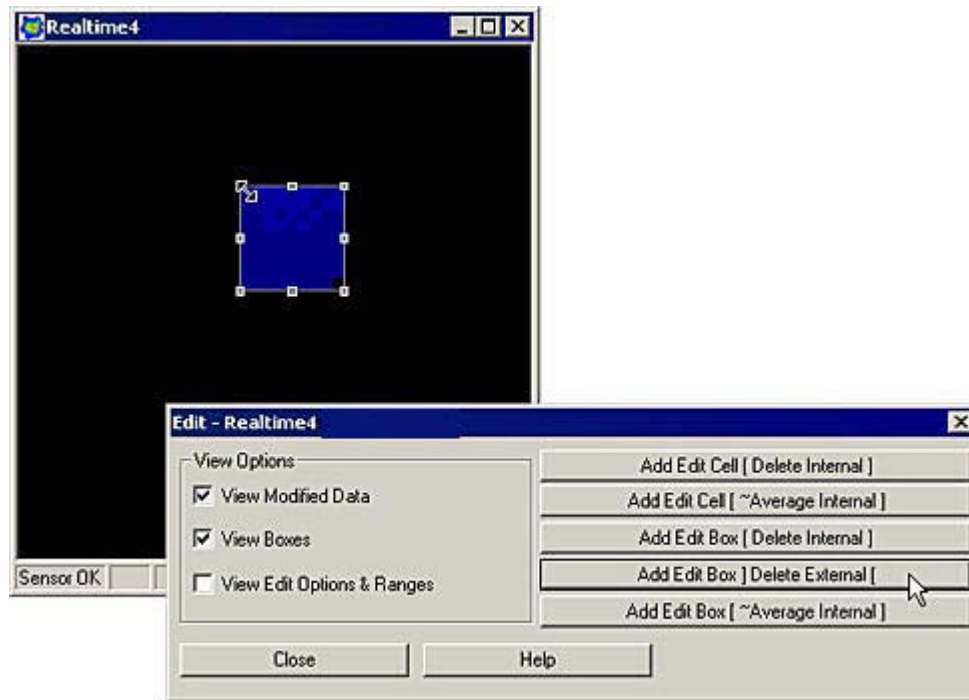
- *** Add Edit Cell [~Average Internal]**: Sets the pressure reading of the edit cell to the average of the pressure readings of the neighboring sensels.



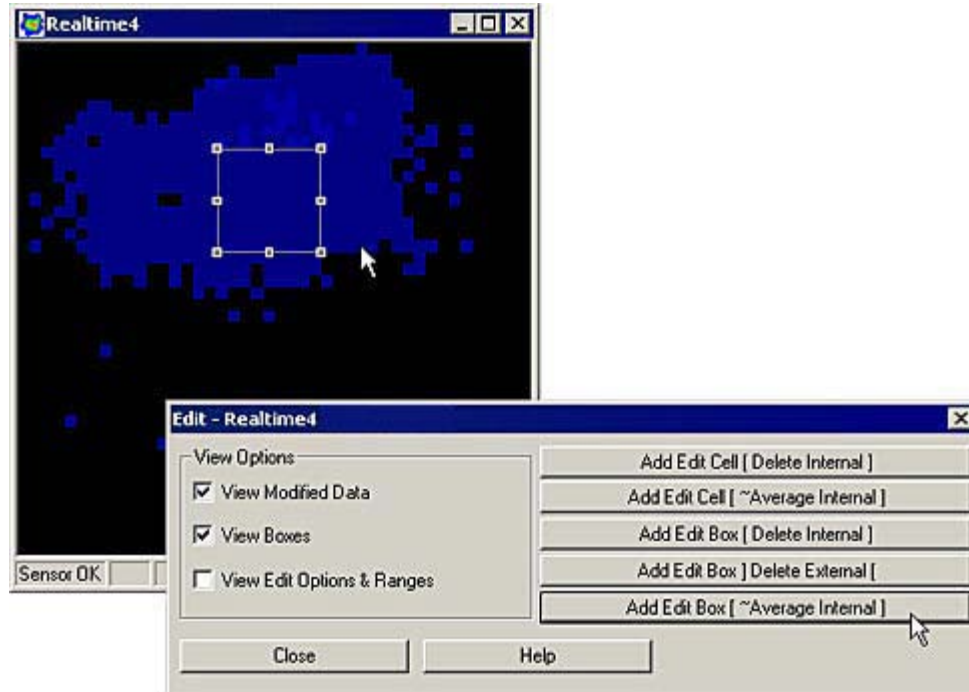
- **Add Edit Box [Delete Internal]**: Sets the pressure reading inside the edit box to a value of 0 (no pressure).



- **Add Edit Box]Delete External[:** Sets the pressure reading outside the edit box to a value of 0 (no pressure).



- *** Add Edit Box [~Average Internal]:** Sets the pressure reading inside the edit box to the average of the pressure readings of the neighboring sensels.



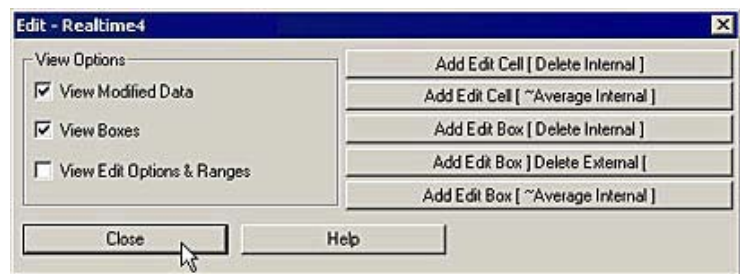
*** Cells are averaged in the following manner:**

If the edit cell/box is:

- >2 sensels wide and >2 sensels high, all sensels inside the edit box will be the average value of the cells inside the box.
- ≤ 2 sensels wide, each sensel value is calculated by averaging the sensels that border the edit box on the left and right.
- ≤ 2 sensels in height, each sensel value is calculated by averaging the sensels that border the edit box on the top and bottom.
- ≤ 2 sensels in width and height, the sensels values are calculated by averaging the sensels that border the edit box on the left and right (edit cells are calculated this way).

1. Choose your edit box view options. Select '**View Modified Data**' if you want to view the edited sensel values in the real-time window. Select '**View Boxes**' if you want to see the edit box outlines; de-selecting this option will hide the boxes.

2. When you have added all the edit boxes/cells you wish to add, click the **Close** button in the Edit dialog box. When you open this movie at a later time, the edited sensels will have their new values. Any further changes to the sensel data can still be performed on the edit cells/boxes. New edit cells/boxes can be added or removed at any time.



Note: Edit boxes/cells are not visible in the Real-time window, unless the Edit dialog box is open, and 'View Boxes' is selected. However, when the cursor is placed over an edit box/cell in the window, the cursor becomes a 'pencil' symbol to show its location, or a two-way or four-way arrow, to allow you to make changes.

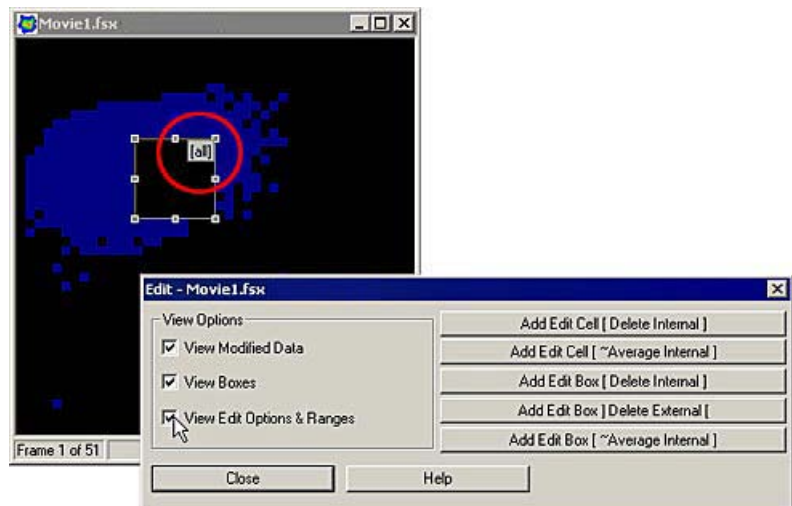
Note: The edit box/cell information (position and edit options) can be saved at any time while a Real-time window is open, using the 'Save Edit File' option. This is helpful if you would like to use the same edit boxes/cells in another window. Refer to the 'Load/Save Edit File' section for more information.

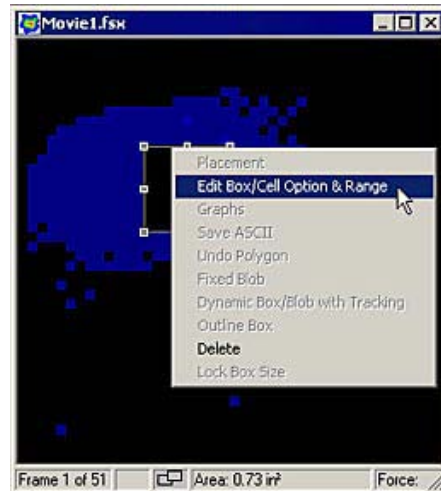
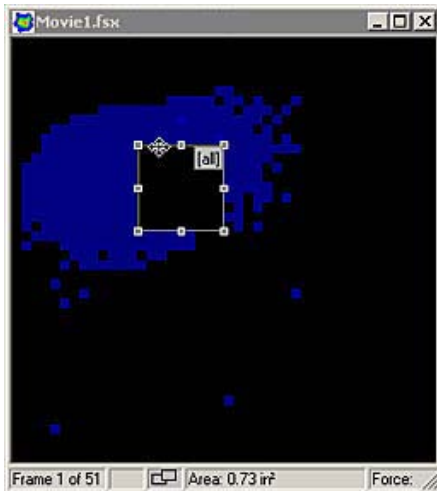


View Edit Options & Ranges:

This option allows you to view the range of frames that are being targeted within a movie. The edit range is displayed in the upper right corner of the edit box, as shown below:

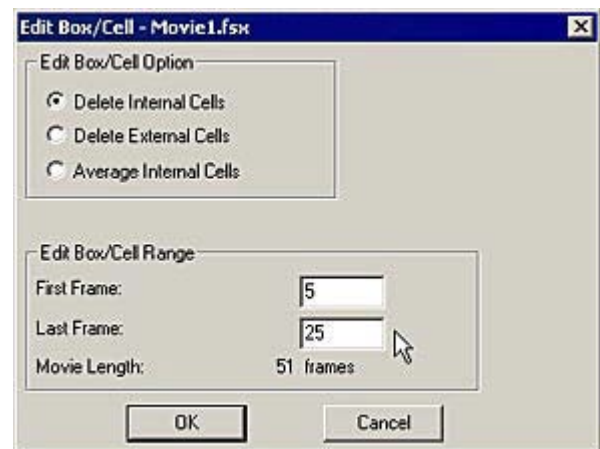
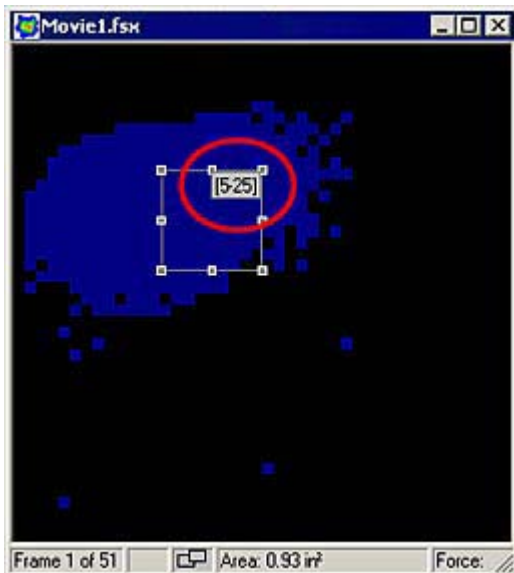
This option becomes extremely useful when you are viewing a movie that has multiple frames. You can set up different edit boxes to affect specific frames within the movie. To target a specific number of frames, highlight the edit box with your cursor (the cursor will change to a 4-sided arrow). Then right-click and select "Edit Box/Cell Option & Range."





The dialog at right opens where you can change the Edit Box/Cell Options, as well as the frames within the movie which are to be affected by the edit option.

When you select a specific frame range, and click OK, the frame range and edit options are displayed within the right corner of the edit box, as shown below:

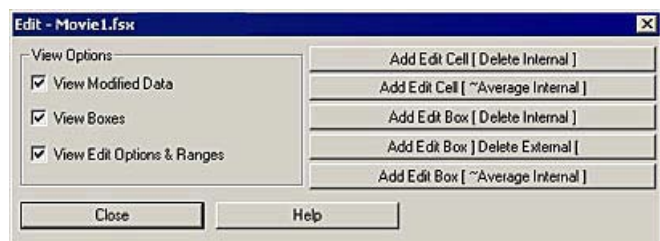


MOVIE EDITING

Movie Editing can be used to improve the data display if the sensor became damaged during use, or if excessive noise appeared outside the main contact area. You may also remove entire frames from a movie, as described in the “Cut Frames” section.

Note: Movie Editing is non-destructive. The movie's original state can always be retrieved by removing any editing cells or boxes from the movie.

1. Open the movie file. Click on the **View** menu, and select **2-D**. In 2-D mode, it is possible to see each individual sensel, making editing much easier.
2. Click the **Edit** menu, and select **Edit Mode**. The Edit dialog box that appears (see image below) will apply only to the currently active Movie or Real-time window (the name of the active window will be shown in the Edit dialog's title).



To place an edit box or edit cell:

1. Select any of the five choices on the right of the dialog box:

- **Add Edit Cell [Delete Internal]**: Sets the pressure reading of the edit cell to a value of 0 (no pressure).
- *** Add Edit Cell [~Average Internal]**: Sets the pressure reading of the edit cell to the average of the pressure readings of the neighboring sensels.
- **Add Edit Box [Delete Internal]**: Sets the pressure reading inside the edit box to a value of 0 (no pressure).
- **Add Edit Box [Delete External]**: Sets the pressure reading outside the edit box to a value of 0 (no pressure).
- *** Add Edit Box [~Average Internal]**: Sets the pressure reading inside the edit box to the average of the pressure readings of the neighboring sensels.

*** Cells are averaged in the following manner:**

If the edit cell/box is:

- **>2 sensels wide and >2 sensels high**, all sensels inside the edit box will be the average value of the cells inside the box.
- **≤2 sensels wide**, each sensel value is calculated by averaging the sensels that border the edit box on the left and right.
- **≤2 sensels in height**, each sensel value is calculated by averaging the sensels that border the edit box on the top and bottom.
- **≤2 sensels in width and height**, the sensels values are calculated by averaging the sensels that border the edit box on the left and right (edit cells are calculated this way).

Note: The edit options can also be selected by placing the cursor over the active window, clicking the right mouse button, and selecting the desired option, as shown below.



- Place the cursor at the desired point in the Movie window (a 'pencil' symbol will be added to the cursor), and click the left mouse button. A gray edit box or edit cell will be placed at that point.
- Once an edit box/cell has been placed, it can be moved in the window. Click on any edge of the edit box/cell with the left mouse button, and a four-way arrow will appear. Hold down the left mouse button and drag the cursor to move the edit box/cell until it is in the sensor area that you wish to edit.

The dimensions of an edit box may also be changed. Click on one of the eight nodes on the box's outline, and a two-way arrow will appear. Hold down the left mouse button and drag the node until the edit box outlines the group of sensels that you wish to edit.

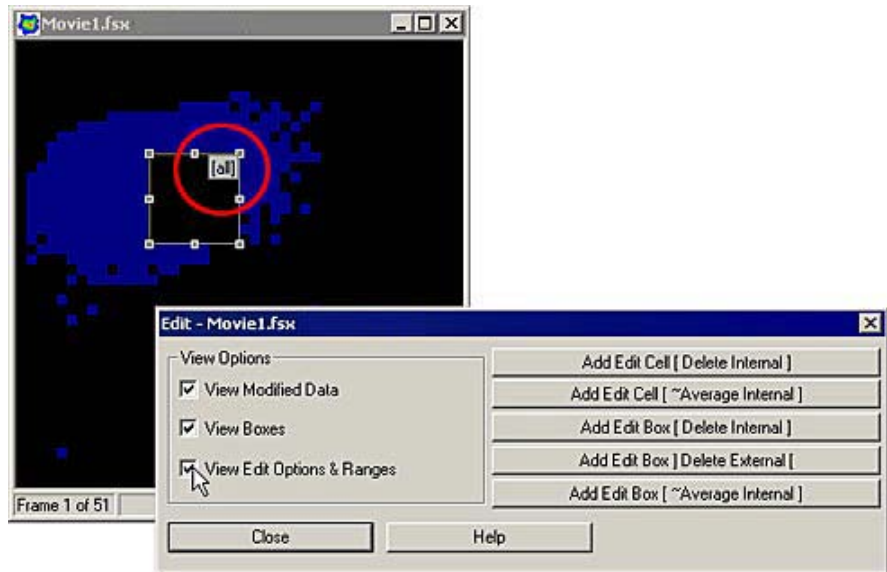
Note: The software automatically aligns the edges of edit boxes and edit cells with the edges of sensor rows and columns.

Note: You can use keyboard shortcuts to perform certain functions once an edit box has been placed in the window. Refer to [Keyboard Shortcuts](#) for more information.

4. Once you have placed the edit box/cell around the desired sensel(s), the selected edit option will be applied to each of the sensor cell values inside the edit box/cell.

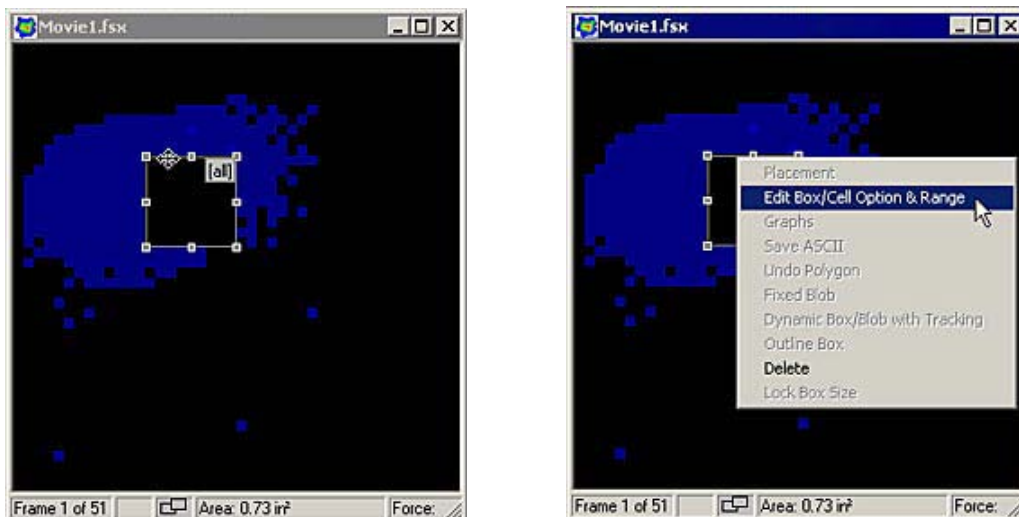
Hint: Using Peak mode allows you to see the entire pressure area while editing; this is especially helpful if using Average Cells mode.

5. Select the "View Edit Options & Ranges" checkbox if you want to display the frame range that is affected by the edit cell/box, as well as the type of edit option being applied. The edit range is displayed in the upper right corner of the edit box, as shown below:

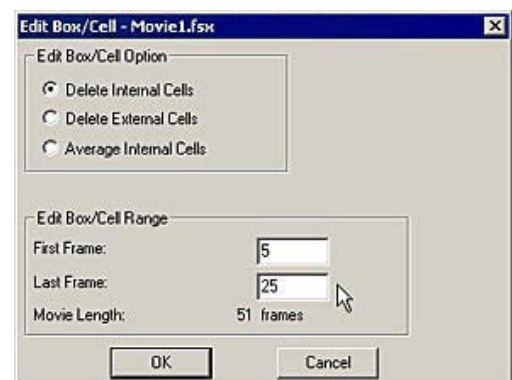


This option becomes extremely useful when you are viewing a movie that has multiple frames. You can set up different edit boxes to affect specific frames within the movie.

To target a specific number of frames, highlight the edit box with your cursor (the cursor will change to a 4-sided arrow). Then right-click and select "Edit Box/Cell Option & Range" (as shown below).



The dialog at right opens where you can change the Edit Box/Cell Options, as well as the frames within the movie which are to be affected by the edit option.



When you select a specific frame range, and click **OK**, the frame range and edit options are displayed within the right corner of the edit box, as shown at right.

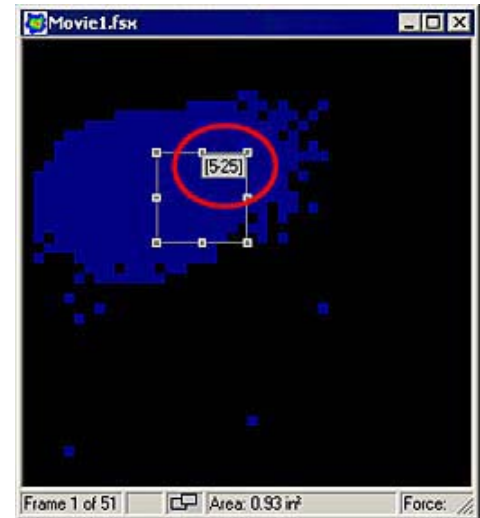
Note: The edit boxes/cells, including all their defining information (position and edit options), can be saved at any time before the "Close" button is clicked, using the 'Save Edit File' option. This is helpful if you would like to perform the same edits in another window. Refer to the 'Load/Save Edit File' section for more information.

6. When you have added all the edit boxes/cells you wish to add, click on the **Close** button in the **Edit Mode** dialog box.

Note: Editing the movie in this manner, does not affect the movie data. You can always delete the edit boxes/cells at a later time.

7. Save the movie, with the edited sensel data, as a new file using the **File**>>"Save Movie" or "Save Movie As" menu item.

Note: If you forget to save an edited recording and attempt to exit, the system will warn you that data will be lost unless you save it.



SAVING/LOADING EDIT FILES

When a recording is saved, the data that was created by editing the sensor data is saved with the file. Edit cells/boxes are saved along with a Movie. There is no need to save the edit cells/boxes separately unless you want to import them into another movie at a later date. This can be advantageous if you are going to use the same sensor at a later time. The resulting edit file can then be loaded into a Real-time or Movie window.

Note: Each time you open a Real-time window which needs the edit information, you must load the edit file. The software does not remember the previous edit information when a new Real-time window is opened.

To save an edit file:

1. Once you have placed the edit boxes and/or cells into a Movie or Real-time window, and are satisfied with their placement and option settings, select **Save Edit File** from the **Edit** pull-down menu. A "Save As" dialog box will be displayed, and the file can be saved as an edit file (with an *.fed extension).



Enter the desired filename and path (location) and click the **Save** button. The default filename will be the name of the active window (e.g. "Movie1.fed" or "Real-time2.fed"). The edit box/cell information, which includes the coordinates and selected Edit and View Options, will be saved in the file.

Note: All edit boxes/cells in the active window will be saved in the edit file.

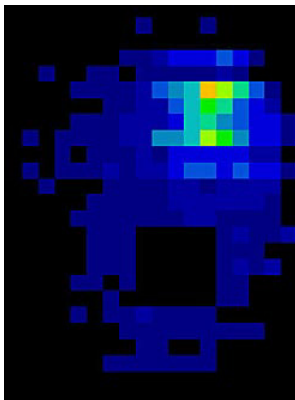


To load an edit file:

1. Open a Movie or Real-time window, and then select **Load Edit File** from the **Edit** pull-down menu (see image at right). An "Open" dialog box will be displayed, and edit files (with the extension *.fed only) can be opened. Edit files are only valid for a particular sensor. If you change sensors while an edit file is loaded, you should load an edit file for that sensor, or create a new file.



2. Enter the desired filename and path (location) and click the **Open** button to load the edit file (see image at left). When the edit file is loaded, the edit boxes/cells are placed at the same coordinates at which they were saved, with the same properties, and the "Edit Mode" dialog box is opened.
3. Make any necessary changes to the placement and option settings of the loaded edit boxes/cells, then click **Close** to accept the changes and close the "Edit Mode" dialog box (shown below).



Note: The example shown at left is an extreme example. For more practical purposes, editing cells would probably not be so pronounced. This example is exaggerated to outline the editing process.

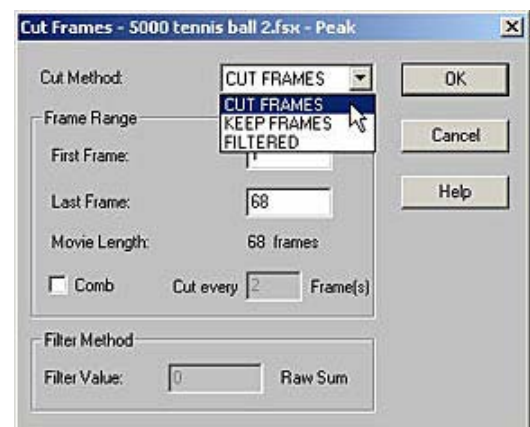
CUT FRAMES

The **Cut Frames** option enables you to discard unwanted frames from a recording using a number of methods. This option is helpful for removing nonessential data from movies, and also for making large movie files more manageable.

The three options in the 'Cut Method' pull-down field are 'Cut Frames', 'Keep Frames', and 'Filtered'. 'Cut Frames' allows you to discard specific frames, 'Keep Frames' allows you to keep specific frames, and 'Filtered' allows you to discard frames whose total applied force is less than a specified minimum value.

To remove frames of data from a movie:

1. Open the movie that you wish to edit. Click on the movie to make it active (edits will only apply to the currently active movie window). Select **Cut Frames** from the **Edit** pull-down menu. The 'Cut Frames' dialog box will appear.

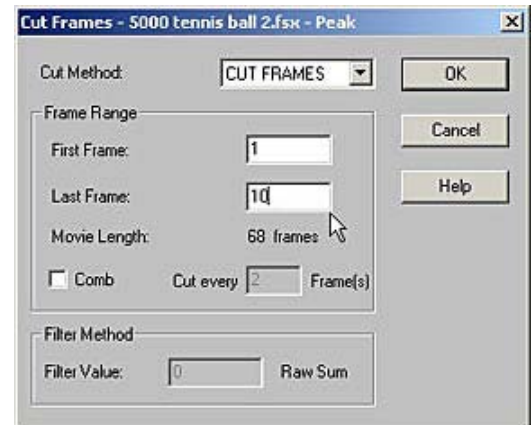
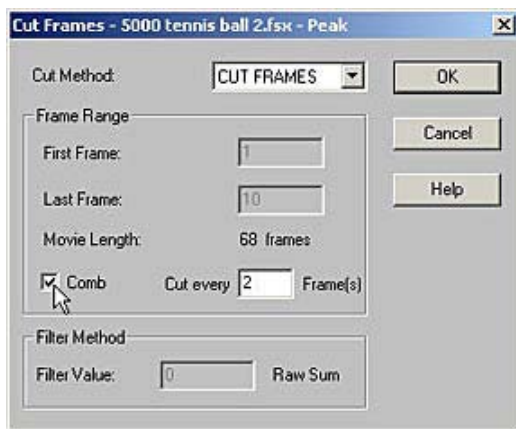


2. Follow the instructions below for one of the three methods to discard unwanted data (frames) from your movie.

i. **'Cut Frames' method:**

- a) Select **'Cut Frames'** in the **'Cut Method'** section of the dialog box (see image at right).

- To cut a range of frames from the movie, enter the first ('First Frame') and last ('Last Frame') frame numbers that you want to remove. The 'Movie Length', in number of frames, is listed below these fields (see image at right).

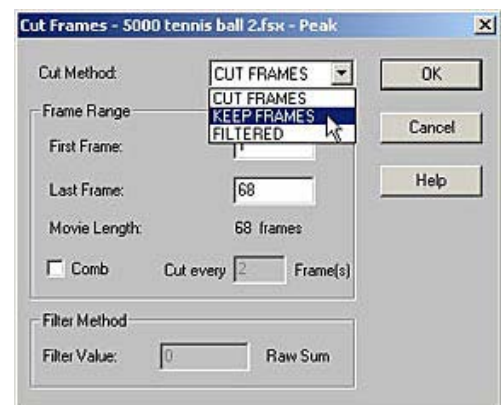


- To remove frames from the movie at specified intervals ("**Comb-Cut**"), place a check mark in the 'Comb' check box, and then enter a number in the "**Cut Every ___ Frames**" field. Each n^{th} frame will be removed from the movie. For example, if you entered a "3", every 3rd frame (i.e. the 3rd, 6th, 9th, and so on) would be removed. The comb-cut feature is useful for reducing the number of frames in a movie while retaining the overall pattern.

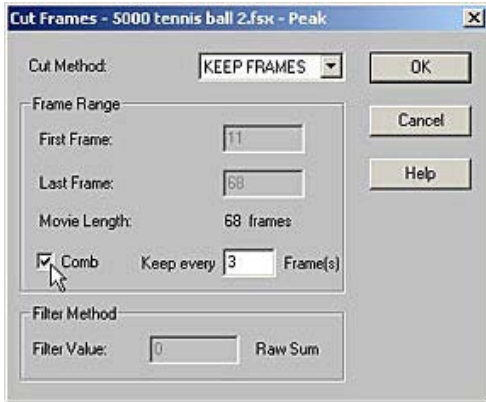
- b) When you are satisfied with your entries, click on **<OK>**. The total frame count, in the Movie Status Bar, will be updated to reflect the change. If a graph is open, it will also be updated.

ii. **'Keep Frames' method:**

- a) Select **'Keep Frames'** in the **'Cut Method'** section of the dialog box (shown at right).



- To keep a range of frames in the movie, enter the first ('First Frame') and last ('Last Frame') frame numbers that you want to keep. The 'Movie Length', in number of frames, is listed below these fields.

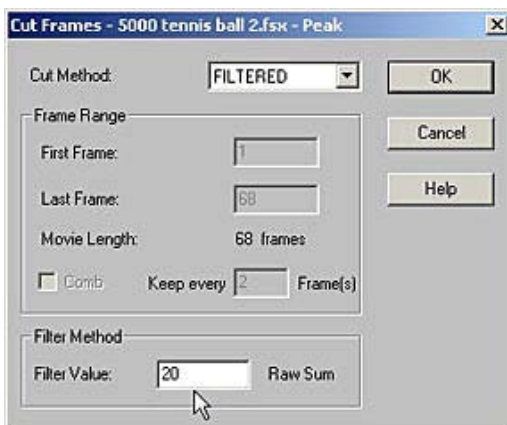
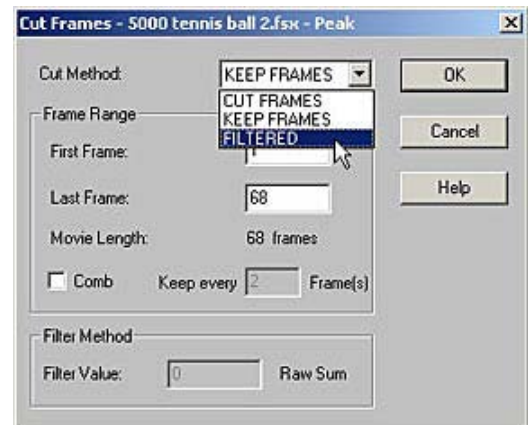


- To keep frames of the movie at specified intervals ("Comb-Keep"), place a check mark in the 'Comb' check box, and then enter a number in the "Keep Every ___ Frames" field. Each n^{th} frame of the movie will be retained. For example, if you entered a "3", every 3rd frame (i.e. the 3rd, 6th, 9th, and so on) would not be removed. The comb-keep feature is useful for reducing the number of frames in a movie while retaining the overall pattern.

- b) When you are satisfied with your entries, click on <OK>. The total frame count, in the Movie Status Bar, will be updated to reflect the change. If a graph is open, it will also be updated.

iii. 'Filtered' method:

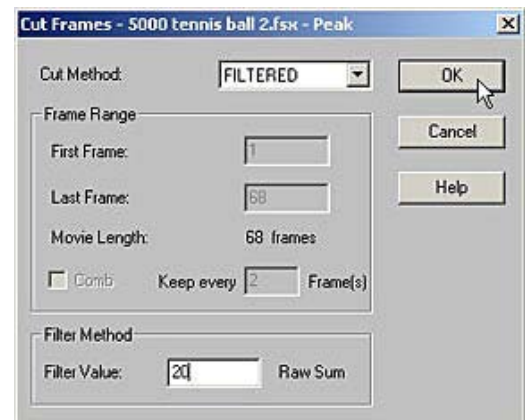
- Select 'Filtered' in the 'Cut Method' section of the dialog box (shown at right).
- In the 'Filter Value' field, at the bottom of the dialog box, enter the minimum force value (for an entire frame) that you want to keep in the movie (see image at right). All movie frames in which the total force is less than this value will be removed.



This force value is in the calibrated units of the recording, if a calibration was done; otherwise, they will be Raw units. A number must be typed into this field.

- When you are satisfied with the selected force value,

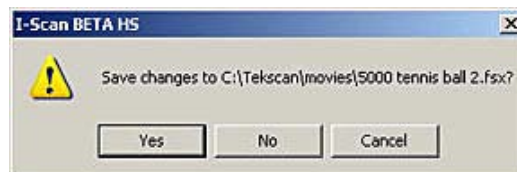
click on <OK> (see image at right). The total frame count in the Movie Status Bar will change to reflect the change. If a graph is open, it will also be updated.



3. Save the movie in order to save the new changes. Select **File>>Save Movie As**, and save it under a different file name, if you wish to save the original movie, without the changes (see image below).



Note: If you forget to save an edited recording and attempt to exit, the system will ask if you want to save the movie (because the movie has changed).

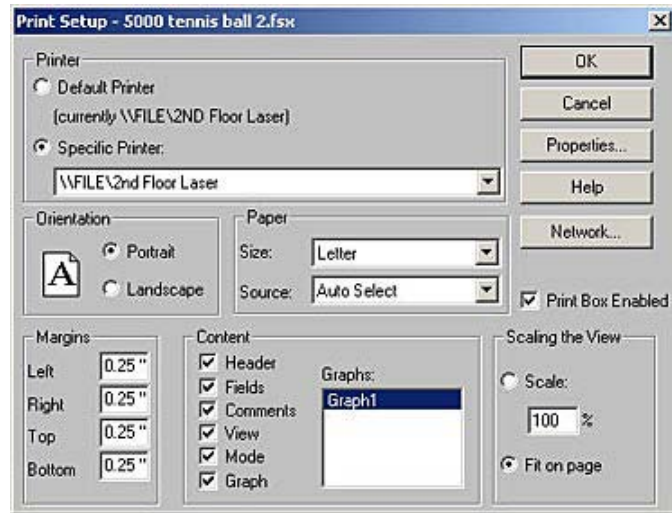


PRINTING

The Print function allows you to print the contents of the currently active Movie or Real-time window, including comments, header information, and graphs, to any of the hundreds of printers which MS Windows supports. The window will be printed exactly as it appears on the screen, and will include the options selected in Print Setup.

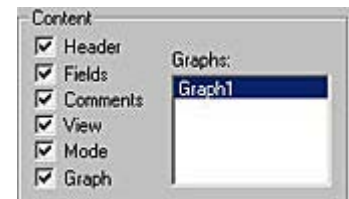
To print:

1. Select **Print Setup** from the **File** menu. Print Setup gives you a number of options on how your printed page will look. Most of the options are standard MS Windows printing options.



2. Select (click on) one or more of the following options, under '**Content**' at the bottom of the dialog box, to control exactly what will be printed:

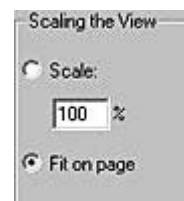
- **Fields:** Prints any fields that were entered under **Comments** in the **File** pull-down menu, as well as the standard patient fields (patient's first, middle, and last names, patient ID, gender, and date of birth). Refer to the 'Including Comments with a Recording' section for more information on entering and changing fields.
- **Comments:** Prints the comments that were entered under **Comments** in the **File** pull-down menu. In addition, if the movie was 'marked' with a procedure, the procedure name will be listed in the header information. Refer to the 'Including Comments with a Recording' section for more information on comments.
- **View:** Prints the contents of the active Movie or Real-time window. If this option is not selected, no sensor data will be printed.
- **Graph:** Prints any graphs that correspond to the active window. When this option is selected, all available graphs (i.e.: 'Graph 1', 'Graph 2') are listed to the right. You can click the name of the graph to select/deselect it for printing. If a graph doesn't print, or isn't visible in the **Print Preview**, check the **Print Setup** options.



***Hint:** If a graph doesn't print, or is not visible in the Print Preview, check all of the Print Setup options. If it still does not print, decrease the size of the printout (enter a percentage that is less than 100) under 'Scaling the view' in Print Setup*

- **Fit on Page or Scale:** These selections are mutually exclusive for a window. If Fit on Page is selected, the size of the image will be set to the largest printable area. If Scale is selected, you may enter the percentage of the actual sensor that is displayed.

Note: The software will automatically rotate an image so it will print as large as possible.

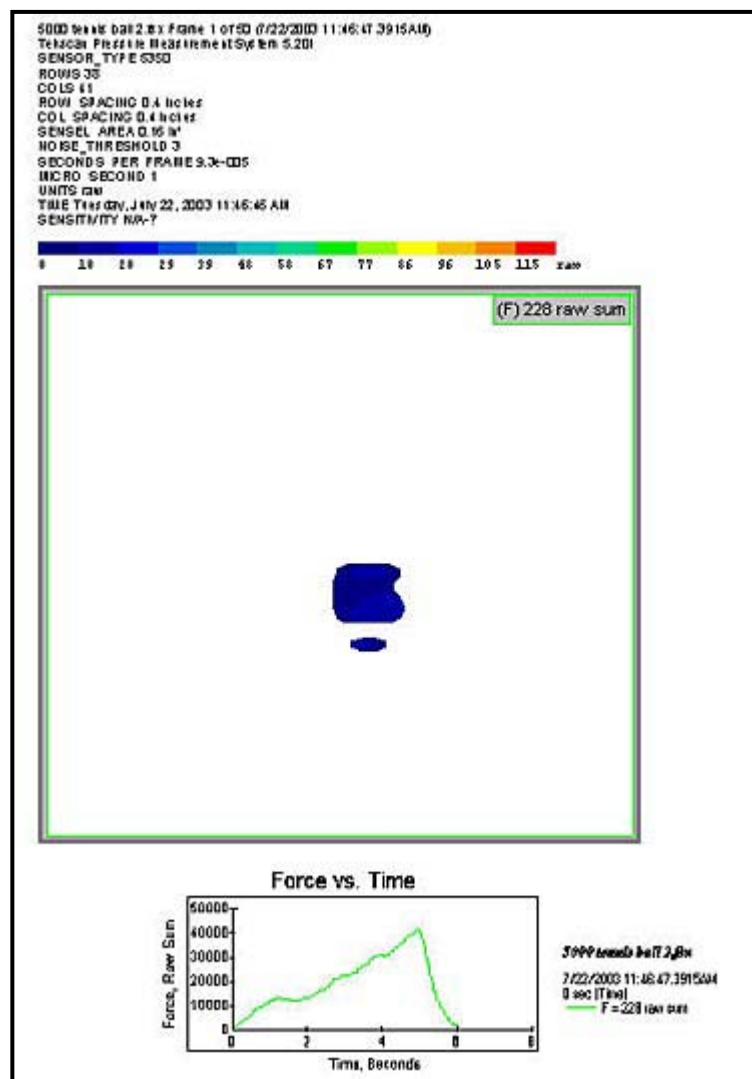


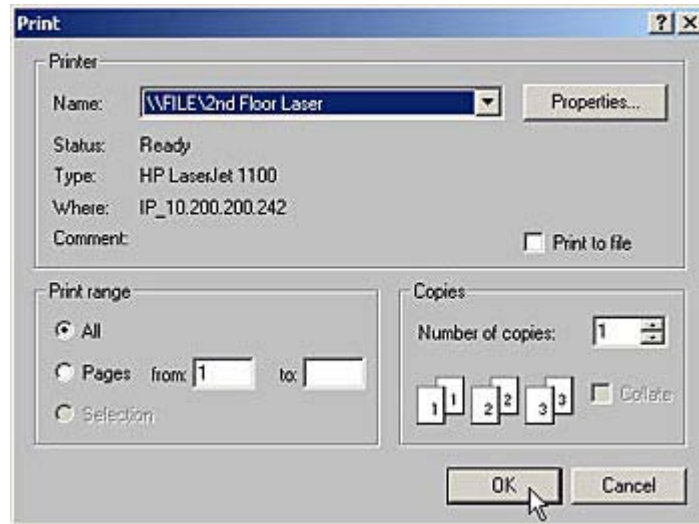
The image below shows a sample of the Print Preview:

- When you are satisfied with the Print Setup, click on <OK>. Select **Print Preview** from the **File** menu, to see how your print out will look. Alternately, you can click on the Print Preview icon from the toolbar (below). Make any necessary adjustments to the Print Setup.



- Select **Print** from the **File** menu, and click <OK>. If you receive an error message while printing, or printing does not begin in a timely manner, close any Real-time windows and/or stop recording data. If printing does not resume immediately, re-send the print job.





PRINTING GRAYSCALE

The ***I-Scan*** software will also allow you to print the color screen display as a grayscale image. When a grayscale (non-color) printer is installed, and is selected as the default printer (in the Print Setup or Print options in the File menu), the image will be printed in grayscale.

In color printouts, the color scale (legend) goes from blue (lowest pressures) to red (highest pressures). The ***I-Scan*** software does not simply print these colors in black and white, because red and blue will both print out dark and appear to be at the same pressure. When printing grayscale, the software automatically adjusts the Legend to divide the 13 possible pressure ranges into 13 shades of gray, going from light (low pressures) to dark (high pressures). The corresponding pressure areas on the printout of the display window will be adjusted to agree with the Legend. In this way, the pressure differences are much more easily discerned.

Note: This option is intended for grayscale printers, and may not work correctly if printing to a color printer, with the 'print grayscale' option selected.

HIGH SPEED I-SCAN

Most *I-Scan* applications use the [Evolution](#) Handle and system. However, *High Speed I-Scan* applications are available with 2 or 4 [VersaTek](#) handles with a 2- or 8-port VersaTek Hub. The sensors available for High Speed applications are the [9500](#) and [9550](#). While other *I-Scan* sensors can be used for High Speed applications, speeds will vary.

The following Table lists the components shipped with a High Speed I-Scan VersaTek System. Refer to the [VersaTek Component Identification](#) and [VersaTek Component Descriptions](#) sections for information on each of these components.

HIGH SPEED I-SCAN VERSATEK SYSTEM (IVHS) COMPONENTS

- (1) VersaTek 2-Port Hub
- (1) VersaTek Handles
- (1) Hub Power Supply (with US AC Cord)
- (1) USB Cable
- (1) I-Scan Software CD with 9500 or 9550 Sensor Map
- (1) 9500 or 9550 Sensor
- (1) System Carrying Case
- (1) Sensor Carrying Case
- (1) System Manual

SCAN SPEED

Scan Speed is defined as **Frequency**, or the number of frames that are recorded each second (frames/second). Scan speed is an important consideration when evaluating dynamic events. Faster scan speeds give greater visibility and insight into rapid dynamic events, and provide finer time resolution.

The following chart outlines the maximum scan speed possible with a variety of sensors for your VersaTek *High Speed I-Scan* system. Scan speeds can vary depending upon the sensor and number of handles used.

Note: This list is not a definitive list of sensors. This list provides Scan speeds for a portion of Tekscan Sensors.

<i>Sensor</i>	<i>Scan Speed with VersaTek High Speed I-Scan System (Hz.) – Valid for up to 4 Handles/Sensors</i>
4201	Up to 1,300
5027	Up to 730
5051	Up to 730
5076	Up to 730
5101	Up to 730
5210N	Up to 630
5250	Up to 730
5260	Up to 530
5270	Up to 630
5315	Up to 680
5400N	Up to 250
5501	Up to 4,400
5510	Up to 650
5511	Up to 810
5515	Up to 550

<i>Sensor</i>	<i>Scan Speed with VersaTek High Speed I-Scan System (Hz.) – Valid for up to 4 Handles/Sensors</i>
5526	Up to 530
5550	Up to 4,400
5555	Up to 530
5570	Up to 4,400
6220	Up to 650
6230	Up to 540
6300	Up to 470
6900	Up to 710
8100	Up to 520
8110	Up to 600
8150	Up to 480
9500	Up to 4,000
9550	Up to 20,000
9801	Up to 160
9830	Up to 490

Note: For comparison purposes, the maximum speed of an I-Scan sensor using Evolution Handles is 100 Hz.

For further information about setting your Scan speed or Frequency, refer to the [About Data Acquisition Parameters](#) section.

CALIBRATION FOR HIGH SPEED APPLICATIONS

There are several methods for Calibrating an *I-Scan* sensor. All of these methods are outlined in the [Calibration](#) section.

Particular attention should be paid to high speed load implications. Impact loads present a particular calibration challenge. It is very difficult to arrange a known calibration impact force with the same duration as a rapid event. If an object of known mass and incident velocity strikes the sensor with known dwell time and known exit speed, Newtonian physics can be used to calculate the peak force.

Impacts may affect the sensor as briefly as a few milliseconds. In calibration, applying a dead weight load to the sensor takes seconds to minutes, so that method does not mimic the elapsed time of the experiment. Some researchers place high-speed load cells in series with a Tekscan sensor to measure events of a short duration. When that is not convenient, consider placing the sensor on a plate at the bottom of the swing of a pendulum. If the mass, impact velocity, and rebound velocity of the pendulum is known, force can be estimated from Newtonian mechanics:

$$Force = Mass * Acceleration$$

Other researchers mount an accelerometer in the pendulum to obtain a direct reading of acceleration.

Impact loads (High Speed) may be calibrated after the recording with [Frame \(Post\) Calibration](#).

Note: For complete guidelines for Calibration, refer to the "Appendix: Equilibration & Calibration Practical Suggestions" located in your software Help File.

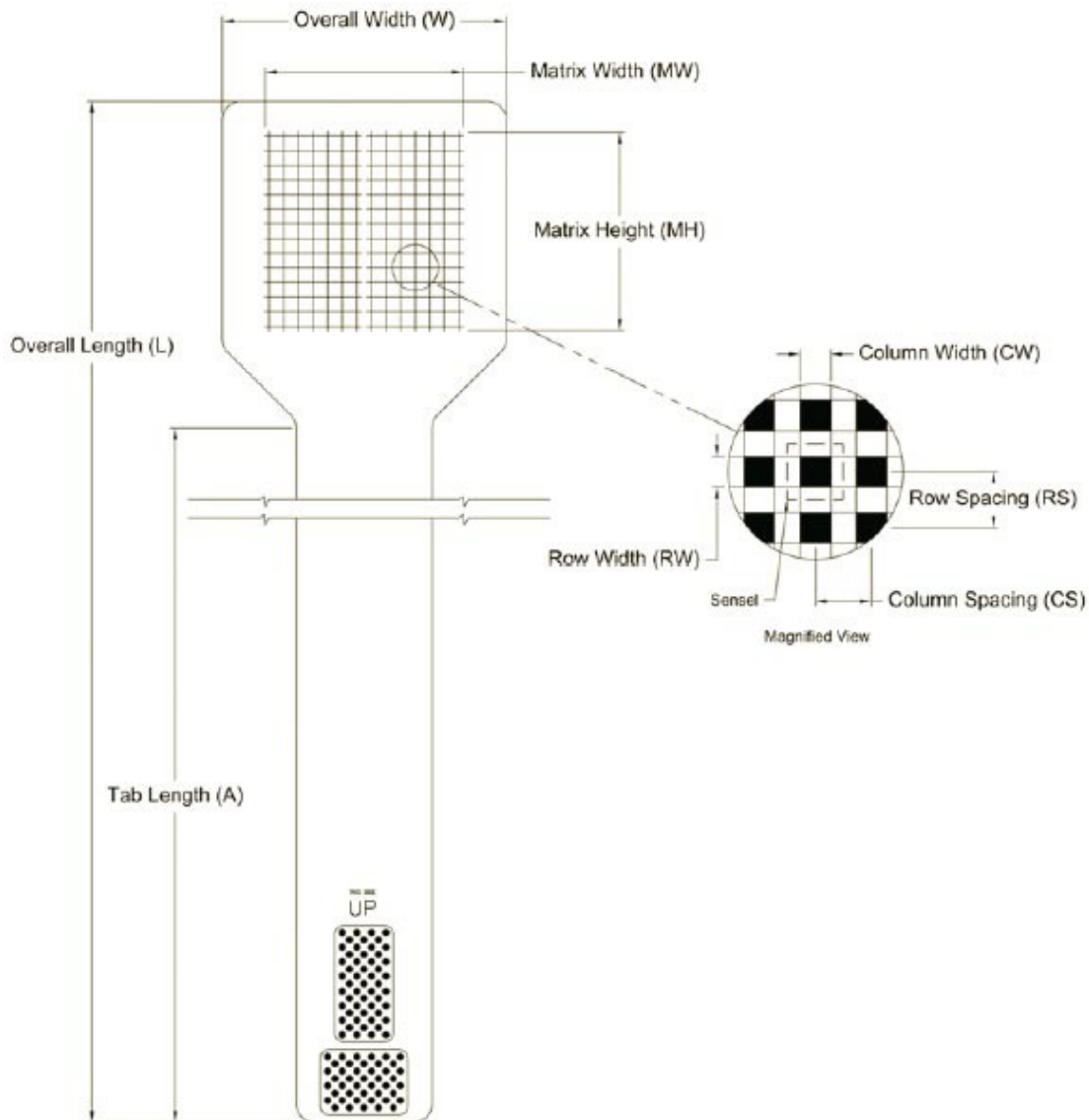
9500 SENSOR

Application Example:

- Knee bolsters, head impacts to “A and B” pillars, airbag and car bumper impacts

Special Features:

- Hi-speed sensor, with sampling speeds at up to 1.4 kHz
- Internal vents



Model	General Dimensions			Sensing Region Dimensions								Summary	
	Overall Length L	Overall Width W	Tab Length A	Matrix Width MW	Matrix Height MH	Columns			Rows			Total No. of Sensels	Sensel Spatial Resolution
	(in)	(in)	(in)	(in)	(in)	CW	CS	Qty.	RW	RS	Qty.		(sensel per sq-in)
US 9500	24.00	3.88	19.53	2.80	2.80	0.130	0.200	14	0.130	0.200	14	196	25.0
	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)		(mm)	(mm)			(sensel per sq-cm)
Metric 9500	609.6	98.7	496.0	71.1	71.1	3.3	5.1	14	3.3	5.1	14	196	3.9

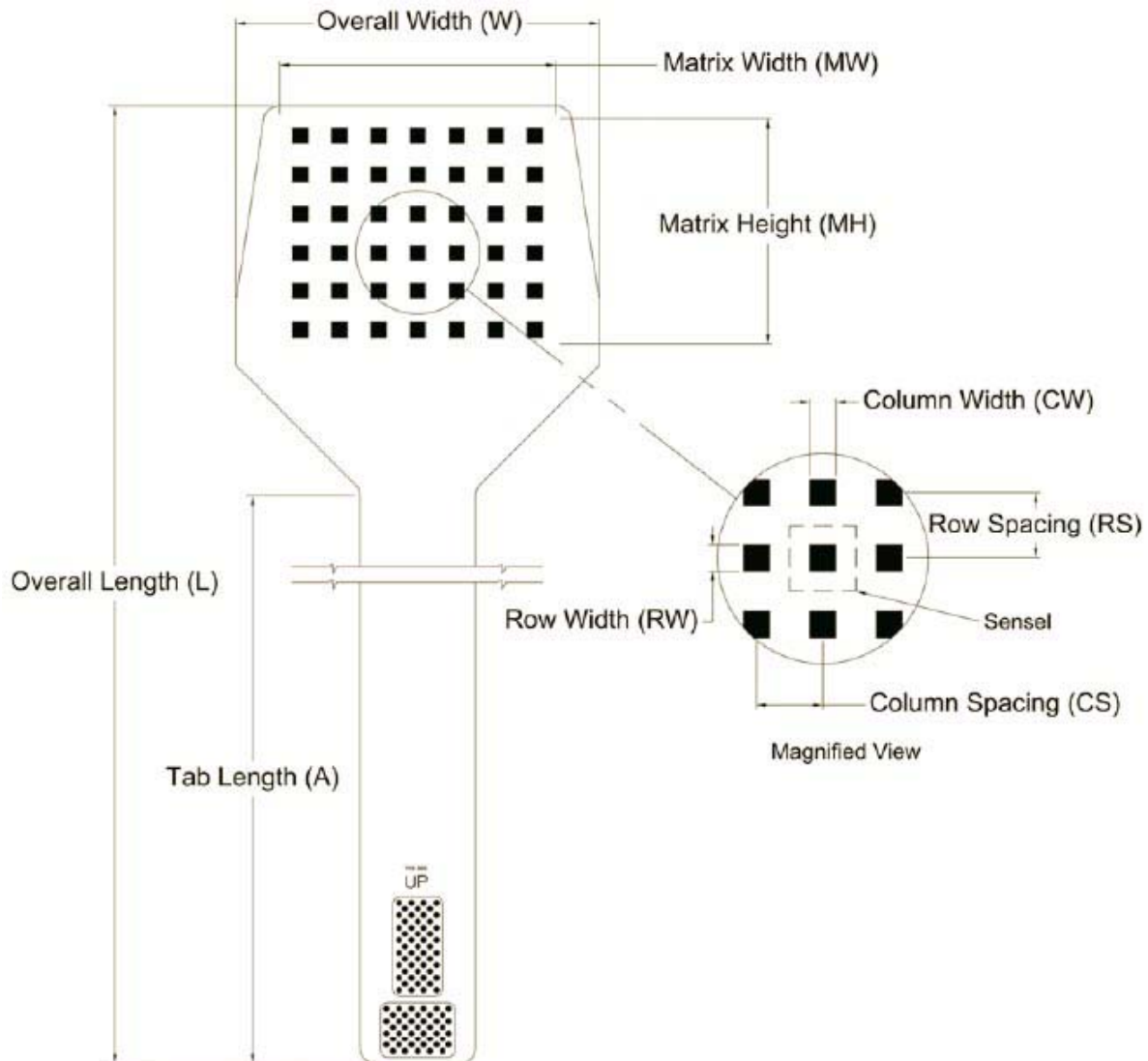
9550 SENSOR

Application Example:

- Head impacts to “A and B” pillars

Special Features:

- Hi-speed sensor, with sampling speeds at up to 10 kHz
- Internal vents

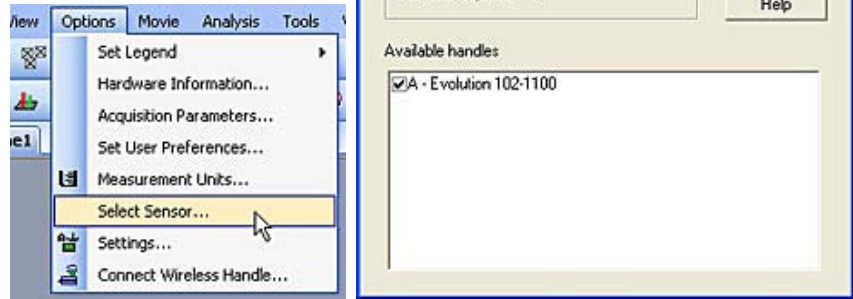


Model	General Dimensions			Sensing Region Dimensions								Summary	
	Overall Length L	Overall Width W	Tab Length A	Matrix Width MW	Matrix Height MH	Columns			Rows			Total No. of Sensels	Sensel Spatial Resolution
US 9550	(in)	(in)	(in)	(in)	(in)	(in)	(in)	Qty.	(in)	(in)	Qty.		(sensel per sq-in)
	18.03	5.86	11.73	4.41	3.78	0.280	0.630	7	0.280	0.630	6	42	2.5
Metric 9550	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)		(mm)	(mm)			(sensel per sq-cm)
	457.9	148.8	298.0	112.0	96.0	7.1	16.0	7	7.1	16.0	6	42	0.4

TROUBLESHOOTING

If your *I-Scan* system is not performing as it should, first check the status of the hardware. To do this, select **Hardware Information** from the **Options** pull-down menu. The '**Hardware Information**' dialog box (shown at right) is displayed.

The Hardware Information dialog box tells you which Receiver (*VersaTek*, *Evolution*, *USB*, or *Super-Receiver*) is connected between the handle and the computer, and the number of Handles that are connected. Make sure that the



correct Receiver(s), and the correct number of handles is detected by the system. If not, check all connections, from the Receiver to the sensor, as well as the power supply (if applicable).

The other information (Address, IRQ Line, and DMA Channel) describes the computer's internal setup, and is intended primarily for service personnel. If an Interface board or a Parallel adapter is connected to the system, the Address and IRQ Line should be found, but the DMA Channel should not be found (or used).

TROUBLESHOOTING TABLE

The following gives some guidelines for troubleshooting problems with your I-Scan @ system:

Symptoms	Probable Cause	Action
System boots up with no real-time display	Handle(s) are not connected to PC	Exit Software, connect handle(s) to PC and re-start software
	Incorrect map(s) and/or handle(s) selected in 'Select Sensor' dialog box.	Select the appropriate map(s) and handle(s) in Options>>Select Sensor.
	Software is unable to identify or locate hardware.	Contact Tekscan Hot Line for technical assistance.
Spurious contacts appear during recording or in the real-time display.	Broken or damaged sensor.	Replace the sensor.
	The lower limit in the Legend is too low.	Increase the lower limit on the Legend.
	EMI or RFI (noise) sources are nearby.	Remove the sources of EMI and/or RFI or relocate the <i>I-Scan</i> equipment. Check to see that clean power is coming into the PC. Raise the 'Noise Threshold' setting under Options>>Acquisition Parameters .
No pressure is indicated in the real-time window when force is applied to the sensor.	Insufficient contact pressure to activate the sensor.	Ensure that the applied load is in the sensor range. Decrease the Legend's lower limit, or the 'Noise Threshold' (Options>>Acquisition Parameters), if necessary.
'Sensor OK' indicator is on.	Alignment may not be correct, even though the 'Sensor OK' indicator is lit.	Remove the sensor tab from the handle and reinsert.
An entire row or column will not respond to applied pressure.	Sensor is misaligned.	Remove the sensor tab from the handle and reinsert.
	The terminal section of the sensor is dirty.	Clean the terminal sections of the sensor carefully with a cloth moistened with alcohol.
	The incorrect sensor type is being used.	Select the correct sensor type under Options>>Select Sensor .
	The sensor is damaged.	Replace the sensor.
	The sensor is defective.	Replace the sensor with a new one. Defective sensors will be replaced free of charge from Tekscan (if unused).

The Real-time Status Bar reads 'MISALIGNED!' instead of 'Sensor OK'.	Sensor is misaligned or was pulled out of the handle.	Remove and re-insert sensor.
	Sensor is inserted upside down in the handle.	Remove the sensor. Make certain that the This Side UP legend is facing the handle latch.
	The terminals on the sensor are dirty.	Clean the terminal sections of the sensor carefully with a cloth moistened with alcohol.
	The contact pins of the sensor are dirty.	Cut a clean piece of drawing Mylar® the same width as the handle contact area. Insert this strip in the handle and latch. Pull the paper out of the handle. Turn the Mylar® over and repeat to clean the other set of contacts.
	Two sensors are stuck together.	Remove sensors and peel apart. Re-insert one sensor.
	Sensor is defective.	Replace sensor with a new one. Any defective sensors will be replaced by Tekscan (if unused).
	Cable is unplugged or not properly plugged in.	Turn power off and remove handle unit cable from the PC. Check to see if any of the cable connector pins are bent or broken. Carefully reconnect the handle unit cable. Turn the power back on.
Legend appears incorrect when using calibrated sensors.	Incorrect calibration file is being used.	Make certain that the correct calibration file was loaded.
	Calibration was performed incorrectly.	Check the saturation pressure in the 'Calibration' dialog box (Tools>>Calibration). It should be a reasonable value. If not, perform another calibration.
Computer runs slow, or hard drive is very active (seen by the hard drive LED), even when you are not doing anything.	The system does not have adequate resources (memory) for all of the currently open applications.	Close all applications other than I-Scan , if possible.
		Increase the amount of memory in your computer.
Software gives message, 'Can not start recording'.	The software does not have enough available memory to record a movie of the length selected.	Reduce the number of 'Frames to Record' in Options>>Acquisition Parameters .
		Increase the amount of memory in your computer.
An error is received while printing, or printing doesn't begin in a timely manner.	Printer will not support simultaneous data acquisition and printing.	Close any real-time windows and/or stop recording data. If printing doesn't resume immediately, re-send the print job.
Contents or Search button under Help menu does not open the Help File.	Known Issue.	Close the Tekscan Application. Right-click on the Tekscan Shortcut icon (application icon in the start menu or on the desktop), and select Properties. Navigate to the "Shortcut" tab, and click on the Find Target button. The Tekscan executable is found and highlighted. Double click this executable file to open the application. You should now be able to access the help file via Help -> Contents or Help -> Search. <i>Note: This should re-establish the link between the Help File and the Application, and you should only have to perform this procedure once. If the link becomes broken once again, and you cannot access the Help File, perform the previous procedure once again.</i>

Tekscan, Inc. will provide technical assistance for any difficulties you may experience using your **I-Scan** system. Write, call or fax us with any concerns or questions. Our knowledgeable support staff will be happy to assist you.

Tekscan Technical Support can be reached between 8:30 and 7:00pm EST by calling:

(617)-464-4500 or (800)-248-3669 in the US and Canada

Or at anytime via fax or email:

Fax: (617)-464-4266

E-mail: support@tekscan.com

Optional Add-Ons

To your Tekscan System

*The following are optional add-ons to your current Tekscan System.
For more detailed information, please reference your Help File located under
the Help menu in your software or contact marketing@tekscan.com.*

The Tek WiFi Wireless System Add-On

Wireless Connection between Sensor Handle and PC

Tek WiFi, Tekscan's Wireless capability add-on, provides real time communications over a (802.11B) wireless link from your sensor/handle to your PC/Laptop, enabling display and recording of pressure data on the local PC/Laptop from a remote handle(s) in real-time. The wireless capability includes a Windows CE Handheld PC (PDA) with a USB Port and Tek WiFi link software. Tek WiFi Link works with one or two Tekscan USB handles!



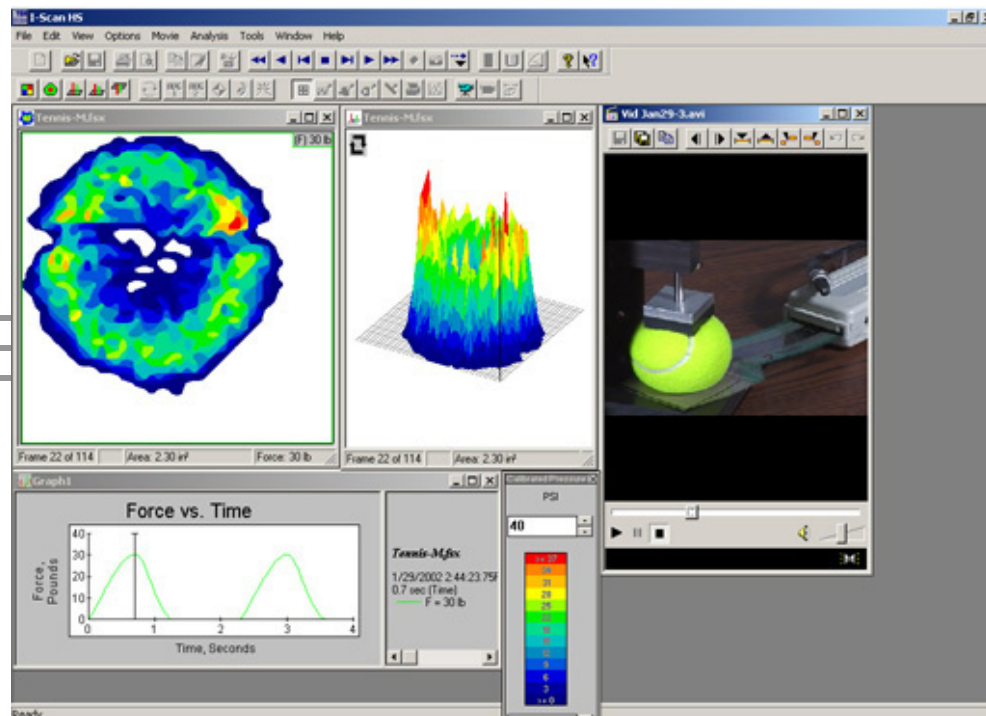
Benefits:

- Increased flexibility in capturing data
- Sampling rates of up to 10 Hz (up to 22,880 sensels/sec) or up to 50 Hz for MatScan
- Includes PDA and Tek WiFi link software
- Lightweight and portable
- Access Point or Computer to Computer (Ad-Hoc) connectivity
- Industry standard protocols

Video Synchronization™ Add-On

Video Synchronization Software

The *Video Synchronization*™ software is available as an optional add-on to Tekscan's pressure and force measurement system. *Video Synch* software allows you to record video sequences and synchronize them with your pressure data. The video, the pressure profiles, and/or the graphs (pressure, force, etc.) can then be simultaneously viewed in the Tekscan software, which truly gives you the whole picture!



Benefits:

- See the position/event (i.e machine state) alongside the pressure pattern
- Improved understanding when video is combined with pressure data
- Turnkey: requires no external video capture programs
- Enhanced data analysis
- Enables pressure and video data collection
- Saves time

Equilibration/Calibration Devices

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.

Model	Active Area in x in (mm x mm)	Maximum Pressure PSI (kPa)	Type	Sensor Model & Pressure Ranges*		
PB5A	4.5 x 12.5 (114 x 318)	5 (34.5)	Pneumatic (self contained w/weights)	4201 @ 5 PSI 6300 @ 10 PSI	5051 @ 7 PSI 9801 @ 5 PSI	6077 @ 5 PSI
PB5B	17.5 x 19.5 (445 x 495)	5 (34.5)	Pneumatic (compressed air)	5315 @ 5 PSI Plus all sensors applicable to PB5A	5350 @ 5 PSI	
PB5D	0.85 x 32.6 (21.6 x 828)	5 (34.5)	Pneumatic (self contained w/weights)	4300 @ 2 PSI 9901 @ 10 PSI	4305 @ 2 PSI 9920 @ 10 PSI	4308 @ 2 PSI
PB15A	17.5 x 19.5 (445 x 495)	15 (103)	Pneumatic (compressed air)	5210N @ 12 PSI 9830 @ 10 PSI Plus all sensors applicable to PB15C	5250 @ 10 PSI 9910 @ 10 PSI	5510 @ 4 & 10 PSI
PB15C	4.5 x 12.25 (114 x 311)	15 (103)	Pneumatic (compressed air)	5051 @ 7 & 20 PSI	5101 @ 6 & 10 PSI	6300 @ 10 PSI
PB100B	17.5 x 19.5 (445 x 495)	100 (689)	Pneumatic (compressed air)	3150 @ 125 PSI 6010N @ 29 PSI 7100 @ 300 PSI Plus all sensors applicable to PB100C & PB100D- Except 8050	5315 @ 30 PSI 6020N @ 29 PSI	5515 @ 500 PSI 6030N @ 29 PSI
PB100C	13.0 x 19.5 (330 x 495)	100 (689)	Pneumatic (compressed air)	3200 @ 300 PSI 5210N @ 28 & 189 PSI 5555 @ 500 PSI Plus all sensors applicable to PB100D	4255N @ 29 PSI 5250 @ 25 PSI 8000 @ 300 PSI	5150N @ 189 PSI 5511 @ 500 PSI 8050 @ 300 PSI
PB100D	4.5 x 12.25 (114 x 311)	100 (689)	Pneumatic (compressed air)	3000 @ 75 & 125 PSI 5027 @ 50 & 500 PSI 5076 @ 50 thru 500 PSI 5550 @ 40 & 200 PSI 6300 @ 50 & 100 PSI 9801 @ 35 PSI	3001 @ 125 PSI 5040 @ 50 & 500 PSI 5101 @ 50 thru 500 PSI 5570 @ 100 & 500 PSI 7501 @ 300 PSI 9855N @ 218 PSI	5026 @ 500 PSI 5051 @ 50 thru 500 PSI 5501 @ 50 PSI 5800N @ 36 PSI 9550 @ 500 PSI 9856N @ 218 PSI
PB500A	13" circumference x 3.5" length, (329 x 89)	500 (3445)	Pneumatic (compressed air)	4000 @ 1500 PSI 5027 @ 500 PSI 5100 @ 500 PSI 6900 @ 1100 PSI	4201 @ 2000 PSI 5051 @ 500 thru 2500 PSI 5511 @ 500 thru 2500 PSI 9500 @ 1200 & 2000 PSI	5026 @ 500 PSI 5076 @ 500 PSI 6300 @ 1000 PSI 9550 @ 500 & 2000 PSI
VB5A	22.5 x 23 (572 x 584)	5 (34.5)	Vacuum (self contained)	5330 @ 5 PSI		
VB5B	37.5 x 26 (953 x 660)	5 (34.5)	Vacuum (self contained)	5400N @ 4 PSI		
VB5C	18 x 26 (457 x 660)	5 (34.5)	Vacuum (self contained)	5360 @ 7 PSI		

Notes:

Pneumatic (compressed air) - 100 PSI (698kPa) minimum

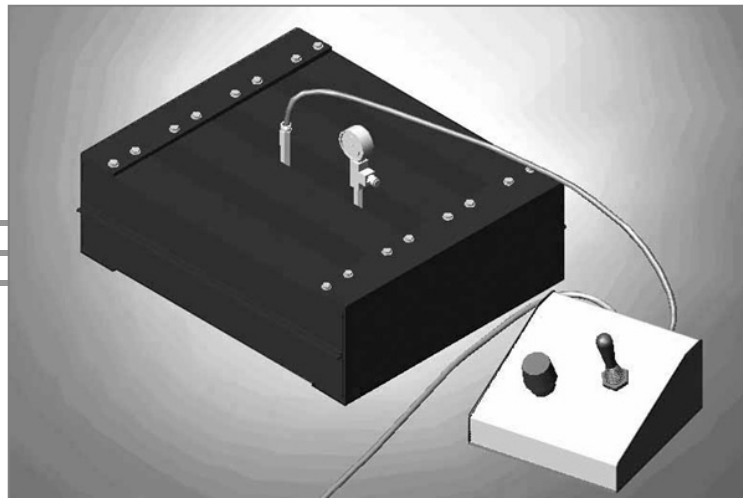
Vacuum (self contained) - 90-240 VAC 50/60 Hz

*(1 PSI = 6.895 KPa = 0.06895 Bar = 0.070 kgf/cm² = 703 kgf/m²)

PB100B Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 17.5" x 19.5"
(445 mm x 495 mm)
Width: 19.0" (483 mm)
Length: 21.8" (554 mm)
Height: 13.0" (330 mm)
- Including Gauge and Fittings
Weight: 70 lbs (31.8 kg)
Frame Material: Aluminum
Bladder Material: Urethane
Gauge Accuracy: 1% F.S.
Pressure Range: 0-100 PSI (0-689 kPa)

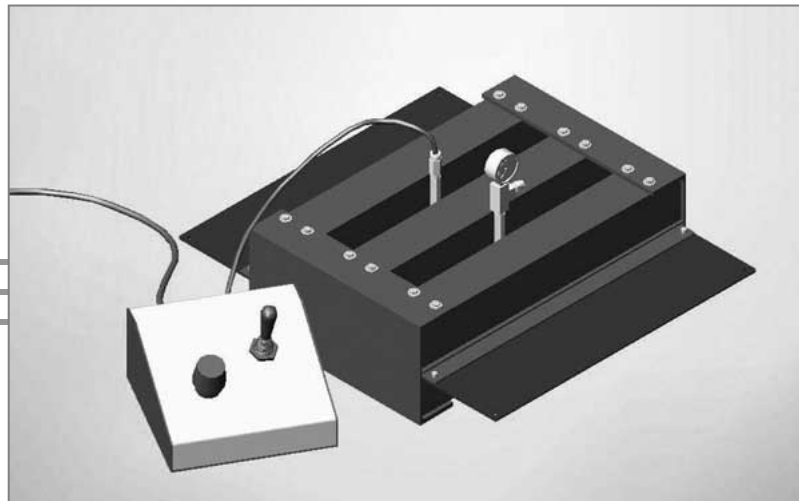
Control Box Specs:

Width: 10.0" (254 mm)
Length: 10.0" (254 mm)
Height: 7.5" (191 mm) - To Tip of Valve Knob
Weight: 7 lbs (3.2 kg)
Material: Aluminum
Air Required: 100-125 PSI (689-861 kPa)
(Supplied by Customer, Clean and Dry, Non Volatile or Flammable)
Connection: 3/8" Flex Tube and 1/4" Industrial Design Disconnect Plug
Features: 3-Way Toggle Valve for Vacuum, Adjustable Regulator

PB100C Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

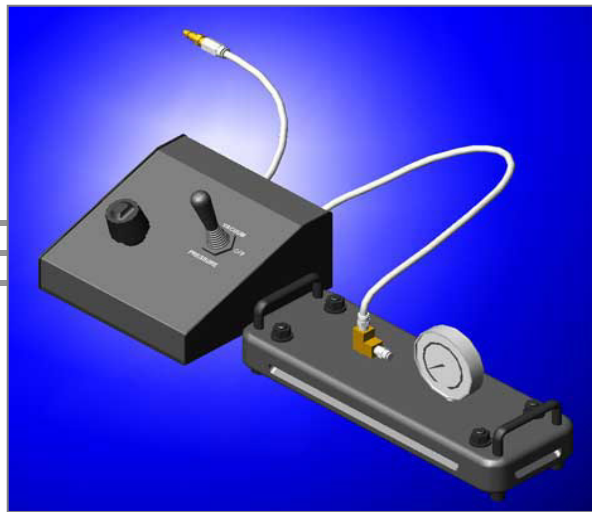
Active Area: 13.0" x 19.5"
(330 mm x 495 mm)
Width: 16.0" (406 mm)
- Not Including Handle Shelves
Length: 21.8" (554 mm)
Height: 13.0" (330 mm)
- Including Gauge and Fittings
Weight: 52 lbs (23.6 kg)
Frame Material: Aluminum
Bladder Material: Urethane
Gauge Accuracy: 1% F.S.
Pressure Range: 0-100 PSI (0-689 kPa)

Control Box Specs:

Width: 10.0" (254 mm)
Length: 10.0" (254 mm)
Height: 7.5" (191 mm)
-To Tip of Valve Knob
Weight: 7 lbs (3.2 kg)
Material: Aluminum
Air Required: 100-125 PSI (689-861 kPa)
(Supplied by Customer, Clean and Dry, Non Volatile or Flammable)
Connection: 3/8" Flex Tube and 1/4" Industrial Design Disconnect Plug
Features: 3-Way Toggle Valve for Vacuum, Adjustable Regulator

PB100D Add-On Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 4.5" x 12.25"
(114 mm x 311 mm)
Width: 6.0" (152 mm)
Length: 17.75" (451 mm)
Height: 7.2" (152 mm)
- Including Gauge and Fittings
Weight: 23 lbs (10.5 kg)
Frame Material: Aluminum
Bladder Material: Urethane
Gauge Accuracy: 1% F.S.
Pressure Range: 0-100 PSI (0-689 kPa)

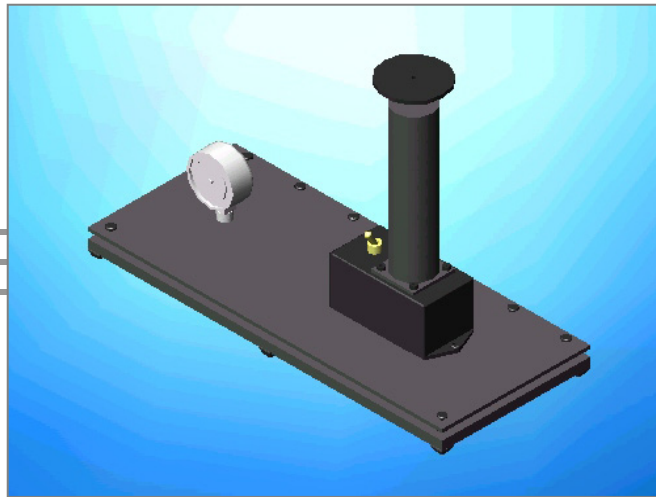
Control Box Specs:

Width: 10.0" (254 mm)
Length: 10.0" (254 mm)
Height: 7.5" (191 mm)
-To Tip of Valve Knob
Weight: 7 lbs (3.2 kg)
Material: Aluminum
Air Required: 100-125 PSI (689-861 kPa Max)
(Supplied by Customer, Clean and Dry, Non Volatile or Flammable)
Connection: 3/8" Flex Tube and 1/4" Industrial Design Disconnect Plug
Features: 3-Way Toggle Valve for Vacuum, Adjustable Regulator

PB5A Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 4.5" x 12.5"

(114 mm x 318 mm)

Width: 6.25" (159 mm)

Length: 15.0" (381 mm)

Height: 14.5" (368 mm)

- Extended Load Platform

Weight: 8.5 lbs (3.9 kg)

Frame Material: Aluminum

Bladder Material: Mylar

Gauge Accuracy: 3-2-3% F.S.

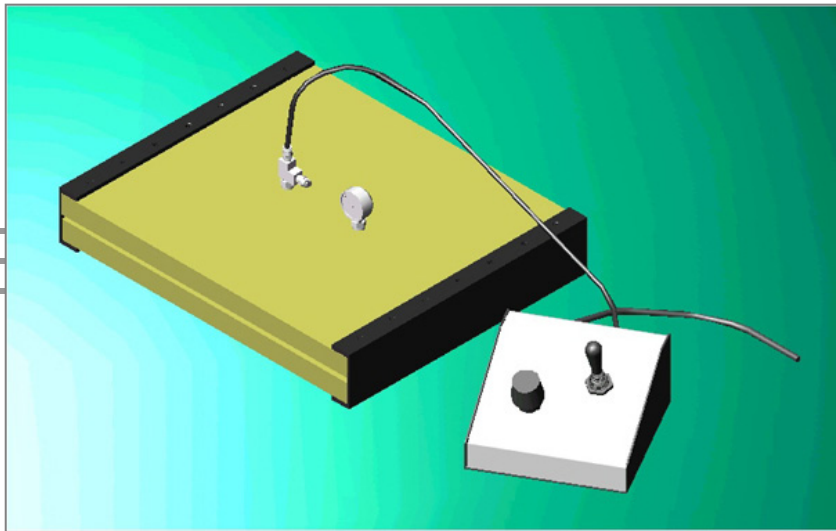
Pressure Range: 0-5 PSI (0-34.5 kPa)

Weight Set: 2100 g - 13 Piece Set Included

PB15A Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 17.5" x 19.5"
(445 mm x 495 mm)
Width: 20.0" (508 mm)
Length: 24.4" (620 mm)
Height: 10.8" (274 mm)
- Including Gauge and Fittings
Weight: 40 lbs (18.2 kg)
Frame Material: Wood and Aluminum
Bladder Material: Urethane
Gauge Accuracy: 3-2-3% F.S.
Pressure Range: 0-15 PSI (0-103 kPa)

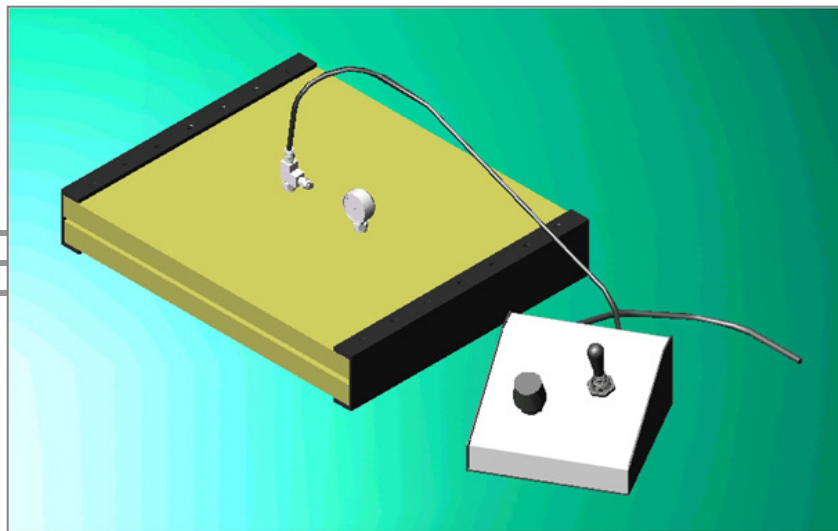
Control Box Specs:

Width: 10.0" (254 mm)
Length: 10.0" (254 mm)
Height: 7.5" (191 mm) -To Tip of Valve Knob
Weight: 7lbs (3.2 kg)
Material: Aluminum
Air Required: 15-125 PSI (103-861 kPa Max)
Supplied by Customer, Clean and Dry, Non
Volatile or Flammable
Connection: 3/8" Flex Tube and 1/4" Industrial
Design Disconnect Plug
Features: 3-Way Toggle Valve for Vacuum,
Adjustable Regulator

PB5B Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 17.5" x 19.5"
(445 mm x 495 mm)
Width: 20.0" (508 mm)
Length: 24.4" (620 mm)
Height: 10.8" (274 mm)
- Including Gauge and Fittings
Weight: 40 lbs (18.2 kg)
Frame Material: Wood and Aluminum
Bladder Material: Urethane
Gauge Accuracy: 3-2-3% F.S.
Pressure Range: 0-5 PSI (0-103 kPa)

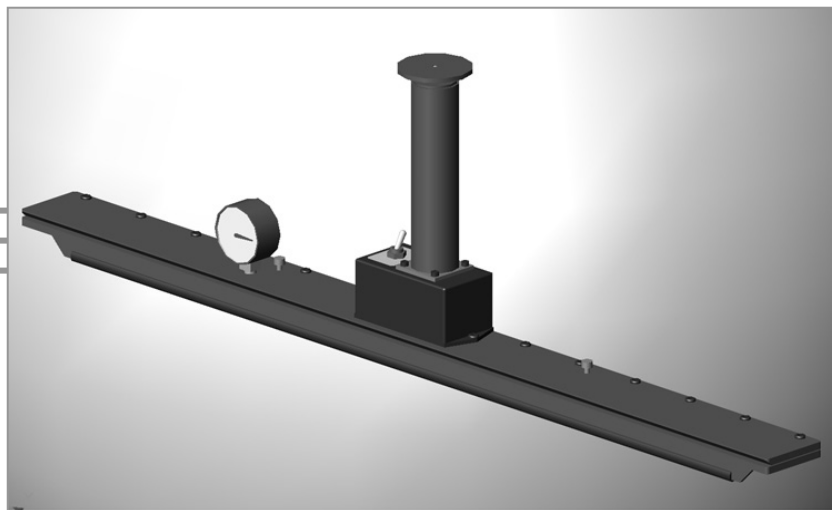
Control Box Specs:

Width: 10.0" (254 mm)
Length: 10.0" (254 mm)
Height: 7.5" (191 mm) -To Tip of Valve Knob
Weight: 7lbs (3.2 kg)
Material: Aluminum
Air Required: 5-125 PSI (34.5-861 kPa Max)
Supplied by Customer, Clean and Dry, Non
Volatile or Flammable
Connection: 3/8" Flex Tube and 1/4" Industrial
Design Disconnect Plug
Features: 3-Way Toggle Valve for Vacuum,
Adjustable Regulator

PB5D Add-On

Equilibration/Calibration Device

Tekscan offers a full line of equilibration/calibration devices to use with your pressure measurement system. These devices apply a uniform pressure to the active area of the sensor to allow the Tekscan software to compensate for variation in the output of individual sensing elements. Tekscan software determines and applies a gain (scale factor) to each sensel so its Digital Output (D.O.) is equal to the average D.O. of all loaded sensels. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing or repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor to confirm that the sensor has an acceptable uniform output, is operating correctly, and has not lost continuity within rows or columns.



Specifications:

Active Area: 0.85" x 32.6"

(21.6 mm x 828 mm)

Width: 2.5" (64 mm)

Length: 34.0" (864 mm)

Height: 15.0" (381 mm)

- Height with Extended Load Platform

Weight: 6.0 lbs (2.7 kg)

Frame Material: Aluminum

Bladder Material: Mylar

Gauge Accuracy: 3-2-3% F.S.

Pressure Range: 0-5 PSI (0-34.5 kPa)

Weight Set: 2100 g Combined Total Weight

External Trigger Synch Add-On Triggering Device

The TS-100 box is used to generate an external trigger to be transmitted to the Tekscan software. Upon activation, the Tekscan Trigger-synch sends a signal that is received by Tekscan software to trigger recording. This can be used to synchronize data captured by external devices (such as force plates, digital and analog video, EMG and similar devices) with the timestamp created on each frame of Tekscan data.



Usage:

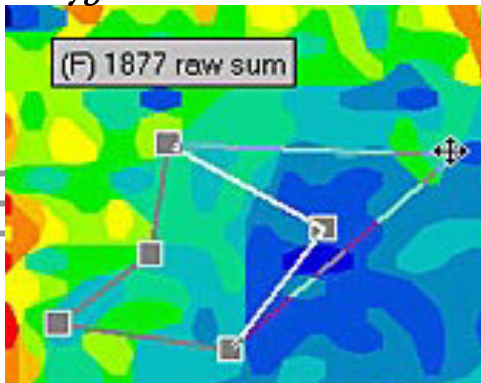
- The trigger switch can be used by hand to trigger a recording.
- External devices (such as an electric eye or light beam) can be connected to generate input that will trigger recording in the software.
- External devices can be connected to receive output from the trigger box for other data gathering purposes. The TS-100 generates a single pulse to connected devices when the triggering event occurs.

Additional Advanced Analysis Add-Ons

Polygon/Blob/Tracking Box

This optional software package includes three additional analysis options: **Add Polygon**, **Add Blob**, and the **Tracking Box**.

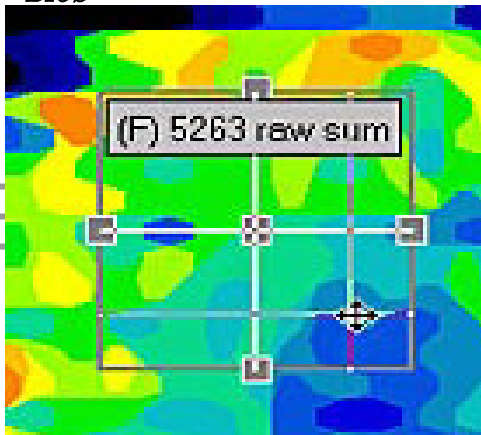
Polygon



Polygon:

- Add Polygon feature allows you to add a custom-shaped box to a real-time or movie window.

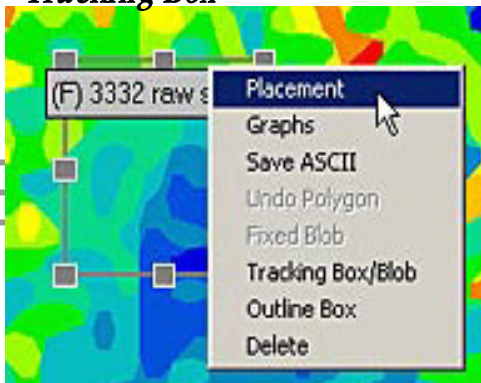
Blob



Blob:

- The Add Blob feature enables you to study a group of loaded sensels separately from the rest of the active window.
- While a Polygon will allow you to study an irregularly shaped area, the Blob will allow you to study only the

Tracking Box



Tracking Box:

- The Tracking Box feature allows you to add a box to the Movie or Real-time window that will “track,” or follow, the loaded sensels in the window.

GLOSSARY

#

- 2-D Contours View:** The 2-D Contours view displays the recording as a two-dimensional, contoured image. The sharp corners of the sensor output are smoothed, making the pressure boundaries easier to discern.
- 2-D View:** The 2-D view displays the recording in two-dimensional form with the color of each sensel location representing the pressure sensed at that point. This display looks the closest to the actual raw output of the sensor, and individual sensels can be identified. 2-D is the default view.
- 3-D Contours View:** The 3-D Contour view displays the recording as a three-dimensional image, in which higher pressures are shown as solid peaks. This display can often provide a view of complex data that is easier to comprehend than the 3-D Wireframe image.
- 3-D Reverse:** The 3-D Reverse view displays the recording as a three-dimensional image, in which higher pressures are shown as inverse peaks. The 3-D Reverse image is simply an inverted, or reversed, 3-D Contour image. 3-D Reverse is especially useful for viewing the exact footprint of the applied force.
- 3-D Wireframe View:** The 3-D Wireframe view displays the recording as a three-dimensional image, in which higher pressures are shown as peaks. The relative peaks provide visual insight into the gradients within each color band. 'Wireframe' means that the image is not a solid image, and the image is represented by a series of interconnected lines. This display can often provide a better view of complex 3-D data than the 3-D Contour and Reverse views.

A

- About I-Scan:** displays the "About " dialog box, which gives basic information about the Tekscan software.
- Acquisition Parameters:** This Option menu item pops up the 'Data Acquisition Parameters' dialog box, which allows you to select the number of frames, duration, frequency and period of your recordings. This dialog box also enables you to add a 'Noise Threshold', and includes your 'Triggering' options. Before you take a recording, you should set the recording parameters to values appropriate for your patient session and your computer system capacity (memory). When setting the movie recording (data acquisition) parameters, you will want to set the frequency and duration based upon the period of time of interest and the rate at which pressures change in your tests.
- Add Blob *:** The Add Blob feature enables you to study a group of loaded sensels separately from the rest of the active window. While a Polygon will allow you to study an irregularly shaped area, the Blob will allow you to study only the loaded sensels in a particular area.
- Add Box:** Add Box allows you to place a box in the active Movie or Real-time window. The data inside that box can then be displayed, graphed (movies only), copied, printed and analyzed separately from the rest of the window's pressure data. When 'Add Box' is selected, a box symbol is added to the cursor, and clicking the cursor at the desired point in a Movie or Real-time window may then place a box. You can also add a box by pressing the <CTRL> key and clicking the cursor in an open window, or by clicking the right mouse button with the cursor over an 'active' window, and selecting 'Add Box'.
- Add Line:** Add Line allows you to place a line in the active Movie or Real-time window, which can be used to measure the distance or pressure difference between two points on the sensor. When 'Add Line' is selected, a line symbol is added to the cursor, and clicking the cursor at the desired point in a Movie or Real-time window may then place a line. You can also add a line by pressing the <SHIFT> key and clicking the cursor in an open window, or by clicking the right mouse button with the cursor over an active window, and selecting 'Add Line'.

Add Polygon *: [Optional Feature] Add Polygons will add points to create irregular shapes within your movie. In order to create your shape, select the Add Polygons command from the menu or Toolbar, and left-click in your movie to add the first point (also called a 'node'). Click on a second location to create the second point in your shape. A line will be drawn from point to point. Continue doing this to add as many sides to the shape as you like. The final step is to close the shape by clicking on the starting point/node. The shape automatically closes, and measurements are calculated for the new polygon.

Adjust Sensitivity: Adjust Sensitivity will adjust the sensor sensitivity to read higher or lower pressures, depending. Lower sensitivity will mean more force/pressure must be applied to the sensor to give a reading. Higher sensitivity means that less force/pressure must be applied to the sensor to give a reading. This fine-tunes the sensor, as it can become less sensitive over the life span of the sensor. It is always wise to Equilibrate and/or Calibrate your sensor first, however, and it is suggested to leave this setting on "default".

Analysis Menu: The Analysis menu option provides all of the tools necessary to analyze the pressure data displayed in the Movie and Real-time windows.

API 2: Tekscan software with the API2 option enabled will be able to export its real-time data/image, saved snapshot image file names and other pressure related data to the client software written by the User.

Append: Append allows you to record additional frames of data at the end of an existing movie. This option is only available when a Movie window (recording) is open, and will be unavailable (grayed out) if a Real-time window cannot be opened. When you select Append, the Select Sensor dialog box will be opened, and you must select which handle you would like to use to record the appended data. Once you select a handle, and click <OK>, a Real-time window is opened (the title bar will say "Append to ...").

Arrange Icons: Arrange Icons arranges the icons for minimized windows at the bottom of the Main Window. The names of the currently open Real-time, Movie, and Graph windows are listed beneath the Arrange Icons option. A check mark is displayed next to the name of the active window.

ASR (Auto-Sequential Recording): This dialog box allows you to set your ASR (Automatic Sequential Recording) options. When ASR is in effect, the software automatically saves and closes the current movie, opens a new Real-time window, and begins a new recording. This will continue until the movie count is reached. You can stop the ASR function at any point by selecting the Stop button. If triggering is in effect, each movie will be triggered separately.

Averaging: Displays the image with each cell's pressure value modified to reflect the value of its neighbors. It does include neighboring cells that have zero load. This results in a smoother image, with an increase in total loading area since certain cells with a previous zero value now have a value. When Averaging is in effect, it has a check mark beside it in the pull-down menu, and its Toolbar icon is depressed.

B

Background White: Background White changes the background areas (where no pressure is sensed) of the active Movie, Real-time, or Graph window from white to black. Changing the background color to black is useful when viewing lighter colors in the display.

C

Calibrated Legend: A calibrated legend can be opened if the corresponding Movie or Real-time window has been calibrated; otherwise, only a raw legend may be opened. A Calibrated Legend will report the units in a specified measurement, such as PSI. If it is a Raw legend, no pressure units will be associated with each range, and the default upper limit will be 255. All calibrated movies that are open simultaneously will be controlled by one calibrated legend. Likewise, all uncalibrated (raw) movies are controlled by one raw legend. <See also "Pressure Legend" and "Raw Legend">

Calibration (Tools Menu): Calibration opens the "Calibration" dialog box, which enables you to calibrate the sensor, uncalibrate the sensor, and load and save calibration files. You can also access the calibration options by clicking on the "Calibration" tab in the "Settings" dialog box. Once a calibration is completed, it can be saved as a file by clicking on the Save button. This file can later be loaded into another window by clicking on the Load button.

Cascade: Cascade arranges multiple opened windows in an overlapped fashion. Use this feature if your application area is becoming cluttered with too many windows.

Close: Close closes the currently active Movie or Real-time window.

Close All: Close All closes all open Movie, Real-time, and Graph windows (you will be prompted to save them if they have been changed).

COF (Center of Force): The COF shows the center of all of the forces on the sensor, as displayed in the Real-time or Movie window. This feature is used to show how the pressures are balanced on the sensor. The center point is represented on the screen by the same red and white icon that represents COF on the Toolbar. As an optional feature, you can also view the COF for each separate box in the Movie or Real-time window. In this case, each COF marker will be the same color as its corresponding box. You can enable or disable these COF features in the 'Properties' dialog box (Analysis pull-down menu).

COF (Center of Force) Trajectory: This displays the movement of the Center of Force for the duration of a recorded movie. The movement of the COF can be tracked by playing a movie one frame at a time, with COF Trajectory selected. The trajectory is represented on the screen by a red and white line that trails the COF marker.

Comments: Comments (Ctrl+M) opens a dialog box that allows you to view, add, and edit the comments associated with the active Movie or Real-time window. The 'Comments' dialog box also displays the information from the Patient Record (i.e. Name, Patient ID, Gender, Date of Birth) for easier movie identification and record keeping.

Contents (Help Menu): Contents displays the "Table of Contents" for the Help information.

Coordinates: The Coordinates command found under the View menu will place the X and Y coordinates along the top and left edges of the movie window. This can be used as a ruler to locate specific places within the movie or locate specific areas of pressure. The coordinates feature is a visual aid to help you map the sensor areas.

Copy: Copy (Ctrl+C) is a powerful tool that allows you to compare movie frames, create reports or documents using actual color movie frames, and analyze pressure data in a spreadsheet or word processing program. This command saves the currently active Real-time window, Movie frame, or Graph to the Windows clipboard as both an image and as a text array of the actual pressure values. This pressure data can then be pasted into other Windows applications as either a graphic (bitmap) or as text by selecting Edit -> Paste Special.

Cut Frames: Cut Frames opens the 'Cut Frames' dialog box, which enables you to discard unwanted frames from a recording using a number of methods. This option is helpful for removing nonessential pressure data from movies, and also for making large movie files more manageable. The three options in the 'Cut Method' pull-down field are 'Cut Frames', 'Keep Frames', and 'Filtered'. 'Cut Frames' allows you to discard specific frames, 'Keep Frames' allows you to keep specific frames, and 'Filtered' allows you to discard frames whose total applied force is less than a specified minimum value.

D

Data Acquisition Parameters: <see "Acquisition Parameters">

Duration: the length of time (seconds) that a recording will take to complete (the first frame is recorded at time=0). Duration is related to 2 other data acquisition parameters in the following way: " $(\text{Frequency} * \text{Duration}) + 1 = \text{Frames to Record}$ " <see also "Frequency"; "Frames to Record"; "Acquisition Parameters">

E

Edit Menu: The Edit Menu contains commands that pertain to the File you are currently working on within the software application.

Edit Mode: Edit Mode displays the 'Edit' dialog box, which enables you to modify a specific area of the pressure data within a Movie or Real-time window. Edit Mode allows you to place an edit box or edit cell in the active window, and then modify the sensels within this box. An edit box or cell can be used to remove data from one or more sensels, or to average sensel reading(s) with those of its neighbors, in either a Movie or a Real-time window.

Equilibration: Equilibration opens the 'Equilibration' dialog box, which enables you to equilibrate a sensor, unequilibrate a sensor, and to save or load equilibration data as a file. Once an equilibration has been performed, a graphic representation of the pressures in the Real-time window (in grayscale) is displayed in the dialog box. This is intended to show the relative high and low spots of the sensor, and the levels of compensation necessary to ensure uniform results. Refer to the 'Equilibration and Calibration' section for more specific information on each menu item.

Exit: Exit closes the software application.

F

File Menu: The File menu drops down a list of commands from the Main Menu. These commands pertain to working with Movie Files, as well as Printing functions and application-wide commands such as Save and Close.

First Frame: First Frame positions the recording at the beginning (Frame 1).

Fit in Window: Setting in the "Zoom To" dialog box, which is selected through the "View" menu. This allows the entire image to be visible in the window.

Fixed Area Averaging: This option displays each cell's pressure value modified to reflect the value of its neighbors. It does not include neighboring cells that have a zero load value. The result is a smoother image. When Fixed Area Averaging is in effect, it will have a check mark beside it in the pull-down menu, and its Toolbar icon will be depressed. Refer to the 'Analyzing Pressure Data' section for more detailed information.

Frames to Record: the total number of frames to be recorded

Frequency: the number of frames that will be recorded each second (frames/second). Frequency is also related to 2 other data acquisition parameters in the following way: " $(\text{Frequency} * \text{Duration}) + 1 = \text{Frames to Record}$ " <see also "Duration"; "Frames to Record"; "Acquisition Parameters">

G

Gain: This dialog box enables you to "fine tune" your sensitivity (gain) setting, which may be necessary if you need a more exact setting than is offered by the choices in the Adjust Sensitivity dialog box. Clicking on the <Fine Tune> button in the Adjust Sensitivity dialog box accesses this feature. The sensitivity (also referred to as adjustable gain) adjusts the data output range from the computer (0 to 255 raw) to the actual force range output from the system handle. The sensitivity should be adjusted whenever necessary to keep the Real-time display in a usable range (i.e. both the high and low pressures are visible), or if the sensor saturates at a lower pressure than desired.

Graph Window: The Graph Window displays a graph of the pressure data from one or more Movie windows. A Graph window is opened when an object (panes or a box) is placed in a Movie window. A separate color-coded graph trace is displayed for each object in the corresponding graph.

Grayscale Printing: The software can allow you to print the color screen display as a grayscale image. When a grayscale printer is installed, and selected as the default printer (in the Print Setup or Print options in the File menu), the image will be printed in grayscale.

Grid Adjustment: The Grid Adjustment affects Box placement. If the Grid Adjustment box is selected (checked) in the Objects dialog box, all four sides of each box will be aligned with the edges of sensor rows and columns. Each side of the box will be moved to the nearest row or column edge, even if this changes the box's size. If this option is not selected, the boxes may be positioned anywhere inside the window. If a box is already in place, and Grid Adjustment is checked, the box will move to the nearest sensor row and column edge. Pane edges are always aligned with sensor rows and columns, and the Grid Adjustment option does not affect their placement.

H

Handle: The Sensor Handle is a hardware device that gathers the data from the sensor and processes it so that it can be sent easily to the computer. There are four types of handles available: Evolution, USB, Parallel, and Accurate.

Hardware Information: Hardware Information displays a dialog box that shows which Tekscan hardware is attached to the system and how it is configured. This dialog box tells you about the connection between the hardware and the computer.

Help Menu: This menu contains information about Tekscan, as well as important information on how to use your software/hardware system.

I

Installation: This section presents instructions for hardware and software installation, and also provides the system requirements necessary to support the pressure measurement system.

Interface Board (SR-2): The Interface Board is a data acquisition card that uses a 16-bit ISA expansion slot of your computer. Refer to the manual supplied with your computer for any specific details on installing a board in your computer's expansion slot.

L

Last Frame: Last Frame positions the recording at the end (final frame).

Legend: The Legend shows the pressure range that corresponds to each of the 13 possible colors in any open Movie or Real-time window. The Options ->Set Legend menu item can be used to place a Raw or Calibrated Legend in the Main Window.

Load Calibration File: This option allows you to use the calibration data from a previously saved file for your current Movie or Real-time window. When this item is selected, an Open dialog box appears, and you can specify the file name and destination (path) of the calibration file you want to load. Calibration data from calibration files (with the extension *.cal) or from I-Scan movies (with the extension *.fsx) may be loaded.

Load Edit File: Load Edit File enables you to load an existing edit file, with the extension *.fed, into the currently active Movie or Real-time window. An edit file is simply the saved edit boxes and/or cells from a previously edited window. When the edit file is loaded, the edit boxes/cells are placed at the same coordinates at which they were saved, with the same properties, and the 'Edit' dialog box is opened.

Load Equilibration File: This option allows you to use the equilibration data from a previously saved file for your current Real-time window. When this item is selected, an Open dialog box appears, and you can specify the file name and destination (path) of the equilibration file you want to load. You can also access this option by clicking on the Load Eq. File button in the Equilibration dialog box. Only I-Scan equilibration files (with the extension *.equ) may be loaded.

Load Object File: Load Object File allows you to load previously saved objects (panes, boxes, and lines) into the current Movie or Real-time window. Clicking the 'Load Object File' button in the 'Objects' dialog box can also access this file. When Load Object File is selected, an 'Open' dialog box is displayed, and 'object files' (with the extension *.fbx only) can be opened. When an object file is loaded into the current window, the objects will be placed at their saved coordinates (location). If the current window is not the same size as the original window, the software will still attempt to place the objects at the correct coordinates.

Lock Objects: Lock Objects will make all the objects within your movie immovable. This is useful to keep the objects held down in case you are prone to accidentally moving them while you work within the movie.

M

Main Menu: The Main Menu (or Menu Bar) provides the pull-down menus used to control the application. It is located between the Title Bar (at the top of the application) and the Toolbar, running lengthwise along the main application screen. The most frequently used items in the Main Menu also have an icon on the Toolbar

Main Status Bar: The Main Status Bar provides information about what is happening on your screen at any given moment. When the cursor is over a Toolbar icon, the left side of the status bar will give a description of that icon; otherwise, it will say Ready. When the cursor is over a Real-time or Movie window, the right side of the status bar will give the cursor position (Row, Col:), and the applied load (Load) on the sensor at that point. Note that the cursor position corresponds to the actual row and column on the sensor (the origin, or zero point, is the top right corner of the window). You will see pressures in finer detail here in comparison to the ranges of pressure shown by the color bands on the display. The load shown may be affected by the display mode; for instance, it will be an averaged value if AVERAGING is selected. It may also be affected by the Legend's lower limit.

Max Area Frame: Max Area Frame advances the current movie to the frame that has the largest contact area. The actual contact area is displayed in the center of the Movie Status Bar. Note that only pressures above the legend's lower limit (threshold) are displayed, and accounted for in the contact area. Max Area Frame is not available in Real-time mode.

Measurement Units: Measurement Units displays the 'Units of Measure' dialog box, which allows you to choose the units that will be displayed for length, force, and pressure, the number of decimal places that will be displayed for force and pressure, and the increment by which the upper and lower legend limits can be scrolled. The default units are inches for Units of Length, pounds for Force, and PSI (pounds per square inch) for Pressure. The default number of decimal places depends on the units that are selected, and the default 'Legend Scroll' value is 1. The units and decimal place settings are applied to all Movie and Real-time windows, and Graphs, regardless of which view is currently active. Once you change these settings, the software will remember them when you start a new patient session or close the application.

Menu Bar: The Menu Bar provides the pull-down menus used to control the Research Foot 5.20 program. Each Menu Bar option is described in detail in the 'Main Menu' section. When no Movie or Real-time windows are open, the Menu Bar contains only the File, Options, and Help menus. When a window is open, the other menus become available. When a Menu Bar item is unavailable, it will be grayed out.

Movie (Frame) Calibration: The I-Scan system enables you to perform two different types of calibration, linear and 2-point power law, on an existing Movie window. Refer to the Calibration section for instructions on how to calibrate a Real-time window. To perform a movie (frame) calibration, you enter the total force applied to the sensor in a specific frame of the movie, in the Movie Calibration Point dialog box. The I-Scan software then performs a linear interpolation between zero load and this known calibration load. To perform a 2-point power law calibration, you enter the total force applied for a different frame of the movie. The software then performs a power law interpolation based on zero load and the two known calibration loads, using the equation $y = ax^b$.

Movie Averaging: Movie Averaging opens the Select Averaging Range dialog box, which allows you to display the averaged pressure value of each cell for the entire recording, or for a selected range of frames, in one composite frame. You can also choose to average the movie into multiple composite frames.

Movie Contact Averaging: Movie Contact Averaging opens the Select Averaging Range dialog box, which allows you to display the averaged pressure value of each cell for the entire recording, or for a selected range of frames, in one composite frame. You can also choose to average the movie into multiple composite frames. This menu item is very similar to the Movie Averaging option, except that it calculates the cell's average pressure value in a slightly different manner. The difference between these two types of movie averaging is that cells with zero load are disregarded in this option, whereas they are factored into the formula with a zero value in the Movie Averaging option.

Movie Menu: The Movie menu items and corresponding Toolbar icons control the movie playback and recording functions. The playback menu items and icons control all open movies in unison, regardless of which movie is active.

Movie Status Bar: The Movie Status Bar is located at the bottom of the Movie Window, and displays the Frame count (or Relative Time or Absolute Date/Time), Area, and Force. The left side of the status bar displays the frame count, which is the current frame number vs. the total number in the movie. You can choose to display the Relative Time or Absolute Date/Time, in place of the Frame count, in the Set User Preferences dialog box.

Movie Window: The Movie Window displays a previously recorded movie.

Multi-Tile Calibration: I-Scan allows you to calibrate a "virtual" or "multi-region" sensor, either as a single window or as separate tiles. You can perform both linear and 2-point power law calibrations, for a Real-time or Movie window, for this virtual sensor.

Multi-Tile Equilibration: I-Scan allows you to equilibrate a "virtual" or "multi-region" sensor, either as a single window or as separate tiles.

N

New Recording: New Recording (Ctrl+N) opens a new Real-time window, in addition to any Movie or Real-time windows that are currently open. The new Real-time window becomes the active window when it is opened. When New Recording is selected, the window size and position, display options (e.g. 3-D, COF, zoom %, rotation), and calibration and equilibration data are all recalled from the previous Real-time window.

New Session: When beginning a new session, selecting this option closes all open windows (asking the user to save unsaved data), open a new Real-time window, and reset all display options, as well as the sensitivity, calibration, and equilibration data. Note: If you close the application, restart the program and open a Real-time window, the display options (e.g. 3-D, COF, zoom %, rotation) and the sensitivity setting are recalled.

Next Frame: Next Frame (Ctrl+X) plays the recording forward one frame (or phase) each time it is clicked.

Noise Threshold: After a period of use or in the presence of EMI, a sensor may indicate noise as low-level pressure. To reduce the effect of such noise on your recordings, you can set a noise threshold. Digital output values that are equal to or below this threshold are set to zero by the software, thereby filtering out unwanted low level force readings (noise).

O

Objects: Objects displays the 'Objects' dialog box, which can be used to control each object (pane, box, and line) that is placed in a Real-time or Movie window. Using this dialog box, objects can be added, deleted, changed, saved as an ASCII (text) data file, or saved/loaded as an 'Object file'.

Open Linked Video: Open Linked Video will open a video and link the movie and video together. When Play is selected, movie and video will play simultaneously.

Open Movie: Open Movie (Ctrl+O) allows you to open previously recorded movies or snapshots (with the *.fsx file extension only).

Options Menu: This menu contains commands for system settings and how the software interacts with the Hardware (Hardware Information). All the settings found under this menu affect the entire software application (Global settings) as opposed to affecting individual files (local settings).

P

Parallel Interface: The Parallel Interface includes a power supply adapter, a parallel interface module (adapter box), a parallel extension cable, and a sensor handle (see User's Manual for a diagram of the completed setup). The parallel interface module is designed to share the printer port of your computer with the I-Scan system, so you can run your printer and the system without swapping cables.

Pause: Pause causes the active Movie or Real-time window to 'freeze', or to ignore all movie playback or recording commands. When a window is paused, the left-most pane of that window's status bar will 'pop out'; clicking on this pane ('Frame Count' or 'Sensor OK') will toggle Pause on and off for the window. When a movie is paused, its corresponding timeline in any graphs is also paused.

PC Interface Board (SR-2): <see "Interface Board (SR-2)">

Peak: Peak displays the maximum pressure value that each sensel reached during the recording, in one composite frame. This allows you to view, simultaneously, the highest pressures experienced by each part of the sensor during the recording. Peak is available in all 2-D and 3-D display modes. When this item is enabled, the Movie Status Bar will say 'Peak', and the movie playback options will be unavailable. If Peak is in effect while COF Trajectory is enabled, the trajectory of the COF for the entire range of frames is displayed. Peak is toggled on and off; when it is enabled, it will have a check mark beside it in the pull-down menu, its Toolbar icon will be depressed.

Period: Period - the amount of time between frames (seconds/frame). Period is related to Frequency by the following relationship: "Period = 1 / Frequency"

Play Backward: Play Backward (Ctrl+8) plays the recording in reverse, starting at the current frame (or phase). To enable 'Continuous Play', press the <SHIFT> key, and simultaneously select Play Forward or Play Backward. The movie will play in the desired direction indefinitely, looping back to the first or last frame each time.

Play Forward: Play Forward (Ctrl+F, F5) plays the recording forward, starting at the current frame (or phase). To enable 'Continuous Play', press the <SHIFT> key, and simultaneously select Play Forward or Play Backward. The movie will play in the desired direction indefinitely, looping back to the first or last frame each time.

Play Speed: Play Speed sets the speed in which the movie plays. Choices are Slowest, Medium Slow, Normal, Medium Fast, and Fastest. It is usually beneficial to play the movie at a Medium Slow speed in order to analyze the movement in proper detail. Otherwise, the movie may be too fast to notice the change in pressures.

Polar Sensor: Sensors with sensels arranged in rows and columns (Cartesian coordinates) are well suited to applications in which the load area is rectangular. In cases where the load area is circular or annular, sensors with sensels arranged in circles and radii (polar coordinates) may be used. Polar sensors have sensels with different areas as a function of the sensel radial location. The I-Scan software accounts for these variations by breaking the polar sensor grid into a finer Cartesian grid. The displayed image is actually an image of these virtual sensels.

Pressure Legend: The Pressure Legend shows the pressure range that corresponds to each of the 13 possible colors in any open Movie or Real-time window. The number shown on each color in the Pressure Legend is the lowest pressure value in that color's range. The values in the fields at the top and bottom of the legend are the upper and lower limits, respectively. The pressure units (or raw) are also displayed at the top of the legend. A legend can be placed in the Main Window by selecting Set Legend from the Options pull-down menu, or clicking the corresponding Toolbar icon.

Previous Frame: Previous Frame (Ctrl+V) plays the recording in reverse one frame each time it is clicked.

Print: Print (Ctrl+P, F9) allows you to print the contents of the currently active Movie or Real-time window, including comments, header information and graphs, to any of the hundreds of printers which MS Windows supports. The window will be printed exactly as it appears on the screen, and will include the options selected in "Print Setup" <see also "Print Setup">.

Print Preview: Print Preview (Ctrl+F9) shows you exactly how your printed page will appear. It is recommended that you preview all printed pages to ensure that your setup is correct. You can also view the Print Preview by clicking its icon in the Main Toolbar.

Print Setup: Print Setup gives you a number of printing options that control what will be printed, and how this printed page will look.

Properties: Properties opens the 'Properties' dialog box, which controls the display attributes of all objects (panes, boxes and lines) in an active Real-time or Movie window, or the X- and Y-axis display attributes of an active graph in the Main Window. You can also open this dialog box by clicking the right mouse button with the cursor over an active window or graph, and selecting 'Properties'.

R

Raw Legend: If no calibration has been performed on the specified sensor, A Raw Legend will open. This is the default Legend, and any sensor measurements will be recorded on a scale from 0 to 255 (255 being the upper default limit). No pressure units will be associated with each range. <See also "Pressure Legend" and "Calibrated Legend">

Real-Time Status Bar: The Real-Time Status Bar is located at the bottom of the Real-time window, and displays the sensor status (or the frame count of a recording in progress), Area, and Force. The left side of this status bar displays the sensor status; either 'Ready' or 'MISALIGNED!'. If this message is 'MISALIGNED!', remove and reinsert the sensor tab into the handle, ensuring that the sensor is right side up, and the dial on the front is in the proper position.

Real-Time Window: The Real-time window displays each sensor's color-coded pressure information as it is happening, or in real time. If no pressure is on the sensor, a blank Real-time window will appear.

Recent Files: The names of the four most recently viewed movies are displayed between Print Setup and Exit on the File menu. You may open any of these files by clicking on its name.

Record: Record (Ctrl+R, F2) is used to make a recording (movie). When Record is selected, the software begins recording frames of sensor pressure data as it appears in the current Real-time windows, using the Acquisition Parameters selected in the Options pull-down menu. You can also begin recording by pressing the <F2> key on your keyboard. If 'Triggering' is in effect, and Record is selected, recording will not begin until the start trigger event occurs. When data is being recorded, '(Rec)' is added to the Real-time window title bar, and the window's status bar displays each frame number as it is recorded. Movies created by the software application are stored as *.fsx files.

Recording: A Tekscan recording captures sensor pressure over a period of time. It is similar to a video recording, since you can rewind your recording, play it and replay it as many times as you wish. These pressure data recordings are also referred to as "movies".

Rotate: Rotate turns the entire view clockwise in 90 degrees increments from its current orientation. All 2-D and 3-D views are rotated clockwise. When a window is rotated, the zero point (which is initially in the upper left corner of each window) is also rotated. If you get confused at any point, put the cursor over a sensel and look at the coordinates displayed in the Status Bar to get the true row and column numbers.

S

Save ASCII: Save ASCII allows you to save the current Movie, Real-time or Graph window data as an ASCII (text) file, with the file name and location of your choice. Specifically, you can save frame sensel pressure data, center of force (COF) data, or an object's graph data. The 'Objects' dialog box also contains a 'Save ASCII' button, which can be used to save graph data ASCII files.

Save Calibration File: Save Calibration File allows you to save your current calibration data (from a Movie or Real-time window) as a file for future use. When this item is selected, a "Save As" dialog box appears, and the calibration file can be saved with the specific file name and destination (path) of your choice. You can also access this option by clicking on the Save button in the "Calibration" dialog box, or in the "Settings" dialog box ("Calibration" tab). Tekscan calibration files must be saved with the extension *.cal.

Save Edit File: Save Edit File enables you to save the edit boxes and/or cells from the currently active Movie or Real-time window as an edit file, with the extension *.fed. When you select Save Edit File, a 'Save As' dialog box opens, and you must enter a filename and path. For a Movie window, the default file name will be the name of the movie (e.g. "SmitJ01L.fbx"). The edit box/cell information, which includes their coordinates and selected 'Edit' and 'View' options, will be saved in the file. These edit boxes/cells can then be placed into a Movie or Real-time window simply by loading the edit file.

Save Equilibration File: This option allows you to save your current equilibration data (in a Real-time window) as a file for future use. When this item is selected, a Save As dialog box appears, and the equilibration file can be saved with the specific file name and destination (path) of your choice. You can also access this option by clicking on the Save Eq. File button in the Equilibration dialog box. I-Scan equilibration files must be saved with the extension *.equ.

Save Movie: Save Movie (Ctrl+S) will save the newly recorded Real-time window as a file, with the extension *.fsx. It will be saved in the "Movies" directory. When Save Movie is selected, the 'Add Movie to Database' dialog box will be displayed, and you will be asked whether or not you want to save the movie to the file management system.

Save Movie As: Save Movie As will save the current movie, with the specific file name and location of your choice. This menu item is also available when a Movie window is active, and can be used to save a movie under more than one file name. When Save Movie As is selected, the 'Add Movie to Database' dialog box will be displayed, and you will be asked whether or not you want to save the movie to the file management system. Tekscan movie files must have the extension *.fsx. Note that your system may limit file names to eight characters.

Save Object File: Save Object File allows you to save the objects (panes, boxes, and lines) from an active Real-time or Movie window as a file. Objects are saved exactly as they appear in the window, with the same properties and coordinates (location). This option is necessary because, when a Real-time window is saved as a movie file, the objects in the window are not saved as part of the file.

Save View as Movie: Save View as Movie allows you to save the current movie frame, or view, as a movie file (with extension *.fsx). This new movie file will consist of one frame, and will be saved exactly as it appeared when saved. Any View menu items that are in effect when Save View as Movie is selected will be retained in the saved file. Note that COF Trajectory will not be displayed by the saved view, since it only affects movies with more than one frame.

Search (Help Menu): Search allows you to query the Help file to obtain the specific information you are looking for. You enter the search parameters, which can be words or phrases, and the Help file will list all help topics that pertain to your search. You can directly click on these Help topics to open them within the main Help window.

Select Sensor: Select Sensor displays the 'Select Sensor' dialog box, which enables you to designate the sensor (map) and sensor handles that are currently installed on your Tekscan system. The system needs this information in order to display the data correctly in the Real-time windows. The dialog box lists the system's 'Available maps' and 'Available handles'.

Sensitivity: This dialog box enables you to select the handle sensitivity from 11 possible settings, ranging from Low-1 to Hi-4. It can be accessed by selecting Adjust Sensitivity from the Options pull-down menu, clicking the Sensitivity button in the Calibration dialog box, or by clicking on the Sensitivity tab in the Settings dialog box. The sensitivity (also referred to as adjustable gain) adjusts the data output range from the computer (0 to 255 raw) to the actual force range output from the system handle. The sensitivity should be adjusted whenever necessary to keep the Real-time display in a usable range (i.e. both the high and low pressures are visible), or if the sensor saturates at a lower pressure than desired.

Sensors: The I-Scan sensor is an ultra-thin (.004", 0.10mm) flexible printed circuit. It has up to 2288 individual pressure sensing locations, which may also be referred to as sensing elements or sensels. The sensels are arranged in rows and columns (Cartesian coordinates) on the sensor in most cases (not in Polar Sensors). The sensor acts as a variable resistor in an electrical circuit. When the sensor is unloaded, its resistance is very high; when a force is applied to the sensor, the resistance decreases. This output resistance is then converted to a digital value (raw sum) in the range of 0 to 255. Because of the nature of the pressure sensitive layer that is the heart of the I-Scan sensors, it is possible to produce sensors of varying sensitivity. Sensors with pressure ranges as low as 0 - 5 PSI (0 - 260 mmHg) and as high as 0 - 25,000 PSI (0 - 172 Mpa) have been produced. The active sensing area of the sensor is surrounded by the substrate material that contains conductive leads. These leads connect the rows and columns of the active sensing area to the terminal section or tab of the sensor. The topside of the sensor is labeled This Side UP near the terminal section. The sensor tab must be inserted into the handle with the proper orientation.

Set Legend: Set Legend enables you to place a Raw or Calibrated Pressure Legend in the Main Window. The Pressure Legend shows the pressure range that corresponds to each of the 13 possible colors in any open Movie or Real-time window, and allows you to alter the colors that are displayed in these windows.

Set User Preferences: Available from the Options drop-down menu, the User Preferences allow you to customize certain operational features of the system according to your preferences. These Preferences are Global in Nature, meaning they affect the operation of the software across the board, not only specific files.

Settings: Accessed from the Options Menu, the Settings displays the 'Settings' dialog box for the currently active Movie or Real-Time window. This dialog box allows you to view the window's general information, comments, and calibration data, all in one place.

Snapshot: Snapshot (Ctrl+A, F3) is used to make a single-frame recording (movie). When Snapshot is selected, the software records a single frame of sensor pressure data as it appears in the current Real-time windows. You can also take a snapshot by pressing the <F3> key on your keyboard. The only Acquisition Parameters that apply to a Snapshot are 'Triggering' and 'Noise Threshold'. When data is being recorded, the word '(Rec)' is added to the Real-time window title bar. Movies created by the software application are stored as*.fsx files.

Status Bar: The Main Status Bar provides information about what is happening on your screen at any given moment, and is located at the bottom of the application window. When the cursor is over a Toolbar icon, the left side of the status bar will give a description of that icon; otherwise, it will say 'Ready'. When the cursor is over a Real-time or Movie window, the right side of the Main Status Bar will give the cursor position ('Row, Col:'), and the pressure ('Load') on the sensor at that point. Note that the cursor position corresponds to the actual row and column on the sensor (the origin, or zero point, is the top left corner of the window).

Status Bar (View Menu): Status Bar is used to display or hide the Main, Movie, and Real-time Status Bars. These status bars can be toggled on and off by clicking this menu item. If a Graph is the active window, this menu item affects the Main Status Bar. If a Movie window is active, this menu item affects the Movie Status Bar. If a Real-time window is active, this menu item affects the Real-time Status Bar.

Stop (Movie): Stop (Ctrl+T, F4) stops either playback or recording. If a recording is in progress, Stop will cause the software to stop recording the active Real-time window, even if the selected number of movie frames has not yet been recorded.

T

Technical Support (Help Menu): Technical Support provides information on how to contact Tekscan for technical assistance.

Tile (Vertically or Horizontally): Tile (Vertically or Horizontally) arranges multiple opened windows in an overlapped fashion, either vertically (side by side) or horizontally (top to bottom). This is useful for arranging many windows on your screen in an orderly fashion.

Title Bar: Each window (Main, Real-time, Movie, and Graph) has a title bar along the top, which displays the name (title) of that window (e.g. Real-time 1 or Movie1.fsx). If that window is paused, (Paused) is added to the title. If a recording is in progress, (Rec) is added to the title. Each title bar has the usual MS Windows control button at the left end and minimize, maximize and close buttons at the right end. Only one window may be active at one time; the active window's title bar will be colored, while the deactivated windows' title bars will be gray. Note: You can also move dialog boxes by dragging their title bars.

Toolbar: The Toolbar contains the most common software functions. It is located at the top of the main application window, directly below the main Menu Bar. The Toolbar provides the same options as the Menu Bar, but with icons replacing pull-down menus. Each icon on the Toolbar has a corresponding pull-down item in the Menu Bar; however, only the most frequently used items on the Menu Bar have an icon on the Toolbar.

Toolbar (View Menu): Toolbar is used to display or hide the Main Toolbar. The Toolbar can be toggled on and off by clicking this menu item.

Tools Menu: The Tools menu items control the sensor Equilibration and Calibration functions. The only Tools menu item that has a corresponding Toolbar icon is the Calibration item.

U

Un-Calibrate: This option simply removes the sensor's calibration data. When this menu item is selected, a dialog box appears, giving the option of canceling the UnCalibrate operation. UnCalibrate cannot be used to uncalibrate a Movie window, and will only be available when a calibrated Real-time window is open. If you want to recalibrate the sensor during the same testing session, you may either use UnCalibrate to reset the calibration of the current window, or select File -> New Session to get a new Real-time window.

Un-Equilibrate: This option discards the current Real-time window's equilibration data. When this item is selected, a dialog box appears, giving the option of canceling the UnEquilibrate operation. This option may be used if you make a mistake in performing your initial equilibration, or if you want to perform another equilibration without exiting the I-Scan program.

Units: This dialog box allows you to choose the units that will be displayed for length, force, and pressure, and the number of decimal places that will be displayed for force and pressure. The default units are inches for Units of Length, pounds for Force, and PSI for Pressure. The default number of decimal places depends on the units that are selected. The units and decimal place settings are applied to all Movie and Real-time windows, and Graphs, regardless of which view is currently active. Once you change these settings, the software will remember them when you start a new session or close the application.

V

Video Capture and Playback: Synchronizing a (video) image of the subject's movements with the Tekscan pressure data recording can clarify what is causing the pressure patterns one sees when reviewing and analyzing the Tekscan pressure data. The Video Capture and Playback Package enables you to view the images seen by your video camera on your PC screen and record a video (avi file) of your test at the same time you record a pressure movie (fsx file) using the Tekscan Pressure Measurement System and sensor. In addition to enabling you to view and record these two types of files on your PC, this software package allows you to synchronize, link, review and then analyze the two files (avi and fsx) within the Tekscan software. The benefit of the Video Capture and Playback Package is that it eliminates the need for you to purchase a third party software package for video capture and videotape conversion. Thus it provides you with a convenient way to view video directly onto your PC and save the live video feed (or real-time video) directly to the computer's hard drive - instead of having to record the video onto a tape and then convert it to a digital video format that the computer can use as a second step. Since both the recorded Tekscan pressure movie and the captured video are saved on the computer hard drive, they are both ready to be played back immediately, without the delay of having to download and convert the videotape from the video camera after making a recording. This is both time saving and convenient.

View Menu: The View menu items and corresponding Toolbar icons control the primary sensor display functions for the Movie or Real-time windows. When one of the 2-D or 3-D items is enabled, it will have a dot beside it in the pull-down menu, and its Toolbar icon will be depressed. Note that the sensor data does not change when changing modes; only the appearance of the display is altered.

W

Window Menu: The Window Menu contains commands that control the way Real-Time and Movie windows are viewed on your screen, and list all the currently open windows at the bottom of the menu. Clicking on these list items will make the corresponding window active.

Z

Zoom To. . .: Zoom To. . . allows you to resize the display inside the active Movie or Real-time window. Percentages refer to the actual size of the sensor. Zoom percentages are available up to 400%, and any size up to this value can be entered in the 'Custom Fit' area. The default size (when 2-D or 2-D Contours is in effect) is "Fit in Window". Zoom To only applies to the active window, and is not affected by the <CTRL> key.

